



AESIEAP GOLDBOOK 2026



CAMBODIA



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Phnom Penh
Area:	181,035 km ²
Population:	17,638,801
GDP 2024¹:	192,005 Billion Riel
Currency:	Riel
Installed Capacity:	5,183 MW ²
Percentage of population electrified:	99.09% ²

Electricity Consumption by Sector

Sector	Total Power Consumption as of 2024 (MWh)	Total Power Consumption as of 2025 (GWh)
Resident	5,703	5,414
Small Business, services Commercial, Industrial and Administration connected at LV	3,032	3,855
Big, Medium, Commercial and Industrial	7,361	8,866
TOU for Commercial and Industrial	915.83	1,118.36
Rural School, Referral Healthcare and Hospital	21.72	23.65
TOTAL	17,033	19,277

¹<https://mef.gov.kh/documents/budget-in-brief-2025/>

²https://www.eac.gov.kh/uploads/salient_feature/english/salient_feature_2025_en.pdf

National Energy Policy

The Royal Government of Cambodia (RGC) has issued four goals on energy sector development policy detailed out as follows:

1. Providing an adequate supply of electricity throughout the country at reasonable and affordable price;
2. Ensuring reliable and secure electricity supply which facilitate investment in Cambodia and development of the national economy;
3. Encouraging exploration of environmentally and socially acceptable energy resources needed to supply all sectors of the Cambodian economy; and
4. Encouraging efficient use of energy to minimize environmental effects resulting from energy supply and use.

To prepare a governing framework for the electric power supply and services throughout the country, the “Law on Electricity” was adopted by the National Assembly on November 6, 2000 and then promulgated by the Royal Decree on February 2, 2001. This law covers all activities related to the supply, provision of services and use of electricity, and other associated activities of the power sector. It helps reform the current electricity sector and is endorsed to boost private investors in the power sector in a fair, just, and efficient manner for the benefit of the Cambodian society.

Overall, this law has key components including:

1. Establishing the principles for operation of the sectors;
2. Establishing favorable conditions for competition, private investment, private ownership and commercial operation of the electric power industry; and
3. Establishing and defining the functions of the (Electricity Authority of Cambodia) EAC and the Ministry of Mines and Energy (MME)

Laws and Governmental Regulations

The electricity sector in Cambodia is characterized by four primary entities that wield significant influence: the Ministry of Mines and Energy (MME), the Electricity Authority of Cambodia (EAC), the Ministry of Economy and Finance (MEF), and the Electricité Du Cambodge (EDC). Among them, the EDC holds the preeminent position as the most influential utility in the electricity market. Additionally, there are other key participants

such as Independent Power Producers (IPPs), Renewable Energy Entities

(REEs), and various licensees who play a role in either importing electricity from neighboring countries or operating their own independent diesel generators.

The Royal Government of Cambodia has adopted the ‘National Policy on Energy Efficiency 2022-2030’ with the aim of transforming Cambodia into a high middle-income country by 2030 and a high-income country by 2050. The policy focuses on adopting energy efficiency to improve the economy and promote sustainable development. It has six main objectives, including establishing the policy framework, defining government roles and responsibilities, setting measurable targets for energy efficiency adoption, outlining measures to overcome barriers, establishing monitoring and evaluation metrics, and prioritizing human capital development and knowledge dissemination. The policy sets a national target to reduce total energy consumption by at least 19% by 2030 compared to the Business-as-Usual scenario. It also sets specific sector targets, including industrial, residential, commercial buildings, public services, and transport. The policy identifies measures and instruments to prioritize, such as energy performance standards for industries, building energy codes and green building guidelines for buildings, standards and labeling programs for residential appliances, electrification of transport, and retrofitting of lighting and generators for public services. Additionally, energy management programs and the rating of Energy Service Companies (ESCOs) are proposed for industries and buildings.

As the electricity sector in Cambodia is fundamentally governed by the Law on Electricity, sub-degrees and other regulations have subsequently been issued by the EAC. To guide the development of the energy sector, MME has continued the implementation of the Power Development Plan, 2022-2040. Rural Electrification Master Plan focusing on the use of renewable energy is also being implemented. There are three main objectives of PDP 2022-2040:

- To fulfil the future demand for power adequacy with supply of electricity in reliable and affordable way across all sector in Cambodia,
- To strengthen energy security by reducing the dependency on energy imports and maximizing the development of domestic energy resources
- To increase the share of clean energy, including

renewable and variable renewable energy, and energy efficiency, without compromising the reliability and affordability of supply, to contribute to the achievement of Cambodia's national environment goals and global commitments to reduce greenhouse gas emission.

Institutions

Under the electricity law, there are two main institutions playing important roles in governing the electric power sector in Cambodia, namely MME and EAC. While MME is mainly responsible for the formulation of policies and strategies, power development plan, technical standard and other duties, EAC is a legal public entity being granted the right from RGC to be an autonomous agency to regulate the electricity services and to govern the relation between the delivery, receiving, and use of electricity.

Ministry of Mines and Energy

As set out in the Law on Electricity, the MME has the following roles:

- ◆ Responsible for setting and administering the government policies, strategies and planning in the energy sector.
- ◆ Guiding the EAC information on policies, strategies, planning of energy sector and its decision on:
 - ▶ Investments in the rehabilitation and development of the energy sector in the short, medium and long term
 - ▶ Restructuring, private sector participation and privatization of public utilities
 - ▶ Promotion of the use of indigenous energy resources in the generation of electricity
 - ▶ Planning and agreements on the export and import of electricity
 - ▶ Subsidies to specific classes of customers and priorities regarding consumers of electricity
 - ▶ Promotion of efficiency in generation, transmission, distribution and consumption of electricity and action taken to create a comprehensive electricity conservation program for Cambodia and
 - ▶ Electricity sector emergency and energy security strategies.

Electricity Authority of Cambodia

The EAC is a legal public entity, being granted the right from the RGC to be an autonomous agency to regulate electricity services and to govern the relation between the delivery, receiving and use of electricity.

The Law on Electricity regulates the roles of the EAC as follows:

- To issue, revise, suspend, revoke or deny the licenses for the supply of electricity services.
- To approve tariff rates and charges and terms and conditions of electric power services of licensees, except where the authority (EAC) considers those rates or charges and terms and conditions are established pursuant to a competitive, market-based process.
- To implement guidance procedures and standards for investment programs by licensees.
- To review the financial activities and corporate organization structure of licensees to the extent that these activities and Organizations directly affect the operation of the power sector and the efficiency of electricity supply.
- To approve and enforce the performance standards for licensees and resolve consumer complaints and contract disputes involving licensees, to the extent that the complaints and disputes relate to the violation of the condition of licenses.
- To approve and enforce a uniform system of accounts for all licensees.
- To prescribe fees applicable to licensees.
- To determine the procedures for informing the public about affairs within its duties, in order to ensure that the EAC complies with the principle of transparency.
- To issue rules and regulations and to make appropriate orders, and to issue temporary and permanent injunction for electric power service.
- To impose monetary penalty, disconnect power supply, suspend or revoke the license for the violations of this Law, standards and regulations of EAC.
- To require the electric power services and the customers to obey the rules relating to the national energy security, economic, environment and other government policies.

Electricity of Cambodia

Electricity of Cambodia (EDC) is a state-owned public utilities entity that has the following functions and responsibilities:

- To develop, generate, transmit and distribute electric power throughout Cambodia.
- To operate as a commercial entity, independently organize production and operation in accordance with market demand and seek to earn a profit, increase the value of its assets, create economic benefits and raise labor productivity.
- To prepare, build, own, finance, lease and operate power generation and sub-stations, transmission lines, distribution networks and other infrastructure necessary.
- Eliminate inefficiencies from operation, reduce unnecessary costs.
- Maximize the output and reliability of the assets, customer satisfaction with higher quality and better services.

- To be polite, receptive, and act promptly with customers' concerns.

Co-owners of the EDC are the Ministry of Mines and Energy (MME) and the Ministry of Economy and Finance (MEF). EDC is currently holding a consolidated license that has the following components:

1. Generation license: To generate electricity for the purpose of supply to its transmission and distribution system.
2. National Transmission License: To transmit electricity for the purpose of supply to any distribution system and bulk power consumers throughout Cambodia.
3. Distribution license: To distribute and supply electricity to any premises in the authorization distribution areas.

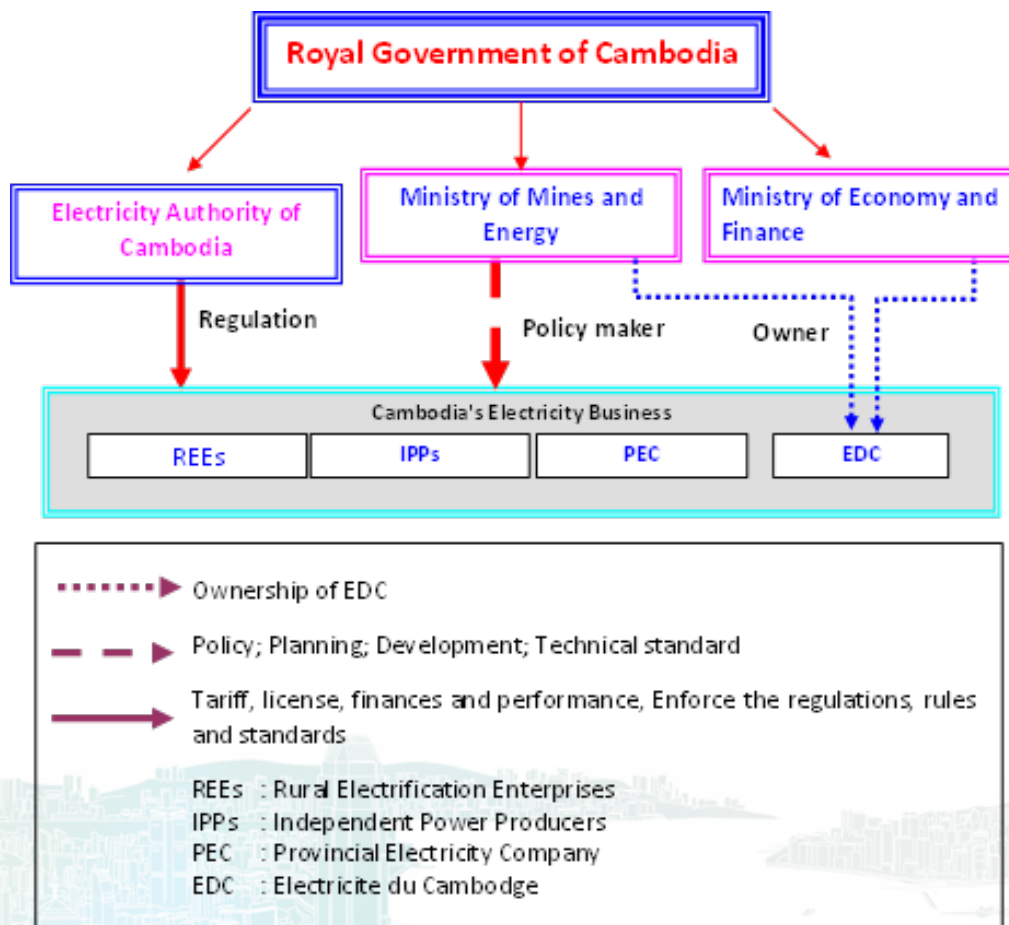


Figure 1. Governance of the Electricity Sector in Cambodia.
 (Source: MME (2017))

Overview of Electricity Market Structure

The supply of electricity is mainly divided into three main types of consumers. First, a supply to consumers with load of 10MW or above and connect from grid substation, Secondly, a supply to the MV consumers with load above 500kVA and connect from MV line of licensee.

Finally, the supply to consumers that connect from MV and LV lines of distribution licensees for supply to retail consumers with capacity below 500kVA and MV consumers with the capacity above 500kV. The

electricity supply has been developed and extended the supply to all areas. More than 3.5 million consumers were supplied at the end of 2025. Figure 2 shows the supplied areas have been divided into three main areas. Firstly, the area covered by the national grid. Secondly, the area that the national grid has not reached means that the supply is through importing power from neighboring countries via the MV line. At last, the area is supplied through mini grids such as diesel generators or other technologies.

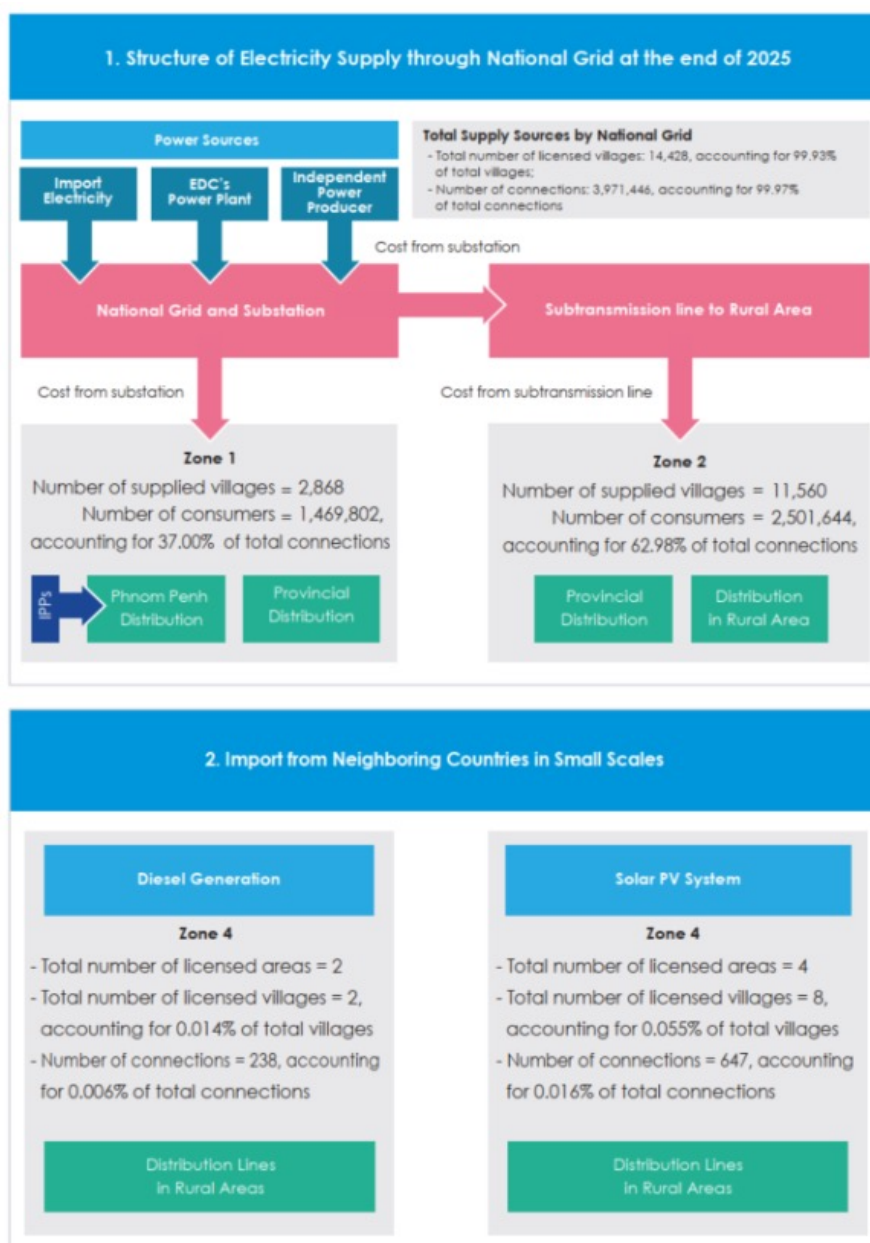


Figure 2. Electricity supply by each area in 2025.
(Source: Salient feature of EAC).

Electricity consumption or demand in 2025-2035 and expected annual growth rate in next decade

According to a significant growth of economics in the recent year and an expected growth in the future, the power demand will be increased accordingly. The demand was around 3,371 MW in 2025 and expected to be around 6,655 MW in 2035.

The majority of this growth will occur in the southern grid which includes Phnom Penh. The Table below depicts the demand for Cambodia.

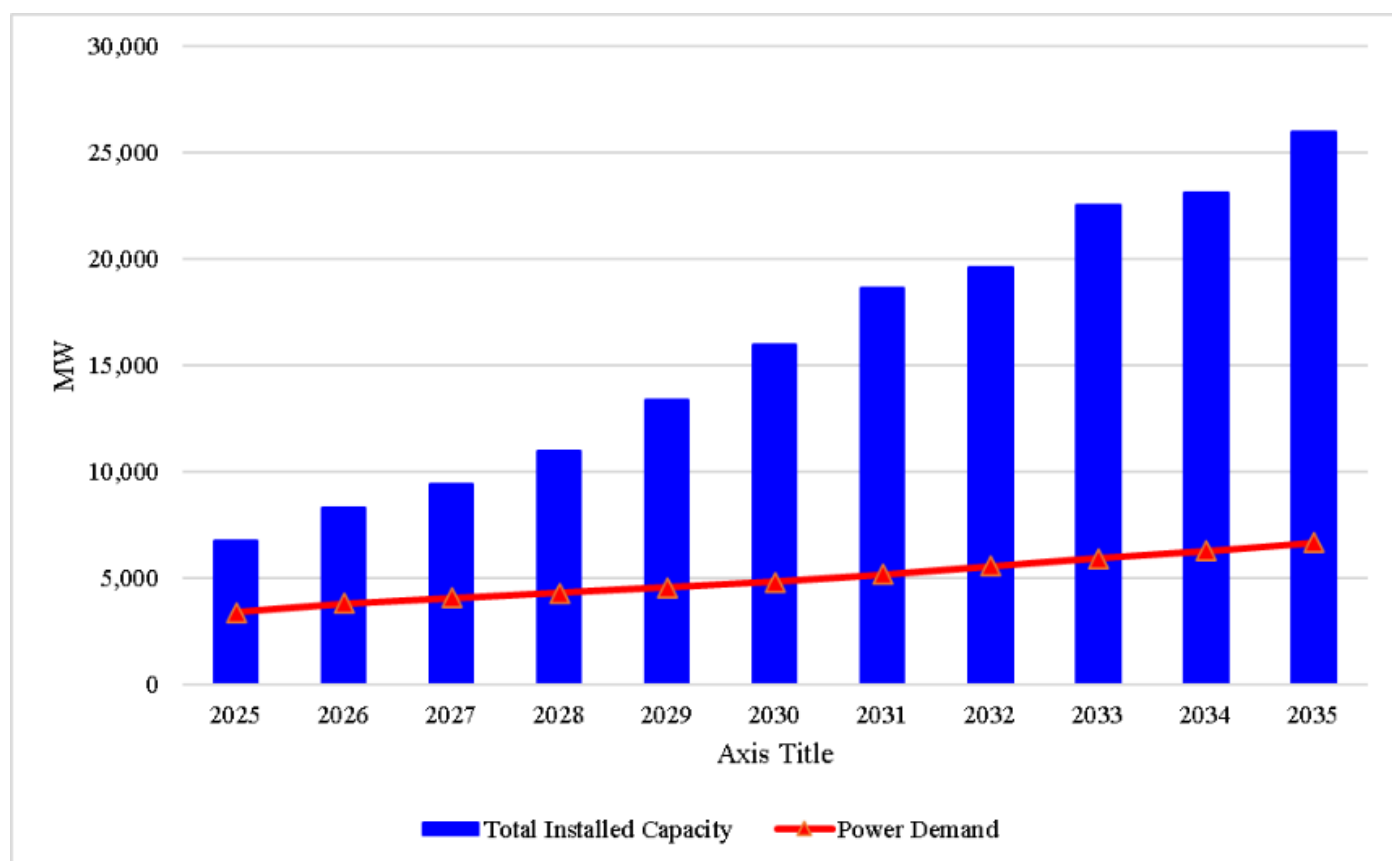
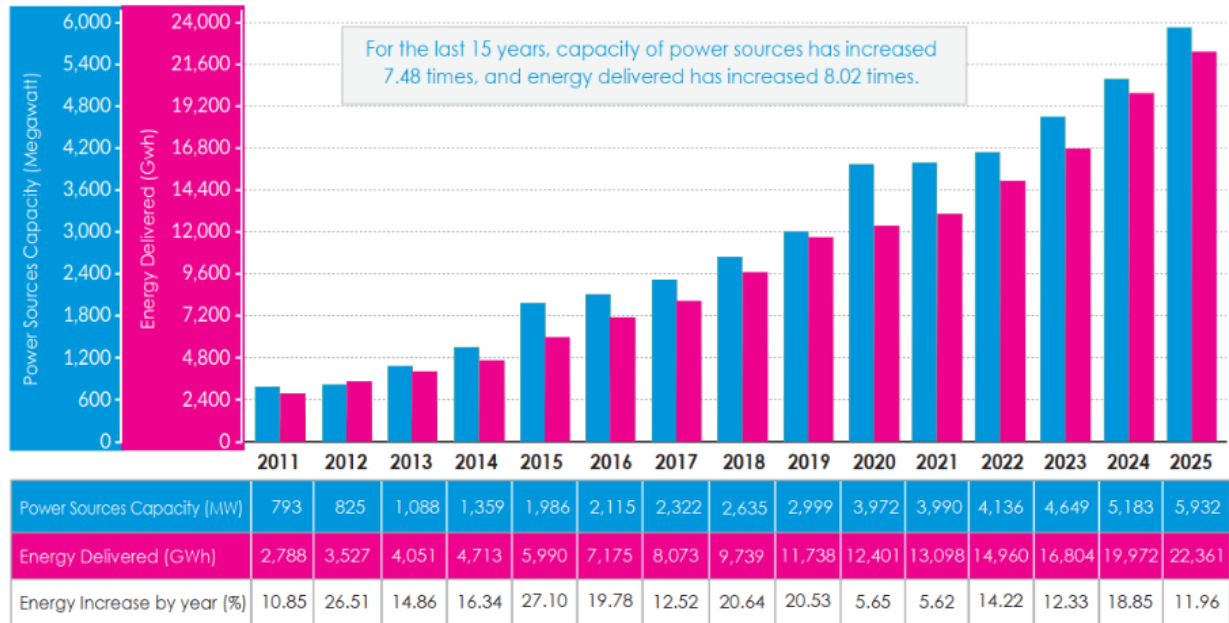


Figure 3. Power demand and energy generation from 2025 to 2035 from PDP (2024-2040)

3

Power Generation of Cambodia

Overview of Electricity Market Structure



Data on Different Source of Power for Cambodia

Power Sources	2024				2025				Plan for 2026			
	Capacity		Energy		Capacity		Energy		Capacity		Energy	
	MW	%	GWh	%	MW	%	GWh	%	MW	%	GWh	%
1. Power Sources directly connected to National Grid												
- Renewable Energy	2,811	62.31%	8,573.39	46.81%	3,325	63.23%	10,317.52	50.09%	4,133	67.40%	11,250.04	48.84%
+ Hydro Power	1,796	39.82%	7,227.78	39.46%	1,801	34.25%	8,167.52	39.66%	2,212	36.08%	8,209.17	35.64%
+ Solar Power	966	21.41%	1,278.96	6.98%	1,476	28.05%	2,097.41	10.18%	1,873	30.53%	2,972.77	12.91%
Solar Power Station	827	18.33%	1,059.78	5.79%	1,307	24.85%	1,831.72	8.89%	1,677	27.35%	2,664.51	11.57%
Rooftop Solar PV Installation	139	3.08%	219.18	1.20%	169	3.20%	265.69	1.29%	196	3.19%	308.27	1.34%
+ Biomass Power	49	1.08%	66.66	0.36%	49	0.92%	52.59	0.26%	49	0.79%	68.09	0.30%
- Non-renewable Energy	1,700	37.69%	9,741.29	53.19%	1,934	36.77%	10,278.56	49.91%	1,999	32.60%	11,782.29	51.16%
+ Coal	1,300	28.82%	9,208.88	50.28%	1,365	25.95%	9,828.08	47.72%	1,430	23.32%	10,017.64	43.49%
+ Fuel Oil	400	8.87%	532.42	2.91%	569	10.82%	450.48	2.19%	569	9.28%	1,764.65	7.66%
Total Power Sources directly connected to National Grid	4,511	100%	18,314.68	100%	5,259	100%	20,596.09	100%	6,132	100%	23,032.33	100%
2. Import Power Sources from Neighboring Countries												
- Thailand	277	41.24%	731.01	44.10%	277	41.24%	391.33	22.17%	-	-	-	-
- Vietnam	339	50.43%	910.04	54.90%	339	50.43%	1,191.28	67.49%	339	85.82%	1,301.05	92.29%
- Laos	56	8.33%	16.68	1.01%	56	8.33%	182.48	10.34%	56	14.18%	108.76	7.71%
Total Import Power Sources from Neighboring Countries	672	100%	1,657.72	100%	672	100%	1,765.09	100%	395	100%	1,409.81	100%
3. Power Sources												
- Total Power Sources directly connected to National Grid	4,511	87.03%	18,314.68	91.70%	5,259	88.67%	20,596.09	92.11%	6,132	93.95%	23,032.33	94.23%
- Total Import Power Sources from Neighboring Countries	672	12.97%	1,657.72	8.30%	672	11.33%	1,765.09	7.89%	395	6.05%	1,409.81	5.77%
Total Power Sources	5,183	100%	19,972.41	100%	5,932	100%	22,361.17	100%	6,527	100%	24,442.14	100%

A brief outlook Plan for next decade, such as Long Term Power Development Plan and T&D Plan

The transmission line in Cambodia consists of 115kV, 230kV and 500kV mainly divided into four locations such as the southern, western, north-eastern and northern system with a total distance around 4,302.32 km. The transmission line was connected as grid to ensure the stability and reliability of the system.

No	Name of System	Transmission Lines	Substations
1	Transmission line 500kV	177.30 (x2)	14 in Phnom Penh, 6 in Kandal, 1 in Takeo, 5 in Kampong Speu, 3 in Kampot, 6 in Preah Sihanouk,
2	Transmission line 230kV	2,754.42 (x2)	4 in Koh Kong, 2 in Prey Veng, 2 in Svay Rieng, 5 in Battambang, 4 in Pursat, 2 in Kampong Chhnang,
		34.30 (x1)	2 in Banteay Meanchey, 3 in Siem Reap, 5 in Oddar Meanchey, 1 in Pailin, 2 in Kampong Cham,
3	Transmission line 115kV	396.63 (x2)	3 in Kratie, 1 in Stung Treng ,1 in Ratanakiri, 1 in Kbong Khmum, 3 in Mondulkiri, 1 in Preah Vihear
		939.67 (x1)	and 1 in Kampong Thom
Total		115-230-500kV = 4,302.32	75 substations supply directly to 25 cities/provinces

Cambodia Climate Change Strategic Plan (CCCSP) 2014-2023

CCCSP has the following strategic objectives:

- Promote the use of renewable energy and energy efficiency to reduce greenhouse gas emissions and health effects:
 - Renewable energy
 - Energy efficiency
 - Appropriate technology transfer
 - Solid/liquid waste management through integrated measures in the capital city, urban areas, and animal waste management.
- Analyze low GHG emissions by sectors (agriculture, energy, transportation, industry, land use and forestry management, and waste management).
- Develop policies, legal framework, and action plans for low carbon development in accordance with national development priorities.
- Promote the transfer of appropriate technologies to promote low-carbon development and facilitate the use of appropriate technologies through:
 - Guidelines for technical assistance and partnership building
 - Financial and tax incentives
 - Carbon marketing mechanism
 - Public and private partnership mobilization
- Establish a system for registering projects and programs to reduce greenhouse gas emissions
- Establish a quality national greenhouse gas inventory system.

National Communication

In line with CCCSP, RGC submitted Cambodia's Initial National Communication (INC) in August 2002 and Second National Communication (SNC) in 2016. INC and SNC highlighted that, though Cambodia is not a high emitting country, the government has committed to reduce GHG emission with the condition of financial and technical support from developed countries

Nationally Determined Contribution (NDC) 2.0

The NDC 2.0 has identified the following five actions to reduce greenhouse gas emissions in the energy sector:

- Maximize the use of renewable energy sources and increase the efficiency of energy consumption in the national power supply system by maintaining an adequate, reliable, and low-cost electricity supply to compete with neighboring countries.
- Connect rural electricity enterprises to the national electricity supply network
- Connect GHG intensive energy consumers to the national grid
- Provide power supply to areas outside the national electricity supply system using clean energy technologies,
- Improve energy efficiency.

Biennial Update Report

In 2020, the first Biennial Update Report has been submitted to the UNFCCC as a reference for preparing policies and strategies to address climate change in the Kingdom of Cambodia. The report includes relevant data and information regarding Cambodia's major sources of GHG emissions and sinks, coupled with necessary mitigation measures already implemented or planned; thus, furthering her contribution to global efforts to reduce GHG emissions.

Nationally Determined Contribution (NDC) 3.0

The NDC 3.0 for the energy sector outlines five key actions aimed at reducing greenhouse gas emissions and promoting sustainable development:

- Expanding renewable energy use through the development of clean energy sources such as solar, hydropower, biomass, and other renewable technologies to reduce dependence on fossil fuels like coal and diesel.
- Enhancing energy efficiency by minimizing energy waste in buildings, industries, transportation, and electrical equipment through the adoption of energy-efficient technologies and appliances.
- Improving transmission lines and distribution systems reduces energy loss and supports the integration of renewable energy into the national power supply.
- Encouraging investment in clean energy technologies by supporting businesses and investors through policies, incentives, and green energy initiatives.
- Strengthening regulations, technical capacity, and public awareness related to climate change, renewable energy, and efficient energy use through training programs and awareness campaigns.

Biennial Transparency Report

The Biennial Transparency Report (BTR) stands as a pivotal mechanism under the Paris Agreement, enabling nations to monitor and communicate their climate progress. In Cambodia, the BTR facilitates the implementation of NDC 3.0 by providing transparent data on greenhouse gas inventories, mitigation measures, adaptation strategies, and climate finance received. Within the energy sector, the BTR tracks critical initiatives such as renewable energy expansion, energy efficiency enhancements, grid modernization, and the adoption of low-carbon technologies. Consequently, the BTR empowers Cambodia to evaluate its national climate targets while fostering accountability and international cooperation.

Energy Sector

In the energy sector, emissions reductions until 2030 come from energy efficiency and conservation through the implementation of the National Energy Efficiency Policy⁴ in buildings, industry, and public services. This will limit growth in energy demand even as demand for energy services rises rapidly. Additionally, policies will be developed and put into place to incentivize switching to electricity for cooking and passenger vehicles, and to coal alternatives in the industrial sector. Major long-term infrastructural changes in the energy sector will be studied and prepared in this first decade. After 2030, energy sector emissions reductions will come from more stringent energy efficiency standards and continued fuel switching to low-carbon sources. Initial steps to decarbonize the electricity supply will be taken, with some investments in renewable energy and gas. With natural gas used as a transition fuel in the power sector, investments in infrastructure for imports, storage and transportation will be necessary. In the last decade of the LTS4CN scenario, between 2040 and 2050, the power sector will start a deeper decarbonization journey by adding more renewable energy in the generation mix, reaching a share of 35 percent in 2050. Within the renewable energy share, 12 percent is assumed to come from solar. Investments in grid modernization, flexibility and storage will be required during this transition. Cambodia is committed to:

- No new coal generation capacity beyond already committed projects
- Use of natural gas as a dispatchable transition fuel
- Investments in liquefied natural gas (LNG) import, storage and infrastructure
- Increase in solar, hydro, biomass and other renewables to 35 percent of the generation mix by 2050, of which 12 percent is from solar
- Investments in grid modernization, flexibility and storage
- Energy efficiency measures in buildings and industry
- Fuel switching to electricity for cooking
- Substitution of coal in the industrial and power sector

5

Development and Deploy of New Technologies

For innovative deployment of EDC is to update the current system to smart grid by electrical metering, but for the initial stage, we are using handheld units and AMR for recording the energy consumption for all customers. We

are understudying the AMI for upgrading our system to become a smart grid, allowing the customer to get the energy consumption every single hour by accessing data from the internet, mobile phone...etc.

6

Major Issues or Transition Influencing

To combat global warming in line with national commitment as well as global policies to achieve greenhouse gas emission reduction, energy transition and energy efficiency. With economic development and increasing the electricity consumption per capita, the accelerating share of renewable energy in the energy mix is one of the major challenges. It is a complex task to transform the power grid to be green with competitive prices and high stability. Although absorbing high renewable energy production is beneficial for less emission, it is also necessary to manage electrical infrastructure that is more agile, justified and modern than conventional network systems for timely data exchange and efficient management.

By phasing out non-committed coal-fired power plants in the plans, the new low-emitted source needs to be built for serving as a base-load in the dry season while the remaining water in the reservoirs can be used for the peak-hour time. The unpredictable nature of renewable technologies such as solar, wind presents a new pathway for strengthening the energy security and reliability which deployments of innovative technologies need to be adopted.

Establishment and development of industrial infrastructure

- Monitor and direct energy supply to major production zones by ensuring the most sufficient and stable electricity supply, which allow factories to plan their production effectively;

- Review the forecast of long-term electricity demand and energy development plan in line with the new vision for economic and industrial development. Energy supply options are also to be reviewed in order to ensure adequate energy supply capacity to major strategic industrial zones

One of four key concrete measures to promote the implementation of the "Industrial Development Policy" In addition to the aforementioned policy measures and action plans, the Royal Government will implement four key concrete measures to be achieved by the end of 2018. These measures are of great significance as they enable the Royal Government to monitor the efforts and progress of the implementation of the IDP. They serve also as an indispensable basis and preconditions for determining the success in ensuring that the vision, objectives, and goals of the policy are realized:

Prepare and implement a plan to reduce electricity tariffs for industrial and commercial purposes including strengthening reliability and expanding coverage of electricity supply:

- Reduce the price of electricity, purchasing from high voltage, medium voltage (in Phnom Penh), medium voltage (province) at USD 11.70 cents/kWh, USD 13.20 cents/kWh and USD 12.7 cents/kWh, respectively.
- Set a different electricity pricing for daytime and nighttime industrial usage;
- Expand the coverage of electricity supply to all the targeted industrial zones;
- Ensure reliability and confidence in the electricity supply by reducing the rate of electricity cut-off to no more than 12 times or 24 hours per annum.

The Royal Government introduced the electricity tariffs reduction and price differentiation schedule for the industrial and commercial purposes toward 2025 as in the **table below**.

Type of Purchase	Unit	Tariff to be applied by year							
		2019	2020	2021	2022	2023	2024	2025	2026
1. Direct Electricity Supply from National Grid Substation									
Purchase from high voltage feeder	\$/kwh	0.1170	0.1170	0.1170	0.1170	0.1170	0.1170	0.1170	0.1170
Commercial consumers connected at LV feeder of substations within Phnom Penh and Takekmao	\$/kwh	0.1300	0.1300	0.1300	0.1300	0.1300	0.1300	0.1300	0.1300
Commercial consumers connected at LV feeder of substations outside Phnom Penh and Takekmao	\$/kwh	0.1220	0.1210	0.1210	0.1210	0.1210	0.1210	0.1200	0.1200
Industrial and agricultural consumers connected at LV feeder of substations within Phnom Penh and Takekmao	\$/kwh	0.1300	0.1300	0.1300	0.1300	0.1300	0.1300	0.1300	0.1300
Industrial and agricultural consumers connected at LV feeder of substations outside Phnom Penh and Takekmao	\$/kwh	0.1220	0.1210	0.1210	0.1210	0.1210	0.1210	0.1200	0.1200
2. Electricity Supply by EDC within and outside Phnom Penh and Takekmao									
Industrial and agricultural consumers connected at medium voltage	\$/kwh	0.1470	0.1370	0.1370	0.1370	0.1370	0.1370	0.1370	0.1370
Commercial, administration and other consumers connected at medium voltage	\$/kwh	0.1090	0.1080	0.1080	0.1080	0.1080	0.1080	0.1080	0.1080
Public administration consumers paid with state budgets connected at medium voltage	\$/kwh				653	653	653	653	653
Industrial and agricultural consumers with meter at low voltage of consumer's transformer	\$/kwh	0.1029	0.14048	0.14048	0.14048	0.14048	0.14048	0.14048	0.14048
Industrial and agricultural consumers with meter at low voltage of licensee's transformer	\$/kwh	0.1609	0.18048	0.18048	0.18048	0.18048	0.18048	0.18048	0.18048
Commercial, administration and other consumers with meter at low voltage of consumer's transformer	\$/kwh	0.1634	0.16432	0.16432	0.16432	0.16432	0.16432	0.16432	0.16432
Public administration consumers paid with state budgets connected at low voltage of consumer's transformer	\$/kwh				679	679	679	679	679
Commercial, administration and other consumers with meter at low voltage of licensee's transformer	\$/kwh	0.1734	0.17232	0.17232	0.17232	0.17232	0.17232	0.17232	0.17232
Public administration consumers paid with state budgets connected at low voltage of licensee's transformer	\$/kwh				712	712	712	712	712
Residents consume from 1 to 10kwh/month	\$/kwh	380	380	380	380	380	380	380	380
Residents consume from 11 to 30kwh/month	\$/kwh	480	480	480	480	480	480	480	480
Residents consume from 31 to 200kwh/month	\$/kwh	610	610	610	610	610	610	610	610
Residents (>200kwh/month), other than residents	\$/kwh	740	730	730	730	730	730	730	730
Schools, Hospitals and Referral Healthcare Centers in rural areas	\$/kwh	610	610	610	610	610	610	610	610
Water pump for agriculture and agricultural consumers and consumers who install Solar PV system from Phu to TAAU, connected at LV and CV (Consumer's transformer or licensee's transformer)	\$/kwh							0.1200	0.1200
Water pump for agriculture and agricultural consumers and consumers who install Solar PV System from Phu to TAAU, connected at CV	\$/kwh	480	480	480	480	480	480	480	480
3. Electricity Supply by Licensees and Sub-transmissions									
Industrial and agricultural consumers connected at medium voltage	\$/kwh	0.1470	0.1370	0.1370	0.1370	0.1370	0.1370	0.1370	0.1370
Commercial, administration and other consumers connected at medium voltage	\$/kwh	0.1090	0.1080	0.1080	0.1080	0.1080	0.1080	0.1080	0.1080
Public administration consumers paid with state budgets connected at medium voltage	\$/kwh				653	653	653	653	653

Industrial and agricultural consumers with meter at low voltage of consumer's transformer	\$/kwh	0.1029	0.14048	0.14048	0.14048	0.14048	0.14048	0.14048	0.14048
Industrial and agricultural consumers with meter at low voltage of licensee's transformer	\$/kwh	0.1609	0.18048	0.18048	0.18048	0.18048	0.18048	0.18048	0.18048
Commercial, administration and other consumers with meter at low voltage of consumer's transformer	\$/kwh	0.1634	0.16432	0.16432	0.16432	0.16432	0.16432	0.16432	0.16432
Public administration consumers paid with state budgets connected at low voltage of consumer's transformer	\$/kwh				679	679	679	679	679
Commercial, administration and other consumers with meter at low voltage of licensee's transformer	\$/kwh	0.1734	0.17232	0.17232	0.17232	0.17232	0.17232	0.17232	0.17232
Public administration consumers paid with state budgets connected at low voltage of licensee's transformer	\$/kwh				712	712	712	712	712
Residents consume from 1 to 10kwh/month	\$/kwh	380	380	380	380	380	380	380	380
Residents consume from 11 to 30kwh/month	\$/kwh	480	480	480	480	480	480	480	480
Residents consume from 31 to 200kwh/month	\$/kwh	610	610	610	610	610	610	610	610
Residents (>200kwh/month), other than residents in provincial towns	\$/kwh	740	730	730	730	730	730	730	730
Schools, Hospitals and Referral Healthcare Centers in rural areas	\$/kwh	610	610	610	610	610	610	610	610
Water pump for agriculture and agricultural consumers and consumers who install Solar PV system from Phu to TAAU, connected at LV and CV (Consumer's transformer or licensee's transformer)	\$/kwh							0.1200	0.1200
Water pump for agriculture and agricultural consumers and consumers who install Solar PV system from Phu to TAAU, connected at CV	\$/kwh	480	480	480	480	480	480	480	480

7

Photos of Development of Energy Sector in Cambodia



Tatay Hydro power Plant



Kampong Chhnang Solar PV



Banteay Meanchey Solar PV (Ray PV)



Battambang Solar PV (Risen PV)



Development of Transmission Line



Development of Grid Substation

CHINESE TAIPEI



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Taipei
Area:	36,197 km ²
Population:	23.30 million
GDP:	US\$ 921.9 billion
Currency:	New Taiwan Dollar (NT\$)
Installed Capacity (in GW):	59.8 GW
Percentage of population electrified:	Nearly 100%

Power Demand by Sector

Sector	Total Power Consumption as of 2025 (MWh)	Projected Change (%) 2025/2026	Projected Power Consumption as of end 2026 (MWh)
Industrial	133,720,831	2.63	137,231,247
Commercial	36,200,884	2.32	37,041,004
Residential	50,934,027	0.90	51,394,028
Other	18,200,848	2.60	18,673,895
TOTAL	239,056,590	2.21	244,340,174

Note: Power consumption in this table is calculated based on electricity sales from Taiwan Power Company's public electricity-retailing utility and renewable energy-based electricity-retailing enterprises.

National Energy Policy

The government's long-term energy policy aims to increase the proportion of renewable and gas-fired power while simultaneously reducing reliance on coal.

In 2024, a secondary energy transition was initiated which consisting of various elements including diverse green energy, deep energy conservation, and advanced energy storage. This will effectively ensure a stable power supply, while also steadily moving towards net-zero.

The policy calls for an energy mix that consists of 60-70% renewable, 9-12% hydrogen, 20-27% thermal with Carbon Capture, Utilization, and Storage (CCUS) by 2050. At that time, renewable energy will have reached a total capacity of 85.7GW. This capacity will be derived from offshore wind power, solar photovoltaics, geothermal, biomass and ocean sources.

Supply and Demand of Primary Natural Resources

(1) Conventional Hydro Power

The installed capacity of hydro power plants in Taiwan was 2,124MW in 2025.

(2) Other Forms of Renewable Energy

The combined installed capacity of other types of renewable energy, including wind power, photovoltaic systems, geothermal and biomass sources, was 20,748MW in 2025.

Laws and Governmental Regulations

Taipower must abide by all relevant regulations and laws, including but not limited to:

- (1) The Electricity Act
- (2) The Renewable Energy Development Act
- (3) The Energy Management Law
- (4) The Government Procurement Act
- (5) The Act Governing the Management of State-Owned Enterprises
- (6) The Budget Law
- (7) The Atomic Energy Law and its related regulations
- (8) The Nuclear Materials and Radioactive Waste Management Act
- (9) The Nuclear Safeguards Agreement

Total amount of energy consumption by sectors (Industrial / Residential / Public & Service) (in MWh and %)

Sector	Total Power Consumption as of 2025 (MWh)	% in Total
Industrial	133,720,831	55.94
Residential	36,200,884	15.14
Residential	50,934,027	21.31
Other	18,200,848	7.61
TOTAL	239,056,590	100.00

Note: Power consumption in this table is calculated based on electricity sales from Taiwan Power Company's public electricity-retailing utility and renewable energy-based electricity-retailing enterprises-based electricity-retailing enterprises.

3

Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (Coal, Liquefied Natural Gas (LNG), Oil, Nuclear and Hydropower) including total installed capacity of generation (in GW) and total production (in TWh) by fuel types (in MWe)

	Installed Capacity (GW)	Year-on-year Growth (%)		Generation (TWh)	Year-on-year Growth (%)
Total	59.84	3.64	Total	252.44	0.40
Hydropower	4.69	0.00	Hydropower	8.28	15.91
Thermal Power (Coal, LNG, Oil)	35.10	3.39	Thermal Power (Coal, LNG, Oil)	213.58	3.51
Nuclear Power	0.00	-100.00	Nuclear Power	3.12	-73.40
Wind Power	4.51	16.51	Wind Power	10.35	18.95
Solar Power	14.80	8.85	Solar Power	13.52	-7.17
Cogeneration	0.63	0.00	Cogeneration	2.58	5.45
Biomass Energy	0.02	1.08	Biomass Energy	0.05	0.02
Geothermal	0.01	0.00	Geothermal	0.02	-7.65
Waste	0.09	168.40	Waste	0.26	55.00
			Battery	0.70	138.38

4

Reduction of Carbon Emission

Policy on Renewable Energy and Major Regulatory Mechanism of Renewable Law

Taiwan's national energy policy centers on a Second Energy Transition that is built upon the existing principles of increasing gas, reducing coal, and expanding renewable energy. Additionally, it pursues three major objectives: ensuring an abundant and resilient power supply, developing low-carbon and self-sufficient green power, and maintaining stable energy prices. Together, these serve as the critical pathway toward achieving the goal of a 2050 Net-Zero Transition. In terms of management mechanisms, the government continues to enforce the Renewable Energy Development Act—which mandates the installation of renewable energy systems on public buildings and simplifies application procedures for projects under 2,000 kW. Amendments to the Electricity Act completed in 2025 also brought a major breakthrough to the green power market by allowing for peer-to-peer trading among renewable energy retailers to significantly enhance green power dispatch flexibility. It also integrates grid-connected energy storage and demand response into the regulatory framework to guide private investment into dispatchable power

resources. With the official launch of the first batch of carbon levy in May 2026, Taiwan has entered the era of carbon pricing, shifting corporate demand for green power from mere compliance to deep “carbon asset management.” Consequently, green power market transactions have fully transitioned from the Feed-in Tariff (FIT) scheme to a corporate power purchase agreement (CPPA) dominated model, while promoting flexible switching between green power procurement and direct supply/wheeling arrangements; meanwhile, for major electricity consumers, the government not only maintains the requirement to install a certain percentage of renewable energy or purchase renewable energy certificates (RECs) but also encourages them to participate in the electricity trading platform to convert energy storage and energy-saving resources into system resilience, thereby meeting the AI and semiconductor industries' urgent demand for “hourly matching” and “dispatchable green power” while collectively expanding the scale of the green economy market.

Production of Renewable Energy (MWh) from various resources (wind, solar, biomass, landfill gas, etc)

Resources	Production Generation in 2025 (MWh) (Jan 1- Dec 31, 2025)
Hydro Power	5,506,149
Wind Power	12,203,288
Solar Power	16,739,153
Waste	3,849,728
Biomass	273,083
Geothermal Power	25,843

Environmental Problems of Electricity Supply (including generation and T&D sectors) and How It Is Being Addressed

As concerns about climate change grow, the global industrial supply chain is facing tighter carbon reduction requirements. Under the framework established by the Paris Agreement, many countries have undertaken reviews of their climate change responses and proposed net-zero emission targets. They have also started working on mitigation and adaptation strategies.

In order to clearly demonstrate Taiwan's determination to reduce greenhouse gas emissions, the Climate Change Response Act was promulgated and implemented on February 15, 2023. The act aims to reduce Taiwan's impact on climate change by establishing a long-term carbon reduction target of net-zero emissions by 2050.

Responsibility for implementing these changes falls to the island's electric power producers. These include the state-owned power enterprise (Taiwan Power Company, or Taipower) and the private companies, Independent Power Producers (IPPs) and Co-generators with whom Taipower has integrating its power generation, transmission and distribution systems. As a state-owned enterprise and a part of the government system, Taipower has the additional responsibility of assisting the government in achieving its carbon reduction targets. In Taiwan, power production is currently dominated by thermal power units, and in consequence, energy producers are a primary contributor to greenhouse gas emissions. As a result, the role of power producers in achieving emission reductions is of paramount importance.

Plans and Strategies to Reduce Greenhouse Gas Emissions

As a state-owned enterprise, Taipower must adhere to the national power development policy of promoting carbon reduction. Electricity consumption in Taiwan has continued to increase with the island's economic growth, while the development of non-carbon and low carbon energy resources has remained limited at present. Consequently, Taipower must implement multiple strategies to maintain the stability of the electricity supply and, at the same time, achieve its carbon reduction goals.

In accordance with the government's energy policy and in keeping with global trends, Taipower has committed to developing high-efficiency generation technologies, adjusting its generation structure, continuously increasing its proportion of renewable energy. With regards to mitigating the effects of thermal power generation, Taipower currently focuses on three main approaches:

- (1) Continuously increasing gas-fired generation while reducing coal-fired generation:
This approach aims to increase the proportion of gas-fired generation and continue the pattern (in place since 2022) of primarily using gas with coal as a support so that gas power generation is proportionally higher than coal power generation.
- (2) Improving the efficiency of gas-fired units:
Efforts are underway to gradually phase out older gas-fired generation units and replace them with new, high-efficiency combined-cycle gas-fired units.
- (3) Investing in the research and development of emission reduction technology, such as Taipower's ongoing research into hydrogen/ammonia co-firing and carbon capture and storage (CCS).

Latest AMI Installations and Smart Grid Projects

Taipower has always approached the development of the smart grid as a systematic engineering process driven by a 'problem-solving' orientation. In accordance with the 'Smart Grid Master Plan,' Taipower is implementing its three major transformation visions in phased stages:

(1) Short-Term (2026) - Enhancing Operational Flexibility:

To fully build a stable power supply network adapted to a high proportion of green energy, and comprehensively elevate flexible dispatching capabilities for supply-demand balance and incident handling.

(2) Mid-Term (2026-2030) - Strengthening Grid Resilience:

To proactively respond to climate change and build a secure, disaster-resistant, and highly adaptive resilient power grid.

(3) Long-Term (after 2030) - Implementing Power Industry Reform:

To deepen the application of low-carbon energy, promote data transparency and fair market trading, and create a secure and reliable electricity market.

To achieve these objectives, Taipower has launched upgrade and deployment initiatives across seven core areas. In terms of technology and management, we are deepening 'Smart Dispatch and Generation' to maximize green energy grid integration, while upgrading 'Grid Management' to comprehensively enhance the automation and self-healing capabilities of the transmission and distribution networks. Regarding systems and demand, we are actively deploying 'Energy Storage Systems' to achieve peak shaving and valley filling for grid stability; furthermore, we are leveraging smart meters to drive 'Demand-Side Management,' guiding the adoption of Time-of-Use (TOU) tariffs and demand response to optimize energy efficiency. On the foundational and institutional front, we are fully implementing 'Information and Communications Infrastructure' to ensure comprehensive cybersecurity monitoring across the

grid, while simultaneously amending 'Regulatory and Institutional Frameworks' in alignment with government policies and the Electricity Act to mature the market mechanism. Ultimately, through this large-scale nationwide infrastructure development, we expect to drive 'Industrial Development,' leading domestic power and ICT sectors to upgrade together and jointly expand into international markets.

Since 2018, Taipower has been deploying low-voltage Advanced Metering Infrastructure (AMI) and, to date, has completed the installation of approximately 4.3 million smart meters, thereby achieving a leading international position in both performance and functionality. To maintain this leadership, Taipower has commenced research, development and prototype manufacturing of a next-generation AMI system. Taipower will evaluate the functional feasibility and performance of Taiwan's next-generation AMI to accommodate future application scenarios, such as real-time management of large-scale renewable energy grid integration, thereby establishing Taiwan's AMI system as a world-class benchmark in terms of functionality and architecture.

In response to an increasingly stringent cybersecurity environment, Taipower will also evaluate the introduction of Public Key Infrastructure (PKI) into its AMI meters and establish relevant certificate management systems, mechanisms, and workflows to fulfill the need for certificate generation, distribution, renewal and revocation for its next-generation meters.

Furthermore, to enhance the manageability of its next-generation meters, the company will develop a comprehensive meter firmware management system alongside Over-the-Air (OTA) update mechanisms and processes, to ensure the feasibility of functional adjustments to future meters. To maximize the benefits of the system, the next-generation AMI will also refine its Route B system and develop related value-added applications.

Challenges faced and how it was solved / addressed

Through electricity transition, green energy maximization and low-carbon power generation policies, the penetration rate of renewable energy is gradually increasing. The intermittent nature of renewable energy increases the difficulty of maintaining stability between generation and load balance within the domestic power system. Consequently, Taipower is actively promoting smart grids, and developing smart functions for power generation, transmission, distribution, and users. Measures include the building of an electricity trading platform, construction of energy storage systems, feeder automation, and introducing AMI data applications. Through the application and integration of various smart grids, the goals of energy transition and net zero emissions will be achieved.

Smart meters play a vital role in modern power systems; however, ensuring their security from cyber threats poses an urgent challenge. Encryption technologies are critical in this regard, with symmetric and asymmetric encryption being the two most common approaches. Symmetric encryption utilizes the same key for both encryption and decryption. It is fast and well-suited to encrypting large volumes of data. However, key management can become a significant challenge, as the security of the entire system is compromised if a key is leaked. Earlier international smart metering standards, prior to IEC 62056 Edition 7, only supported symmetric keys; consequently, major electric utilities worldwide—including Taipower—currently base the cybersecurity of their AMI systems on symmetric encryption. Conversely, asymmetric encryption utilizes a pair of public and private keys for encryption and decryption, where the public key can be openly shared while the private key is kept strictly confidential. Although asymmetric encryption is computationally slower, it provides stronger key-management and authentication capabilities, thereby enhancing overall security. Since cybersecurity risks continuously evolve, the IEC 62056 standard established by the

International Electrotechnical Commission (IEC) is updated regularly. Beginning with IEC 62056 Edition 8, the standard introduced support for Public Key Infrastructure (PKI), marking a major shift toward more secure communications through asymmetric encryption. The distinct differences in cybersecurity capability that emerge with newer versions of IEC 62056—specifically the transition from lacking PKI support to incorporating it—reflect an ongoing focus on and improvement in addressing cybersecurity issues. As smart meter systems continue to develop and expand, the adoption of more secure encryption technologies has become indispensable. Leveraging the evolution of standards like IEC 62056, Taipower will pilot the integration of PKI systems into its next-generation meters with the aim of better countering increasingly severe cybersecurity threats, thereby ensuring stable system operations and the security of user data.

AMI Roll out plan

In order to reduce consumer electricity consumption and increase the efficiency of energy utilization, Taipower has begun deploying smart meters in support of demand-side management. In 2013, Taipower completed a nationwide deployment of 24,000 smart meters for high-voltage customers that were responsible for 60% of power consumption in Taiwan. Following the approval of amendments to the “Smart Grid Master Plan” on February 18, 2017 (1st edition), and March 27, 2020 (2nd edition), Taipower undertook and completed the installation of 4.1 million low-voltage AMI meters by 2025. Currently, the 2026 target is to deploy 4.6 million smart meters. In the future, Taipower will be able to install 6 million smart meters that, by 2030, will be able to handle around 85% of Taiwan's power system usage information.

AMI offerings to customers and its benefit

The adoption of an AMI will provide customers with the following benefits:

(1) Improved Power Savings

AMI meters will provide customers with faster access to their usage data. This will make it easier for them to understand how they are using power. This will, in turn, help customers identify old appliances that are good candidates for replacement and foster power saving habits.

(2) Increased Customer Convenience

AMI meters can automatically read, record and report on meter data. This will allow the Company to collect crucial data remotely without inconveniencing customers. In addition, AMI also allows the Company to remotely monitor. This function decreases the times needed to process mid-month billing. It also increases customer service efficiency.

(3) Speedy Restoration of Service Following Outages

AMI meters can report outage information to Taipower's Outage Management System. This mechanism will help maintenance staff determine outage areas and allow for quicker response times. This should lead to speedy service restoration and reduced power outage times for customers.

Advanced Metering Infrastructure (AMI) enhances Taipower's modern grid management while delivering rich and highly efficient electricity services to customers. The "Taipower App" features various smart meter applications. Beyond enabling users understand their own electricity consumption habits, it offers diverse value-added services. These include "Residential Electricity Analysis," which provides information on high-energy-consuming appliances, and "Electricity Usage Pattern Indicators," which allows users to keep an eye on the well-being of elderly family members, yielding multi-faceted benefits.

The primary customer benefits of the Advanced Metering Infrastructure (AMI) include:

- (1) Leveraging AMI data analysis and behavioural science principles to provide users with more persuasive insights, encouraging them to adjust their electricity usage habits and achieve effortless energy savings.
- (2) Using big data analysis to identify potential customers for Taipower's demand response programs and proactively recommending participation. This approach enhances user engagement in demand response initiatives, leading to improved energy efficiency and cost savings for users.
- (3) By applying big data analysis, Artificial Intelligence (AI) and Machine Learning (ML) algorithms to analyse users' primary electricity loads. This enables demand side management (DSM) applications, such as energy-saving strategies, load shifting, and value-added services like monitoring home electricity usage patterns for elderly residents.

Currently, Taipower R&D units utilize real-time AMI data for value-added electricity consumption analysis. Additionally, Taipower is employing big data analytics to explore innovative research on automated demand response business models and advanced demand response mechanisms. By providing diverse user interfaces and transparent information, Taipower aims to increase user participation and enhance customer experience through innovative applications. These efforts encourage users to modify their electricity consumption behaviours, achieving load adjustments, energy savings, and carbon reduction. Ultimately, this will improve customer satisfaction and strengthen users' understanding and trust in Taipower.

Taipower has deployed battery energy storage systems (BESS) at its Luyuan, Longtan, and Dongshan Extra High Voltage Substations (E/S). These facilities, with capacities of 20MW/10MWh, 60MW/30MWh, and 60MW/30MWh, were commissioned in December 2022, January 2024, and November 2024, respectively.

In addition, regarding regional voltage regulation, the company has installed Static Synchronous Compensators (STATCOMs) at the Nanke E/S, Zhuyuan E/S, ZhangGong E/S, and Yongxing S/Y, each with a capacity of ± 200 MVar.

● Feeder Automation

The increasing use of renewable energy and the growing load demand have created a need for more grid connections while maintaining a stable, high-quality power supply. This has necessitated the expansion and automation of distribution feeders, primarily through the construction and upgrading of distribution network switches equipped with Information and Communication Technology (ICT). By the end of 2025, the total number of automated feeders had reached 10,172, and the penetration rate was about 100%. Another 38,214 feeder automatic switches have also been included.

As the technology underlying automated distribution systems continues to mature, domestic manufacturers have steadily increased their capacity to produce key modules and technologies. Moving forward, the company plans to add 1,500 automatic feeder switches have been included annually to enhance the power dispatch center's situational awareness.

● Battery Storage

In the past, government agencies and important power users frequently addressed post-disaster power outages by installing their own emergency generator sets to maintain power supply. However, developments in technology have made progress in overcoming the vulnerability of the overreliance of traditional power systems on a "centralized power grid" which easily triggers blackout risks. In alignment with this trend, Taipower along with power providers in various other countries are actively promoting the transformation of power systems toward disaster-resilient "decentralized power grids."

Taipower has, during the 2024 to 2025 period, completed the construction of related equipment for Regional Power Grid Energy Storage sites at eight secondary substations (S/S) across Taiwan. The project has installed a total of 20.5 MW of energy storage systems and 4 MW of diesel generators. The effort is a pioneering demonstration of independent power supply and the "island operation" capabilities of single distribution-level feeders when the power supply through the grid is interrupted.

When the upstream transmission system experiences an accidental blackout or power outage, these systems can rely on their energy storage to establish voltage (black-start capability) and their fast response charging and discharging regulation capabilities, and cooperate with the microgrid energy management system (EMS) to integrate the solar photovoltaic solar systems and diesel generator sets within the region. The resulting system can serve as a stable power source in the early stage of reestablishing the distribution feeder grid, and can proactively disconnect from the main grid to operate in island mode, thereby ensuring the continuous power supply to critical loads (such as communications, medical care, and other basic electricity needs) within the region. Once the upstream transmission lines are repaired, it can then synchronize with the restored grid system, and restore the power supply from the original grid.

● Supercritical Power Generation Technology

Taiwan has installed five 800 MW supercritical coal-fired power generating units, deploying advanced pollution control technologies such as Selective Catalytic Reduction (SCR), Particulate Removal Systems (PRS), Seawater Flue Gas Desulfurization (SWFGD), dry bottom ash systems, and enclosed coal storage and transport system. Through a comprehensive three-pronged strategy—encompassing source reduction, process optimization, and end-of-pipe treatment—these projects successfully deliver a stable power supply while demonstrating a strong commitment to corporate social responsibility (CSR).

• Grid Scale Energy Storage (ESS)

1. Strategies and plans for the development of a battery energy storage system

As Taiwan moves towards its low-carbon and climate goals, it is actively developing green power and pursuing the installation of an energy storage system (ESS). Upon completion, the system will not only smooth green power generation, but also maintain frequency stability in the power system. In addition to those advantages, the system will increase the capacity value of renewable energy and improve power dispatching.

By 2025, Taiwan will have greatly increased its use of renewable energy. Consequently, Taipower is working to add ancillary service capacity to its grid to accommodate the increased renewable energy connections. Moreover, Taipower is preparing diverse responsive strategies for power dispatch, including the installation of combined cycle generators with quick ramp rates, changing pumping/storage operating modes, and adopting energy storage equipment.

Taipower has achieved the target for 1000MW Battery Energy Storage System (BESS) by 2025, including 160MW of equipment at its own sites and 840MW of qualified participation capacity on Taipower's Energy Trading Platform.

2. The status of battery energy storage equipment installation

Taipower has completed four grid scale energy storage battery systems in 2024, with a total capacity of 160MW. In 2021, the Energy Trading Platform was established. Moving forward, Taipower will obtain frequency regulation ancillary services from the Day-Ahead ancillary services market. The qualified participation capacity provided by BESS was 1893.4MW as of May 2026.

• Ultra-Supercritical Generation Technologies

The ultra-supercritical generation units in Taipower have been in commercial operation since 2016, and the boilers have adopted a variety of advanced creep materials. TPRI has integrated the on-site sampling and laboratory analysis results before the unit is commissioned and incorporated into the original constructed Boiler Life Assessment System, Failure Analysis Systems. In addition, for the materials and welding issues, the researchers have been conducted including the derivation of micro-structure and mechanical behavior degradation, the influence on the material properties and improvement technology aroused by extraordinary post-weld heat treatment, mitigation technology of oxide film proliferation, welding procedures for dissimilar metals, etc. Various creep parameters for super304H and t91 material are carried out constructing the life assessment for heat exchange tubes progressively. The first commercial boiler is about to operate 10 years, and we plan a detailed inspection plan to identify areas of boiler equipment deterioration.

• Applying XR (Extended Reality) to the Tutoring on Substation O&M

In recent years, Taipower Company has faced a major wave of retirements, leading to an aging workforce and a looming gap in manpower and knowledge transfer. We have also confronted challenges such as time-consuming and labor-intensive equipment inspections, remote communication barriers, and high digital transformation costs. To address these issues, Taipower has introduced the XR Remote Communication & Collaboration System and the XR Troubleshooting Education and Training System. Furthermore, we have integrated development achievements including "69kV GCB Maintenance and Internal Inspection," "OLTC Inspection and Maintenance," and "23kV GIS Switchgear Maintenance and Internal Inspection" into the "VR Substation Training System" at the Taipower Training Institute. This creates a one-click, multi-functional integrated service that fulfills cross-regional collaboration needs, offers a more intuitive and immersive operation flow through the fusion of virtual and real environments, and achieves a sustainable, easily updatable digital transformation for traditional knowledge content.

• **A Typhoon-Induced Power Outage Early Warning and Decision Support**

To address the increasing frequency and intensity of extreme weather events caused by climate change, and to enhance the power system's early warning capability, resource dispatch efficiency, and disaster prevention decision-making quality during typhoons. We have promoted the "Application Study of a Power Outage Early Warning Model for Power Systems under Typhoon Disasters." By integrating generative AI technology, the project is a typhoon power outage early warning and decision support system, assisting power disaster prevention operations in shifting from passive post-disaster repair to proactive pre-disaster prediction and intelligent decision-making.

The overall system functions and benefits include:

- (1) Integrating historical outage records, utility pole structural data, and weather forecast data to establish a data foundation for typhoon outage risk prediction.
- (2) Applying AI models to predict high-risk outage areas during typhoons, helping regional offices identify potential disaster hotspots in advance.
- (3) Combining generative AI with outage prediction results and historical preparedness meeting records to generate recommendations for pre-typhoon preparedness, protective measures, and resource allocation.
- (4) Assisting in the assessment of priority repair areas to improve pre-disaster deployment, emergency response, and post-disaster recovery efficiency.
- (5) The system can be further extended to distribution system maintenance, disaster risk management, and decision support for power grid resilience.

• **The Interactive Visualization Platform for Solar Photovoltaic (PV) and GIS Grid-connection**

To respond to the government's renewable energy policy, Taipower is actively thinking about how to improve the possible hosting capacity of distributed energy resource (DER) in distribution systems while maintaining power quality and grid security. Taipower has built the interactive visualization platform for Solar photovoltaic (PV) and GIS grid-connection based on the core of the distribution planning information system.

Using OpenDSS as computing core, the system can calculate the possible hosting capacity of the feeder sections, thereby integrate the computing results to the Geographic Information System (GIS) technology and display the possible hosting capacity of each feeder section on the map with colors. The system has been launched on the Taiwan power company official website on January 1st, 2019(<http://service.taipower.com.tw/hcweb/>). The system platform is an integrated solar photovoltaic grid-connected capacity visualization platform, which provides the grid-connectable capacity of the feeders, the potential site land area of the solar photovoltaic inventoried by the Energy Bureau, and the wild bird reservation area drawn by the Wild Bird Society. By means of multiple map overlayer, the user can flexibly select the map layers of interest to watch, thereby increasing the intention to apply for renewable energy installation. Yet, there could be no feeders nor transmission lines nearby for a renewable energy field to connect to the power grid. To mitigate the decrease of interest in investment in renewable energy fields due to higher cost under the previously stated situation, Taipower encourages people to invest in self-built substations to share with more power companies, with which more renewable energy fields can be connected, near the substations-to-share, into the power grid. A geospatial layer of those substations-to-share is therefore created and integrated into the system for users to search and view in 2022.

• **AI Customer Service System for the Power Industry**

The power utility frequently faces a massive and highly repetitive volume of customer inquiries (e.g., billing questions, power outage reports). During unexpected events, the system is particularly vulnerable to overload due to sudden surges in message volume. Implementing an AI text-based customer service system can break through the limitations of staff scheduling, providing 24/7 real-time service. This effectively diverts routine inquiries, allowing professional staff to focus on complex cases. Consequently, this optimizes operational costs and significantly enhances crisis response capabilities and customer satisfaction. To build an AI smart customer service system focused on text-based interactions, the following key technologies must be integrated:

- (1) Natural Language Processing (NLP/NLU): Accurately parses user-input text, identifies true intent, and extracts key information.
- (2) Retrieval-Augmented Generation (RAG): Retrieves professional documents and the latest announcements in real-time, ensuring AI-generated answers are accurate and reliable.
- (3) FAQ Knowledge Base Management: Structurally manages high-frequency common questions, ensuring the AI can quickly retrieve standard answers and improving response consistency.
- (4) Backend System Integration: Deeply integrates with the company's backend systems, empowering the AI with automatic query and ticketing capabilities while ensuring data security.
- (5) Human-AI Collaboration: The AI automatically summarizes dialogues, assisting human agents with a seamless transition during transfers and simplifying post-service documentation. This effectively reduces staff burden and allows them to focus on elevating service quality.

The implementation of an AI text-based customer service system is not merely a technological upgrade but a crucial milestone toward smart grid services. The system can provide continuous, around-the-clock service and rapidly triage massive demands during message peaks, such as sudden power outages, effectively alleviating front-line pressure. Automating routine business processes helps optimize the workforce structure, concentrating resources on high-value, professional cases. Furthermore, the analysis of interaction data can pinpoint user pain points accurately, assisting in the continuous optimization of service processes. Through automation and human-AI collaboration, the company will be able to significantly enhance service resilience, achieving an optimal balance among operational efficiency, cost control, and user experience.

With the development of renewable energy, a significant amount of clean energy, primarily derived from solar photovoltaics and wind power, has been integrated into the power system. Simultaneously, battery energy storage systems (BESS), which support system operations, have been introduced to the grid through electricity trading platforms or integrated solar-plus-storage projects.

When the proportion of these Inverter-Based Resources (IBR) increases and they operate in Grid-Following (GFL) mode, there is an increased risk of issues associated with system frequency and voltage stability because they lack inherent rotational inertia and have highly limited fault currents. However, advanced control technologies now allow IBRs to operate in Grid-Forming (GFM) mode, which can provide virtual inertia, damping, and improve system strength, thereby benefiting grid operations.

Relevant technical specifications and grid codes are currently under development and being formalized. Furthermore, testing capabilities are being developed to verify the characteristics of GFM IBRs, ensuring that their performance upon grid integration meets expectations.

In response to climate change and to align with the 2050 net-zero carbon emissions target, Taiwan Power Company has actively been promoting solar power generation and wind power generation. Taiwan Power Research Institute (TPRI) has conducted energy storage demonstration experiments for smoothing renewable energy generation and controlling peak power demand. To successfully achieve these goals, TPRI has introduced an all vanadium redox flow battery's energy storage system from Sumitomo Electric Industries Ltd. in May 2016. This 125kW/750kWh system was installed at Shulin campus of TPRI in February 2017. In addition, a 250kW/400kWh and a 1.MW/1.5MWh container type lithium-ion battery energy storage system from local company has constructed in this area in 2019. Furthermore, PV integrated with storage system was applied in smart microgrid which has been completely tested for planned islanding and unplanned islanding mode at Shulin testing field in 2021. To further investigate and assess the viability of long-duration energy storage technologies, Taipower is scheduled to integrate renewable energy, hydrogen, and battery energy storage systems (BESS) by 2027. This project will involve the deployment of commercial-scale water electrolysis systems to construct a demonstration site, aimed at verifying technical feasibility and gathering operational experience through empirical testing. Based on the concept of a megawatt-scale (MW-scale) hybrid energy system architecture, the initiative will evaluate both the technical feasibility and economic viability of power-to-hydrogen storage solutions utilizing surplus electricity.

(1) Intermittency and Dispatch Pressure: Managing the integration of large-scale renewable energy while ensuring stable power supply is a critical challenge. Since the Electricity Act stipulates priority grid connection and dispatch for renewable energy, power grid planning must be adjusted to accommodate this influx. Future planning should move toward a diversified energy mix, decentralization, and regionalized grid development. By utilizing information and communication technology combined with energy storage systems, smart meters, and demand-side management, a smart grid can enhance energy efficiency, improve grid stability, and reduce peak loads, thereby creating a green, low-carbon power grid.

(2) Slowdown in Renewable Energy Installation Speed: Ground-mounted solar photovoltaics (PV) currently face the largest shortfall. The original target was to achieve 12 GW of ground-mounted solar by 2025. However, due to challenges in land acquisition and frequent conflicts over site selection regarding agriculture and ecological conservation, Taiwan tightened its environmental impact assessment (EIA) regulations at the end of 2025. These tighter controls on ground-mounted solar installations will delay the timeline for achieving this target.

(3) Technical Talent and Marine Engineering Costs: As offshore wind power enters large-scale development, shortfalls remain in related operation and maintenance (O&M) equipment, offshore construction vessels, and specialized technical talent. Furthermore, developing deeper maritime areas has caused costs to rise sharply. Some local supply chain vendors have incurred significant losses during the localization process, negatively affecting their willingness to participate in subsequent bidding.

As concerns about cybersecurity grow, Taiwan Power Company is continuing to expand and develop the cybersecurity defenses of its critical infrastructure by:

(1) Accelerating Scale-Up: Upgrading of a foundational 23 sites was completed in 2025. For 2026, TPC is expanding deployment to 37 critical sites. Real-time threat intelligence from these sites is now directly integrated into a central SOC (Security Operations Center) console.

(2) Three-Tier Defense in Depth: The defense perimeter now covers all three tiers of the smart grid—"Central Dispatch → Regional Transmission & Supply → Local Distribution"—forming a fully unified, multi-layered cybersecurity shield.

(3) Proactive Threat Mitigation: By implementing real-time monitoring of critical network nodes and precision traffic analysis, TPC now has an early warning and precision response capacity through fully realized, integrated smart grid security that can ensure an uninterrupted and stable power supply.

Renewable energy is being integrated into the grid, to allow for the gradual replacement of conventional generating units. The reduction of conventional generators, accompanied by a decrease in system inertia, inevitably brings about frequency stability challenges. To mitigate these risks, Taiwan has procured corresponding ancillary services—such as E-dReg—and implemented an under-frequency load shedding (UFLS) scheme that is reviewed and updated annually to cope with system supply-demand imbalance events.

The short-circuit current on the system is primarily provided by conventional generating units. As a result, system inertia and grid strength are closely interrelated—a decrease in system inertia leads to a reduction in grid strength. Unlike the frequency stability issues caused by reduced inertia, which affect the system as a whole, voltage stability issues associated with grid strength are localized in nature. The greater the grid strength, the easier it is for local voltage to stabilize after a disturbance.

Measures to improve grid strength can be categorized into three areas:

- (1) Maintaining a minimum level of synchronous generating units in operation, and restricting Inverter-Based Resource (IBR) output or limiting the deployment of additional inverter-based resources when necessary.
- (2) Adjusting inverter control characteristics under low grid strength conditions for areas with a high penetration of renewable energy, utilizing Electromagnetic Transient (EMT) software alongside manufacturer parameters, while considering the coordinated response between diverse IBRs and auxiliary equipment.
- (3) Installing synchronous condensers.

The deployment of Distributed Energy Resources (DERs) in smart grids introduces new technologies and services that improve operational efficiency, while simultaneously expanding cybersecurity risks and posing serious threats to grid operations. By examining DER-related cybersecurity standards, this study assists organizations in developing DER cybersecurity risk assessment frameworks and mitigation strategies, enabling systems to rapidly recover from cyberattacks, fulfilling DER network communication security requirements, and ensuring reliable, secure, and real-time information transmission. The specific approaches include:

- (1) Conducting a comprehensive inventory of system assets to identify potential cyber-security threats and vulnerabilities, and pinpointing critical risk points that may impact assets, systems, and processes.
- (2) Referencing power and industrial control system cybersecurity standards — IEC 62351, IEC 62443, and NISTIR 7628 — to develop targeted risk mitigation and pre-vention strategies.
- (3) Deploying an intrusion detection system to provide real-time alerting on anomalous network behavior within the field network.



Tainan Salt Field Solar-Plus-Storage System



Power Construction



Power Construction



Routine Inspection of Transmission Lines



Routine Inspection of Hydropower Equipment



Electrical Equipment Servicing



Offshore Wind Power Phase I Project



Guangming Pumped Storage Hydro Power Project on the Dajia River

HONG KONG SAR



1

Country Information

General Information (as of Dec 2025 or latest data)

Area:	1114.57 km ²
Population:	7.510 million
GDP:	US\$ 426.9 billion
Currency:	HK Dollar
Installed Capacity (in GW):	9258MW (CLP), 3083MW (HK Electric) (as at year-end 2025)
Percentage of population electrified:	100%

Power Demand by Sector

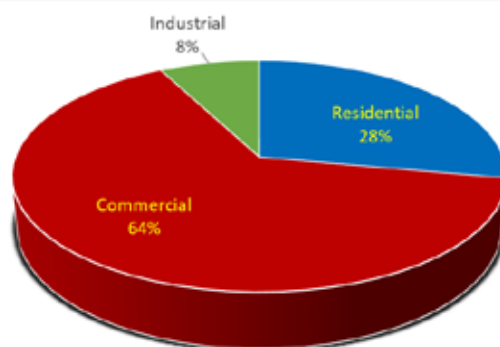
CLP Power

Sector

Total Power Consumption as of 2025 (GWh)

Industrial	2,747.124
Commercial*	23,046.515
Residential	9,966.020
TOTAL	35,759.65812

(*) also include public lighting

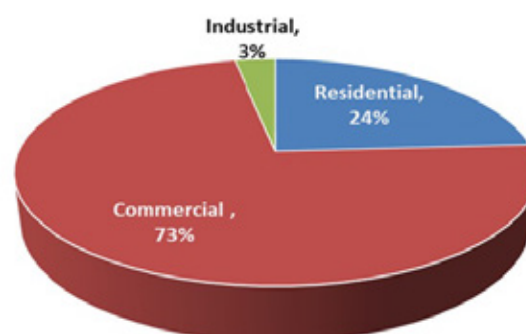


HK Electric

Sector

Total Power Consumption as of 2025 (GWh)

Industrial	277
Commercial	2387
Residential	7252
TOTAL	9916



Energy Policy

According to the HKSAR Government's website (<https://www.eeb.gov.hk/en/energy/energy.html>), the objectives of Hong Kong's energy policy are:

- To ensure that the energy needs of the community are met safely, reliably, efficiently and at reasonable prices; and
- To minimise the environmental impact of energy production and use and promote the efficient use and conservation of energy.

In keeping with Hong Kong's free market economic philosophy, the HKSAR Government (the Government) intervenes only to safeguard the interests of consumers where necessary, ensure public safety and protect the environment. The oil and gas companies, under a voluntary code of practice with the Government, maintain strategic reserves of gas oil and naphtha respectively. The Government monitors the financial and operating performance of the privately-owned power companies through the Scheme of Control Agreements (SCAs). The Government has entered into an Information and Consultation Agreement with the Hong Kong and China Gas Company Limited to make the Towngas tariff adjustment mechanism more transparent. The Government also promotes renewable energy (RE), energy efficiency and energy saving measures in consultation with the power companies.

In January 2017, the Government announced Hong Kong's Climate Action Plan 2030+, outlining the Government's longer-term action in combating climate change and setting out the carbon emissions reduction target for 2030. In 2020, the Government announced that Hong Kong would strive to achieve carbon neutrality before 2050 and then in October 2021 announced Hong Kong's Climate Action Plan 2050, focusing on achieving carbon neutrality before 2050 with an interim target to cut carbon emissions by half by 2035 (compared to 2005 levels). The Government has outlined four major decarbonisation strategies, namely net-zero electricity generation, energy saving and green buildings, green transport and waste reduction.

Supply and Demand of Primary Natural Resources

Hong Kong has no indigenous energy resources. In 2025, the primary energy requirements for coal products (including steam coal and other coal; wood charcoal; and anthracite), oil products (including aviation gasoline and kerosene; unleaded motor gasoline; gas oil, diesel oil and naphtha; fuel oil; LPG; and natural gas) and electricity were 105,121 TJ, 329,169 TJ and 47,075 TJ respectively.

<https://www.censtatd.gov.hk/en/wbr.html?ecode=B11000022023AN23&scode=90>
(Source: Hong Kong Energy Statistics C&SD: Hong Kong Energy Statistics Annual Report.)

Overview of Electricity Market Structure

Hong Kong's electricity demand is met by two vertically integrated power companies, namely CLP Power Hong Kong Limited (CLP Power) and The Hongkong Electric Company Limited (HK Electric), which are both investor-owned. They do not have a franchise. CLP Power supplies electricity to Kowloon, the New Territories, Lantau Island and most of the outlying islands, whereas HK Electric supplies electricity to Hong Kong Island and Lamma Island.

The electricity market in Hong Kong is governed by the Scheme of Control Agreements (SCAs), which define the companies' role as an electricity provider, and provide a regulatory framework for the Government to monitor their operating performance and financial affairs.

Under the regulatory regime, power companies have obligations to provide sufficient and reliable electricity supply in their service areas. Customers obtain quality electricity supply at a reasonable price and in an environmentally responsible manner, while the power companies earn a return which is reasonable in relation to the risks involved and the capital invested. The SCAs also provide an effective and stringent regulatory framework for the Government to monitor power companies' operating and financial performance. Operating performance covers supply reliability, operational efficiency, customer services and energy efficiency. Financial performance covers power companies' capital investment, operating expenditure, permitted rate of return and tariff adjustment.

Laws and Governmental Regulations

The current SCAs with CLP Power and HK Electric have come into effect since 1 October 2018 and 1 January 2019 respectively, and both will run until 31 December 2033.

The current SCAs carry key features including a permitted rate of return for investment, performance targets for supply reliability, operational efficiency, customer services, supply restoration, energy efficiency, demand response and RE development to enhance the service level.

- A permitted rate of return at 8% of the total value of the average net fixed assets.
- The agreements run for 15 years.
- The Fuel Cost Adjustment (FCA) in total tariff is adjusted more frequently to once a month.
- The Feed-in Tariff (FiT) Scheme is in place to encourage the development of RE systems in the community. Renewable Energy Certificates (RECs) offer an alternative for both residential and business customers to support renewable energy without installing their own systems. Business customers can declare their use of renewable energy and reduce the carbon footprint associated with their electricity consumption.
- The incentive and penalty mechanism continues to apply in the current SCAs, but with more stringent performance targets in supply restoration and a new penalty scheme for large-scale electricity supply interruption.
- A new mechanism has been introduced to provide special tariff relief in the event of energy crisis to help targeted customers mitigate the impact of the sharp rise in fuel costs.
- Various funds are in place to promote energy saving for buildings and use of higher energy efficient electrical appliances based on a fraction of incentives ploughed back from the power companies when achieving higher targets on energy audit and energy saved from the initiatives.

The two power companies are subject to a wide range of laws and regulations relating to environmental protection, safety, personal data protection, employment and corporate governance. The key legislations include:

- The Electricity Ordinance (Chapter 406) and the Gas Safety Ordinance (Chapter 51) regulate the safe supply of electricity and gas and the safety of household electrical products and gas appliances. Among other things, they cover registration of generating facilities, contractors and workers for electrical and gas installations, wiring and gas installation standards and safe distribution and use of electricity and gas.
- In relation to environmental protection, the Air Pollution Control Ordinance (Chapter 311), the Waste Disposal Ordinance (Chapter 354), the Water Pollution Control Ordinance (Chapter 358), the Noise Control Ordinance (Chapter 400) and the Environmental Impact Assessment Ordinance (Chapter 499) are the major environmental-related laws and regulations that power companies must abide by.
- In relation to personal data protection, the Personal Data (Privacy) Ordinance (Chapter 486) is established to protect the privacy of individuals in relation to personal data. It stipulates six data protection principles which must be adhered to when dealing with personal data, including employees, contractors, customers, etc.
- In relation to employment, the Employment Ordinance (Chapter 57) sets out the framework for a comprehensive code of employment. It governs the payment of wages, the termination of employment contracts and the operation of employment agencies, as well as provides employment protection for employees. Employees have statutory protection against anti-union discrimination under the Employment Ordinance and against discrimination on the ground of disability, race, sex and family status under the respective legislation, including the Disability Discrimination Ordinance (Chapter 487), the Race Discrimination Ordinance (Chapter 602), the Sex Discrimination Ordinance (Chapter 480) and the Family Status Discrimination Ordinance (Chapter 527).

- The Companies Ordinance (Chapter 622) regulates the incorporation and operation of companies in Hong Kong. It covers the formation and registration of local companies and overseas companies; accounting and auditing provisions; the deregistration of defunct, solvent private companies; the prosecution of companies and their officers for breaches of the various regulatory provisions of this Ordinance; the provision of facilities to inspect and obtain company information.

Amount of Electricity Exported and Imported (in GWh)

In 2025, CLP Power imported 12,681 GWh of electricity from the Mainland China. There was no export to the customers in Mainland China in 2025.

Tariff Structure of Region

Electricity tariff paid by an electricity user comprises two major components: Basic Tariff and Fuel Cost Adjustment (FCA). The Basic Tariff covers the total costs of electricity supply, including operating cost, standard cost of fuels and SCA return; while the FCA reflects either a charge or rebate to cover the difference between the actual cost of fuels spent and the standard cost of fuel collected through the Basic Tariff. The Net Tariff of Hong Kong has been maintained at a relatively stable level under the SCA regime. In addition, Hong Kong's electricity tariff is reasonable and competitive when compared to that of other key major cities in the world.

CLP Power

CLP Power has six tariff categories, namely:

1. Residential Tariff (Residential customers)
2. Non-Residential Tariff (Small and medium enterprises customers)
3. Bulk Tariff (Large businesses and public services with monthly consumption not less than 20,000 units)
4. Large Power Tariff (Largest businesses and public services with monthly demand not less than 3,000 kVA)
5. Electric Vehicle Tariff (fixed-rate for EV charging)

6. Electric Vehicle Residential Time of Use Tariff (offer flexible options for EV charging at residential parking lots)

The tariff structure is designed to be fair and cost reflective for each tariff group of customers, and it therefore avoids cross-subsidies between the customer groups.

The cost of electricity supply to each tariff group takes into account the investment and resources needed to supply them, and the efficiency with which these resources are used. In general, fixed operating costs like metering, billing, customer services, etc. are lower per unit for higher-consuming customers.

For Residential Tariff, an inclining block structure is applied. Under this structure, there are seven blocks with different rates. Higher consumption is charged at a progressively higher unit rate. This encourages the efficient use of energy by residential customers. The lower blocks provide protection for the residential customers with lower household incomes and encourage energy saving. Inclining tariff structures for residential customers are common in many cities world-wide.

CLP Power also offers a Concessionary Tariff for the Elderly aged 60 or above who live alone or with other similarly qualified elderly, and those who are relying on or entitled to Comprehensive Social Security Assistance. The successful applicants will be offered half-price for the first 400 units of electricity consumed in each two-month billing period plus an exemption of the minimum charge per bill. This tariff applies to electricity used for residential purposes only.

Unlike Residential Tariff customers, inclining tariff structures for businesses and public services are uncommon in other cities in the world.

CLP Power's Non-Residential Tariff adopts a fixed rate for all units consumed. High consumption customers under Bulk Tariff and Large Power Tariff categories have a declining tariff structure of two blocks. They have two tariff features:

- They have to pay a Demand Charge in addition to the cost of the energy units they consume. The Demand Charge reflects the capacity of the supply customers draw from CLP Power's network based on their maximum energy demand.

- In addition, under a Time-of-Use tariff feature, they also pay a premium for energy used at peak times but are able to reduce costs if they can move this to off-peak periods. This facilitates demand side management and better utilisation of power generation facilities.

CLP Power has also introduced dedicated tariffs for electric vehicle (EV) charging, including a fixed-rate Electric Vehicle Tariff and a newly launched Electric Vehicle Residential Time of Use Tariff (EV Residential ToU), which offers discounted electricity charges during off-peak period to encourage smart charging behaviour and enhance resource utilisation of the power grid.

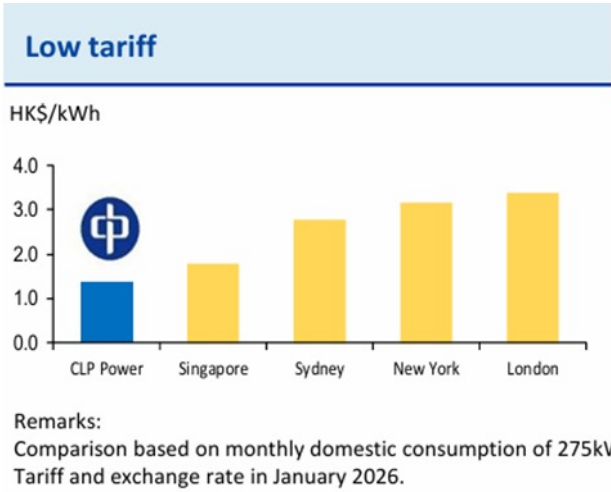
Energy Saving Rebate has been introduced since 2013 to help low consumption residential & small business customers reduce their electricity expenses and encourage energy saving. Residential and small business customers who use 400 units or less per bill can enjoy the rebate.

Total consumption per bill	Rebate Rate (HK cents/Unit)
1 to 200 units	17.2 cents per unit on total consumption
201 to 300 units	16.2 cents per unit on total consumption
301 to 400 units	15.2 cents per unit on total consumption

CLP Power's tariff is highly competitive compared to that of other key metropolitan cities in the world. Without government subsidies, CLP Power continues to provide a highly reliable electricity supply at reasonable tariff levels to support Hong Kong's development. The Average Basic Tariff for January 2026 is 140.6 cents per unit of electricity with a downward adjustment of 2.6% compared with that of 2025.

	January 2026 Rates (HK cents/kWh)
Average Basic Tariff	101.2
Fuel Clause Charge	39.4
Average Net Tariff	140.6

Source: 20251118_en.pdf



HK Electric

HK Electric offers three types of tariffs for different categories of customers, namely Residential Tariff, Non-Residential Tariff and Maximum Demand Tariff. The Net Tariff charged to customers is mainly made up of Basic Tariff and Fuel Clause Charge (FCC). The Basic Tariff is used to recover total operating costs with fuel costs at a standard rate, depreciation and the permitted return. The FCC is used to help recover the actual fuel costs, and the actual fuel costs are passed through to customers without any profits incurred as stipulated by the SCA. Starting from 2019, the Fuel Cost Clause (FCC) has been adjusted on a monthly basis, based on the previous three month average of actual fuel costs, to reflect changes in fuel costs more promptly.

HK Electric had introduced a progressive block tariff structure to its residential customers since the early 1990s to encourage environmental protection through energy conservation and efficiency improvement.

Upon successful implementation in the residential sector, this progressive block tariff structure had been extended to its Non-residential Tariff customers since the early 2000s.

HK Electric introduced a “Super Saver Discount” in 2013. Residential customers with consumption no more than 100 units of electricity in any month will be given a 5% discount on their electricity bills. The discount serves two purposes: (1) to provide incentive for customers to save energy; and (2) to reduce the electricity expenses for low- consumption families.

HK Electric offers the following concessionary tariff schemes for those receiving or eligible to receive Comprehensive Social Security Assistance (CSSA). Successful applicants are entitled to receive a 60% discount for the first 200 units of electricity consumed in a month plus the exemption of the payment of deposit and minimum charge.

- Concessionary Tariff Scheme for the Elderly
- Concessionary Tariff Scheme for the Disabled
- Concessionary Tariff Scheme for Single-Parent Families
- Concessionary Tariff Scheme for the Unemployed

HK Electric announced in the 2026 Tariff Review that, Average Net Tariff for January 2026 will be 163.3 cents per unit of electricity, down by 3.7 cents, or decrease of 2.2% compared to January 2025.

The Net Tariff is composed of the Basic Tariff and the Fuel Clause Charge. After considering factors like capital expenditures and operating expenses, the Basic Tariff for 2026 will be adjusted to 127.9 cents per unit of electricity.

Meanwhile, Fuel Clause Charge for January 2026 will decrease by 8.7 cents to 35.4 cents per unit of electricity, in light of the current situation of the fuel market.

Adjustments from the period of January 2025 to January 2026 are shown below:

Tariff Component	Change in HK cents/unit
Basic Tariff	122.9 → 127.9
Fuel Clause Charge	44.1 → 35.4
Average Net Tariff	167.0 → 163.3

(Source: https://www.hkelectric.com/documents/en/MediaResources/PressReleases/Documents/20251118_pre_TR%202026_final_website.pdf)

In 2026, HK Electric sets aside more than HK\$80 million in 2026 for “Smart Power Services”. Initiatives include continuing to distribute cash coupons to families in need, funding energy efficiency improvements for buildings and supporting underprivileged households in acquiring energy-saving appliances. The company will also launch a Renewable Energy Trial Programme on Po Toi Island.

Since the outbreak of military conflicts between the United States, Israel and Iran from end February 2026 and the resulting escalation of tensions in the Middle East, global energy markets have been highly volatile, driving a sharp increase in fuel prices. As of mid-May 2026, the time of preparation of this report, marine vessels remain unable to transit safely through the Strait of Hormuz, thereby affecting the global fuel supply. Such market situations have been posing challenges to HK Electric and its peer utilities around the world. Due to an inherent deferred effect of the adjustment mechanism, the higher fuel costs are expected to take effect in HK Electric’s Fuel Clause Charge starting from mid-2026, at which point the charge may rise significantly. The actual developments are however highly uncertain and need to be observed.

For context, HK Electric’s power generation relies primarily on natural gas and coal. Given the ongoing global uncertainties, the Company continues to closely monitor energy markets and supply conditions, manage fuel inventory, generation operations and emissions prudently amid the challenges. A range of contingency measures have been put in place, including adjustments to the fuel mix and activation of standby coal-fired units as necessary, to ensure that electricity supply to more than 590,000 customers on Hong Kong Island and Lamma Island remains stable and reliable.

Electricity Subsidies Among Generation, Transmission and Distribution Sectors in the Region (Including Renewable Electricity)

To alleviate the impact of tariff increase on households during the transition to a low-carbon future, the Government has been granting each eligible residential electricity account a maximum electricity charges relief of HK\$1,200 over 24 months (i.e. HK\$50 per month) from January 2024 to December 2025 under the Electricity Charges Relief Scheme (2024-2025). The Scheme ended in end 2025. Any unused relief can be carried forward to offset electricity charges under the same account until 31 December 2026 or the close of the account, whichever is earlier.

Source: Environment and Ecology Bureau - Electricity Charges Relief Scheme (<https://www.eeb.gov.hk/en/energy/ecrs.html>)

CLP Power

Investments	Description	Fuel Type
Castle Peak Power Company Limited (CAPCO)	CAPCO owns and CLP Power operates (as of Dec 2025): Black Point Power Station (3,850MW) One of the world's largest gas-fired combined-cycle power stations comprising eight 337.5MW units and two new units with capacities of 550MW and 600MW respectively	Gas
Coal	Castle Peak Power Station (3,058MW) Comprising four 677MW coal-fired units, two of which can also use natural gas. One 350 MW coal-fired unit is available for emergency purposes. All units can use diesel as a backup fuel.	Coal
Oil	Penny's Bay Power Station (300MW) Three diesel-fired gas turbine units of 100MW each, mainly for backup purposes.	Oil
Gas	West New Territories Landfill Power Generation Units (14MW) Comprising seven 2MW units which utilise landfill gas produced locally at the landfill site as fuel for power generation	Landfill Gas

Note: CLP Power purchases its power from CAPCO, PSDC and GNPS. In 2025, CLP Power has a total installed capacity of 9,258MW available to serve the Hong Kong electricity business.

Investments Gross / Equity MW	Description	Fuel Type
Guangdong Nuclear Power Joint Venture Company, Limited (GNPJVC) 2,052 / 513 MW	GNPJVC, a joint venture between CLP and China General Nuclear Power Corporation, owns the Daya Bay Nuclear Power Station (Daya Bay) at Guangdong Province. Daya Bay is equipped with two 1,026MW pressurised water reactors. Currently, around 80% of Daya Bay's electricity generated is supplied to Hong Kong.	Nuclear
Hong Kong Pumped Storage Development Company, Limited (PSDC) 1,200 / 600 MW	PSDC holds the right to use half of the 1,200MW pumped storage capacity of Phase 1 of the Guangzhou Pumped Storage Power Station until 2034.	Hydro

In support of Hong Kong Government's decarbonisation strategy to increase the use of natural gas for power generation as part of the city's energy transition, CLP Power and HK Electric have jointly developed an offshore liquefied natural gas (LNG) terminal in Hong Kong waters which went into service in July 2023. This infrastructure provides a long-term alternative source to meet Hong Kong's need for fuel supply, which is crucial for enhancing the security of the city's natural gas supply. It also gives the two power companies direct access to international LNG markets, strengthening their ability to source competitively-priced gas. The technology of Floating Storage and Regasification Unit (FSRU) is used to regasify LNG before it is supplied to CLP Power's Black Point Power Station and HK Electric's Lamma Power Station through two separate subsea gas pipelines.

HK Electric

The maximum demand in HK Electric's system was 2,190 MW in 2025. HK Electric has a total installed generating capacity of 3,083MW as at end 2025, consisting of three coal-fired units, five oil-fired simple-cycle gas turbine units, and four gas-fired combined-cycle units, which are 1-on-1 single shaft machines commissioned in 2006, 2020, 2022 and 2024 respectively.

There is also a single 800 kW wind turbine, the Lamma Winds, located on Lamma Island a few kilometres from the power station. The Lamma Winds is the first grid-connected commercial wind turbine in Hong Kong. Since its official opening in February 2006, the turbine and its adjacent exhibition centre have become a major landmark on Lamma Island, attracting numerous visitors from professional, environmental, educational and community groups.

Following the development of Hong Kong's first commercial wind turbine in 2006, HK Electric marked another milestone in renewable energy by commissioning the city's largest solar power system at Lamma Power Station in July 2010, with an installed capacity of 550 kW. In view of its remarkable performance, the system was expanded from 550 kW to 1 MW in 2013. With further upgrades and replacement of individual modules in recent years, the installed capacity of solar power system at Lamma Power Station has been further increase to 1.5 MW in 2025.

In addition to Lamma Winds and the solar power system mentioned above, HK Electric has also commissioned smaller-scale renewable energy installations across various HK Electric's premises, with a combined capacity of around 1.0 MW as of the end of 2025.

After nearly two decades of service, Lamma Winds has reached the end of its service life and ceased operation in the first quarter of 2026, marking the completion of its mission as HK Electric continues to advance renewable energy development. Given that the equipment had reached the end of its designed service life, and in consideration of safety and risk management, Lamma Winds has been permanently closed with effect from 11 May 2026.

Looking ahead, HK Electric will continue to support and promote renewable energy and work closely with the Government in exploring the introduction of zero-carbon electricity to our supply areas through

regional cooperation.

Coal is the primary fuel of the three coal-fired units while oil is only used during start-up, shutdown or flame stabilization. In 2025, coal consumption was around 1.3 million metric tonnes (i.e. around 70% decrease compared with around 4.3 million metric tonnes in 2005 when no gas-fired combined-cycle unit was yet available). The oil-fired simple-cycle gas turbines serve as peak lopping units and also as back-up power supply during emergencies.

The single shaft gas-fired combined-cycle units, designated as Lamma Unit L9 to L12, are located at Lamma Power Station Extension site. Commissioned in October 2006, 2020, 2022 and 2024, they are the first combined-cycle unit in Hong Kong adopting re-gasified liquefied natural gas (LNG), one of the cleanest fuels available today. In 2025, the power output generated from natural gas increased to around 69% of HK Electric's total output, and carbon dioxide emissions decreased by about 42% as compared with 2005 just before natural gas was introduced at Lamma Power Station. To prepare for the retirement of coal-fired units and further decarbonise its fuel mix, HK Electric has commissioned two more new gas-fired generating units, namely L11 and L12, at Lamma Power Station in April 2022 and end March 2024 respectively. HK Electric has also been constructing one more new gas-fired generating units, namely L13, at Lamma Power Station, which is planned to be commissioned in early 2029.

To enhance the reliability of natural gas supply to Lamma Power Station, HK Electric has partnered with CLP Power to develop an Offshore LNG Terminal located in Hong Kong Waters about 4 km east of Tau Lo Chau using the Floating Storage and Regasification Unit (FSRU) technology was commercially operated in July 2023. The Terminal is of double-berth design which can accommodate a Q-Max class FSRU and another LNG carrier. A new 18 km subsea pipelines connecting HK Electric's Lamma Power Station with the Terminal provides a new channel for receiving cost-competitive LNG from different parts of the world, thereby enhancing the security of HK Electric's gas supply sources. The terminal went into service in July

2023.

Re-gasified natural gas is being supplied from the Guangdong Dapeng LNG Terminal located in Shenzhen via a 92-km submarine gas pipeline connecting the LNG terminal and Lamma Power Station Extension since 2006. In July 2023, the Floating Storage and Regasification Unit (FSRU), Bauhinia Spirit, and the Hong Kong Offshore LNG Terminal were successfully commissioned and began commercial operations. Re-gasified natural gas is being supplied from the Hong Kong Offshore LNG Terminal via an 18-km submarine gas pipeline connecting the terminal to the Lamma Power Station Extension, as the second source of supply of natural gas.

Following the use of high-quality fuels combined with the effective performance of emission reduction facilities, the sulphur dioxide (SO₂), nitrogen oxides (NO_x) and respirable suspended particulates (RSP) emissions from Lamma Power Station in 2025 were significantly reduced by 85% to 97% as compared with 2005.

CLP Power

400kV
132kV
22kV
11kV
380kV

HK Electric

275kV
132kV
22kV
11kV
380kV

**A Brief on Outlook Plan for the Next Decade,
Such as Long Term Power Development Plan,
Transmission Plan and Distribution Plan**

CLP Power

CLP Power strives to provide customers with a world-class supply reliability, an improving environmental performance, and reasonable tariffs. The company remains dedicated to supporting Hong Kong to achieve carbon neutrality before 2050, despite the challenges of extreme weather, uncertain global economic outlook, and volatile international fuel markets.

The current SCA took effect on 1 October 2018.

CLP Power is committed to working with customers to put the new SCA into effect and to promoting RE and greater energy efficiency. The FiT Scheme and RECs were introduced in October 2018 and January 2019 respectively to support RE development. The former shortens the payback period for RE projects, while the latter provides an alternative way for customers to support the development of local RE without installing their own systems. The Eco Building Fund, originally established in 2014 for residential buildings to carry out energy efficiency improvement works in the communal area was extended to cover eligible commercial and industrial buildings with effective from 1 October 2018.

The Community Energy Saving Fund (CESF) also commenced operations in January 2019, with the aim of launching various community support programmes that address the evolving needs of the community each year. In 2025, CLP Power has allocated HK\$270 million from the CESF to roll out programmes in 2026 that support the needy, stimulate the local economy and promote energy conservation. Key initiatives include an allocation of HK\$79 million for the Community Green Programme. This programme comprises a range of tailored initiatives aimed at promoting energy efficiency and decarbonisation among different target groups. Measures include providing subsidies to support underprivileged households in replacing high energy-consuming appliances, such as window-type inverter air conditioners, electric water heaters, refrigerators, and washing machines, with models that carry Grade 1 energy labels.

Another HK\$60 million is dedicated to the CLP Retail and Catering Coupons to boost consumer spending and economic momentum, while HK\$50 million will go towards electricity subsidies to alleviate the economic burden on 73,000 underprivileged households including those living in subdivided units.

The 2024-2028 Development Plan focuses on a range of investments to underpin Hong Kong's accelerating economic and infrastructural development, continued delivery of a world-class reliable electricity system, the transition to a smart and resilient city, and continuation of the decarbonisation journey.

In a fast-changing world, CLP Power will capitalise on technological advances to continually enhance its operations and to provide better value for customers. Examples include a mass rollout of smart meters, intelligent management system Grid-V and making use of data analytics, drones and robotics. The company also continues to work closely with various stakeholders and support the community in practical and meaningful ways, providing environmental education, supporting youth development, and helping disadvantaged people in the community. As Hong Kong becomes a world-class smart city, it needs a smart utility to support and facilitate the transformation. CLP Power is ready to take that role.

In view of the growing demand for power expertise arising from large-scale infrastructure developments, CLP Power launched the CLP Power Academy (the Academy) in 2017 to establish industry-recognised professional training courses for those who aspire to develop their career in power industry.

The Academy works in partnership with various educational institutions in Hong Kong, the Chinese Mainland and overseas to offer a variety of part-time accredited electrical and mechanical engineering programmes, ranging from diplomas and professional diplomas to Bachelor's and Dual Master's degrees. The Academy aims to provide an alternative for young people and working adults to pursue professional knowledge in power engineering to enhance their career prospects and professional standing.

In 2022, the Academy launched the first-ever Hong Kong-Guangzhou training courses in partnership with Vocational Training Council and Guangzhou Industry and Trade Technician College (GITTC). The training courses for high voltage and low voltage electrical workers help local young talents and industry practitioners obtain work permits in the Greater Bay Area.

In addition, CLP established an education fund in 2009 to provide scholarships for financially disadvantaged students in the Department of Electrical Engineering at Tsinghua University, supporting both academic research and practical training. In 2025, CLP became Tsinghua University's first "Undergraduate Practice Training Base" in Hong Kong, setting a benchmark for deep integration of industry, academia, and research, and contributing to the training of talent for the nation's energy sector.

HK Electric

HK Electric will continue with clean energy transition while investing to ensure the highest standards of supply reliability under the new 2024-2028 Development Plan. Key capital projects under the 2024 - 2028 Development Plan include the replacement of an ageing coal-fired unit with a new gas-fired generating unit, L13, to maintain sufficient generation capacity while reducing carbon and other emissions further. With appropriate future modifications and repurposing, our gas-fired units are expected to be capable of operating on both natural gas and hydrogen, enabling us to explore the use of green hydrogen as a zero-carbon energy source in Hong Kong. To ensure reliability, three oil-fired open-cycle generating units with fast start-up capability will be constructed to replace four similar units which have been in service for nearly 50 years. In addition, the service life of two coal-fired units will be extended to optimise capital investment pending the possible import of zero-carbon electricity into our system.

In 2025, HK Electric continued to fully comply with the emission allowances for SO₂, NO_x and RSP specified by the Government.

For 29 consecutive years since 1997, HK Electric has upheld a supply reliability rating exceeding 99.999%, including achieving 99.9999% or above in multiple years since 2017. In 2025, HK Electric achieved an exceptional reliability rating of over 99.9999% again, with an average unplanned Customer Minutes Lost of less than half a minute. HK Electric mainly uses underground and submarine cables for transmission and distribution of electricity thereby minimising disruption during typhoons and lightning. In addition, the main power plants and coastal transmission and distribution stations are equipped with flood-resilience measures.

4

Reduction of Carbon Emission

The Policy of Renewable Energy and Major Regulatory Mechanism of Renewable Law

The HKSAR Government is committed to the development of renewable energy (RE) in Hong Kong with a view to further improving Hong Kong's air quality

(Source: https://www.eeb.gov.hk/en/energy/renewable_energy.html).

The Government estimated that Hong Kong has about 3-4% of realisable renewable energy potential arising from wind, solar and waste-to-energy that can be exploited by 2030. In the Hong Kong's Climate Action Plan 2050 published in October 2021, a RE target of 7.5-10% by 2035 was proposed by the Government, which would be further increased to 15% subsequently

(Source: <https://www.climate-ready.gov.hk>).

Production of Renewable Energy (MWh) from various resources (wind, solar, biomass, landfill gas etc.)

Given Hong Kong's densely populated urban environment and the nature of its terrain, there is limited potential for Hong Kong to launch significant land-based renewable energy projects.

CLP Power

In 2010, CLP Power implemented Hong Kong's first commercial-scale standalone RE generation (Solar & Wind: 192kW) and storage system (battery: 1MWh) on Town Island. The Project was named one of the "Hong Kong People Engineering Wonders in the 21st Century" in a prestigious public vote organised by the Hong Kong Institution of Engineers.

Despite limited RE resources and land scarcity in Hong Kong, CLP Power has been actively exploring practicable RE opportunities. CLP Power has developed the WE Station at the West New Territories (WENT) Landfill. It is the largest landfill gas power generation project in Hong Kong, which makes use of landfill gas produced locally at the landfill site for power generation and electricity generated is transmitted to CLP's grid. The first phase of WE Station, comprising five generation units with a total generation capacity of 10MW, went into operation in March 2020. The second phase, comprising two additional generation units with a total capacity of 4MW, commenced operation in October 2024.

Under the current SCA, CLP Power has introduced Feed-in Tariff (FiT) Scheme and Renewable Energy Certificates (RECs) to promote community participation in RE development. The FiT Scheme incentivises people and businesses to develop renewable energy projects by shortening the payback period of their investment. CLP Power remains committed to facilitating easy grid connection for these projects.

At the end of 2025, 449MW of capacity had been approved under the FiT scheme, up from 404MW a year earlier. Meanwhile, over 1800 solar panels were installed on the rooftops of some buildings in CLP Power's generation plants and stations.

RECs allow customers to support the local RE development in a simple and flexible way. In 2025, close to 363GWh units of RECs were sold, a significant increase from the 340GWh units sold in the past year. Some large commercial customers including AirTrunk Hong Kong Holding Limited and Citi Hong Kong Limited have signed multi-year REC agreements.

HK Electric

In 2025, the solar power systems at Lamma Island and the Lamma Winds generated a total of 1.83 million units of green electricity.

In response to stakeholders' expectation for a wider application of renewable energy (RE) in Hong Kong, HK Electric launched a series of Smart Power Services including the Feed-in Tariff (FiT) Scheme and Renewable Energy Certificates (REC) Scheme, which target to increase the uptake of RE in the supply territory of HK Electric.

The FiT Scheme was rolled out in August 2018 for application, under which HK Electric will purchase electricity generated by grid-connected renewable energy power systems (REPS) (solar photovoltaic and/or wind power systems) of customers at FiT rates determined by the Government. The FiT rate will be fixed from the date when the customer participates in the FiT Scheme throughout the project life of the REPS or until 31 December 2033, whichever is earlier. By the end of 2025, over 750 RE systems with a capacity of about 15 MW have been successfully connected to the grid and started receiving FiT payment.

HK Electric also offers RECs for customers to purchase for achieving RE/environmental targets while supporting the local RE development and take advantage for offsetting their GHG emissions generated from their electricity use. REC represents all environmental attributes associated with the electricity generated from local solar photovoltaic and wind systems which are either owned by HK Electric or its customers. REC is priced at \$0.5 per unit of electricity.

Under the current SCA, HK Electric allocates about HK\$5.0 million every year to its “Smart Power Education Fund” to promote energy efficiency and conservation (EE&C), low-carbon lifestyle and the application of renewable energy (RE) through education and promotional activities. These include the anchor green education programme, Happy Green Campaign, community-based energy efficiency education and sponsorship of various EE&C initiatives organised by the Government, professional bodies and green groups.

For over 20 years, Happy Green Campaign has promoted EE&C, RE and low-carbon lifestyle through different activities including competitions, visits and workshops with emphasis on engaging the younger generation. To date, more than 1.3 million people have participated in its programme and over 800 schools across Hong Kong have joined the campaign’s “Happy Green School” network, taking part in various Other Learning Experience (OLE) activities such as story-telling workshops on decarbonisation, SDG-themed aerobics exercise workshops and upcycling workshops, all designed to promote green messages in a fun and interactive way.

The annual Green Energy Dreams Come True Competition showcases the innovative spirit of local students. The competition encourages secondary and tertiary students to realise their green energy dreams, promoting decarbonisation, EE&C and RE. Winning teams are invited to join a study tour to Chinese Mainland starting in 2025 to explore the latest developments in green energy and the application of new technologies, while fostering cultural exchange and gaining a deeper understanding of the nation’s advancements. An Alumni Club has been set up to encourage experience sharing and exchange of ideas.

In addition to promoting green education among students, HK Electric, in collaboration with The Conservancy Association, organises the “Green Hong Kong Green” Eco-Heritage Tour for the general public. Through guided tours and experiential activities, the programme showcases green initiatives across different sectors of Hong Kong society, while promoting cultural preservation and raising awareness of the city’s rich biodiversity. By connecting participants with both the natural and cultural heritage of Hong Kong Island, the tours encourage sustainable living and foster a deeper appreciation of nature within its urban landscape.

Environmental Problems of Electricity Supply (Including Generation, Transmission and Distribution Sectors) and How They are Being Addressed

At present, the Government monitors the ambient concentrations of the following pollutants: sulphur dioxide (SO₂), nitrogen oxides (NO_x), respirable suspended particulates (RSP/PM₁₀), fine suspended particulates (FSP/PM_{2.5}), Ozone (O₃) and carbon monoxide (CO).

As part of the measures intended to improve local air quality in Hong Kong, the Government has been steadily tightening the emissions levels allowed for power stations in Hong Kong.

So far, ten Technical Memoranda (TMs) were issued. The First TM set the emission allowances for the emission years between 2010 and 2014. The emission allowances for the Second to Ninth TMs took effect from 1 January 2015, 1 January 2017, 1 January 2019, 1 January 2020, 1 January 2021, 1 January 2022, 1 January 2024, 1 January 2026 respectively while those for the Tenth TM will take effect from 1 January 2030. Despite an increase in the projected electricity demand from 2010 to 2026, the emission allowances of SO₂, NO_x and RSP set in the Ninth TM and informed, in early 2024, that the emission caps set by the Ninth TM for power plants would be still applicable for 2028 and 2029. The Government initiated the review of the Ninth TM in November 2024 with an aim to promulgate a Tenth TM in 2025 for allocation of emissions allowance in 2030 and beyond to further tighten the emissions of power sector.

The 24-hour AQO for SO₂, the 24-hour and annual AQOs for PM₁₀ and PM_{2.5} were tightened from the prevailing IT-2 level to the more stringent IT-3 level of the WHO AQGs. Three new AQOs were introduced, namely 24-hour AQO for NO₂ and peak season AQO for ozone set at IT-1 level, and 24-hour AQO for CO set at AQG level of the WHO AQGs.

To preserve the integrity of Environmental Impact Assessment (EIA) system, when implementing the new AQOs, a transitional arrangement would be included for designated projects with Environmental Permits (EPs) granted under the Environmental Impact Assessment Ordinance before the commencement of the new AQOs. For these projects, the prevailing AQOs which took effect on 11 April 2025, will apply to an application for a variation of an EP lodged within 36 months of the commencement of the new AQOs.

In November 2020, Hong Kong pledged to achieve carbon neutrality before 2050. The Chief Executive of the HKSAR announced that a variety of measures will be implemented to meet the carbon neutrality target, such as exploring new environmental-friendly technologies and advocating the use of electric vehicles (EVs). The Government announced the Hong Kong Roadmap on Popularisation of EVs (the Roadmap) in March 2021, setting out the long-term policy objectives and plans to promote the adoption of EVs and their associated supporting facilities in Hong Kong to attain zero vehicular emissions before 2050. In February 2026, the Government announced an Updated Version of the Updated Roadmap) outlining the latest strategies and measures to further accelerate EV popularisation.

In October 2021, the Government released its Climate Action Plan 2050 which sets out a clear roadmap for achieving neutrality before 2050, and commits to a more aggressive medium-term target to reduce Hong Kong's total carbon emissions by half against 2005 levels before 2035. The 2050 carbon neutrality target will have major implications not just for the electricity industry but also for Hong Kong as a whole.

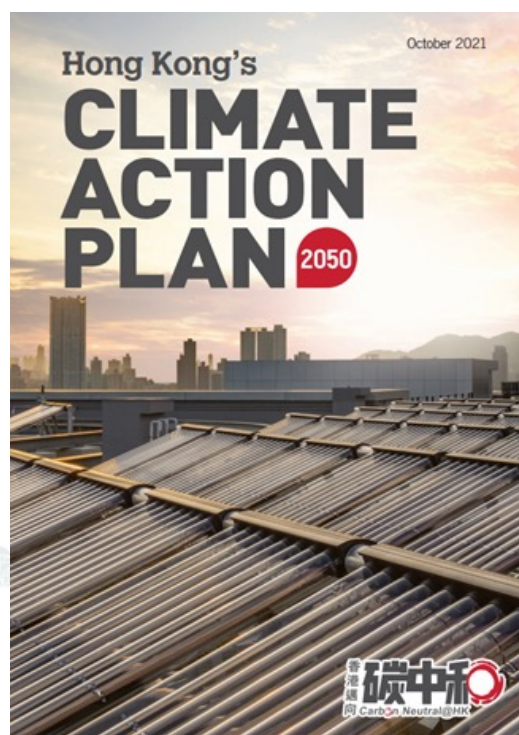
The Plan outlined the strategies and targets for combating climate change and achieving carbon neutrality before 2050. Leading Hong Kong towards the goal of carbon neutrality, the new plan outlines the four major decarbonisation strategies and measures,

namely net-zero electricity generation, energy saving and green buildings, green transport and waste reduction. Along the 2050 carbon neutrality pathway, vigorous interim targets are set to reduce Hong Kong's carbon emissions by 50% before 2035 as compared to the 2005 level.

Local electricity generation was Hong Kong's largest source of carbon emissions (61%) in 2024, followed by transport (18%) and waste (8%). As such, the local power sector must take lead in facilitating Hong Kong in timely achieving decarbonisation goals.

For the net-zero electricity generation, the Hong Kong's Climate Action Plan 2050 outlines the following objectives:

- Cease using coal for daily electricity generation by 2035;
- Increase the share of RE in the fuel mix for electricity generation to 7.5 % to 10 % by 2035, and to 15 % subsequently;
- Explore the use of various zero-carbon energy and strengthen regional co-operation to achieve the long-term target of net-zero electricity generation before 2050; and
- Seek investment opportunities and participate in & operate zero-carbon energy projects near Hong Kong



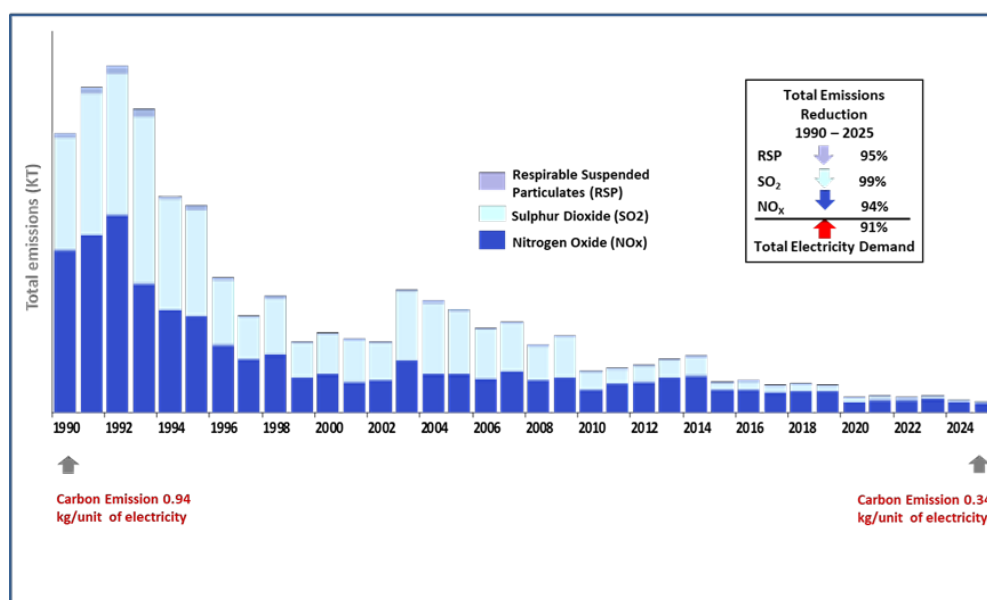
The Hong Kong's Climate Action Plan 2050 was published in October 2021

CLP Power

CLP Power manages the environmental impact of electricity generation responsibly, adopts the world's best practices and has established effective environmental management systems which conform to the globally recognised ISO 14001 standard. Over the years, CLP Power has diligently made effort to manage carbon and air emissions in its operations.

The emission allowances of CLP's power stations have been progressively tightened over time. The emission caps for 2024 and 2025 require CLP Power to further reduce air pollutant emissions by 26% to 49%

compared with the permitted level in 2022. CLP Power has successfully met the increasingly stringent emissions caps for its power stations set by the Government through a combination of emission reduction technologies and changes to the fuel mix, including the introduction of natural gas, nuclear power, low-emission coal and the addition of sophisticated emissions control facilities. CLP Power's emissions in 2025 have been reduced by over 90% since 1990, while electricity demand has grown by over 90% during the same period.



The second new gas-fired generation unit (Unit D2) at Black Point Power Station went into operation in April 2024, further enabling CLP Power to support the Government's target of increasing natural gas use to around 50% of Hong Kong's fuel mix for power generation from 2020 in order to achieve the carbon reduction target and improve air quality. In 2025,

gas generation accounted for over 50% whilst coal generation was maintained at less than 15% of our fuel mix. Furthermore, the landfill gas power generation units at the West New Territories Landfill also contributes to Hong Kong's transition to a low-carbon electricity supply.

HK Electric

HK Electric installed three new gas-fired combined cycle generating units L10, L11 and L12 which were commissioned in 2020, 2022 and 2024 respectively, to replace the retiring coal-fired and gas-fired units. The three units, each equipped with a Selective Catalytic Reduction System with nitrogen oxide removal efficiency of 90%. After the commissioning of L11 in 2022, HK Electric's gas generation ratio increased and met the Government's fuel mix and environmental targets. After the commissioning of unit L12 in 2024, the total gas-fired electricity generation further increased to around 69% in 2025.

To maintain supply reliability while increasing gas-firing generation capacity, it is imperative to secure the supply chain. Hong Kong at present lacks storage facilities for natural gas imported by sea, which also affects the local power industry's negotiating power with international gas suppliers. HK Electric commissioned an offshore LNG terminal using Floating Storage and Regasification Unit (FSRU) technology together with CLP Power in July 2023 to help enhance the security and cost competitiveness of gas supply.

Plans and Strategies to Reduce Greenhouse Gas Emissions

In 2017, the Government published the Hong Kong's Climate Action Plan 2030+ which set out ambitious carbon intensity target of 65% to 70% by 2030 using 2005 as the base, that is equivalent to 26% to 36% absolute reduction and a reduction to 3.3 - 3.8 tonnes on a per capita basis.

From June to September 2019, the Council for Sustainable Development (SDC) was appointed by the HKSAR Government to convene a Public Engagement (PE) on Hong Kong's Long-term Decarbonisation Strategy. The PE aims to strengthen public awareness of the impact of carbon emission and gauge public views of the community in development long-term decarbonisation strategies. In November 2020, SDC announced that over 70,000 views collection forms received and were used to formulate a comprehensive Council Report. The Council Report put forward a total of 55 recommendations in eight key areas, including:

A. Target; B. Lifestyle; C. Education, Training and Research; D. Built Environment; E. Energy; F. Transport; G. City Planning and Management; and H. Finance.

(Sources: Council for Sustainable Development – Report on Public Engagement on Hong Kong's Long-term Decarbonisation Strategy.
Links: [lds_sdc_e.pdf](#))

In 2021, the Government announced Hong Kong's Climate Action Plan 2050, which outlined four major decarbonisation strategies, namely “net-zero electricity generation”, “energy saving and green buildings”, “green transport” and “waste reduction”, that would lead Hong Kong towards the goal of carbon neutrality before 2050. It also sets out a more vigorous interim decarbonisation targets to reduce Hong Kong's carbon emissions by 50% before 2035 as compared to the 2005 level.

CLP Power

The decarbonisation of Hong Kong's power supply took a major step forward with the launch of a new offshore LNG terminal in the summer of 2023. The floating storage and regasification unit (FSRU) vessel, named Bauhinia Spirit, which is moored at the jetty of the terminal is the largest of its kind in the world, increasing our access to diversified supply of competitively priced natural gas from international markets. In addition, the two gas-fired generation units (Units D1 and D2) at Black Point Power Station were commissioned in 2020 and 2024, playing a key role in facilitating the gradual retirement of coal-fired generation units at Castle Peak A Power Station. CLP Power also completed the upgrade works of the Clean Energy Transmission System (CETS), which will enhance the reliability and transmission capacity of the existing cross-border transmission overhead line circuits connecting CLP Power's grid and Shenzhen, providing Hong Kong with greater flexibility to utilise more zero-carbon energy and supporting Hong Kong's long-term decarbonisation.

Separately, the 2024-2028 Development Plan allows for CLP Power to establish a grid-scale battery energy storage system (BESS) to help reduce peak system demand and facilitate the integration of energy from non-fossil fuel sources. Additionally, we a pilot project is planned to explore the feasibility of using hydrogen-blended natural gas for power generation at Black Point Power Station.

HK Electric

HK Electric acknowledges the urgency of reducing greenhouse gas emissions to mitigate the worst effects of climate change. Currently, HK Electric continues to switch from coal-fired to gas-fired generation as a transitional decarbonisation measure and increases local renewable energy generation. HK Electric will continue to phase out coal-fired generation progressively, with the objective of ceasing daily coal-fired generation entirely by 2035. In 2022, HK Electric updated carbon reduction target and committed to reducing Scope 1 greenhouse gas emissions by 68.4% per kWh of electricity generated by 2035 from a 2019 base year. This mid-term carbon intensity target has been validated and approved by the Science Based Targets initiative (SBTi). According to the latest available information, Hong Kong's total greenhouse gas (GHG) emissions in 2024 decreased by 1.3 million tonnes as compared with the previous year. In the same period, HK Electric's GHG emissions dropped by more than 0.59 million tonnes, contributing to more than 45% of the city's total reduction in GHG emissions. In long-run, HK Electric targeted to achieve net-zero electricity generation before 2050.

The Policy and Regulations of Energy Conservation and Efficiency

The Government's policy objective related to energy efficiency and conservation is to increase and sustain conservation of energy in order to reduce the growing trend of energy use. To achieve this objective, the Energy Efficiency Office (EEO) was set up under the Electrical and Mechanical Services Department (EMSD) in 1994 to spearhead and coordinate Government's efforts to promote energy efficiency and conservation. Over the years, the Government has developed standards and guidelines and implemented different programmes to promote the community's understanding and awareness and to take appropriate action in achieving energy saving. Some key initiatives include the implementations of the Mandatory Energy Efficiency Labelling Schemes, the Buildings Energy Efficiency Ordinance and the Scheme on Fresh Water Cooling Towers, the launch of the Energy Consumption Indicators and Benchmarks as well as the Energy Saving for All Campaign, etc.

(Source:https://www.eeb.gov.hk/en/energy_efficiency.html)

In 2015, the Government unveiled the first-ever Energy Saving Plan for Hong Kong's Built Environment 2015~2025+, which sets a new target of reducing Hong Kong's energy intensity (energy demand per unit of economic output) reduction by 40% by 2025 using 2005 as the base (or about 6% actual reduction in energy consumption in 2012). It analyses energy use in Hong Kong and sets out the policy, strategy, target and key actions that can help Hong Kong achieve the new target.

(Source:<https://www.eeb.gov.hk/en/energysavingplan.html>)

CLP Power

CLP Power is firmly committed to energy efficiency and conservation. We encourage our residential and business customers and the Hong Kong community to use energy more efficiently. We provide tools and organise activities for residential customers to raise their environmental awareness and change their energy consumption habits. On the other hand, we have introduced a series of measures for business customers to help them reduce energy use and save costs in their operations, hence accelerating their decarbonisation journey, and together creating a greener and smarter city.

CLP Power encourages residential customers to save energy through an integrated approach combining digital tools, incentives and behavioural engagement. Through the CLP App, customers can monitor their electricity usage with smart meter data, access personalised energy-saving tips and manage their bills, making consumption more visible and actionable. At the same time, the Eco-Grade programme gamifies energy saving by rating households (Green, Silver, Bronze) based on their consumption and green actions, with higher grades linked to greater rewards, encouraging continuous improvement. Customers are further incentivised through Domeo points, which are earned by reducing energy use, adopting digital services or participating in programmes, and can be redeemed for vouchers, lifestyle offers and energy-efficient products. These are reinforced by initiatives such as Power Connect and various campaigns that combine education, activities and rewards to promote low-carbon habits. Overall, CLP drives energy saving by making usage transparent, rewarding behavioural change and creating an engaging, gamified customer journey towards a more energy-efficient lifestyle.

For business customers, CLP Power provides an energy audit service to assess their energy efficiency performance. This helps them identify savings opportunities and areas in need of improvement. Under the SCA starting from 1 October 2018, CLP Power targets to offer 600 energy audits a year to business customers, with total electricity saved expecting to reach 48GWh each year.

Under the current SCA, two important energy efficiency funds are in place. The CLP Eco Building Fund promotes energy saving for buildings to carry out energy efficiency improvement works in communal areas, in which around 48 GWh of electricity was saved by over 600 residential and commercial buildings in 2025. Whereas the Electrical Equipment Upgrade Scheme under CLP Community Energy Saving Fund aims to subsidise business customers, in particular small and medium enterprises, to replace or upgrade the lighting and air-conditioners to more energy-efficient models.

In 2025, the total energy saving from Electrical Equipment Upgrade Scheme was around 24GWh.

Since the launch of the CLP Electrical Equipment Upgrade Scheme in 2019, over HK\$180 million in subsidies has been offered to C&I customers for replacing or upgrading their electrical equipment to more energy-efficient models.

Electrification is transforming transport in Hong Kong and CLP Power continued to step up service innovation and strengthen partnerships in the EV ecosystem. To encourage off-peak charging and optimise power grid utilisation, CLP Power introduced the EV Residential Time of Use Tariff in May 2025 with discounted energy charge during off-peak period.

CLP Power continued to leverage its power expertise to facilitate the implementation of the Government's EV-charging at Home Subsidy Scheme (EHSS). Comprehensive technical support and services from CLP Power including preliminary power supply capacity assessments, advice on electricity meter installations and technical seminars for EHSS consultants and contractors have enabled more private residential estates in Hong Kong to install EV charging facilities.

Following the launch of the Government's Fast Charger Incentive Scheme which aims to install 3,000 fast chargers across the city by 2028, CLP Power has identified around 8,000 potential locations to facilitate interested charge point operators to accelerate their installation of fast chargers. Also as part of the scheme, CLP Power is also working closely with bus operator to upgrade power supply to facilitate the installation of over 170 additional EV charging guns at bus depots.

CLP Power also provided technical advice and power supply solutions to enable EV charge point operators to set up new fast charging hubs, and support the retrofit of petrol filling stations into EV fast charging facilities. CLP Power introduced an upgraded Fleet Electrification Advisory Service to support commercial fleet operators in their transition to EVs. In addition, CLP Power is supporting EV battery swapping companies to establish battery swapping stations for eTaxis.

CLP Power continued to expand its eMobility Grid Management Platform (eGMP), a data analytics system that monitors EV charger activities to optimise power system planning. By the end of 2025, eGMP's coverage increased to over 5,000 EV chargers across more than 400 sites in Hong Kong.

CLP Power also promotes electrification on construction sites and encourages the industry to switch from diesel generators to Battery Energy Storage System (BESS) to power their equipment and reduce carbon emissions. CLP Power further developed a 'General Guideline on BESS Adoption for Construction Sites' which provides practical guidance for the industry on the installation, application, and maintenance of BESS.

Furthermore, CLP Power introduced the city's largest BESS to the Airport Authority Hong Kong for storing electricity produced by the existing generators during routine testing. Currently, it is the largest emergency backup power supply system in Hong Kong, which effectively help enhance the Hong Kong International Airport's energy efficiency and reduce carbon emissions. We look forward to cooperating with more customers to further expand the adoption of BESS.

CLP Power has in recent years supported and encouraged property developers to promote low carbon property developments as well as 'All-Electric Homes'. We introduce innovative energy-saving technologies and energy-efficient appliances such as induction cookers and water heaters to property developers for improving the premises' energy efficiency. We also provide support for installing solar panels and EV charging facilities at the buildings, with an aim to reduce the environmental impact and improve the quality of life for communities.

Commercial and industrial customers with multiple operating locations often face challenges when it comes to evaluating their business's overall energy performance and taking steps to improve energy efficiency. CLP's Smart Energy Online (SEO) energy management platform, provides half-hourly electricity consumption data and offers consumption prediction and alerts, allowing business customers to effectively monitor and analyse the consumption patterns of multiple business premises so as to enhance efficiency and reduce operating costs. The SEO user base had increased to over 2,800 C&I customers at the end of 2025, an increase of 12% compared with 2024, reflecting growing adoption among C&I customers.

CLP Power embraces innovation and integrates sustainable initiatives in its operation through application of new technologies. CLP Power is introducing a more systematic and innovative approach for the design of green substations, introducing sustainable features such as increased ratio of greenery, rainwater recycling system and automatic dripping irrigation system to save more water for irrigation, as well as photovoltaic (PV) panels to maximise the harvest of the solar energy.

BEAM Plus is Hong Kong's leading initiative offering independent assessments of building sustainability performance. The assessment covers a wide range of sustainability issues relating to the planning, design, construction, commissioning, fitting out, management, operation and maintenance of a building.

The Hong Kong-Zhuhai-Macao Bridge Substation, Queen's Hill Substation, Shing Kai Road Substation, Ma Sik Road Substation, Kwu Tung North Substation, Ho To West Substation, and Tuen Mun Eco Park Substation were awarded Final Platinum rating under BEAM Plus New Buildings, while Yuen Long Industrial Estate Substation was awarded Provisional Platinum rating.

On the power generation side, the first new gas-fired generation unit (Unit D1) in Black Point Power Station attained the Final Platinum rating in 2021 under BEAM Plus New Buildings V1.2. The green and sustainable architectural features include natural lighting, energy-efficient electrical fixtures and appliances, an extensive rainwater harvesting system, roof-top solar panels and vertical greening. These features support energy saving and promote sustainability.

HK Electric

The vast majority of Hong Kong's energy-saving opportunities arise in the built environment. HK Electric's policy is to engage regularly with various stakeholders in the community, including building owners, district council members and property managers, to promote EE&C.

HK Electric provides comprehensive Smart Power Services (SPS) and engagement platform assisting customers and the community in decarbonisation through adoption of renewable energy and electric vehicles, promotion of energy efficiency and conservation, electrification and low-carbon lifestyle as well as providing care to the disadvantaged along Hong Kong's journey towards a smart zero-carbon caring city.

Various funding/service schemes have been devised to cater for different sectors of the community to help promote energy efficiency and support the local development of renewable energy. SPS Schemes on energy efficiency include:

- **Smart Power Building Fund** – subsidies (10%-50% of project costs and capped at HK\$250,000 to HK\$500,000 per building) for building owners to carry out retrofitting, Retro-Commissioning or building-based smart technologies projects to enhance the energy efficiency of communal building services installations (e.g. lighting, air-conditioning, lift, escalator, electrical installation, etc.). A total of 77 applications, covering 161 buildings, involving subsidies of about \$17 million were approved in 2025.

- **Smart Power Energy Audit** – free energy audits service to help non-residential customers identify energy saving potential for improving energy efficiency at their business premises. Written reports providing recommendations on energy saving will be provided for the customers. In 2025, we completed over 200 energy audits.
- **Smart Power Loan Fund** – subsidies for loan interests of the loans provided by collaborating banks for customers who have completed Smart Power Energy Audit or successfully applied for Smart Power Building Fund to implement energy efficiency enhancement projects.
- **Smart Power Care Fund** – services, advice and financial assistance helping the disadvantaged, customers and the community adopt low-carbon and smart lifestyles and improve living environment while providing care to the community.
- **Smart Power EV Charging Solution** – one-stop service to assist customers in implementing EV charging solutions for reducing carbon emissions and improving roadside air quality.
 - **EV Charging Solution Service**
One-stop service to assist customers in implementing EV charging solutions for reducing carbon emissions and improving roadside air quality.
 - **Power Public EV Charging**
Assessment of electricity supply capacity and provision of electricity supply solutions for public car parks to establish EV charging stations.
 - **Zero-carbon EV Charging**
Renewable Energy Certificates for EV customers to purchase for realising zero-carbon electric mobility.

Up to 2025, HK Electric provided technical support for EV charging-enabling infrastructure projects under the Government's EV-charging at Home Subsidy Scheme ("the EHSS") for more than 470 cases. A total of 95 EV charging-enabling infrastructure projects were completed in 2025.

- **Smart Power for Data Centre** – one-stop service to provide power supply solutions for data centres on Hong Kong Island and decarbonise the construction and operation of data centres.
- **Smart Power for Construction Site** – one-stop service to timely provide grid-electricity supply for construction sites to eliminate the use of diesel generators to provide electricity; and thereby substantially reduce carbon emissions, air and noise pollution during construction stage while saving energy cost.
- **Smart Power Education Fund** – with an annual budget of \$5 million, the Fund supports programmes and activities to promote energy efficiency, renewable energy and low-carbon living. Key initiatives include the Company's signature programme, Happy Green Campaign, and donations and sponsorships supporting various educational and promotional activities on environmental protection and conservation.

HK Electric also set up a Smart Power Gallery, a 5-storey experience and education centre in Sheung Wan, to showcase the important role of low-carbon energy transition on Hong Kong's journey towards a smart zero-carbon caring city under the following themes:

- ◆ HK Electric's History and Power System
- ◆ Climate Change and Carbon Neutrality
- ◆ Transition to Zero-carbon Electricity
- ◆ Electrification and Energy Management
- ◆ Smart Power Services

The gallery features multimedia displays and interactive games and provides renewable energy and green transport related STEAM workshops for students and talks for stakeholders. Apart from free physical guided tours, the gallery also provides virtual tours and online learning activities.

Latest AMI Installations and Smart Grid Projects

CLP Power

Smart meters are a key component for Advanced Metering Infrastructure, helping to further improve network and asset performance. They also enable CLP Power to engage customers on their energy use, empowering them to adopt more energy-efficient behaviours. To support Hong Kong's development into a smart city, CLP Power has largely completed a seven-year programme to install smart meters for residential and small-to-medium enterprise customers, while providing a range of digitalised services and solutions to encourage energy saving and further improve supply reliability.

HK Electric

Having started the installation of smart meters in 2020, HK Electric completed a full deployment of smart meters at the end of 2025 in support of Hong Kong's transformation into a smart city. Customers already connected with smart meters can access to a suite of practical functions for optimising energy use through the corporate website or the HK Electric App, such as reviewing more detailed consumption, setting consumption alert and obtaining information on projected consumption for the next bill.

Apart from supporting energy management by customers, AMI also helps to improve the efficiency of operations by saving manpower for manual meter readings and by enhancing data accuracy.

Challenges Faced and How It Was Solved / Addressed

CLP Power

Hong Kong has a unique cityscape – a high-density urban area with many tall buildings which brings complex technical challenges for installing and connecting smart meters as well as effectively

collecting smart meter data. Most of the meters and electricity supply equipment in urban area are installed in meter rooms on each floor and some meters for electric vehicle chargers are installed in basement car parks, isolated by concrete walls and doors. It proved challenging both to establish a field area network capable of penetrating multiple steel-reinforced concrete walls in meter rooms, and to secure a sufficiently strong mobile signal for backhaul communication. The mitigation measures focused primarily on extensive site surveying to identify optimal locations within meter rooms for installing the data collector in at where signal is stronger, alongside exploring alternative connectivity solutions.

With strong commitment on supply reliability and customer caring, CLP Power liaises with Estate Management Office and commercial customers for mutually-agreed power outage schedule and seeks to minimise the impact to their residents and their business operations.

AMI Roll-out Plan

CLP Power

By the end of 2025, more than 2.88 million smart meters had been connected, marking a significant milestone in the digital transformation of the city's power infrastructure. This city-wide rollout strengthens grid resilience, enhances supply reliability, and supports Hong Kong's goal of achieving carbon neutrality by 2050. It also lays the foundation for smarter energy services by transforming how electricity is supplied, monitored, and managed to meet the needs of a more digital and low-carbon future.

HK Electric

The roll-out was completed successfully at the end of 2025 as planned.

AMI Offerings to Customers and its Benefit

CLP Power

Smart metering provides a wide range of benefits in the aspects of demand side management, system reliability, asset performance as well as operational efficiency. The following are some of the new services and benefits of smart meter.

Through the CLP One mobile app, customers can monitor their hourly electricity consumption anytime and anywhere, estimate upcoming charges, and receive alerts when usage is higher than usual. These features help households and businesses better understand their energy consumption patterns, identify periods of higher usage, and make informed adjustments to improve energy efficiency and manage electricity costs.

Smart metering also enhances safety and operational efficiency by eliminating manual meter reading, thereby reducing associated safety risks. In addition, it minimises bill estimation resulting from inaccessible meters in locked premises.

Access to near real-time usage data also facilitates participation in demand response programmes, which encourage reduced electricity consumption during peak periods. Since 2020, CLP Power has offered the Summer Saver Rebate Programme for residential customers with smart meters. In 2025, three events were held between June and July, engaging more than 1.6 million households. Nearly 80% of participating households successfully reduced their electricity consumption, saving over 600,000 kilowatt hours (kWh) of electricity within a total of six hours and reducing around 230 tonnes of carbon emissions, setting a new programme record.

Smart meters also enable efficient remote supply disconnection and reconnection, providing a stronger deterrent against overdue payments while allowing power to be restored more quickly once outstanding bills are settled.

CLP Power launched the Community Watch & Care Service Pilot Programme in 2025. By integrating smart meter data with artificial intelligence technology, the programme analyses the electricity consumption patterns of elderly people living alone, enabling social workers and caregivers to identify unusual situations in a timely manner and provide appropriate support, thereby enhancing community care for the elderly.

HK Electric

The corporate mobile app “HK Electric App” and the online platform “Account-On-Line” (AOL) enable customers to monitor and optimise their electricity use for better energy management. Customers with smart meter accounts can enjoy the following services via the App or the AOL:

- Track and download monthly, daily and half-hourly consumption data up to 12 hours ago. Such data will be kept and made available for up to 25 months.
- Set and receive consumption alert when the consumption reached the selected percentage of the consumption of the last billing cycle.
- Check projected bill amount and consumption for the next billing cycle.

HK Electric has also leveraged the smart metering system and its data to enhance customer services such as handling of customer enquiries and move-in/move-out requests. There are also operational benefits in terms of safety associated with manual meter reading and prompt notifications of power outages and restorations.

CLP Power

Building Information Modeling (BIM)

The application of Building Information Modelling (BIM) at CLP Power initially started with new projects, including the construction of transmission substations and the second new gas-fired generation unit (Unit D2), to facilitate design verification and construction planning. The technology can also be applied to enhance the management of site construction works, improve site coordination, and reduce wastage. CLP Power plans to further leverage the advantages of BIM for operations and maintenance and is therefore kick-starting the creation of BIM models for existing assets to enhance asset visibility.

For the transmission grid, a BIM platform has been set up and tested to effectively support engineers in their operations and maintenance. The platform integrates BIM models together with asset data as well as documents for operations and maintenance such as checklists, records, and manuals. The up-to-date construction progress is made available on the platform for direct comparison with the BIM model. In addition, BIM models for existing substations are being generated. As a result, architects and engineers can save time for searching for drawings and expedite site verification tasks can be streamlined, while BIM models also offer a higher accuracy. Further applications of the BIM technology are being explored, empowering teams to enhance work efficiency.

Asset Health Management

Asset Health Management, which refers to the monitoring, assessment and prediction of asset health, is an important cornerstone in ensuring the efficient and reliable supply of electricity to CLP Power's customers. As CLP Power continues its digitalisation journey, the application of data analytics is one of the key components driving its digital asset management capabilities for generation, transmission, and distribution primary equipment forward. This ensures cost-effective asset utilisation, maintenance, and replacement.

For the power grid, the asset health management system provides a powerful tool for early detection of any abnormality, just-in-time intervention to prevent potential issues as well as replacement planning. It computes health and criticality indices for each asset from a large range of asset types down to the distribution level. This system is supported by data from different analytics and monitoring systems, for example real-time information from partial discharge and switchgear monitoring. For high voltage transmission transformers, live health status information is considered based on dissolved gas analysis, with the system's scope currently being expanded to a subset of other transmission transformers.

Grid-V for enhancing power grid resilience

To further enhance the resilience of the power grid amidst increasingly extreme weather events, CLP Power has developed an advanced, centralised intelligent management system known as Grid-V. Grid-V integrates various surveillance and information systems and leverages artificial intelligence (AI) technology. It provides real-time monitoring of the condition of key power facilities, such as transmission overhead lines, substations, high-voltage switchgears, and construction sites. The AI component of Grid-V can detect external disturbances such as hill fire, smoke, and flying objects, and alert operation staff to take prompt action. The AI feature also enhances physical security by detecting intrusions, thereby bolstering the protection of substations. Ongoing enhancements to Grid-V focus on the development of sophisticated AI-driven algorithms and adaptive training models, further advancing its capabilities in proactive risk identification and mitigation.

Smart Site Safety System (4S) for Enhanced Project Management

Smart Site Safety System (4S) is an innovative centralised management platform that integrates various smart devices, enabling real-time monitoring of construction sites and early identification of potential safety hazards. CLP Power is one of the pioneers in the local industry to adopt the 4S for the development of WE Station (West New Territories Landfill Gas Power Generation Project) Phase 2. By leveraging advanced data analytics, the implementation of 4S not only enhances safety performance but also boosts productivity and quality control, ultimately uplifting overall project governance.

Efficiency Enhancement on Operations and Maintenance through New Ways of Working

With an aim to pursue continuous improvement on its work performance and efficiency, CLP Power has been exploring new ways of working in various operations and maintenance activities.

In power stations, for example, a cutting-edge innovation known as the cage drone is adopted for indoor inspection. The cage drone is small and uses a Light Detection and Ranging (LiDAR) system and sensors to reach high and inaccessible spaces flexibly and quickly. This not only makes inspections more thorough and comprehensive but also saves time and reduces cost associated with erecting scaffolding. It enhances work safety by mitigating the potential risks of working at height and in confined spaces, and improves the accuracy of inspections. Various kinds of robots are also adopted, including the crawler robot for inspecting underground cooling water culverts. The crawler robot has a retractable top and is equipped with lights and rotation lenses, enabling comprehensive inspection and capturing detailed footage in narrow and dimly lit culverts. This has significantly improved the safety, accuracy, and efficiency of culvert inspections. In addition, CLP Power has collaborated with The Hong Kong Polytechnic University to jointly develop a generator inspection robot. The invention won two awards at the 51st International Exhibition of Inventions Geneva. The robot is equipped with retractable legs and wheels, which can flexibly navigate the narrow air gaps between the rotor and stator of the generator, and cross ventilation baffles to automatically complete major inspection tasks without the need to remove the rotor. The robot significantly enhances inspection efficiency and operational safety.

For the distribution grid, AML data is utilised for enhanced grid operation and maintenance in addition to the added customer services previously described. This includes the pre-locating process for any broken 11 kV overhead line connectors resulting in a faster repair or replacement. AML data is also utilised to enhance digital low voltage network models, e.g. regarding phase connection information, and to verify their topology.

Furthermore, various use cases for Artificial Intelligence for the grid operation and maintenance are explored. This includes for example anomaly detections for overhead lines and substations.

These initiatives underscore CLP Power's dedication to innovation and its commitment to delivering reliable and sustainable power supply to the customers.

Application of Cleaner Energy

To support the decarbonisation of Hong Kong, CLP Power has been exploring the use of emerging technologies, including hydrogen and carbon capture and storage, to support the Hong Kong Government's target of achieving carbon neutrality before 2050. For instance, a grid-scale battery energy storage system of around 100MW will be established to help reduce peak system demand, facilitate the integration of clean energy sources and enhance system flexibility. In addition, a pilot project is planned to explore the feasibility of using hydrogen-blended natural gas for power generation at Black Point Power Station.

Low-Voltage Smart Grid

The Hong Kong's Climate Action Plan 2050 outlines a clear pathway towards achieving carbon neutrality before 2050, including net-zero electricity generation. In support of the strategy, CLP Power has started its Low-Voltage Smart Grid (LVSG) initiative to enable the integration of distributed energy resources (DERs) into the distribution network. The LVSG enhances network resilience, particularly in areas with high DERs penetration, through the deployment of an AI-enabled LVSG Controller, Battery Energy Storage Systems (BESS), Line Voltage Regulators (LVR), and micro-weather stations. Concurrently, CLP Power has developed a diversified BESS portfolio, comprising Centralised BESS for rural networks with increasing DERs penetration and peak demand, and Distributed BESS, including Pole-Mounted BESS (P-BESS) and Residential BESS (R-BESS) to strengthen local voltage control and enable dynamic load management in space-constrained networks. Leveraging this foundation, CLP Power is progressing the integration of BESS into grid to enhance system flexibility, by incorporating behind-the-meter BESS into a coordinated resource pool that enables real-time monitoring, aggregation of distributed flexibility, and dispatch control to support grid operations and Virtual Power Plant (VPP) development.

Advanced Distribution Management System (ADMS) and Distribution Energy Resource Management System (DERMS)

CLP Power

CLP Power has recently enhanced the distribution grid management through the implementation of Advanced Distribution Management System (ADMS). It provides system operators with more comprehensive visibility of the distribution network, generation and load forecasting, network constraint management, and operational functionalities such as outage and switching order management.

At the same time, CLP Power is dedicated to optimising the existing capacity of the power grid and efficiently perform constraint management through optimally managing the distributed energy resources and customer load. CLP Power is therefore in the process of establishing a Distributed Energy Resource Management System (DERMS) which is the core of Virtual Power Plant (VPP) system. This system enables the location-based and economically optimised activation of flexibility resources, including demand response programmes, electric vehicles, and battery storages. DERMS can provide flexibility for system load management and, by integrating with ADMS, also manage distribution network constraints. Through its optimisation algorithm, DERMS can identify the most suitable portfolio to provide the required flexibility and automatically dispatch the resources.

HK Electric

In order to enhance our emergency services for customers which are sensitive to noise and emission, we have procured two 500-kWh 250-kW Mobile DC Energy Storage and Power Conversion Systems as a pilot scheme for provision of back feed supply to our customers under emergency situation. Unlike the traditional mobile generator which mainly powered by internal combustion engine or gas turbine, the system converts DC power stored in the internal batteries to 380-V AC supply without any noise and emission. The system consists of two sets of Energy Storage Truck to sustain continuous back feed supply to our customers. The green energy ensures minimum disturbance to the stakeholders even under power outage.

To tackle unforeseeable district-wide unplanned outage, we have commissioned a 380V/11.7kV 2,000-kVA Mobile Transformer Module in 2024. The Mobile Transformer Module can convert the power from Mobile Generators of 380 V to 11 kV to back feed the 11-kV network or HV customers. With the availability of this mobile transformer module, 11kV supply can be applied not only for contingency but also for soaking of 11kV cables if required.

To enhance the reliability of the plants in transmission network, all 275-kV and 132-kV gas insulated switchgears are monitored by online partial discharge monitoring systems. In addition, the coverage of online partial discharge system is gradually extended to cover the 275-kV and 132-kV gas insulated transformers to monitor their conditions. For the oil-insulated transmission transformers, they are monitored by online dissolved gas analysis or total combustion gas systems.

With the completion of retrofitting of online partial discharge monitoring systems for all 11-kV air insulated switchgears at primary substations, as a further step, online partial discharge monitoring systems are being extended to cover the primary 22-kV zone substations for monitoring the conditions of the 22-kV gas insulated switchgear.

A tailor-made solution for remote circuit breaker monitoring (RCBM) system was successfully developed and used to continuously monitor the health of circuit breakers at primary 11-kV zone substations by capturing trip/close coil current profile information. Additionally, the statistical model was also developed for identifying and classifying any abnormalities of the circuit breakers. This method can prioritize our maintenance work and prevent failures of the circuit breakers.

An innovative portable equipment for capturing the current profile of the DC control circuit during switchgear operation was developed for monitoring the mechanical integrity of distribution switchgear. A number of sticky switchgear panels were identified by this tool for relubrication during switchgear operation test. The tool has widely been adopted as a preventive measure to identify sticky switchgear for urgent repair. For this innovation initiative, a Hong Kong Short Term Patent regarding "Switchgear Operation Time Sensor" was granted to HK Electric.

In addition to monitoring of transformers and switchgears, Very Low Frequency (VLF) Tangent Delta (TD) measurement and partial discharge (PD) mapping have been adopted as the diagnostic testing for 11-kV distribution cables since 2010 with an aim to assessing the cable circuit conditions and pin-pointing weak components before failure. To avoid overstressing the cable circuit under tested while maintaining the effectiveness in identifying possible weak components, an improved VLF monitored withstand test regime was developed and implemented in HK Electric's 11-kV distribution network. Under this regime, test voltage and duration were fine-tuned according to cable types, preliminary diagnostic test results and other factors. Based on the diagnostic test results, recommendations were made for timely replacement of weak components to enhance supply reliability.

In addition to the AI tool for evaluating the failure risks of 11-kV distribution cables, software to determine the selection priority for proactive cable testing was being developed in-house. The 11-kV distribution cable circuit failure rates have been on a decreasing trend since the introduction of VLF diagnostic testing. With relentless efforts of improving the reliability of aged distribution cable assets, 2025 marked the third-best result we have ever achieved despite aging of cable assets.

With the practical experiences gained in VLF testing of 11-kV cables, we have extended the VLF testing to low voltage (LV) cable in 2017, in particular on LV cables with low insulation found during cable fault repair. This test will not only provide the general health condition of the LV cable after fault repair but also identify those possible weak components for timely replacement. Through this test method, the availability of LV cables is improved. The number of LV cable multiple faults has been on a decreasing trend after the application of VLF testing on LV cable.

The Hong Kong's first Low Voltage Direct Current (LVDC) System was commissioned at Ocean Park in August 2025, enhancing the supply reliability to the "Giant Panda Adventure" exhibit, the home to the city's giant pandas and adjacent facilities. The system is capable of operating in complementary and substitutive power supply modes, enhancing overall operational flexibility and supply reliability. It leverages DC technology to significantly enhance the transmission capacity of existing low-voltage cables, reaching 700 kVA, which is 2.5 times of the current carrying capacity of existing LV cable running on AC. In the event of extreme or unexpected circumstances that may affect power supply, the LVDC System could provide sufficient backup power to ensure stable operations throughout the facility.

Automatic Identification System (AIS) has been used since 2015 to prevent submarine cable damage at East Lamma Channel Cable Area and Hung Hom Fairway Cable Reserve. The AIS detects stationary marine vessels with potential of causing cable damage by anchorage in the cable reserve zone and initiates alarm to alert an appropriate action to prevent cable damage.

Perfluorocarbon Tracer (PFT) Leak Location System has been used since 2015 for detection of leakage points on fluid-filled transmission cables. Cable Fluid Pressure Remote Monitoring System has been used since 2023 for online pressure monitoring of selected fluid filled 275-kV and 132-kV cables. On the other hand, online Partial Discharge Monitoring System has been used for monitoring the condition of selected 275-kV and 132-kV XLPE insulated cables.

HK Electric is committed to electrifying its operation fleet and deployed 191 operation EVs (over 56% of its fleet) as at 2025. In 2020, we completed the replacement of all operational sedan cars with EVs or hybrid vehicles.

A smart robotic solution for real time inspection has been deployed in one of the cable tunnels as a trial through 5G technology since 2023. This remote control robot has equipped with high-definition cameras, infrared cameras, sensors and gas detectors which can capture real time video footage and environmental parameters inside cable tunnel with the aims to enhance inspection quality and eliminate the risks associated with human entry into a confined space. Engineers can also control the robot remotely to carry out ad-hoc inspection and retrieve instant environmental parameters of cable tunnel without physical access to the tunnel in case of emergency.

In Dec 2024, HK Electric launched SmartLink @ HK Electric, an in-house LoRaWAN-based communication network, to establish a comprehensive wireless platform supporting a wide range of Internet of Things (IoT) sensor applications. These applications encompass the condition monitoring of equipment and apparatus within power plants, substations, and customer premises. The LoRaWAN network boasts substantial coverage across Hong Kong Island and Lamma Island, with over 200 sensors strategically deployed. As the network continues to expand, it accumulates critical data that holds significant potential. By seamlessly integrating artificial intelligence and data analytics at the backend, HK Electric can continuously monitor the health status and condition of various assets within the power network. This proactive approach enables timely interventions to prevent failures and optimise overall performance.

The SmartGrid @ HK Electric is a digital platform designed to enhance the monitoring, control and reliability of power transmission and distribution networks by providing real-time visibility and smarter operational decision-making. It integrates cloud computing, communication networks, edge devices and field sensors into a unified "cloud-pipe-edge-device" architecture, enabling continuous surveillance through AI-powered CCTV, IoT sensors and intelligent equipment. Building on this foundation, SmartGrid @ HK Electric incorporates advanced condition monitoring and diagnostic technologies to support predictive maintenance, early fault detection and optimized asset management, while innovative solutions such as mobile energy storage systems, mobile transformers and robotic inspection enhance emergency response with low-noise, low-emission supply and safer inspections. At the same time, integration with IoT platforms and AI analytics improves situational awareness, allowing proactive management of risks from extreme weather, ageing assets and external interference, ultimately delivering a resilient, cost-effective and future-ready power network.

Unit L10 and L11 will adopt a bunch of advanced technologies so that the efficiency is better than L9 by 3%.

Looking forwards, HK Electric will continue to explore opportunities for decarbonisation through adopting advanced technologies, such as hydrogen fuel, carbon capture and storage, energy storage, etc., as long as they are commercially viable.



CLP Power

Building Resilience to Physical Climate Risks

In 2025, Hong Kong experienced a record-breaking 14 tropical cyclones, including two No.10 hurricane signals issued under Super Typhoon Ragasa and Typhoon Wipha. A record total of five Black Rainstorm Warnings were issued within July and August. Despite the extreme weather, CLP Power's power supply system maintained a high overall performance, demonstrating the effectiveness of our preparedness efforts.

CLP Power has also implemented a range of measures to strengthen its resilience to climate change, tailored to different geographies, asset types and locations:

To address extreme heat and increased temperature:

- Operational guidelines have been established to ensure safe and reliable operations under high temperatures

To address flooding:

- Flood risk assessments are conducted for intensified heavy rainfall scenarios, with mitigation measures implemented for new and existing substations.

To address tropical storms:

- Continuous reinforcement of transmission overhead line tower structures.
- Strengthening of transmission tower foundations, and stabilisation of adjacent slopes.
- Enhancement of automatic detection and isolation of faulty sections within distribution overhead line circuits.
- Implementation of predictive vegetation management to minimise risk from overgrown vegetation.

To strengthen Hong Kong's resilience to extreme weather, CLP Power and the Hong Kong Quality Assurance Agency have jointly launched the "Climate Resilience Care" platform. Built on international standards and tailored to local conditions, the platform provides a practical framework to help businesses and building owners systematically identify climate risks, implement response measures, and integrate climate resilience into ESG management. Amid rising extreme weather threats, the platform supports a more forward-looking and structured approach to manage climate risks. CLP also provides professional audits on

power quality and energy efficiency, complemented by partners' green solutions and insurance offerings to support broader societal resilience.

Decarbonising Our Portfolio for Hong Kong

CLP Power offers Feed-in Tariff (FiT) for its customers, in support of energy decentralisation and renewable energy. The Renewable Energy Feed-in Tariff (FiT) Scheme in Hong Kong enables customers to earn FiT payments by installing a renewable energy system on their premises and connecting the system to the CLP grid. From the commencement of the FiT Scheme in mid-2018 to the end of 2025, CLP Power received over 28,605 applications. Approximately 95% of the applications, representing a total capacity of about 450MW, equivalent to the annual electricity usage of more than 100,000 residential customers, have been approved. About 25,859 applications have been completed and connected to the grid. The FiT Scheme has attracted customers from a variety of sectors, including commercial and industrial sectors, schools, urban households and rural village houses.

Renewable Energy Certificates (RECs) offer an alternative way for customers to support local RE development. Each unit of a REC represents the environmental attributes of electricity produced by local renewable energy sources, generated or purchased by CLP Power. Since the launch of the RECs in January 2019, commercial and industrial customers have shown continual interests in decarbonisation services and solutions including RECs, energy audits, training and subsidies for installing energy-efficient appliances. Compared to 2024, the REC sales in 2025 increased by 6.8% to around 363GWh.

Opportunities to Apply New Technologies

CLP Power continues to adopt the advanced and most relevant technologies to improve the performance of its power systems. CLP Power is one of the power companies worldwide which develops smart grid in a vertically integrated approach, covering all aspects including power generation, transmission and distribution, as well as customer service.

In light of this, CLP Power has introduced the Advanced Distribution Management System (ADMS), a digital solution designed to enhance network visibility, supply reliability and optimise resource utilisation.

Battery Energy Storage Systems (BESS) have a critical role in transforming power systems to be cleaner, more efficient and sustainable. They generate multiple benefits of integrating variable distributed renewable energies and EV, improving supply reliability and mitigating network congestion. On this, CLP Power is also pursuing a series of BESS pilots, taking steps to identify the most suited business case for its grid development.

To combat climate change and to operate environmentally friendly power systems of the future, CLP Power is aware of the world trend in tightening climate policies to eliminate future emission of greenhouse gases. We are therefore introducing pilot projects to assess the performance before making decisions on transiting to green T&D equipment in addition to decarbonising power generation.

Some of the advanced smart features incorporated in the power systems include:

- A real-time monitoring system has been developed as a home-grown IoT platform, comprising intelligent sensors deployed across key generation facilities to enable continuous remote monitoring of critical parameters such as temperature, current, and vibration. Integrated with the Asset Health Management System (AHMS), the platform supports predictive maintenance and advanced asset analytics, enabling early fault detection and improving plant reliability, operational efficiency, and maintenance effectiveness;
- AI-enabled visual monitoring systems have been deployed for turbine hall crane operations, alongside drone-based AI analytics to support efficient civil inspections;
- ADMS provides system operators with comprehensive visibility of the distribution network, generation and load forecasting, network constraint management, and operational functionalities such as outage and switching order management.
- 11kV overhead line automatic restoration system which automatically isolates the faulty section of the overhead line, and shift to other sources for immediate supply restoration;
- Transmission Overhead Line Condition Monitoring System developed to conduct online surveillance of transmission overhead lines at locations with higher risk of external interference.
- AI images and video analytics utilised to detect any defects of overhead line components.
- Virtual Reality applications to enhance staff training
- Smart transmission substations which have installed the most advanced and fully automated equipment including self-healing systems, which can significantly shorten the power restoration time from several minutes to less than one second.
- Online condition monitoring systems adopted at transmission transformers and switchgears for conducting round-the-clock health checks.
- A single centralised platform for monitoring asset health across multiple asset types, including transformers, switchgears, cables, and overhead lines.
- Condition monitoring system on transmission cable network to help detect incipient fault and enhance supply reliability for critical transmission circuits.
- Trial implementation of more environmentally-friendly transformer insulation oil has begun at both transmission and distribution levels.
- Switchgear insulated with non-SF6 gas is currently being evaluated through field trials at the distribution level.
- Drone deployment for efficient and safe asset inspection in areas that are difficult to reach.

CLP Power's Customer Interaction Centre (CIC) has integrated advanced algorithms powered by big data and AI to drive continuous improvements in operational efficiency and service quality. Handling over 2,000 calls daily on average, the CIC faces substantial demand. By applying intelligent analytics to identify call patterns and peak service hours, the centre can optimise workforce deployment, enhancing service responsiveness and significantly reducing customer waiting time.

Beyond call management, AI is also used to analyse customer enquiries and complaints, while supporting frontline teams in drafting responses and enhancing the overall customer experience.

HK Electric

In response to the growing demand for convenience, particularly with the advent of AI and cutting-edge technologies, HK Electric has continuously enhanced its online platforms to offer customers more efficient self-service options for managing account-related matters. In 2025, a new online enquiry form was launched, allowing customers to select from a list of key topics, thereby simplifying the enquiry process and reducing the need to search for different contact channels. Besides, the electronic forms for move-in applications were upgraded to enable residential customers to apply for bank autopay during the move-in process. To further promote e-payment and autopay services, a series of step-by-step instructional videos was produced and published to guide customers on making e-payments through the HK Electric App and applying for autopay.

HK Electric continued its collaboration with local universities in the area of artificial intelligence and big data analytics to evaluate asset condition, including the development of software platforms for anomaly detection and health condition classification of circuit breakers and cables.

Major challenges

Global warming and the effects of climate change have become a major challenge in our world. In recent weather events, Hong Kong was struck by a super typhoon Mangkhut in 2018 and a historic torrential rainstorm in 2023 which caused serious regional flooding to the coastal areas and geographical low-lying areas on Hong Kong Island. Thanks to the well-equipped anti-flooding systems in our transmission and distribution stations as well as the rapid response of our engineers, the typhoon and rainstorm had only caused minor disturbance to our system and customers. To maintain the electricity supply reliability to our customers, HK Electric has been periodically reviewing the system design and reinforces the infrastructure to prepare for the challenges caused by climate change.

Other related activities / initiatives

HK Electric actively engages diverse community groups to support and create shared value across the community. Through the "CAREnJOY" community programme, the company collaborates with local NGOs, as well as District and Legislative Council members, to deliver a range of initiatives. These include Electrical Safety Talks and Fun Days, which promote the safe and efficient use of electricity, as well as energy saving, through interactive games and engaging activities. The U3A (University of 3rd Age) Network of Hong Kong, co-founded by HK Electric and the Hong Kong Council of Social Service in 2006, provides elderly participants with learning opportunities and avenues for active community engagement through volunteering. Since its establishment, the programme has delivered more than 170,000 learning and engagement opportunities for local retirees. Looking ahead, HK Electric is committed to supporting the Government's initiatives to advance the silver economy by expanding choices for seniors, as well as promoting gerontechnology and smart living. Volunteering is also deeply rooted in HK Electric's culture, reflecting its commitment to caring for the community. In 2025, the company's volunteer team contributed a total of 4,849 hours across a wide range of services, from supporting major sporting and cultural events to visiting elderly people and preparing meals for people in need.

Employee wellness and development remain a key focus at HK Electric. The company's Employee Wellness Programme offers a broad spectrum of activities and plans continuing to promote physical and intellectual wellbeing, emotional and social wellbeing, and good family relationships. HK Electric has various talent development programmes in place to ensure that its workforce has the necessary skills and expertise to excel in all core areas of its business.

The HK Electric Institute continued to nurture engineering experts, ensure knowledge transfer and provide advanced power engineering training programmes for our technical staff to cope with the demanding and ever-changing operating environment of the electricity industry. Since the signing of the first Memorandum of Understanding (MoU) with The Hong Kong Polytechnic University (PolyU) in 2015, the Institute has continued to integrate the field knowledge of power engineering into the curriculum and nurture power experts through postgraduate education programmes. The MoU has

been renewed with PolyU in 2023 for another five years. As one of the collaborations with PolyU, HK Electric continues to co-teach High Voltage Engineering and Modern Generation Technologies and Grid Integration in Master of Science (MSc) course.

In addition, HK Electric and Hong Kong Baptist University (HKBU) signed an MoU on 15/07/2025, to strengthen collaboration in artificial intelligence (AI), marking a new chapter in industrial-academic partnership. The first initiative under the MoU was a joint project titled “Anomaly Detection for Low-voltage Distribution Network Using Artificial Intelligence”, which focuses on improving anomaly detection and in-time maintenance of LV cable circuits across the supply grid of HK Electric for improving supply reliability. This collaboration between industry and academia exemplifies the powerful synergy that transforms pioneering research into real-world solutions.

In 2025, over 400 participants attended 11 modules conducted by experienced staff of HK Electric or retirees from various disciplines. The total training man-hours were up to 1,700 hours.

We have also worked with industry experts to deliver lectures to enrich the syllabus of the Institute. HK Electric, The Hong Kong Polytechnic University (PolyU), Xi'an Jiaotong University (XJTU) and the State Grid Corporation of China (SGCC) signed a Memorandum of Understanding (MoU) in 2024, to extend their collaboration on the “Belt and Road Advanced Programme in Power and Energy” for another four years. Now in its eighth consecutive year, the programme offers a comprehensive curriculum on the design and operation of power systems. The Programme is designed for academics and management staff in the power industry across “Belt and Road” countries and regions.

The theme of the 2025 Programme was “Promoting Collaborations on Green Energy; Building New Type of Power System” aiming to broaden participants' perspectives and equip them to tackle future global green energy transition challenges. Over the 13-day course, 28 professionals from 6 Belt and Road countries or regions took part in a series of professional workshops, field studies and technical discussions in Jinan, Xi'an and Hong Kong. They also participated in activities including lectures, seminars, interactive discussions, and experience-sharing sessions.

As part of the Programme, participants visited several cutting-edge power facilities, including HK Electric's Marsh Road Station Building, SGCC's Vehicle-Grid Interactive Charging Station Demonstration Project in Jinan, and XJTU's State Key Laboratory of Electrical Insulation and Power Equipment in Xi'an. These visits offered participants firsthand insights into advanced technologies and modern management theories, while enabling them to exchange ideas and gain invaluable experience from experts from HK Electric, PolyU, XJTU and SGCC.

Since its launch, the programme has engaged over 850 professionals and facilitated over 10,000 hours of both online and offline training exchanges.

Virtual Reality (VR) Training Centre

The VR Training Centre has been established in 2020 to improve the effectiveness of operational and safety training. Potential risks and emergency site conditions are simulated using imseCAVE VR technology for training purpose. Under the virtual but zero-risk environment, our engineers are able to practice hands-on operations and benefit from experiential learning. Also, the training effectiveness has been further improved by using live-streaming of first-person 3-D view of the trainee to a 2-D Smartboard to facilitate large group training and post-training evaluation.

The following VR training contents have been developed for training of our employees and contractors:

- 11-kV Switching Operation
- Safety Precautions for Working in Zone Substations
- Safety Precautions for Working Inside Distribution Substations
- Application of Portable Fire Extinguishers
- Safety Precautions for Work Above Ground
- Safety Precautions for Lifting Operations

Safety Experience Training Centre

In order to enhance the safety awareness to our employees and contractors, HK Electric has established a Safety Experience Training Centre in which the following training set-up and simulator have been installed:

- Capacitor Bank Training setup
- Fall Arresting Training setup
- Manual Handling Operation Training set-up
- Safety Shoes and Helmet Accident Simulator

By using these training facilities, the participants can gain more hands-on safety practices as well as understand the potential risks in their daily working environment and the importance of using personal protective equipment.

CLP Power



Two new gas-fired generation units (Units D1 and D2) at Black Point Power Station were commissioned in 2020 and 2024 respectively. This enables an increase in the use of natural gas to over 50% of CLP Power's generation fuel mix in Hong Kong, representing a significant reduction of carbon emissions and contributing to Hong Kong's decarbonisation goal.



A new offshore LNG terminal using floating storage and regasification technology went into service in July 2023. The terminal enhances Hong Kong's fuel supply stability by adding a new supply source for natural gas and enables Hong Kong to source competitively priced natural gas from international markets.



Since its commissioning in 1994, Daya Bay Nuclear Power Station has been providing Hong Kong with non-carbon, safe, reliable power supply at a stable price for over 30 years. Currently, CLP imports 80% of Daya Bay's electricity generated to Hong Kong, meeting about a quarter of the city's electricity needs.



To support the Government's policy of promoting waste-to-energy and renewable energy initiatives, CLP Power has installed power generation units at the West New Territories (WENT) Landfill. The units make use of landfill gas produced locally at the landfill site for power generation and the electricity generated will be transmitted to CLP Power's electricity grid.



The enhancement works of Clean Energy Transmission System (CETS) were completed in March 2026, providing Hong Kong with greater flexibility to utilise more zero-carbon energy and supporting Hong Kong's long-term decarbonisation.



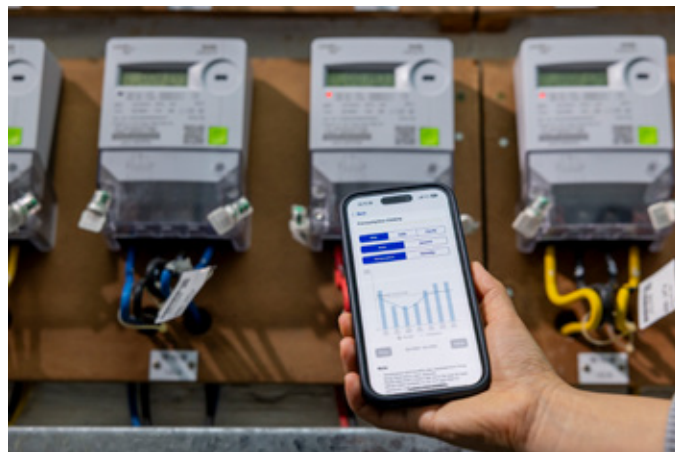
CLP Power adopted cage drone for indoor inspection at power stations, which makes inspections more thorough and comprehensive. It also helps mitigate the potential risks associated with working at height and in confined spaces, as well as saving time and cost of erecting scaffolding.



To support the HKSAR Government's Fast Charger Incentive Scheme, CLP Power has identified around 8,000 potential locations within its supply area suitable for the installation of fast chargers, enabling more effective planning and deployment of charging facilities.



CLP launched a new “Electric Vehicle Residential Time of Use Tariff” in May 2025. Customers can enjoy a discounted energy charge during “Off-peak Period”, which helps enhance resource utilisation of the power grid and reduce peak load.



Residential customers can better manage their electricity consumption through the CLP One mobile app, which allows them to monitor hourly usage anytime and anywhere, set personalised usage alerts, enhance energy management, and participate in demand response programme.



CLP Power has completed its Smart Meter Replacement Programme, which began in late 2018, enabling customers to better understand and manage their electricity consumption. By the end of 2025, more than 2.88 million smart meters had been connected for residential and small-to-medium enterprise customers, marking a significant milestone in the digital transformation of Hong Kong's power infrastructure. Smart meters give customers greater control over their electricity use by providing timely consumption data and digital services to encourage energy savings and enhance supply reliability.

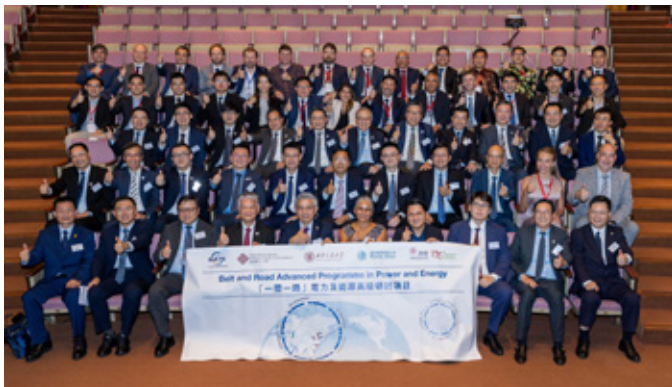


To support the government's promotion of geotechnology and ageing in place policies, CLP Power has launched the Community Watch & Care Service Pilot Programme. By analysing the electricity usage data collected through smart meter, the programme identifies abnormal electricity usage among the elderly, enabling social workers and carers to better understand their living conditions and provide an additional layer of care and protection.



To enhance Hong Kong's overall resilience to extreme weather, CLP Power and the Hong Kong Quality Assurance Agency have jointly announced to launch the city's first "Climate Resilience Care" platform to strengthen climate resilience management among businesses and building owners and promote forward-looking risk management and response strategies.

HK Electric



28 participants from 6 Belt and Road countries or regions joined the Graduation Ceremony of Belt and Road Advanced Programme in Power and Energy 2025.



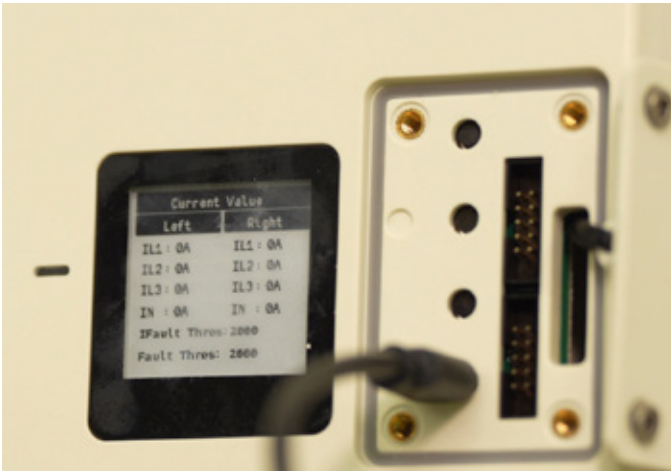
By 2025, Lamma Power Station had an installed capacity of 3,083 MW, with gas-fired units contributing 1,475 MW. The three new gas-fired units (L10 – L12) are equipped with Selective Catalytic Reduction systems to reduce emissions, supporting Hong Kong's progress toward decarbonisation.



HK Electric has also been actively promoting the application of renewable energy. Since 2010, a large-scale solar power system has been installed in phases on open areas and rooftops at the Lamma Power Station (LPS).



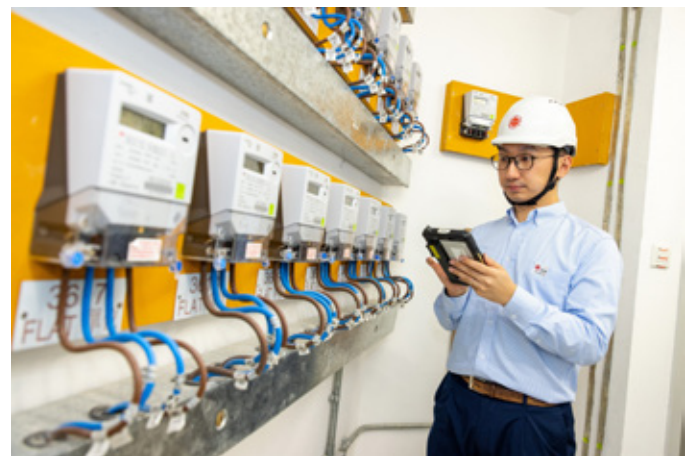
The use of cable tunnels makes HK Electric's network less susceptible to adverse weather conditions.



HK Electric launches a new Low-Voltage Management System (LVMS) which delivers near real-time visibility of the low-voltage network, helping to improve operational efficiency, minimise fault impacts, and accelerate supply restoration.



Power restoration drills are conducted periodically to maintain emergency preparedness.



With the completion of the rollout of smart meters, together with the associated network and IT infrastructure across our customer base of more than 599,000, the Advanced Metering Infrastructure helps customers better manage energy use while strengthening HK Electric's overall grid management.



HK Electric's Intelligent Distribution Solution integrates AI video analytics and edge computing to enhance grid visibility and resilience. The system enables real-time power quality monitoring and hotspot detection, reducing fault detection time and equipment failure risk.





HK Electric advances transport electrification through Smart Power Services, supporting electric buses, ferries, EV charging and petrol-station upgrades



Under the Feed-in Tariff (FiT) Scheme, 750 customer-side FiT installations are now linked to HK Electric's grid with a combined capacity of 15.1 MW. An International School expanded its BIPV system under the support of FiT Scheme, strengthening local renewable generation.



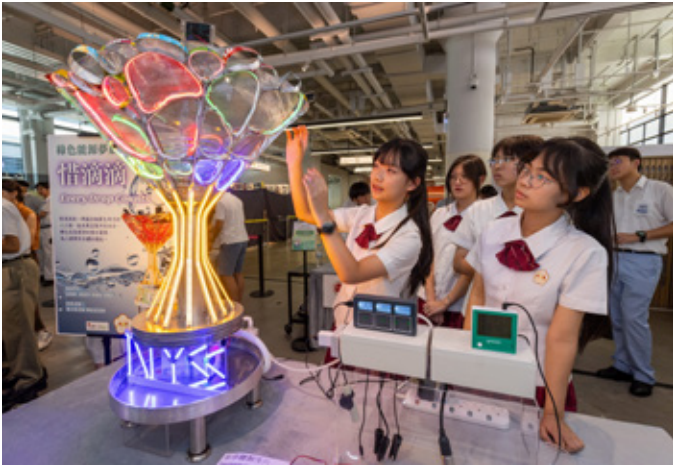
Over 100 industry leaders and partners from the political, business, engineering, and academic sectors, attend HK Electric 135th Anniversary Engineering Conference to explore future directions in innovation, energy transition, and sustainable development.



Children enjoy interactive games at the Smart Power Gallery while learning about renewable energy and sustainable living.



HK Electric organizes Electrical Safety Fun Day to educate seniors and their caregivers on safe and smart use of electricity.



Through the Green Energy Dreams Come True competition, HK Electric encourages students from secondary schools and tertiary institutions to propose innovative ideas to help make Hong Kong a more sustainable city.



HK Electric actively facilitated the electrification of various public transport modes, including buses and ferries. In addition, as a member of the Green Rehabilitation Bus Pilot Scheme Steering Committee, HK Electric assisted social welfare organisations in transitioning to electric rehab buses. One such initiative involves the Aberdeen Kai-fong Welfare Association Social Service, where HK Electric provided a tailored power supply solution to support EV charging facilities, enabling greener transport and improved mobility services for people with disabilities.



The Green Hong Kong Green programme offers eco-heritage tours to enhance public knowledge of local eco-heritage resources and foster an appreciation for nature.



HK Electric Volunteer Team supports cultural events, deliver environmental programmes, and assist underprivileged communities.





HK Electric supported different stakeholders through the Energy-efficient Community Subsidy, providing up to HK\$3 million to aided schools, non-governmental organisations and Community Living Rooms to enhance energy efficiency and promote low-carbon living. This included upgrades to high-efficiency air-conditioning, LED lighting and electric equipment at Tung Wah Group of Hospitals Fong Shu Chuen Social Service Building; the installation of energy-efficient appliances at two Community Living Rooms, and the provision of electric laundry facilities for a secondary school providing a Special Education Needs (SEN) programme.

CLP Power



CLP Power has collaborated with The Hong Kong Polytechnic University to jointly develop a generator inspection robot, which can flexibly navigate the narrow air gaps between the rotor and stator of the generator to automatically complete major inspection tasks without the need to remove the rotor. It significantly enhances inspection efficiency and operational safety. The invention won two awards at the 51st International Exhibition of Inventions Geneva.



CLP Power's eMobility Grid Management Platform (eGMP) is an innovative solution designed to analyse EV charging data, such as the utilisation and usage patterns of charging stations across different locations in Hong Kong, enabling the power company to optimise grid planning and resource allocation in support of the expansion of the EV charging network.



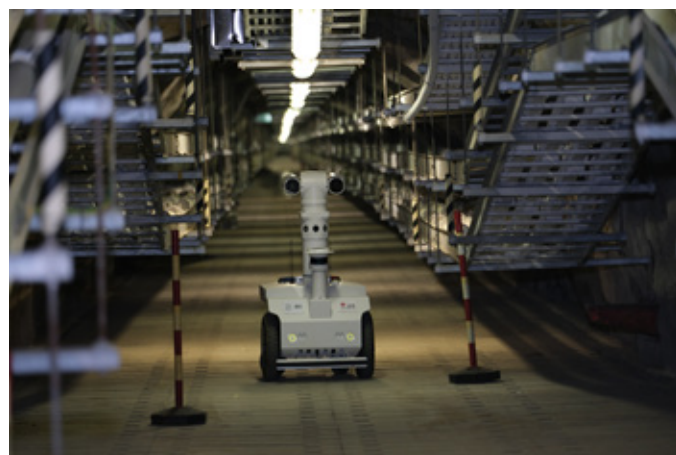
CLP Power adopted Beyond Visual Line of Sight (BVLOS) operations to inspect transmission towers and overhead lines as one of the first batch of LAE Regulatory Sandbox Pilot Projects, significantly enhanced the efficiency of inspections.



To further enhance the reliability of power system, CLP Power has introduced an intelligent management system Grid-V to monitor the conditions of key power facilities in real time.



HK Electric has developed a Fish Deterrent System at Lamma Power Station which has achieved an impressive 99.5% reduction in fish intrusion into its water-cooling system. This innovation not only ensures the stable and reliable operation of the power plant but also plays a crucial role in conserving the marine environment and protecting fisheries resources.



HK Electric has upgraded its Cyberport cable tunnel to a "Smart Cable Tunnel", using robots and a 5G network for round-the-clock automated monitoring. This innovation enhances the safety and efficiency of inspection in confined spaces.



HK Electric has commissioned Hong Kong's first Low-Voltage Direct Current (LVDC) System at Ocean Park, establishing a hybrid alternating current/direct current (AC/DC) microgrid to serve the "Giant Panda Adventure" exhibit — home to the city's giant pandas — and adjacent facilities. In the event of system anomaly, devices within the microgrid can swiftly switch power sources, ensuring stable supply while minimising disruption.

INDONESIA



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Jakarta
Area:	1,904,569 km ²
Population:	≈ 284.4 millions (2025 estimate)
GDP:	US\$ 1.44 trillion
Currency:	Indonesian Rupiah (IDR)
Installed Capacity (in GW):	75.9 GW
Percentage of population electrified:	99.83%

Power Demand by Sector

Golongan Tarif / Tariff Category	Pelanggan / Customers	Tersambung / Connected (MVA)	Terjual / Sold (MWh)	Pendapatan *) (ribu Rp) / Revenue *) (thousand Rp)
S-1	2.164.021	6.680,20	8.465,67	6.628.854.677,15
S-2	7.786	2.824,81	4.685,89	4.266.307.630,61
S-3	29	22,50	41,53	56.193.143,64
R-1	84.522.694	77.793,67	115.516,70	110.872.144.904,25
R-2	2.365.289	10.155,41	12.173,80	20.679.599.904,08
R-3	511.665	6.229,62	6.873,35	11.661.965.233,63
B-1	4.679.126	8.635,69	11.288,80	10.818.552.249,34
B-2	912.708	18.862,83	25.927,14	37.607.526.970,92
B-3	9.889	13.112,04	24.760,25	28.544.182.692,93
I-1	228.266	877,31	509,77	540.534.771,31
I-2	54.779	5.399,09	7.943,47	8.571.198.932,32
I-3	17.084	26.807,44	62.403,93	70.613.026.350,60
I-4	119	7.191,87	24.246,21	22.511.795.192,53
P-1	261.776	2.476,01	3.057,90	4.918.989.557,03
P-2	2.393	2.104,92	2.301,54	3.613.521.395,39
P-3	395.265	1.463,04	3.502,51	5.980.296.773,16
Traksi T-1	99	711,79	486,23	386.198.046,40
Curah C-1	3.324	1.032,36	1.586,48	1.677.992.578,96
Layanan Khusus	37.922	240,01	1.920,68	3.544.493.047,67
Total	96.174.234	192.620,59	317.691,86	353.493.374.051,90

*) Data diambil dari Aplikasi Pengelolaan Pelanggan Terpusat (AP2T) / Data is taken from AP2T Application

National Energy Policy

PLN is prioritizing sustainable business practices by balancing economic, environmental, social, and governance. This practice is followed by considering affordability of electricity prices and the sustainability of corporate finance and national fiscal, emissions reduction, and reliability of electricity supply. The development of electricity aims to ensure the availability of electricity in sufficient quantities and reliability to improve the welfare and prosperity of the people and to ensure sustainable development is achieved.

PLN follows National General Energy Planning (KEPMEN ESDM 143 K/20/MEM/2019 RUKN 2019-2038) :

i. Strategy of Generation Expansion

- The location of power plants must be as close as possible to the source of primary energy.
- Every power plant must supply the electricity to the surrounding demand.
- Renewable energy resources are the first priority, then mine-mouth coal power plants and well-head gas power plants are the next priority.
- In 2025, 23% of the source of electricity is renewable energy.
- Improve the operational flexibility of power plants, so that it can be operated both as base load and load following power plant.
- Implementing environmentally friendly technologies in operating the coal-fired power plant.
- Using pumped storage as the load factor correction and as the storage of the excess energy sourced from variable renewable energy sources.
- Implementing close cycle and combined cycle in gas power plants.
- Implementing the co-firing of biomass and coal in operating coal-fired power plants.

ii. Strategy of Transmission Expansion

- Inefficient power evacuation from power plant or substation to the load must be transferred using a distribution network.
- Reducing losses and voltage drop.
- Improving system interconnections.
- Developing HVDC in system interconnections.
- Ensure that the current capacity lasts for at least 30 years.
- Developing interisland connections.

iii. Strategy of Distribution Expansion

- The development of the distribution network aims to achieve a 100% electrification ratio.
- Improvement of electricity quality and reliability.
- Prioritise energy transfer to national strategic area, special economic area, prioritized tourism destinations, and industrial area.
- Developing smart grid infrastructure and micro grid infrastructure.

iv. Net Zero Emissions Commitment

- Indonesia is committed to reaching Net Zero Emissions (NZE) by 2060, with interim targets that include a substantial reduction in greenhouse gas emissions by 2030.

v. Energy Security

- The policy emphasizes energy security through diversification of energy sources, investment in infrastructure, and development of domestic resources.

Supply and Demand of Primary Natural Resources

i. Supply of primary energy sources

1. Coal:

- a. Supply: Indonesia is one of the largest coal producers globally, with significant reserves. Coal remains a dominant source of energy for electricity generation.
- b. Demand: The demand for coal is gradually declining due to environmental regulations and the shift toward renewable energy. With the phase-down program for coal-fired power plants (PLTU) as outlined in PLN's RUPTL, the demand for coal is expected to decline significantly.

2. Natural Gas:

- a. Supply: Indonesia has substantial natural gas reserves and is a major exporter. The government is promoting natural gas as a transition fuel to reduce carbon emissions.
- b. Demand: The demand for natural gas is increasing, particularly in power generation and industrial sectors.

3. Renewable Energy:

- a. Supply: Indonesia has abundant renewable resources, including geothermal, solar, hydro, and biomass.
- b. Demand: There is growing demand for renewable energy sources, driven by both domestic policies and international commitments to sustainability.

4. Biomass and Waste-to-Energy:

- a. Supply: Biomass resources are abundant due to the country's agricultural sector, and waste-to-energy initiatives are being developed.
- b. Demand: Demand for biomass and waste-to-energy solutions is expected to rise as part of the commitment to renewable energy and sustainability.

Laws and Governmental Regulations

- Law Number 30 of 2007 concerning Energy
- Law Number 30 of 2009 concerning Electricity has been amended by Law Number 6 of 2023 concerning the Determination of Government Regulations in Lieu of Law Number 2 of 2022 concerning Job Creation into Law;
- Government Regulation Number 14 of 2012 concerning Business Activities for Providing Electricity has been amended by Government Regulation Number 23 of 2014 concerning Amendments to Government Regulation Number 14 of 2012 concerning Business Activities for Providing Electricity;
- Government Regulation Number 42 of 2012 concerning Cross-Border Sales and Purchase of Electric Power;
- Government Regulation Number 62 of 2012 concerning Electric Power Support Services Businesses;
- Regulation of the Minister of Energy and Mineral Resources No. 15 of 2010 concerning the List of Fast-Track Power Plant Projects Utilizing Renewable Energy, Coal, and Gas and Related Transmission, as amended by Regulation No. 21 of 2013;
- Government Regulation Number 79 of 2014 concerning National Energy Policy;
- Presidential Regulation No. 4 of 2010 concerning the Assignment to PT Perusahaan Listrik Negara (Persero) to Accelerate the Development of Power Plants Using Renewable Energy, Coal, and Gas, as last amended by Presidential Regulation No. 194 of 2014;
- Regulation of the Minister of Energy and Mineral Resources No. 32 of 2008 concerning the Supply, Utilization, and Distribution of Other Fuels, as last amended by Regulation No. 12 of 2015;

- Government Regulation No. 14 of 2015 concerning the National Industrial Development Master Plan (RIPIN) for 2015–2035;
- Regulation of the Minister of Energy and Mineral Resources No. 9 of 2016 concerning Procedures for the Provision and Pricing of Coal for Mine-Mouth Power Plants, as amended by Regulation No. 24 of 2016;
- Presidential Regulation Number 4 of 2016 concerning the Acceleration of Electricity Infrastructure Development has been amended by Presidential Regulation Number 14 of 2017 concerning Amendments to Presidential Regulation Number 4 of 2016 concerning the Acceleration of Electricity Infrastructure Development;
- Presidential Decree No. 6 of 2017 concerning the Designation of Outermost Small Islands;
- Regulation of the Minister of Energy and Mineral Resources No. 19 of 2017 concerning the Utilization of Coal for Power Plants and Purchase of Excess Power;
- Presidential Regulation No. 22 of 2017 concerning the National Energy Plan;
- Government Regulation No. 46 of 2017 concerning Environmental Economic Instruments;
- Decree of the Head of the Investment Coordinating Board on behalf of the Minister of Energy and Mineral Resources No. 2 of 2018, concerning the Amendment of the Electricity Supply Business License of PT PLN (Persero);
- Regulation of the Minister of Energy and Mineral Resources No. 10 of 2017 concerning Main Provisions in Power Purchase Agreements, as amended by Regulation No. 49 of 2017 and Regulation No. 10 of 2018;
- Presidential Regulation No. 35 of 2018 concerning the Acceleration of the Development of Waste-to-Energy Power Plants Based on Environmentally Friendly Technology;
- Regulation of the Minister of Energy and Mineral Resources No. 37 of 2018 concerning the Offering of Geothermal Working Areas, Issuance of Geothermal Licenses, and Assignment of Geothermal Operations;
- Regulation of the Minister of Energy and Mineral Resources No. 58 of 2017 concerning the Sale Price of Natural Gas via Pipelines in Downstream Oil and Gas Activities, as amended by Regulation No. 14 of 2019;
- Regulation of the Minister of Environment and Forestry No. 15 of 2019 concerning Emission Quality Standards for Thermal Power Plants;
- Regulation of the Minister of Energy and Mineral Resources No. 27 of 2017 concerning Service Quality and Costs Related to the Distribution of Electricity by PT PLN (Persero), as amended by Regulation No. 18 of 2019;
- Decree of the Minister of Energy and Mineral Resources No. 143 of 2019, concerning the National Electricity General Plan for 2019 to 2038;
- Regulation of the Minister of Energy and Mineral Resources No. 28 of 2016 concerning Electricity Tariffs Provided by PT Perusahaan Listrik Negara (Persero), as last amended by Regulation No. 3 of 2020;
- Regulation of the Minister of Energy and Mineral Resources No. 50 of 2017 concerning the Utilization of Renewable Energy Sources for Electricity Supply, as last amended by Regulation No. 4 of 2020;
- Regulation of the Minister of Energy and Mineral Resources No. 45 of 2017 concerning the Utilization of Natural Gas for Power Generation, as amended by Regulation No. 10 of 2020;
- Regulation of the Minister of Energy and Mineral Resources No. 7 of 2017 concerning Procedures for Determining Benchmark Prices for the Sale of Metallic Minerals and Coal, as amended by Regulation No. 11 of 2020;
- Regulation of the Minister of Energy and Mineral Resources No. 25 of 2018 concerning the Management of Mineral and Coal Mining, as last amended by Regulation No. 17 of 2020;

- Decree of the Minister of Energy and Mineral Resources No. 34 of 2020 concerning the Determination of Allocation and Utilization of Natural Gas for Electricity Supply by PT Perusahaan Listrik Negara (Persero).
- Presidential Regulation No. 40 of 2016 concerning the Determination of Natural Gas Prices, as amended by Presidential Regulation No. 121 of 2020;
- Regulation of the Minister of Energy and Mineral Resources No. 5 of 2021 concerning Business Activity Standards and Products in the Implementation of Risk-Based Business Licensing in the Energy and Mineral Resources Sector;
- Minister of Energy and Mineral Resources Regulation Number 11 of 2021 concerning Implementation of Electricity Businesses.
- Government Regulation Number 25 of 2021 concerning the Implementation of the Energy and Mineral Resources Sector;
- Regulation of the Minister of Energy and Mineral Resources No. 26 of 2021 concerning Rooftop Solar Power Plants Connected to the Electricity Grid of Public Electricity Supply Business License Holders;
- Regulation of the Minister of Energy and Mineral Resources No. 16 of 2022 concerning Procedures for Implementing Carbon Economic Value in the Power Generation Subsector;
- Presidential Regulation Number 112 of 2022 concerning the Acceleration of Renewable Energy Development for the Supply of Electric Power;
- Regulation of the Minister of Energy and Mineral Resources No. 10 of 2023 concerning Procedures for the Preparation, Submission, and Approval of Work Plans and Budgets as well as Reporting of Mineral and Coal Mining Business Activities

Total amount of energy consumption by sectors (Industrial/ Commercial/ Residential) (in MWh and %)

Golongan Tarif / Tariff Category	Pelanggan / Customers	Tersambung / Connected (MVA)	Terjual / Sold (MWh)	Pendapatan *) (ribu Rp) / Revenue *) (thousand Rp)
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*) Data diambil dari Aplikasi Pengelolaan Pelanggan Terpusat (AP2T) / Data is taken from AP2T Application

3

Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (Coal, LNG, Gas, Nuclear-fired and Hydro) including total installed capacity of generation (in MWe) and total production (in MWh) by fuel types (in MWe)

Dec 25	Installed Capacity (GW)	Year-on-year Growth (%)
Total	80,189	5.6
Hydropower (PLTA, PLTM, PLTMH)	6,344	6.98
Thermal Power (PLTU, PLTG, PLTGU, PLTD, PLTMG, PLTBm)	70,929	5.41
Nuclear Power (PLTN)	-	-
Wind Power (PLTB)	0.13	0
Solar Power (PLTS)	0.318	30.86
Other (PLTP)	2,468	5.34

	Generation (TWh)	Year-on-year Growth (%)
Total	354,928	3.21
Hydropower	24,449	13.66
Thermal Power	311,841	2.33
Nuclear Power	-	-
Wind Power	0.383	-18.51
Solar Power	0.516	22.86
Other (PLTP)	17.74	5.85

4

Reduction of Carbon Emission

Policy on Renewable Energy and major regulatory mechanism of Renewable Law

Indonesia's renewable energy development is guided by a combination of national planning documents and sector-specific regulations that collectively provide the strategic direction, implementation roadmap, and commercial framework for renewable energy deployment.

The National Electricity General Plan (RUKN) serves as the overarching policy framework for the power sector. It establishes the long-term direction for electricity development, emphasizing energy security, affordability, sustainability, and the gradual transition toward a low-carbon electricity system. Building

upon the RUKN, PLN's Electricity Supply Business Plan (RUPTL) translates national policy into an actionable investment roadmap for the electricity sector. The 2025-2034 RUPTL prioritizes renewable energy as the primary source of new generation capacity, including solar PV, hydropower, geothermal, wind, and bioenergy, while also expanding battery energy storage systems (BESS), pumped hydro energy storage (PHES), and transmission infrastructure to improve system flexibility and facilitate higher renewable energy integration. The RUPTL therefore serves as the principal implementation instrument for Indonesia's electricity transition over the coming decade.

Complementing these planning documents, Presidential Regulation 112/2022 on the Acceleration of Renewable Energy Development for Electricity Supply (Perpres 112/2022) establishes the key regulatory mechanisms to support renewable energy deployment. The regulation introduces a dedicated framework for renewable energy procurement, establishes benchmark pricing mechanisms for renewable electricity, prioritize renewable energy utilization in electricity planning, and provides the policy basis for the gradual retirement of coal-fired power plants under specific conditions while maintaining system reliability and affordability. The regulation also strengthens the investment environment by improving regulatory certainty for renewable energy developers and aligning electricity sector planning with Indonesia's long-term decarbonization objectives.

Together, the RUKN, RUPTL, and Perpres 112/2022 form the core policy architecture governing Indonesia's renewable energy transition, combining strategic planning, infrastructure development, and regulatory support to accelerate renewable energy deployment while ensuring reliable electricity supply.

Production of Renewable Energy (MWh) from various resources (wind, solar, biomass, landfill gas, etc)

In 2025, electricity generated from renewable energy sources are as follows. It should be noted that these numbers are total electricity production from PLN and IPP.

Power Plant	Electricity Production (Gwh)
Hydro	24.448,47
Wind	382,462
Geothermal	17.740,33
Solar	515,93
Biomass	2.542,46
Biofuel	4.155,583

Plans and Strategies to Reduce Greenhouse Gas Emissions

PLN's roadmap to achieve NZE by 2060 is built on three key pillars and two critical enablers. The first two pillars focus on transitioning from carbon-intensive power generation through three strategic objectives: accelerating renewable energy development, decarbonizing the generation portfolio, and strengthening system infrastructure and flexibility. The first pillar spans short-term (2025-2026) and medium-term (2026-2030) goals to support the NDC. The second pillar is dedicated to the long-term (2031-2060) goal of achieving NZE aspirations. The third pillar emphasizes developing new technologies and business ecosystem to enable sustainable growth. Complementing these three pillars are two critical enablers: developing human capital and securing financing and investment. Human capital development involves upskilling and empowering the workforce to drive innovation and implement solutions across the three pillars. Meanwhile, securing adequate funding ensures the financial resources necessary to realize various initiatives and projects under the three pillars.



Latest AMI Installations and Smart Grid Projects

1. AMI Managed Services currently serves 1,220,630 customers spread across 8 UID/UIW (Distribution/Regional Units), with the breakdown as follows:

- UID Sumatera Utara (Sumut): 42,208
- UID Banten: 143,512
- UID Jaya: 181,856
- UID Jawa Barat (Jabar): 132,998
- UID Jawa Tengah (Jateng): 52,073
- UID Jawa Timur (Jatim): 83,665
- UID Bali: 528,827
- UID Sulselrabar(SSTB): 55,501

2. PLN has initiated Smart Grid development as part of Indonesia's power system modernization and energy transition. One of the flagship projects is the Smart Microgrid in Nusa Penida Island, Bali, which integrates renewable energy, battery energy storage systems (BESS), and advanced monitoring and control technologies to improve system reliability and renewable energy utilization.

Challenges faced and how it was solved / addressed:

The initial challenge in AMI deployment was the accuracy of customer and network data, which is critical for ensuring reliable meter operations, network visibility, and future Smart Grid applications. To address this issue, PLN conducted comprehensive restructuring and cleansing of customer and network databases (PDPJ), strengthened telecommunications infrastructure in deployment areas, and implemented

standardized integration between AMI, distribution management, and enterprise systems. In addition, pilot projects and phased deployment strategies were adopted to validate technologies, improve operational readiness, and ensure scalability. These initiatives have significantly improved data quality, system interoperability, and operational efficiency, establishing a strong digital foundation for broader Smart Grid deployment.

Building on the experience gained from AMI dan Smart Grid implementation, PLN has identified several key challenges for Smart Grid development, including the integration of renewable energy resources, limited grid visibility, varying levels of network infrastructure maturity across regions, increasing data management complexity, interoperability between legacy and new systems, and cybersecurity risks associated with greater digitalization.

Another challenge in Smart Grid implementation is the regulatory and financial framework required to support large-scale deployment. The transition to Smart Grid involves significant upfront investments in digital infrastructure, advanced equipment, communication networks, and cybersecurity systems, while existing regulations may not yet fully accommodate emerging technologies such as distributed energy resources (DER), demand response, and energy storage systems. Thus, PLN is working closely with government agencies and stakeholders to develop supportive policies, establish appropriate regulatory frameworks, and prioritize investments through a phased and cluster-based implementation approach to ensure cost-effectiveness and long-term sustainability.

PLN has developed a Smart Grid Implementation Roadmap to address these challenges, with implementation planned to begin in 2026. The roadmap adopts a cluster-based approach to align Smart Grid solutions with the characteristics and readiness of each distribution system:

- a. **Large Cluster** – Metropolitan and highly interconnected systems requiring advanced solutions such as Advanced Distribution Management Systems (ADMS), Distributed Energy Resource Management Systems (DERMS), Demand Response Management Systems (DRMS), large-scale AMI deployment, and advanced grid automation.
- b. **Medium Cluster** – Urban and regional systems focusing on AMI expansion, distribution automation, renewable energy integration, and asset optimization.
- c. **Small Cluster** – Remote, rural, and island systems emphasizing smart microgrids, renewable energy integration, Battery Energy Storage Systems (BESS), remote monitoring, and energy management systems.

This phased and cluster-based strategy leverages the digital foundation established through AMI deployment to enhance grid reliability, accelerate renewable energy integration, improve operational efficiency, and support Indonesia's transition toward a low-carbon, resilient, and digitalized power system.

AMI Roll out plan

The AMI roll-out plan targets the following number of customers per year:

- 2026: 1,316,038 customers
- 2027: 2,711,949 customers
- 2028: 2,605,220 customers
- 2029: 2,602,856 customers
- 2030: 2,793,472 customers

AMI offerings to customers and its benefit

Through AMI, PLN delivers a range of value-creation services to customers, with the following key offerings and benefits:

- **Remote Meter Reading (EOB):** meters read remotely and automatically, improving billing accuracy and reliability.
- **Electricity Theft & Misuse Detection (P2TL evaluation):** remote detection of irregularities to reduce losses and recover energy.
- **Remote Connect/Disconnect (Tusbung):** connection and disconnection orders executed remotely, speeding up service.
- **Energy Management System (EMS):** supports demand-side management.
- **Early Warning System (EWS):** faster outage response and restoration (outage management).
- **Loss Analysis:** distribution loss analysis across regional units.
- **Substation Inspection:** remote monitoring and inspection of substations.
- **Beyond-kWh services:** control of distributed energy resources (DER) and demand response management (DRMS).
- **Smart Prepayment:** flexible prepaid scheme for customers.
- **New Connection / Power Upgrade / Migration (PB/PD/Migration):** uses the power limiter on the AMI meter, set remotely, so connections, power up/down-grades, and migrations are processed faster.

One of the most significant transformations impacting the electricity supply industry in our country is the large-scale digital transformation being implemented across all business functions. The organization is progressively adopting digital technologies to modernize its operations, improve service delivery, and enhance decision-making capabilities.

As variable renewable energy sources such as solar and wind become more prominent, maintaining system reliability becomes increasingly challenging due to their intermittent nature. To address this issue, Indonesia's RUPTL 2025-2034 includes the development of energy storage systems, particularly Battery Energy Storage Systems (BESS) and Pumped Storage Hydropower (PSH). While several projects are already under construction, these technologies require substantial investment and careful integration into the power system.

The primary objectives of this transformation are to increase transparency, improve operational efficiency, and establish a single source of truth for data across the enterprise. Digital initiatives have been introduced in various areas, including asset management, customer services, financial processes, workforce management, and data analytics. By integrating previously siloed systems and processes, the organization aims to create a more connected, data-driven, and agile operating environment.

However, as digitalization expands and operational technologies become increasingly interconnected with information technologies, cybersecurity resilience has emerged as a critical priority. The growing reliance on digital platforms, cloud services, smart grid technologies, and real-time data exchange has significantly increased the organization's cyber risk exposure. Protecting critical infrastructure, ensuring operational continuity, and maintaining public trust have therefore become essential considerations alongside digital innovation.

Another key challenge is the uneven geographical distribution of renewable energy resources across the country. Regions with abundant renewable energy potential are often located far from major demand

centers. Therefore, expanding and strengthening transmission interconnections between islands and regions is essential to enable efficient power transfer, optimize renewable energy utilization, and support the country's long-term energy transition objectives.

To strengthen cybersecurity resilience, the organization has implemented several best-practice initiatives, including:

- Establishing a comprehensive cybersecurity governance framework aligned with international standards such as ISO/IEC 27001 and the NIST Cybersecurity Framework.
- Conducting regular cybersecurity risk assessments and integrating cyber risk management into enterprise risk management processes.
- Implementing a Security Operations Center (SOC) with continuous monitoring and threat detection capabilities.
- Enhancing network segmentation and security controls between Information Technology (IT) and Operational Technology (OT) environments.
- Deploying multi-factor authentication (MFA), privileged access management (PAM), and zero-trust security principles for critical systems.
- Performing regular vulnerability assessments, penetration testing, and security audits.
- Strengthening incident response and cyber crisis management capabilities through exercises and simulations.
- Developing cybersecurity awareness and training programs for employees and contractors.
- Improving third-party and supply chain cybersecurity risk management practices.
- Establishing business continuity and disaster recovery capabilities to support resilience against cyber incidents and operational disruptions.

Through these efforts, the organization seeks to ensure that digital transformation not only delivers greater efficiency and transparency but also maintains a robust level of cybersecurity resilience to support the reliable and secure delivery of electricity services.





KOREA



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Seoul
Area:	100,460 km ²
Population:	51.6 million
GDP:	US\$ 1.87 trillion
Currency:	KRW
Installed Capacity (in GW):	137.5
Percentage of population electrified:	100%

Power Demand by Sector

Sector	Total Power Consumption as of 2025 (MWh)	Projected Change (%) 2024/2025	Projected Power Consumption as of end 2025 (MWh)
Industrial (Production)	271,596	97.97	271,596
Residential (Households)	85,832	101.66	85,832
Public & Service	191,989	102.04	191,989
TOTAL	549,417	99.83	549,417

National Energy Policy

In February 2025, the Korean government officially confirmed the 11th Basic Plan for Power Supply and Demand (2024–2038), which aims to ensure flexibility, energy security, and carbon neutrality through 1) demand management, 2) facility plan, 3) power system, and 4) electricity market.

1) Demand management

The 11th Plan targets a peak demand reduction of 13.4GW by 2038 through digital-based efficiency programs, expanded demand response (DR), and real-time metering infrastructure. Electricity demand is projected to grow at 1.8% annually, reaching 129.3GW by 2038, mainly driven by EV expansion, AI industries, data centers, and hydrogen utilization.

2) Facility Plan

Nuclear power capacity will increase to 35.2% by 2038, including SMR, Shin-Hanul Units 3 and 4. In particular, coal power generation will be drastically reduced to resolve the issues of fine dust and greenhouse gas and will be converted to LNG. Also, regarding the pace of renewable energy supply, it is expected to revise its Intermediate goal upward by 2025, according to Korea's Green New Deal policy.

Renewables will expand to 121.9GW by 2038, including 117.9GW from solar and wind power according to the 11th Plan. LNG capacity will be moderately maintained. Hydrogen/ammonia co-firing will be introduced. Coal capacity will reduce to 22.2GW by 2038. The Power Sector will target a 45.9% GHG reduction by 2030 compared to 2018, in alignment with Korea's NDC.

Dispersed energy resources (decentralized generation) will be actively expanded, especially small-scale units near demand centers, in order to reduce dependence on long-distance transmission infrastructure.

3) Power System

The 11th Plan reinforces the transmission and distribution system by expanding high-voltage lines, regional interconnections, and flexibility resources to support large-scale renewable integration.

In addition, it will build backup facilities such as pumped ESS, which can adjust the output within a short time, for stable operation of the power system and quick response to output changes from the increasing new & renewable energy.

It plans to build the foundation for rationalizing the compensation system for expanding dispersed energy and for establishing new laws and systems to improve the utilization of dispersed resources.

4) Electricity market

The 11th Plan outlines reforms to enhance price signals, support flexible resources like DR and ESS, and enable integration of renewables under increasing variability. Forecasting obligations will be strengthened for large-scale renewable generators to improve reliability and system operation. The 11th Plan expands compensation for flexibility resources, including DR and ESS, to maintain system stability under high renewable shares.

Supply and Demand of Primary Natural Resources

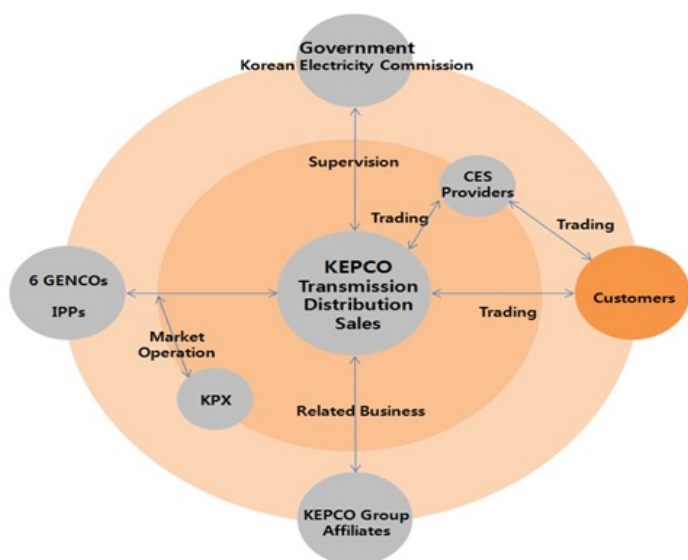
Resource	Capacity (MW) as of end 2025	Generation (GWh) for 2025
Hydro	4,700	4,423
Nuclear	26,050	184,693
Renewables	38,747	68,117

(Source : Monthly Report on Major Electric Power Statistics, KEPCO, January 2026)

Laws and Governmental Regulations

Korea heavily depends on overseas markets for energy (97%) and its geographically-isolated grid system makes it impossible to import electricity from neighboring countries. In addition, long-distance transmission is required because production is concentrated in the south while most of the power is consumed in the northern part of the country. Furthermore, demand is very difficult to control, and considerable time and massive investment is required to secure supply capacity.

The six power generation companies (GENCOs), independent power producers (IPPs) and community energy system (CES) providers generate electric power and sell it to KEPCO on the Korea Power Exchange (KPX). KEPCO, in turn, sells electricity to general customers through the transmission and distribution network. Competition in the sales sector of the industry is increasing as large-volume customers (more than 30,000 kW) are allowed to buy electric power directly from producers, and each CES provider is allowed to supply electric power in a specific district.



**Total amount of energy consumption by sectors
(Industrial/ Commercial/ Residential) (in MWh and %)**

Sector	Total Power Consumption as of 2025 (MWh)	% in Total
Industrial (Production)	271,596	49.43%
Residential (Households)	85,832	15.62%
Public & Service	191,989	34.95%
TOTAL	203,579,000	100%

(Source : Monthly Report on Major Electric Power Statistics, KEPCO, January 2026)

3

Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (coal, LNG, Gas, Nuclear-fired and hydro) including total installed capacity of generation (in MWe) and total production (in MWh) by fuel types (in MWe)

Fuel Types	Installed Capacity (MW)	Production (GWh)	Fuel Types	Production (GWh)	Year-on-year Growth (%)
Hydro	4,700	4,424	Hydro	4,424	-5.409%
Coal	40,767	170,828	Coal	170,828	2.195%
Oil	628	1,138	Oil	1,138	-5.482%
Gas	45,204	163,111	Gas	163,111	-2.448%
Nuclear	26,050	184,693	Nuclear	184,693	-2.151%
Renewables	38,747	68,118	Renewables	68,118	7.858%
Others	467	3,337	Others	3,337	-2.255%
Total	156,563	595,649	Total	595,649	0.014%

(Source : Monthly Report on Major Electric Power Statistics, KEPCO, January 2026)

(Source : Monthly Report on Major Electric Power Statistics, KEPCO, January 2026)

4

Reduction of Carbon Emission

Policy on Renewable Energy and Major Regulatory Mechanism of Renewable Law

Regarding the “Policy on Renewable Energy and major regulatory mechanism of Renewable Law”, Korea announced the “6th Basic Plan for Renewable Energy Technology Development, Use and Distribution” in December 2023. Its goal is to increase the share of new & renewable energy to 33% of power generation by 2038, and to this end, it has established six policy tasks. The six policy tasks are shown in the table below.

Task	Key Details
Promoting Public Participation	▶ Expanding private use, small scale, farms, etc. to supply 48.7GW of new & renewable energy facilities ▶ Enhancing profit stability of small business operators and simplifying procedures through Korean FIT
Operating Market-Friendly System	▶ Increasing RPS mandatory ratio and adjusting REC weight considering the characteristics of each energy source ▶ Introducing a two-way REC trading system and expanding the opening cycle of spot market
Expanding Entry Into Overseas Markets	▶ Support for joint venture between large (public) companies and small and medium-sized companies through consultative bodies for overseas entry ▶ Establishing customized strategies considering the industrial features of the target country
Creating new markets	▶ Implementing the mandatory system for mixing new & renewable energy fuels for transportation ▶ Support for large-scale projects centered on private and public institutions, etc.
Enhancing R&D capacity of new & renewable energy	▶ Providing support for strategic R&D and demonstration for solar and wind power to lead the future market ▶ Creating clusters to strengthen capabilities of region-centered renewable energy ecosystem
Expanding the base for systematic support	▶ Introducing 44 international standards of new & renewable energy into KS by 2018 and internationalizing industrial standards ▶ Mitigating regulations such as rental standards of national and public property and building a foundation for waste disposal

Production of Renewable Energy (GWh) from various resources (wind, solar, biomass, Fuel Cell, etc)

Resource Types	Production (GWh) (2025)	%
Wind	3,664	5.39
Solar	37,798	55.64
Biomass	11,854	17.45
Fuel Cell	8,417	12.39
IGCC	1,510	2.22
Hydro	3,804	5.60
Other Renewables	888	1.31
Total	67,935	100

(Source : Monthly Report on Major Electric Power Statistics, KEPCO, January 2026)

Environmental Problems of Electricity Supply (including generation and T&D sectors) and How It Is Being Addressed

KEPCO has set up an integrated response system for our group companies to respond to climate change. We have established the 'Basic Plan for GHG Reduction of KEPCO Group Companies 2030' and the 'RE3020 Basic Plan for KEPCO Group Companies' to take the lead in reducing GHG emissions and promoting new & renewable energy. In order to achieve our GHG reduction target (a reduction of 47 million tons compared to BAU by 2030), construction of new coal-fired power plants has been prohibited in principle. And a decision was made to decommission a total capacity of 3,345 MW, 10 old coal-fired power plants

Plans and Strategies to Reduce Greenhouse Gas Emissions

and a plan was confirmed to convert four coal-fired power plants to LNG. Furthermore, in September 2019, we submitted an additional LNG conversion plan for fourteen coal-fired power plants(7 GW) to the government. In addition, KEPCO is promoting large-scale new & renewable energy business development, and is expanding research and development in the field of new & renewable energy with the aim of achieving a clean and safe energy mix. We are also striving to make the entire power supply value chain eco-friendly, from production to sales, by reducing coal-fired power generation, collecting and purifying SF6 gas, and improving energy efficiency

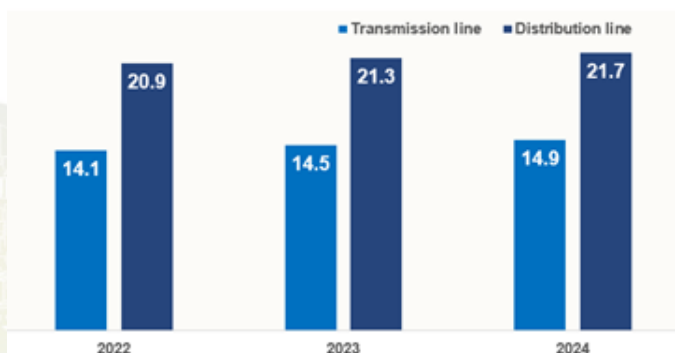
KEPCO recognizes that the construction of power facilities is an important issue directly impacting the residential environment of citizens. Accordingly, in order to increase social acceptance, we are striving to build eco-friendly and resident-friendly power facilities while also expanding our underground power facilities.

When constructing a transmission network, areas close to living environments such as villages and schools are selected as priority targets for undergrounding. Separately, in order to strengthen safety in case of disaster, the company has identified vulnerable facilities and has been continually conducting undergrounding work on its own.

Looking forward, KEPCO plans to continue developing new and advanced underground facility models to minimize ground equipment for the construction of power facilities that coexist with local residents.

Status of Underground Transmission and Distribution Rates

(Unit : %)



Increasing renewable share through solar Korea declared "Carbon-neutrality 2050" in December 2020, and established related strategies. Carbon neutrality means that it will make zero emission of greenhouse gases such as carbon dioxide emitted by individuals, companies, and organizations and the Korean government built its three policy directions such as 'low carbonization of the economic structure', 'creating a new promising low-carbon industrial ecosystem', and 'fair transition to a carbon-neutral society' as strategies for promoting carbon neutrality, and proposed '3+1' strategies that added the reinforcement of a carbon-neutral institutional foundation to the three policy directions. In addition, top ten tasks according to the three policy directions are presented as follows: 'accelerating energy transition', 'innovating high-carbon industrial structure', 'transition to future mobility', 'low-carbonization of cities and national territory', 'nurturing promising new industries', 'establishing the base of innovative ecosystem', 'promoting a circular economy', 'protecting vulnerable industries and groups', 'realizing region-centered carbon neutrality', and 'raising public awareness of a carbon-neutral society'.

The main contents of the 3+1 strategies are as follows.

1) Low-carbonization of economic structure

It is driving the acceleration of energy transition by converting the current fossil fuel-centered energy system to new & renewable energy. In the case of coal and LNG power generation, it accelerates the policies of greenhouse gas reduction through the development of CCUS (carbon dioxide capture, utilization, and storage) technology and it expands the transmission and distribution network and spreads the distributed energy system of regional production and local consumption to solve the volatility problem of renewable energy. It promotes low-carbon transition through CCUS, energy efficiency improvement, and green hydrogen utilization technologies for industries that emit large amounts of greenhouse gases such as steel, petrochemical, refining, and cement to innovate the high-carbon industrial structure. And it will accelerate the transition of eco-friendly vehicles by expanding the production and distribution of hydrogen and electric vehicles, technology development and infrastructure expansion. Accordingly, it will fully prepare three infrastructure facilities such as

residence-centered electric vehicle chargers (20 million households nationwide), hydrogen charging stations for each city center and base, and a green hydrogen production system. In addition, it makes zero-energy construction mandatory for new buildings, and considers carbon neutrality when establishing national land plans, and strengthens the function of carbon absorption using ecological resources such as forests and tidal flats.

2) Creating the ecosystem for a new promising low-carbon industry

It secures (leads) the global market by fostering low-carbon industries such as secondary batteries and biotechnology, and industrializes innovative technologies and services to accelerate carbon neutrality early. Also, it will intensively discover and support companies with promising technologies in the eco-friendly, low-carbon, and new energy industries to foster them as green pre-unicorns. And it will expand regulation-free special zones in low carbon and eco-friendly fields such as utilizing carbon dioxide as a resource, green energy, and hydrogen. In addition, it prepares a roadmap for the circulation of major raw materials to maximize the use of recycled raw materials, and it promotes the reduction of raw materials by setting and reinforcing the target rate for the use of renewable resources by industry. Moreover, it will drive the development of innovative materials (carbon fiber, graphene, etc.) that replace steel and plastics.

3) Fair transition to a carbon-neutral society

In the case of coal power generation and internal combustion locomotive industries, which are shrinking due to structural transition, it actively supports their business transition to alternative and promising fields. And based on the changes in the industry and process caused by changes in the industrial structure, it will identify the demand for new jobs, and strengthen its support for customized vocational training and reemployment. Also, it builds the foundation for strengthening and supporting carbon-neutral capacity of local governments, and establishes the foundation for promoting region-led carbon neutrality. Moreover, it will strengthen environmental education and publicity for the public through various media and channels such as schools, broadcasting, and social networking service (SNS), and establish a carbon-neutral culture through the spread of climate action by actors such as civil society, industry, and small and medium enterprises.

4) Strengthening the institutional (systematic) foundation for carbon-neutrality

It creates a new 'climate response fund' to support the transition to a carbon-neutral ecosystem. And it will reconstruct the pricing system by comprehensively reviewing the means for imposing carbon prices such as taxation, levy, and emission trading system. And in the process of budget deliberation by the National Assembly in 2021, it will promote an increase of 300 billion won in related business areas such as support for energy transition and development of carbon reduction technology. In addition, it will promote voluntary corporate participation in reducing carbon emissions through tax incentives for corporate activities that contribute to achieving carbon-neutrality. And, it will increase the proportion of financial support in the green sector by financial institutions and back up corporate support for the transition to a low-carbon industrial structure. Also, it intensively supports the development of core technologies (CCUS, energy efficiency, etc.) for carbon neutrality. Moreover, it will strengthen international cooperation by establishing a cooperative system with major leaders in terms of climate change such as the EU and the United States and by promoting cooperation between Korea, China and Japan for carbon-neutrality.

A joint public-private 2050 Carbon Neutral Committee is established under the direct control of the President to promote carbon neutral strategies, and the committee is responsible for the deliberation and resolution of national carbon neutral strategies and major policies and plans, and checking the status of implementation. Also, it establishes an office (secretariat) to support the Carbon Neutral Committee, coordinates disagreements between ministries to build strategies, checks and evaluates strategies and GHG reduction performance, and conducts internal and external public relations.

KEPCO has prepared the "Basic Plan for GHG Reduction by Electric Power Group Companies 2030" to contribute to achieving the reduction of 58 million tons, which is the GHG reduction target in the national transition sector. Accordingly, it established the goal of reducing greenhouse gas emissions of 47 million tons in 2030 compared to BAU and it plans to reduce 22 million tons by improving power generation infrastructure such as phased abolition of coal-fired units based on the 11th plan and LNG conversion and by reinforcing eco-friendly power generation mix such as increase in the proportion of renewable energy generation,

and suspension of operation in spring. Additional 29 million tons will be reduced through the expansion of self-reduction measures such as additional abolition of old coal power plants (based on the 11th Basic Power Plan), domestic and external projects, and overseas CDM projects. Meanwhile, the power group companies are expanding investment in large-scale offshore wind power and solar power projects with the goal of increasing to 121.9GW of renewable energy by 2038, and recognizing the climate crisis as a new business opportunity, they are promoting new energy businesses such as establishing electric vehicle charging infrastructure (EVC), and distributing the comprehensive energy management system (K-BEMS, KEPCO Building Energy Management System).

Policy on Energy Conservation and Efficiency

The Korean government announced the “Innovation Strategy of Energy Efficiency” in August 2019 to increase energy efficiency and foster related industries. The strategy included measures such as raising the average fuel economy of passenger cars to 1.7 times the current level by 2030, removing fluorescent lamps that consume a lot of electricity, and exempting companies that have reduced energy use from mandatory energy diagnosis. The government plans to reduce final energy consumption by 15.6% compared to existing demand by 2030 by increasing consumption efficiency while maintaining the annual average economic growth rate of 2%. In other words, as of 2017, the final energy consumption is 176 million TOE and if this energy consumption pattern continues, the demand is expected to increase to 205 million TOE in 2030, but it means that it will be reduced to 173 million TOE. In this way, it will be possible to save about 1.5 trillion won in energy imports in 2030. To this end, this strategy included measures to increase energy efficiency in the industrial, building, and transportation sectors. In the industrial sector, which accounts for 61.7% of total energy consumption, a voluntary energy efficiency target system is introduced for businesses that consume a lot of energy, such as steel and petrochemicals. It will review measures to provide companies that set goals and achieve them benefits such as exemption from mandatory energy diagnosis, certification of excellent business operators, and reimbursement of contributions from the power business base fund for the implementation year in order to voluntarily induce energy efficiency. In addition, it will expand subsidies for the installation of the factory energy management

system (FEMS) to small and medium-sized and middle standing companies. In the building sector, to save energy, the ‘best-efficient home appliance’ is selected every year among ten home appliances subject to efficiency grade management such as TV, refrigerator, air conditioner, and washing machine, and a certain percentage of the purchase price is refunded to consumers who purchase it. Also, market sales of newly manufactured or imported fluorescent lamps will be banned from 2027. In addition, it will give energy-efficient buildings the Energy Star mark to provide benefits such as exemption from mandatory diagnosis.

In the transportation sector, the average fuel economy of passenger cars has improved significantly from 16.8km/ℓ in 2017 to 28.1km/ℓ in 2030 through technology development and expansion of eco-friendly vehicles. As for mid- to large-sized vehicles with an energy consumption of five times that of passenger cars per vehicle, the average fuel economy standard will be introduced by 2022. And the roads for building intelligent transportation systems (ITS) will be expanded from 15,743 km in 2018 to 17,112 km in 2023 to improve the convenience in using public transportation. Also, it will develop region-specific transportation services and establish and operate a test bed for autonomous cooperative driving by conducting a demonstration project of the next-generation intelligent transportation system (C-ITS) that can share traffic information in real time.

KEPCO also intends to expand EERS and a comprehensive energy management system (K-BEMS) to improve energy efficiency.

In order to reduce energy consumption, EERS is a system that assigns energy saving targets proportional to the amount of electricity sold to power providers, and allows them to achieve these targets through various energy efficiency improvement investments. As the representative power supplier in Korea, KEPCO implemented the EERS pilot project for the first time in Korea in May 2018, taking the lead in reducing GHG emissions by improving energy efficiency. We operate dedicated departments such as the EERS Planning and Operation Team at the KEPCO Headquarters and the Energy Management Team at 15 regional headquarters nationwide, and directly support the supply of high-efficiency equipment such as heat pump boilers, LED lighting, and inverters. In 2019, we supplied high-efficiency equipment for the first time in the field of agricultural use, and the scope of EERS recognition

was expanded from the performance measurement method focusing on equipment supply to K-BEMS and ESCO* agency projects. Consequently, a total of 1,019 GWh of energy was saved as of 2019, resulting in a reduction of 470,000 tons of greenhouse gases. In addition, we have established a long-term roadmap with the aim of reducing energy consumption by 1% of annual electricity sales and achieving 45,000GWh of reductions by 2031, through an investment of KRW 4.2 trillion. We are committed to research, analysis and system development to achieve successful EERS reduction goals. Moving forwards into a low-carbon era, KEPCO will continue to lead innovation towards a highly efficient energy consumption structure and strive to promote the sustainable development of the power industry.

KEPCO's self-developed K-BEMS (KEPCO Building Energy Management System) is a system that collects and analyzes customer energy use information in real

time to control the operation of building facilities. Through successful establishment within 120 company buildings, we have presented a standard model for energy efficiency and usage reduction. We have also promoted K-BEMS-based energy efficiency projects for 79 energy-consuming buildings, universities, factories, etc. In particular, KEPCO has successfully carried out a pilot project for overseas export to the Dominican Republic by forming a consortium with KEPCO Energy Solutions Co., Ltd., and is progressing with a KEPCO smart building project that combines energy efficiency and renewable energy to demonstrate the smart building business in KEPCO Yeongjong Branch. Not only will KEPCO continue the business for multi-use facilities, such as companies occupying national industrial complexes, large retailers, but we will also expand overseas demonstration projects in Southeast Asia (Indonesia, Vietnam). This will enable us to contribute to reducing GHG emissions through establishing profit models for new businesses in the energy sector and reducing energy usage.

5

Advanced Metering Infrastructure (AMI/Smart Grid Deployment)

Latest AMI Installations and Smart Grid Projects

Our smart city energy platform aims to create a carbon-neutral high-tech future city by providing improved services in the areas of energy efficiency, safety, and convenience for citizens through the integrated management of urban energy resources based on renewable energy, smart grid, and ICT technology. We are promoting a smart city energy platform construction project for local governments in places such as Sejong, Siheung, and Magok district in Seoul. In particular, we are participating in national R&D involving 3D virtual city simulation development and AI-based data analysis. Moreover, we are conducting a research project in order to demonstrate integrated meter readings combining five types of energy (electricity, water, gas, hot water, heating) in Siheung smart city. Going forward, we will continue to actively participate in the government's smart city development projects by expanding the energy platform construction business of local governments and concentrating our capabilities on developing new services based on our technology and experience accumulated through energy platform development.

AMI Roll out plan

We are successfully carrying out overseas projects in the fields of engineering, procurement, construction (EPC), and consulting, based on our technological competitiveness and domestic transmission and distribution business experience. Moving forward, we plan to expand from small consulting and EPC projects to largescale investment and operation projects, such as BOT (Build, Own, Transfer) and M&A, and develop these into long-term, profitable businesses utilizing our core transmission and distribution capabilities. In addition, we aim to leverage our experience in developing and demonstrating new business technologies, such as Smart-Grid and AMI, in order to promote our entry into the new overseas energy business market.

AMI offerings to customers and its benefit, Challenges faced and how it was solved/addressed

“Power Check Mobile,” developed by KEPCO via the convergence of the analysis of power-related big data and the artificial intelligence technology of our Intelligent Power Metering System (AMI*), is a service that provides information, by mobile, on electricity quality and electrical safety necessary for the operation of large-scale power facilities such as buildings, factories, and shopping malls, etc. Electrical safety managers are able to operate power facilities more safely and prevent electrical failures through timely inspections by utilizing various pieces

of electricity quality information, such as information pertaining to high frequency, power factors, and unbalanced phase voltage, provided through Power Check Mobile and a preemptive power outage risk warning notification function. Further, electrical safety managers facing difficulties conducting on-site visits and facility inspections due to the prolonged COVID-19 crisis can remotely monitor electricity quality and inspect facilities 24 hours a day, thereby enabling them to perform their tasks more conveniently and efficiently.

6

Latest Technologies or Innovations

World's first offshore wind power batch installation technology

We developed the world's first offshore wind power batch installation technology*, and held a launching ceremony for the Multi-Purpose Mobile Base (MMB) at Gunsan Port in July 2021. The MMB was built over a period of 60 months with an investment of KRW 34 billion and was specially designed to transport and install 5MW-class wind turbines that weigh up to 1,500 tons and stand at 140m tall. Through the MMB, we can reduce installation costs by approximately KRW 3.7 billion when constructing one wind turbine and shorten the offshore installation period from 90 days to 10 days. Going forward, we also plan to apply this MMB technology to lower foundation transport installation at the offshore wind power facilities in southwestern Jeollabuk-do and Shinan, and we also expect to utilize the MMB in small-scale offshore wind farms and multi-purpose offshore works mainly promoted by the private sector.

World's first offshore wind power batch installation technology

Here at KEPCO, we contribute to the expansion of a social safety net through the development of a new public service by implementing various social welfare solutions that integrate big data. Our single-person household care service is a social welfare service solution that analyzes electricity consumption, mobile phone calls, and location data in order to detect signs of anomalies in terms of the safety of single-person households and provide emergency alerts to social welfare officials. We began with a pilot project in Usan-dong, Gwangju in 2019, are currently providing the service in Seogwipo and Siheung, and plan to expand the number of local governments that introduce the service nationwide. We have been recognized, both domestically and internationally, for our achievements in relation to developing social welfare solutions using power-related big data, for example we were awarded the Presidential Award at the 20th Korea Digital Management Innovation Awards.

Development of SF6 Gas Purification & Reduction Technology and Alternative Insulating Gas



SF6 gas is one of the six major greenhouse gases, and its effect on global warming is 23,900 times that of carbon dioxide (CO2). Here at KEPCO, we use SF6 gas, which exhibits excellent performance in terms of insulation, as an insulating material for transmission and distribution facilities. Accordingly, SF6 gas accounts for about 62% of our total greenhouse gas emissions. We thus established a climate change response center at the Shinokcheon substation in 2019 and continue to reduce emissions by recovering and refining SF6 gas emitted from switchgear for power distribution and recycling it for use in the substation field. In addition, we are working to essentially reduce the greenhouse gas emissions generated during the power distribution process by developing eco-friendly switchgear using substitute materials, such as solid insulation and nitrogen, instead of SF6 gas.

Promoting the Development of Intelligent Digital Power Plant (IDPP)

Here at KEPCO, we are promoting the construction of an Intelligent Digital Power Plant (IDPP) that utilizes 4th industrial revolution technologies such as IoT, big data, and AI to maximize power plant facility operation efficiency through digital transformation. The IDPP construction project primarily consists of platform construction and intelligent app development. This project is capable of optimizing power plant operation and maintenance efficiency by building AI that can share and utilize various types of operating data generated from power plants, and by building a standard platform tailored for power plants equipped with big data technology, while also developing and operating intelligent apps capable of performing various functions such as power generation facility monitoring, diagnosis, and failure warning. Through the establishment of an IDPP, we are able to maximize power generation revenue, minimize maintenance costs, and secure power plant operation flexibility. Currently, we are conducting a pilot operation of the platform at six plants in cooperation with electric power group companies, and are seeking to demonstrate the capabilities of the apps by developing five types of apps with functions such as AI-based boiler diagnosis and early warning. In the future, whilst expanding our development of intelligent apps, we also plan to collaborate with private sector companies to develop a commercialization model based on power generation data in the fields of digital power plant construction, power plant operation management, and facility improvement and maintenance.

A brief Outlook plan for the next decade, such as Long Term Power Development Plan and T&D Plan

KEPCO's plans for power facilities for the next 10 years are as follows. Renewable energy is expected to expand to 29.2% (205.7TWh) of power generation and to 121.9GW of facility capacity in 2038. For coal-fired power generation, all generators will be abolished after 30 years of operation and it will be converted to LNG fuel. As for nuclear power generation, it plans to expand new & renewable energy while maintaining its reduction target. It is promoting energy transition by specifying the 2030 national target, 193 million tons of greenhouse gas emissions.

For transmission and substation facilities, it is driving(running) a 500kV HVDC construction project on the east and west coasts of Korea and it has established its plan for the existing transmission and substation facilities, aiming to build reinforced grid upgrades and flexible operation investments, including 150kV HVDC that connects Jeju Island and land in a timely manner. In addition, according to the trend of expanding renewable energy, the construction of transmission and substation facilities will be minimized in order to improve the grid capacity and will focus on optimizing facilities such as reinforcement of the existing grid and application of ESS. Also, it is planning to introduce micro-grids (micro power grids) using MVDC in stages in order to solve the problem of the distribution network due to the expansion of decentralized power generation.



Offshore transmission towers, SIHWA lake in Korea

Voltage Line length(C-km)	765kV	345kV	154kV	66kV	180kV (DC)	TOTAL
2024	552	9,620	23,265	93	130	33,660
2023	538	9,478	23,132	97	152	33,397
2022	524	9,210	22,985	106	195	33,020

(Source : KEPCO Electric Power Statistics (2022-2024))

MACAU SAR



1

Country Information

General Information (as of Dec 2025 or latest data)

Area:	33.4 km ²
Population:	688,900 million
GDP:	US\$ 75,617
Currency:	MOP
Installed Capacity:	408MW
Percentage of population electrified:	100%

Power Demand by Sector

Sector	Total Power Consumption as of 2025 (MWh)	Projected Change (%) 2024/2025
Industrial	137.2	-0.8%
Commerical	4759.2	0.5%
Residential	1358.4	0.1%
EV	9.4	51.7%
TOTAL	6,263	0.4%

2

Energy Policy and Electricity Market (as of end 2025 or latest data)

National Energy Policy

Macau is a Special Administrative Region of the People's Republic of China. CEM - Companhia de Electricidade de Macau, SA, (CEM) prepares the electrical infrastructure mostly to comply with the importation from Mainland China.

Metering, Electrical Installations (Design & Installation, Licensing, Responsibilities, Safety), Fuel Installations, Environmental Protection (Acoustics, Water, Ozone, Sulphur, CO2, Particulates, etc), Occupational, Health and Safety, Labour and Social Security, as well as with Penal, Administrative and Commercial Codes.

Supply and Demand of Primary Natural Resources

CEM uses fuel oil and natural gas for generation but the most part is imported from Mainland China

Laws and Governmental Regulations

CEM has to abide with the concession contract law and to follow all laws and regulations related with the electricity service such as the Supervised Entity Law, the General Conditions of Supply for MV and LV, the Quality of the Supply, Tariffs and Charges,

Total amount of energy consumption by sectors (Industrial / Commercial / Residential) (in MWh and %)

Sector	Total Power Consumption as of 2025 (MWh)	% in Total
Industrial	137.2	2.19%
Commercial	4,759.2	75.97%
Residential	1,358.4	21.69%
Other	9.4	0.15%
TOTAL	6,263	

3

Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (coal, LNG, Gas, Nuclear-fired and hydro) including total installed capacity of generation (in MWe) and total production (in MWh) by fuel types (in MWe)

Fuel Types	Installed Capacity (GW)	Fuel Types	Generation	Year-on-year Growth (%)
Total	408MW	Total	583 GWH	8.6%
		Solar Power	2.89 GWH	25%

4

Reduction of Carbon Emission

Policy on Renewable Energy and Major Regulatory Mechanism of Renewable Law

The Macau SAR Government launched the "Regulation for Safety and Installation of Solar Energy PV Interconnections" in January 2015, with an on-grid feed in tariff higher than normal tariff to purchase.

Production of Renewable Energy (MWh) from Various Resources (wind, solar, biomass, landfill gas, etc)

Total electricity consumption reaching 6,262.6 GWh, PV generation reaches 0.046% in energy source mix (2.88 GWH from PV system)

Environmental Problems of Electricity Supply (including generation and T&D sectors) and How It Is Being Addressed

Macau SAR adheres to the Kyoto Protocol since it was applied in Macau in 2008. It shows that Macau has been working alongside the global community on global energy-saving and emission reduction projects.

Plans and Strategies to Reduce Greenhouse Gas Emissions

CEM's green development master plan is an extension of CEM's corporate vision. Responding to the national call to fully support the Macau SAR to reach peak carbon emissions and carbon neutrality, as well as explore the development of green and low-carbon power services is the core value and mission of CEM in fulfilling its corporate social responsibility.

CEM's green development goal is to reach peak carbon emissions by 2030 and achieve carbon neutrality by 2050. CEM's low-carbon development path will be steadily advanced and implemented in three phases. Phase 1 (2022-2030): Reach peak carbon emissions. Optimize and adjust the power and energy development strategy, import low-carbon and clean energy from China Southern Power Grid, and use natural gas for local power generation. Phase 2 (2031-2040): Rapid reduction of carbon emissions. After reaching the peak, the main ways to reduce emissions are energy saving, consumption reduction and electric energy substitution, replacing traditional fuel vehicles with electric vehicles. Phase 3 (2041-2050): Deep decarbonization, participation in carbon sequestration and achieving the goal of carbon neutrality.

Policy on Energy Conservation and Efficiency

China launched the National 15th Five-Year Plan (2026–2030) with renewed commitment to its dual carbon goals of carbon dioxide emission peak before 2030 and carbon neutrality by 2060. The Plan sets a binding target to reduce carbon intensity by 17% over the five-year period, while raising the share of non-fossil energy in total energy consumption to 25% by 2030. It advances the formation of industrial structures, manufacturing methods, lifestyles and spatial landscapes grounded in resource efficiency and ecological stewardship, deploying green and low-carbon technologies, constructing zero-carbon factories and parks, and strengthening energy efficiency governance across key sectors. Through these measures, China continues to provide energy corporations with clear directional guidance for innovative, high-quality green development in the years ahead.

Latest AMI Installations and Smart Grid Projects

Smart meters cover all of Macau. By deploying advanced smart meters and Advanced Metering Infrastructure (AMI), online electricity monitoring and management are realized, optimizing business processes and enhancing operational efficiency, while bringing a smarter electricity usage experience to customers, aiding assisting the Macau SAR Government in building a smart city.

Challenges faced and how it was solved / addressed

Fiber backbone deployment limited by trench work, requires better coordination with construction departments and government departments.

- The complex and extremely densely living environment and electricity network environment is one of the biggest challenges for last-mile communication. Two different communication technologies PLC and RF will be tested in the AMI pilot project for performance verification.
- Introduction of AMI would have great impact to existing business processes which requires extensive discussion and reengineering on business processes.

AMI Roll out plan

CEM has completed full smart meter coverage across the city.

AMI offerings to customers and its benefit

With the complete coverage of smart meters and the continuous improvement of the system, CEM actively applies smart meter data and functions in various business operations. This includes remote power disconnection and reconnection to enhance the efficiency of power supply services, significantly shortening the response times for new power supply applications (with pre-installed meters), termination of supply applications, and the restoration of power services due to late payments, greatly improving work efficiency. On the other hand, for non-residential users, smart meters not only effectively utilize system alerts and daily electricity consumption records for monitoring but also introduce electricity theft detection models and line loss detection methods based on big data, helping to investigate electricity theft, ensure electrical safety as well as secure revenue.

Moreover, electricity consumption data is also widely applied in customer installation analysis, detection of non-compliant electricity usage, forecasting and planning of transmission and distribution network loads, and other areas. This provides a reliable basis for data-driven decisions in grid planning, operational dispatch, and customer service for CEM, and lays a solid foundation for the construction of an advanced, transparent power grid.

With the help of smart meters, there is no longer a need to estimate electricity bills. CEM has added a daily consumption analysis feature to the CEM APP, allowing customers to view their daily consumption for the last 30 days at any time, get a clear view of their power usage, and thus adjust their electricity usage habits, save on electricity, and reduce electricity bill expenses.

6

Latest Technologies or Innovations

CEM has deployed a range of AI-powered intelligent protection solutions, alongside asset management and infrastructure upgrades for the power data network, situational awareness systems, and backup infrastructure replacement to reinforce its overall cybersecurity resilience.

In intelligent operations and maintenance, CEM introduced robotic dog inspection systems for the first time, equipped with image recognition algorithms, thermal imaging cameras, and partial discharge detectors powered by AI and deep learning — delivering precise equipment state identification and round-the-clock automated substation inspections. Data center optimization efforts have achieved measurable improvements in Power Usage Effectiveness (PUE), laying the groundwork for

future digital twin applications, while private cloud deployment and AMI data warehouse construction continue to advance in parallel, with AI-powered intelligent voice bot capabilities introduced to enhance customer service scenarios.

To modernize core business operations, CEM has progressed the planning of a next-generation ERP system integrating AI and mobile functionalities to standardize and automate business workflows. A unified OA platform, together with a newly launched warehouse management system and digital procurement platform, collectively drive end-to-end digitalization across internal operations and supply chain management, enhancing transparency, efficiency, and sustainability.

7

Major Issues or Transformation Impacting the Electricity Supply Industry

CEM has made significant progress in power data network modernization, with over 1,400 customer substations completing fiber-optic communication upgrades. Pilot projects on telemetry data collection from customer substations are underway, with ongoing research into integrating this data into the SCADA system and transmitting it to the AMI system for further analysis and application. The OTDR system for real-time fiber-optic quality monitoring continues to expand, strengthening network asset management and communication stability. A new integrated network management system has also been put into operation, enhancing monitoring capabilities across the power data network, assets, and terminals. In parallel, CEM continues to advance distribution network automation upgrades and has deployed a relay protection information management system to achieve systematic control over relay protection equipment and grid fault information, elevating the professional management of grid operation and ensuring safe, stable grid performance.

In support of China's "dual carbon" goals and the Macau SAR Government's net-zero emissions target, CEM has initiated a project to install a floating photovoltaic (FPV) system on the reservoir surface at the Coloane Power Station to harness solar energy and partially offset electricity consumption during the power generation process. CEM also conducts regular flue gas testing at the Coloane Power Station to ensure atmospheric emissions remain in full compliance with the prescribed standards.

On the cybersecurity front, the situational awareness system has been fully deployed, further strengthening security protection for industrial control networks. CEM is also actively researching and testing domestic database management systems, operating systems, hyper-converged infrastructure (HCI) hardware, and hypervisor software to ensure the stability and resilience of equipment supply and services. Building on its AI and big data capabilities, CEM continues to leverage smart meter data, partial discharge detection, and image recognition technologies to enhance real-time monitoring, predictive maintenance, and operational efficiency across the distribution network.



MALAYSIA



1

Country Information

General Information (as of Dec 2025 or latest data)

Country:	Malaysia
Capital:	Kuala Lumpur
Area:	329,960.22 km ² ⁴
Population:	34.2 million ⁵
GDP (as of Q4 2025):	RM 532.9 billion ⁶
Currency:	Ringgit (MYR)
Installed Capacity:	Peninsular Malaysia: 27,333.5 MW ⁷ East Malaysia: 1,177.3 MW ⁸
Percentage of population electrified:	100%

Power Demand by Sector

Sector	Total Power Consumption as of 4Q 2025 (GWh)	Y-o-y Change (%) 2024/2025
Industrial	39,221	-5.8
Commercial	51,164	10.0
Residential	36,792	1.1
TOTAL	133,895	2.3

(Source: Analyst Briefing 4Q FY2025)

⁴The Government of Malaysia's Official Gateway's website: <https://www.malaysia.gov.my>

⁵Department of Statistic Malaysia, OpenDOSM

⁶Single Buyer

⁷Energy Commission Annual Report, 2023

⁸The World Bank

National Energy Policy & Market Evolution

Malaysia's energy transition is anchored on its commitment under the Paris Agreement, where the country pledged to reduce its carbon emission intensity by 45% by 2030 relative to 2005 levels (conditional). This commitment has driven a series of national policies and market reforms aimed at transforming the electricity supply industry into a more sustainable and competitive ecosystem.

The evolution began with foundational frameworks such as the Green Technology Master Plan (2017) and was further strengthened under the 11th and 12th Malaysia Plans, which embedded green growth and sustainability as national priorities. The Malaysia Renewable Energy Roadmap (MyRER), launched in 2021, established a structured pathway to increase renewable energy (RE) penetration, focusing on solar, hydro, bioenergy, and emerging technologies.

Building on this, the Government introduced the National Energy Policy (NEP) 2022–2040 and subsequently the National Energy Transition Roadmap (NETR) in 2023. NETR sets more ambitious targets, aiming for 31% RE capacity by 2025, 40% by 2035, and 70% by 2050, while supporting Malaysia's aspiration of achieving net-zero emissions by 2050. Complementary to this, the Carbon Capture, Utilisation and Storage (CCUS) framework introduced in 2025 further reinforces Malaysia's long-term decarbonisation strategy.

In parallel with policy development, Malaysia has progressively introduced market-based mechanisms to enhance efficiency and competition within the electricity sector as well as facilitating integration of renewable energy into the system. Initiatives such as the Feed-in Tariff (FIT), Net Energy Metering (NEM), and Large Scale Solar (LSS) programmes have successfully catalysed RE deployment. The introduction of the New Enhanced Dispatch Arrangement (NEDA) marked a significant shift towards short-term generation competition, enabling non-PPA participants to enter the market and improving dispatch efficiency.

More recently, the market has evolved to enable direct participation from corporate consumers. Programmes such as the Corporate Green Power Programme (CGPP) and the Corporate Renewable Energy Supply Scheme (CRESS) allow businesses to procure renewable energy through virtual and physical arrangements, reflecting a gradual transition towards a more liberalised and consumer-driven electricity market. The introduction of the Community Renewable Energy Aggregation Mechanism (CREAM) in 2025 further expands participation by enabling distributed generation and aggregation of rooftop solar assets. Other initiatives rolled out by the Government include Green Electricity Tariff (GET), which allows customers to opt for certified bundled green power and the Energy Efficiency and Conservation Act (EECA), which mandates energy management and efficiency across key sectors. In addition, the Hydrogen Economy and Technology Roadmap, launched in 2023 positions hydrogen as a key enabler of the energy transition, supporting energy security and industrial decarbonisation.

The establishment of Energy Exchange Malaysia (ENEGEM) in 2024 represents a key milestone in extending Malaysia's electricity market beyond domestic boundaries. ENEGEM facilitates cross-border renewable energy trading, beginning with exports to Singapore, and serves as a platform for price discovery, regional integration, and monetisation of surplus green generation. Beyond its initial 100 MW export capacity, ENEGEM is strategically positioned to support the development of the ASEAN Power Grid and enhance Malaysia's role as a regional clean energy hub.

Overall, Malaysia's energy transition reflects a gradual shift from a traditionally vertically integrated, single-buyer model towards a more hybrid structure incorporating competitive mechanisms, corporate participation, and cross-border electricity trade. Continued investments in grid modernisation, energy storage, and system flexibility will be critical to support higher penetration of variable renewable energy and ensure long-term system reliability.

Supply and Demand of Primary Natural Resources for Malaysia

Primary Production by Fuel Type



Total Primary Energy Supply by Fuel Type



Source: National Energy Balance 2022, Suruhanjaya Tenaga (published on 5/12/2025)

Overview of Electricity Market Structure

The electricity supply industry in Malaysia consists of three main vertically integrated power utilities, namely Tenaga Nasional Berhad (TNB) in Peninsular Malaysia, Sabah Electricity Sdn. Bhd. (SESB) in Sabah and Sarawak Energy in Sarawak. SESB is an 83% owned subsidiary of TNB and is a vertically integrated utility providing electricity generation, transmission, distribution and supply services in the state of Sabah

and the Federal Territory of Labuan. The three main utilities undertake electricity generation, transmission (rebranded to Grid for TNB only), distribution (parted into distribution network and retail for TNB only) in their respective areas and are investor-owned entities with the Government as the major shareholder.



Map of Malaysia

Laws and Governmental Regulations

The electricity supply industry in Malaysia is principally governed by the Electricity Supply Act 1990 and regulated by the Energy Commission of Malaysia, established under the Energy Commission Act 2001.

The Electricity Supply Act provides the legislative framework governing the generation, transmission, distribution, supply, and use of electricity in Malaysia, including licensing requirements, operational reliability, safety standards, consumer protection, and regulatory oversight of electricity supply activities.

The Energy Commission is responsible for overseeing licensing, technical and operational standards, tariff regulation, market conduct, and compliance across the electricity sectors. All licensees are subject to regulatory conditions imposed by the Commission, while system operations are governed by the Grid Code and other relevant technical standards to ensure system reliability, security, stability, and efficiency. Recent amendments to the Electricity Supply Act have further expanded the regulatory framework to facilitate and enable cross-border electricity trading arrangements, supporting renewable energy exports and regional electricity market integration through initiatives such as ENEGEM.

Overall, Malaysia's regulatory and policy framework has progressively evolved to support three key objectives i.e. energy security, sustainability, and market efficiency, in line with the country's broader energy transition agenda and aspiration towards a low-carbon energy future.

Policy

Malaysia's energy policies have transitioned from ensuring security of supply towards enabling decarbonisation and market liberalisation:

- **National Energy Policy (1979)** – Established the foundation for secure, reliable, and efficient energy supply.
- **Four-Fuel Diversification Policy (1981)** – Strengthened energy security through diversification of fuel sources (oil, gas, hydro, and coal).
- **Five-Fuel Policy (2001)** – Introduced renewable energy as the fifth fuel, promoting biomass, solar, and mini-hydro.
- **National Renewable Energy Policy and Action Plan (2008)** – Formalised long-term strategies to scale up renewable energy deployment and reduce dependence on fossil fuels.
- **National Energy Policy (2010)** – Focused on increasing renewable energy contribution, developing the RE industry, and enhancing environmental sustainability.
- **National Energy Efficiency Action Plan (2015)** – Established a structured approach to improve energy efficiency across industrial, commercial, and residential sectors.
- **National Energy Policy (2022–2040)** – Provides the overarching framework guiding Malaysia's energy transition, balancing economic growth with sustainability and resilience.
- **National Energy Transition Roadmap (NETR) (2023)** – A comprehensive strategic plan to accelerate decarbonisation, targeting up to 70% renewable energy capacity by 2050, while enabling new market mechanisms, electrification, and low-carbon technologies.
- **Hydrogen Economy and Technology Roadmap (HETR) (2023)** – Positions hydrogen as a key future energy carrier, supporting deep decarbonisation, sector coupling, and potential cross-border energy trade in the long term.

- **National Carbon Market Policy (NCMP) (2026)** – Provides the legislative, governance, and technical foundation to establish a credible and transparent carbon market ecosystem to accelerate decarbonisation

Act

The following key legislations govern the electricity and energy sectors:

- **Electricity Supply Act 1990** – Governs the licensing, generation, transmission, distribution, and supply of electricity, ensuring safe, reliable, and cost-effective operations.
- **Gas Supply Act 1993** – Regulates gas supply activities, including importation, regasification, transportation, and distribution.
- **Energy Commission Act 2001** – Establishes the Energy Commission and empowers it to regulate the energy supply industry and enforce compliance with energy laws.
- **Renewable Energy Act 2011** – Provides the legal basis for renewable energy development, including mechanisms such as the Feed-in Tariff (FIT).
- **Sustainable Energy Development Authority Act 2011** – Establishes the Sustainable Energy Development Authority (SEDA) to administer and promote renewable energy initiatives.
- **Energy Efficiency and Conservation Act 2024** – Introduces a regulatory framework to improve energy efficiency across sectors, reduce energy wastage, and support Malaysia's carbon neutrality goals.
- **Carbon Capture, Utilisation and Storage (CCUS) Act 2025** – Establishes the legal and regulatory framework for carbon capture and storage, supporting deep decarbonisation and the development of a low-carbon economy.
- **Electricity Supply Act 1990 (amendment 2025)** – The amendments have expanded the regulatory scope to facilitate cross-border electricity trade, including export and import of electricity via the national grid. This provides the legal foundation for initiatives such as ENEGEM and supports Malaysia's participation in regional power market integration.

Amount of Electricity Exported and Imported (in MWh)

Overview of ASEAN Power Grid (APG) and Cross-Border Initiatives

The ASEAN Power Grid (APG) initiatives have been ongoing since the last 20-year and TNB has been actively supporting the realisation of the ASEAN Power Grid (APG) through cooperation with our ASEAN counterparts and pursuing interconnection initiatives with Malaysia's bordering countries.

The integration of the electricity supply system in Malaysia with the neighbouring countries has been carried out since the 1980s to meet the needs for secure, stable, and reliable supply among the countries involved. This includes the interconnection between Peninsular Malaysia and Singapore in the south and Peninsular Malaysia and Thailand in the north.

ASEAN Power Grid (APG) Initiatives

Aligned with ASEAN cooperation efforts, TNB as Malaysia's electricity utility, plays a key role in connecting the national power grid with its neighbouring countries. Currently, several cross-border interconnections are in progress with Singapore, Thailand, Indonesia, and Sarawak. The following are the overview of interconnection with utilities in neighbouring countries/region:

1. Interconnection between Peninsular Malaysia-Singapore:

The interconnection between Peninsular Malaysia and Singapore was initiated in 1983 until today, which acts as a mutual energy transfer in times of need and allows sharing of spinning reserves. Realising the opportunities of power trade in the future as part of achieving APG aspirations, both TNB and Singapore Power have agreed to upgrade the existing 250 MW HVAC lines connecting both countries to 2 x 550 MW HVAC lines. The cable upgrading work was completed in August 2022.

To further explore the potential of development between Peninsular Malaysia and Singapore, TNB, through the MOU signed on 25 August 2023 with SP Power Assets (SPPA), Singapore's prominent electricity

transmission and distribution company, is actively exploring the technical feasibility of a second power interconnection facility connecting Peninsular Malaysia and Singapore. This facility, upon completion, will further bolster energy trading between the two regions, enhancing energy supply reliability.

2. Interconnection between Peninsular Malaysia-Thailand:

The interconnection between Peninsular Malaysia and Thailand acts as a mutual energy transfer involving bilateral sale and purchase of commercial energy between TNB and EGAT. Currently, there are two existing interconnections namely 132 kV AC Bukit Ketri – Sadao with capacity of 80 MW and 300 kV HVDC Gurun – Khlong Ngae with capacity of 300 MW.

There are plans by TNB and EGAT to enhance the existing 300 kV HVDC interconnection capacity considering it has been in operation for more than 20 years.

3. Interconnection between Indonesia – Peninsular Malaysia:

TNB and Indonesia's PT PLN (Persero), in collaboration with the ASEAN Centre of Energy (ACE), are currently conducting a feasibility study to assess the potential for a new electricity interconnection between Sumatra and Peninsular Malaysia. This study is a result of a Memorandum of Understanding (MOU) signed by TNB, PLN, and ACE on August 25, 2023, and is presently ongoing.

4. Interconnection between Peninsular Malaysia – Sarawak:

Malaysia's Government has proposed a project to establish an electricity interconnection to transmit at least 1,000 MW of electricity from Sarawak to Peninsular Malaysia. The initiative is currently being explored and is at a preliminary study stage by Sarawak Energy Berhad (SEB) and TNB.

Cross-Border Power Trade Initiatives

Malaysia has several cross-border power trade initiatives to enable and facilitate the sales of electricity between Peninsular Malaysia and its neighbouring countries as well as supporting multi-lateral power trade between ASEAN countries.

1. Cross-Border Electricity Sales (CBES) Renewable Energy (RE) and non-RE Schemes:

In March 2020, the Government of Malaysia approved the implementation of CBES in which a third party can generate electricity and sell it to Singapore and Thailand. Through this mechanism, Power Plant Developers (PPD) can generate and sell electricity to Singapore and/or Thailand using the existing or new interconnection facilities subject to the terms and conditions. Following the Government's decision, the Energy Commission of Malaysia has published the first guide for CBES in December 2020 with the following objectives:

- To prepare a basic framework for the cross-border sale of electricity between the Peninsula and neighbouring countries.
- To determine the conditions and requirements for the cross-border sale of electricity between PPDs in the Peninsula and buyers in neighbouring countries.
- To determine the roles, functions, and responsibilities of relevant parties.
- To facilitate the cross-border sale of electricity.

In line with Malaysia's energy transition aspiration and commitment to support regional power integration, the Government has established the Energy Exchange Malaysia (ENEGEM) in April 2024. Subsequently, ST has revised the Guide for CBES published in April 2024 to incorporate the following:

- Guideline for the CBES RE Scheme, facilitating the cross-border sales of green electricity to neighbouring countries.
- Enhanced function, role and responsibilities of the Single Buyer (SB) as the operator of the ENEGEM.

ENEGEM was established on 15 April 2024 as a platform to facilitate the sales of green electricity between Malaysia and its neighbouring countries. It serves as a marketplace for trading and exporting green electricity through a bidding mechanism. A total of 300MW is allocated for RE export under the CBES RE Scheme, with cross-border green electricity sales starting with a pilot run. For the pilot phase, TNB has signed a RE Supply Agreement (RESA) with Sembcorp Power Pte Ltd on 9 December 2024 to supply 50MW of green electricity from Malaysia to Singapore.

2. Lao PDR–Thailand–Malaysia Power Integration Project (LTM-PIP) (2018-2021):

LTM-PIP is the first multilateral interconnection project in ASEAN, this project has continued to be seen as a critical milestone to enhance multilateral trade in ASEAN Power Grid (APG) and the regions towards energy security, sustainability, and cost efficiency.

A tripartite agreement known as the Energy Purchase and Wheeling Agreement (EPWA) was signed on 27 September 2017 among Electricite Du Laos (EDL), Electricity Generating Authority of Thailand (EGAT) and Tenaga Nasional Berhad (TNB) for a purchase of up to 100 MW from EDL, Lao PDR via existing transmission grid of EGAT, Thailand. The term of EPWA is for two (2) years from 1 January 2018 until 31 December 2019 and the total import by Malaysia from EDL is 30.2 GWh.

A Supplementary Agreement to EPWA was signed on 27 December 2019 between EDL, EGAT and TNB for the extension of another 2 years (1 January 2020 until 31 December 2021) with capacity up to 300 MW and committed energy up to 50 GWh. Due to the COVID-19 pandemic and low demand, the energy transaction until 31 December 2021 is approximately 2.61 GWh.

3. Lao PDR–Thailand–Malaysia–Singapore Power Integration Project Phase 1 (LTMS-PIP Phase 1) (2022–2024):

LTM-PIP was later expanded to include Singapore as agreed in January 2020 during the Senior Officials Meeting on Energy (SOME) in Chiang Mai. Under LTMS-PIP Phase 1, Singapore acts as the buyer of the energy. This allows Lao PDR to sell electricity (minimum 30 MW, maximum 100 MW) to Singapore via existing grids in Malaysia and Thailand.

TNB (Malaysia) and EGAT (Thailand) act as wheelers to deliver the energy from Lao PDR to Singapore. A total energy of 266.4 GWh has been transacted under LTMS-PIP throughout its effective years from 22 June 2022 until 21 June 2024.

4. Lao PDR–Thailand–Malaysia–Singapore Power Integration Project (LTMS-PIP Phase 2) (2024–2026):

Following the expiry of LTMS-PIP Phase 1, LTMS-PIP Phase 2 has commenced for a capacity of up to 200MW, with the additional 100MW coming from Malaysia. The project is implemented in two (2) parts as follows:

Part 1: Bilateral energy sales of up to 100MW from Malaysia to Singapore for 2 years effective September 2024 via the signing of Energy Supply Agreement between TNB and Keppel Electric Pte. Ltd. (Keppel).

Part 2: Energy sales from Lao PDR to Singapore where Malaysia and Thailand act as wheelers for up to 100 MW. EDL, EGAT and TNB have signed the Energy Wheeling Agreement (EWA) Phase 2 on 14 January 2026 to enable the full implementation of LTMS-PIP Phase 2.

3

Power Generation and T&D Sectors Total Installed Capacity of Generation by Fuel Types

Peninsular Malaysia

As of 31st December 2025, the total installed capacity for Peninsular Malaysia stands at 27,333.5 MW, with the breakdown as follows:

Fuel Types	Capacity (MW)
Coal	12,180.0
Gas	10,917.6
Hydro	2,536.1
Transmission-connected Solar	1,699.8
Total	27,333.5

(Source: Single Buyer)

Sabah



(Source: Energy Commission Annual Report 2023)

Electricity generation in Sabah consists of power stations using gas, diesel, hydro and renewable energy (RE) sources including solar, biomass and biogas. In 2023, the dependable capacity in the state recorded a decrease of 7% to 1,177MW from 1,268MW in 2022. This drop in capacity is due to changes in declared capacities at power stations such as Tenom Pagi, Tawau, Sandakan, Patau-Patau and Serudong, attributed to prolonged outages and aging facilities.

In Sabah, the JPPPET Meeting No. 1/2023 was held on 12 June 2023, jointly chaired by the Minister of Natural Resources, Environment and Climate Change and the Chief Minister of Sabah. To address supply adequacy issues, the meeting agreed to implement several short-term generation capacity programmes to immediately increase generation capacity. In this regard, the Commission directed SESB to implement several initiatives which include a 100MW short-term capacity rental.

In Sabah, the electricity generation mix consists of gas, diesel, hydro and renewable energy (RE) such as biomass and biogas as well as generation from LSS. In 2020, the state recorded a dependable capacity of 1,171 MW in comparison to 1,277 MW in 2019.

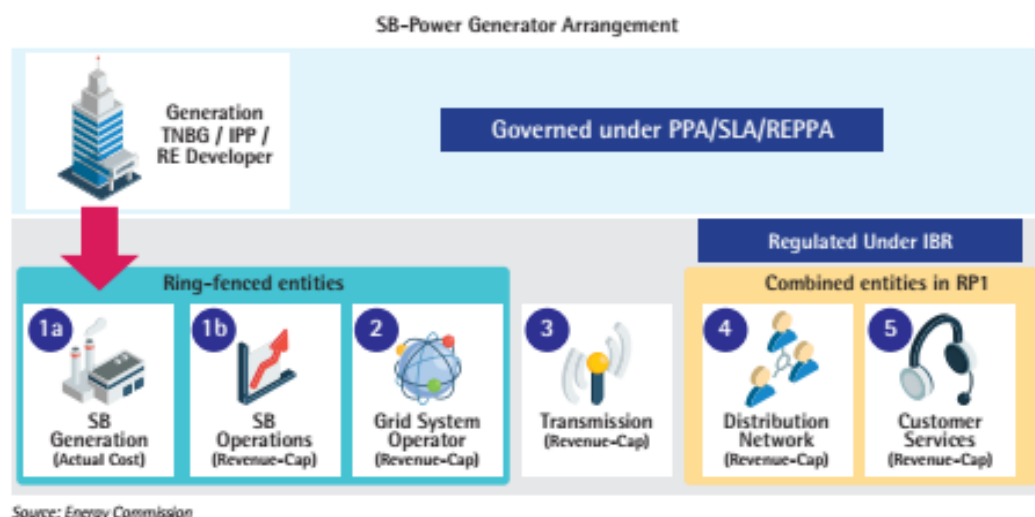
The decrease in total dependable capacity was largely due to the reliability review conducted by the Grid System Operator (GSO) on the Tenom Pangi Hydro Power Station, the Tawau Power Station (Caterpillar Genset) and the Sabah Electricity Sdn. Bhd. (SESB) Power Station. Several generation projects with a total capacity of 101 MW, namely One River, SJ Serudong, SJ Libaran, GT Melawa and Tawau DG Rehab, were also postponed in 2020 due to the enforcement of the MCO in Sabah.

Also, as of 1 February, the dependable capacity of hydro units at the Tenom Pangi Hydro Power Station was reduced to 67 MW from 72.55 MW. This came as a result of the Tested Annual Availability Capacity (TAAC) testing and contributed to the decrease in the total dependable capacity from 1228.76 MW to 1223.21 MW. Similar trends were seen at the SESB Power Station, namely units SJ Patau-Patau, SJ Kubota and SJ Tawau, with dependable capacity decreasing from 1,223.21 MW to 1,174.96 MW after the enforcement of the TAAC testing on 1 July.

Several units at the Tawau Power Station (CAT) and SPRE Power Station also experienced a drop in total dependable capacity, from 1,174.96 MW to 1,174.52 MW after testing on 15 October. The final round of TAAC testing was carried out on 30 November, after which several units at the Tawau Power Station (CAT) saw a reduction in total dependable capacity from 1,174.52 MW to the current total of 1,170.88 MW.

Structure of Installed Generating Capacity by Producers (Utilities and IPPs)

Power generation in Peninsular Malaysia is currently in the hands of national utility company TNB and Independent Power Producers (IPP). Power generation was first deregulated in the mid-1990s in the interest of energy security in a rapidly industrialising Malaysia. This led to the birth of IPPs who invested heavily in large power plants and sold energy to TNB according to the commercial terms specified in their respective Power Purchase Agreements (PPA), which typically run over a period of 21 to 25 years. In 2012, the Commission introduced the concept of the Single buyer (SB) to manage the impending renegotiation of expired PPAs. SB is a ring-fenced entity within TNB that is the sole off-taker of electricity and related services from the power producers.



List of Power Plants by Fuel Types

List of Power Plants by Fuel Types (Peninsular Malaysia) as of 30 September 2024

Plant	Type	Installed Capacity (MW)
Jimah East Power Sdn. Bhd	Coal	2000
Jimah Energy Venture	Coal	1400
TNB Janamanjung Sdn. Bhd (JMJG)	Coal	2070
TNB Janamanjung Sdn. Bhd (M4)	Coal	1010
TNB Janamanjung Sdn. Bhd (M5)	Coal	1000
Kapar Energy Ventures Sdn. Bhd	Coal	1600
Tanjung Bin Power	Coal	2100
Tanjung Bin Energy	Coal	1000
Edra Energy	Gas	2242.13
Genting Sanyen Power Plant (KLPP)	Gas	675
TNB Connaught Bridge Sdn. Bhd (New CBPS)	Gas	375
TNB Prai Sdn. Bhd	Gas	1071.43
Pengerang Power Sdn Bhd	Gas	600
Pulau Indah Power Plant (PIPP)	Gas	1200
Segari Energy Ventures	Gas	1303
Southern Power Generation (SPG)	Gas	1440
S.J Tuanku Jaafar (PD1)	Gas	703
S.J Tuanku Jaafar (PD2)	Gas	708
Kapar Energy Ventures Sdn Bhd (Gas)	Gas	600
S.J Bersia	Hydro	72
S.J Chenderoh	Hydro	40.5
S.J Hulu Terengganu	Hydro	265
S.J Kenering	Hydro	120
S.J Kenyir	Hydro	400
S.J Lower Piah	Hydro	54
S.J Pergau	Hydro	600
S.J Sultan Idris	Hydro	150
S.J Sultan Yusof	Hydro	100
S.J Temenggor	Hydro	348
S.J Ulu Jelai	Hydro	372
S.J Upper Piah	Hydro	14.6
Grid-connected Large Scale Solar (LSS)	Solar	1699.8

A Brief Outlook Plan for The Next Decade, Such as Long-Term Power Development Plan and T&D Plan

The Peninsular Malaysia Generation Development Plan that is governed by the Ministry of Energy Transition and Water Transformation, outlines the long-term generation development plan, that includes projection of electricity demand growth and future generation capacity requirement. It is formulated based on several planning criteria to ultimately achieve optimal balance on the energy trilemma: Energy Security, Sustainability and Energy Affordability.



Capacity Mix

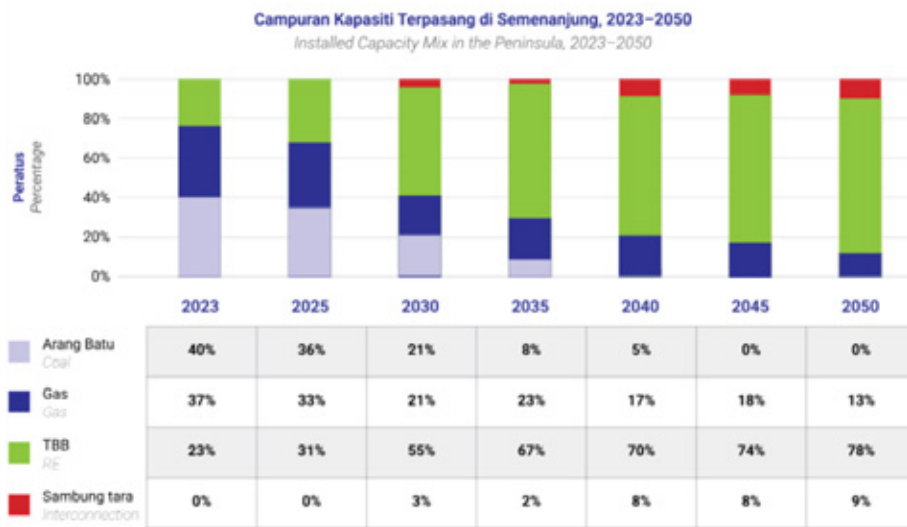
Based on Energy Commission 2023 Annual Report, a significant decrease in fossil fuel-based generation is anticipated in the coming 27 years, going from 77% in 2023 to 13% in 2050. Coal-based power generation will see the largest reduction from 40% in 2023 to 36% by 2025 before completely phased out by 2050.

On the other hand, generation from RE sources is expected to dominate the capacity mix in the coming years with the installed capacity for RE in the Peninsular is predicted to increase from 23% in 2023 to 31% by 2025 and reach 78% by 2050. The potential increase in interconnection has been considered to meet HHI planning criteria.

An annual average of 2.5GW RE capacity needs to be developed starting from 2026 to ensure the 78% RE target in the Peninsula is achieved by 2050. The introduction of energy storage technologies such as battery energy storage systems (BESS) is expected to assist in keeping the grid system stability.

Fossil fuel-based capacity is expected to see a significant decline over the next two decades, dropping from 82% of Peninsular Malaysia's installed capacity in 2022 to 62% by 2041. The most substantial reduction will come from coal, which is projected to decrease from 38% in 2022 to just 7% by 2041. Natural gas, meanwhile, will continue to play a dominant role in the capacity mix.

Installed Capacity Mix (%) (2023-2050)



(Source: Energy Commission Annual Report 2023)

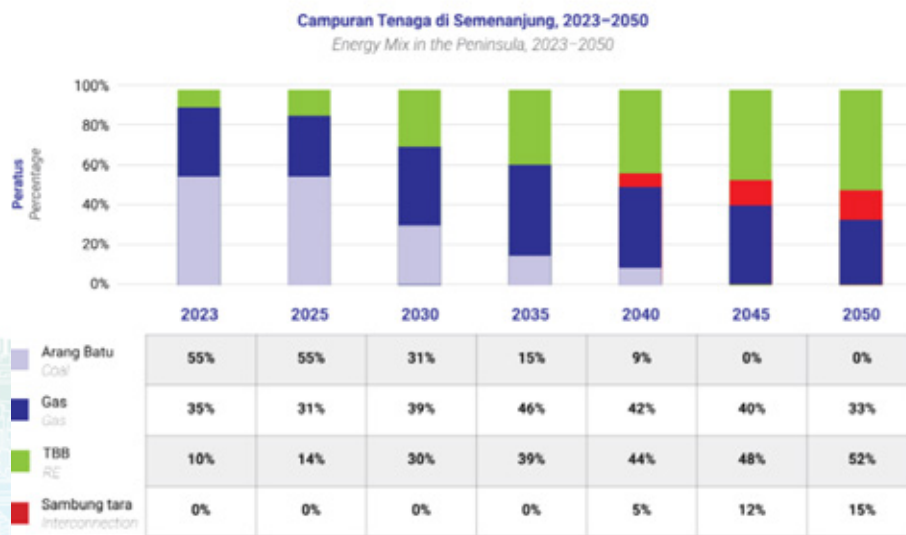
Although hydrogen is expected to make up a modest share of the capacity mix at 3% by 2041, its introduction supports in diversifying energy sources, decarbonising the power sector, and improving the overall Herfindahl-Hirschman Index (HHI) score. Meanwhile, the share of RE - comprising solar, hydro, biomass, and biogas are projected to increase steadily from 18% in 2022 to 35% by 2041.

Energy Mix

Energy mix from Energy Commission Annual Report 2023 shows coal leading other fossil fuel generation at 55% compared to gas at 31% and 14% RE by 2025, despite RE dominating the capacity mix until 2025, due to its lower cost projections. This is expected to change over the next 20 years, as coal-fired power plants are driven by early retirement plans by 2040 which will see the declining coal composition in the energy mix. However, RE will dominate the energy mix at 52% followed by gas at 33% and interconnections at 15% by 2050.

While gas is expected to remain the dominant source in the energy mix through 2041, coal will continue to play a major role in the near term, making up 44% of the energy mix by 2030, compared to gas at 40% and renewable energy (RE) at 16%. However, coal's share is expected to decline steadily over the next decade. By 2041, gas is projected to supply 69% of the energy mix, followed by coal at 13%, RE at 15%, and hydrogen at 3%.

Energy Mix (%) (2023-2050)



(Source: Energy Commission Annual Report 2023)

4 TNB Sustainability Journey

As a leading provider of energy solutions, we are committed to steering towards sustainability locally and internationally. With our eyes set on the TNB Sustainability Pathway, we are driving towards becoming a Net Zero company by 2050, propelling ourselves towards ESG excellence.

Considering the pressing need for environmental action, it has become more evident than ever that adopting sustainable practices is crucial. TNB acknowledges this imperative and is taking proactive measures to tackle these challenges head-on, guided by the TNB Sustainability Framework and Policy, which were approved by the TNB Board of Directors (BOD) in 2024.

The TNB Sustainability Policy is driven by a commitment to ethical, responsible, and sustainable practices. This policy sets a course for environmental, social, and governance excellence,

aligning with legal requirements and international standards. Our aim is to drive positive change in the communities we serve, ensuring that our actions are sustainable and have a positive impact through fourteen (14) key focus areas in the Environmental, Social and Governance pillars in the TNB Sustainability Framework.



Energy Transition and Innovation

Malaysia's energy transition efforts to achieve Net Zero by 2050 will shift the nation's generation mix from fossil fuel-driven energy to 70% installed renewable energy capacity. The announcement of ten (10) National Energy Transition Roadmap (NETR) flagship projects by the government has strengthened the nation's commitment. TNB is entrusted to champion three (3) of these NETR projects, including hybrid hydro floating solar PV, solar parks as well as co-firing of hydrogen and ammonia. We remain committed to supporting other key energy transition projects, such as residential solar, EV charging stations and carbon capture storage projects.

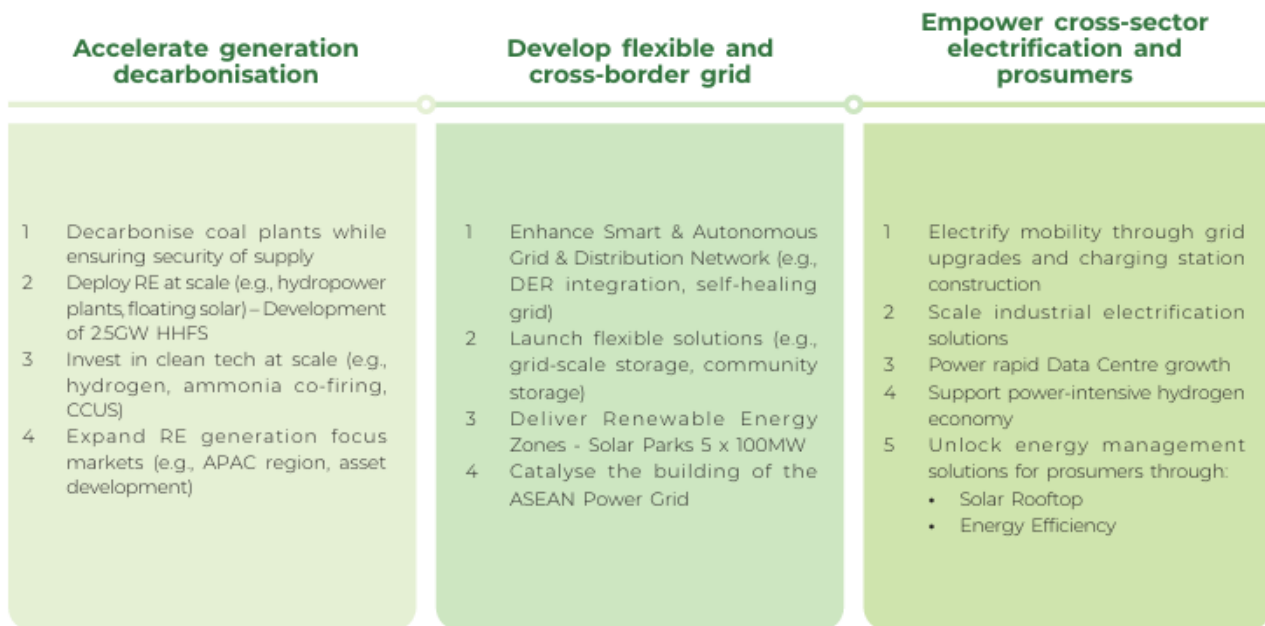
TNB Energy Transition Plan which cuts across the electricity value chain focuses on three pillars, backed by initiatives to shift from fossil fuels to greener sources, aligning with its broader Reimagining TNB 2.0 (RT 2.0) programme. The three (3) pillars are accelerating

generation decarbonisation, developing a flexible and cross border grid, and empowering cross-sector electrification and prosumers. TNB Energy Transition Plan continues to support balancing the energy trilemma and Malaysia's Net Zero target by 2050.

TNB continues to allocate resources into high impact projects to innovate and advance electricity value chain. Innovation and strategic partnership are keys to achieving the highest sustainable impact and business growth through advancement in technology, process, and business model.

Our Energy Transition Plan cuts across the energy value chain, which includes transitioning to clean energy sources, developing the energy transition network, and innovative customer solutions that promote efficient energy usage.

TNB Energy Transition Plan



(Source: TNB Sustainability Statement 2024)

Dynamic Energy Solutions

Consumers play a crucial role in driving a successful energy transition. In 2024, we made significant strides in creating a sustainable transportation ecosystem, encouraging the adoption of renewable energy, and empowering consumers through digital solutions

Electrification of Mobility

TNB plays a vital role in developing the Battery Electric Vehicle (BEV) ecosystem by ensuring that the grid and network are equipped to meet the increasing demand for charging infrastructure. Reinforcing our commitment to this sector, TNB has allocated RM90 million for investments from 2022 to 2025.

As a founding member of the Malaysia Zero Emission Vehicle Association (MyZEVA), TNB collaborates with the Malaysian government and regulatory bodies to shape the nation's EV landscape. MyZEVA unites 57 stakeholders to advocate for BEV policies and initiatives. It serves as a central hub for research and industry insights, supporting policy proposals such as EV charging system safety guides, incentives for EV manufacturers, and inputs for the Garis Panduan Perancangan Petak Pengecasan Kenderaan Elektrik (GPP EVCB). TNB's involvement in MyZEVA strengthens its commitment to advancing sustainable transportation in Malaysia.

BEV Charging Infrastructure

01

Supplying electricity Infrastructure to Charge Point Operators (CPOs) via the EV Green Lane Initiative. The EV Green Lane Initiative was designed to streamline the process and reduce the lead time for supply. 109 applications have been processed, with an average time of 20 days without substation and 50 days with substations.

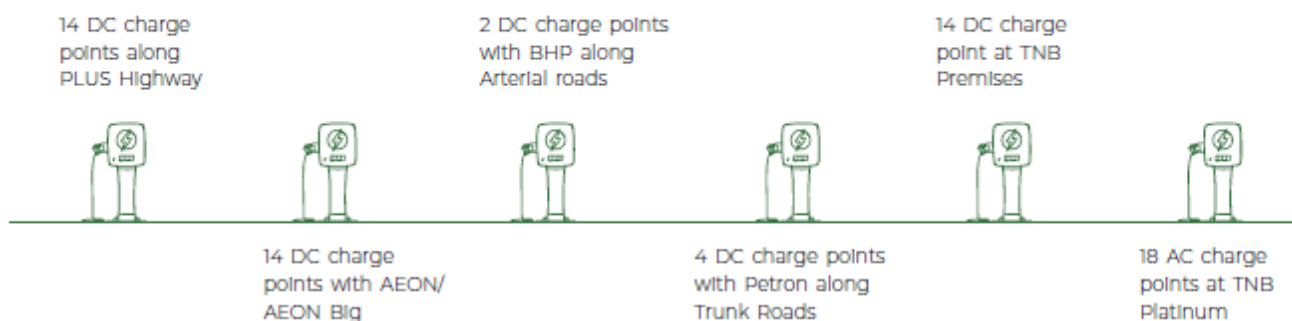
In supporting the growing demand for charging points, TNB focuses on two key roles



Installing EV charge points under the brand name TNB Electron. A budget of RM90 million has been allocated, with 66 charge points installed in 2024 and another 150 planned for installation by the end of 2025.

02

The 66 charge points are as follow:



(Source: TNB Sustainability Statement 2024)

EV Fleet Conversion

We target to electrify 30% of our fleet by 2030, in support of NETR's goal of 20% EV adoption by 2030. Our EV fleet conversion covers a wide range of pickups, vans, passenger cars and specialised units. We aim to replace over 1,000 units to significantly reduce our Scope 1 emissions. We have already deployed 157 EVs, including 138 technical operational vehicles and 19 corporate vehicles, marking a strong start towards our fleet electrification goals.

On 12 September 2025, PETRA announced a new solar rooftop program known as the Solar Accelerated Transition Action Program (Solar ATAP), serving as a continuation of NEM, which has expired on 30 June 2025. Solar ATAP applies the core principle of NEM which allow excess energy to be exported to the grid as an energy offset based on the system marginal price (SMP). Solar ATAP has officially launched since 1 January 2026

Rooftop Solar Prosumers

In 2024, PETRA launched Net Energy Metering (NEM) 3.0 and the Solar for Rakyat Incentive Scheme (SolarIS) to encourage domestic customers to install solar PV systems. NEM 3.0 enables consumers to generate their own electricity from solar energy and export excess power back to the grid with a quota of 2,400MW, where installed capacity is 1,468MW in 2024. Leveraging this initiative, we continue to offer solar rooftop solutions through our subsidiary, GSPARX Sdn. Bhd.

The SolarIS scheme is open to Malaysian citizens on a one-time basis per individual. In 2024, 19,081 customers have received the SolarIS rebate totalling RM76.2 million, administered by TNB.

Enhancing Customer Experience Through MyDigital ID

MyDigital ID is a government-backed digital identity initiative aims to enhance online service authentication by providing individuals with a secure and reliable digital identity. Aligning with this initiative, TNB has successfully implemented MyDigital ID as an additional login option, enhancing customer experience and cybersecurity.

This implementation received positive feedback from our customers, where a survey finding shows that over 90% of new and existing users would start using MyDigital ID as their main login method and would recommend MyDigital ID to other TNB customers. All the surveyees also agreed that the login option provides better and more convenient login experience.

Corporate Renewable Energy Supply Scheme (CRESS)

The Corporate Renewable Energy Supply Scheme (CRESS), introduced by the Ministry of Energy Transition and Water Transformation (PETRA), is designed to enhance the adoption of green energy by allowing participants to supply or purchase green electricity through open access to the grid network with a pre-determined System Access Charge. CRESS enables Green Consumers to contract directly with Renewable Energy Developers (REDs) for green electricity and associated with Renewable Energy Certificates (RECs). By participating in CRESS, Green Consumers can meet their sustainability objectives by offsetting their carbon emissions.

As Malaysia's national utility and the owner of the grid network, TNB plays a significant role in ensuring the success of this initiative. TNB is committed to maintaining the national grid remains reliable, resilient, and capable of accommodating the growing volume of applications under CRESS, as more solar players participate in this initiative.

TNB will also continuously serve as the Supplier of Last Resort (SOLR), providing an essential safety net for Green Consumers during the unavailability of RE Developers.

Community Renewable Energy Aggregation Mechanism (CREAM)

CREAM aims to encourage local community participation in the energy transition by allowing homeowners to lease their rooftops for Solar PV installations by LEGA, aligning with Flagship Catalyst Project under NETR.

Using similar open access concepts, CREAM facilitates the aggregation of residential rooftop solar by Local Energy Generator and Aggregator (LEGA) for selling green energy to the Local Green Consumers (LGC). The framework enables LEGA to access local distribution network by paying a pre-determined Community Access Charge. ST published the CREAM Guidelines on 28 March 2025 to lay out the implementation details.

TNB sees this as another opportunity for GSPARX to expand their generation footprint. This is proven as GSPARX has already established a joint venture with Sime Darby Property in July 2024 to collaborate and participate in CREAM initiative via Sime Darby townships.

Green Lane Pathway

Data centres are energy intensive facilities, requiring substantial and uninterrupted power to run the server and cooling systems. The Green Lane Pathway initiative emphasises efficient and sustainable solutions for data centres operations, aligning with global trends towards greener technology. It facilitates the smooth setup of data centres in Malaysia through:

- **Fast-Track Electricity Supply:** Data centres can connect to the power grid three times faster than usual, reducing setup time from 36-48 months to 12 months.
- **One-Stop Centre (OSC):** This facility provides dedicated support for data centre investors, simplifying the onboarding process and expediting related approvals.

TNB has experienced a notable increase in demand for electricity connections from data centres with 37 Electricity Supply Agreement (ESA) for a total demand of 5,765MW signed in 2024.

In 2024 alone, we completed nine data centres electricity supply projects, with a total energy demand of 1,236MW. Since the establishment of the Green Lane Pathway, we have successfully completed 18 supply connections, meeting a demand of 1,856MW.

Reducing Our Carbon Emissions Intensity

The TNB Carbon Management Strategy and Renewable Energy Capacity Expansion was approved by the SETC (chaired by TNB President/CEO) and was acknowledged by BSRC and TNB BOD. The initiatives intend to reduce carbon emissions intensity from our operations and set the stage for long-term sustainability by integrating renewable energy and innovative carbon management technologies.

Policy on Renewable Energy and Major Regulatory Mechanism of Renewable Law

The Malaysian Government continues to advance the nation's aspiration to achieve **net zero greenhouse gas (GHG) emissions by 2050**, supported by a strengthened policy and institutional framework.

A key institutional development was the restructuring of the energy and environmental ministries. The **Ministry of Natural Resources and Environmental Sustainability (NRES)** was established in **December 2023**, taking over the environmental and natural resource portfolio. Energy-related functions were placed under the Ministry of Energy Transition and Water Transformation (**PETRA**), reflecting a more focused governance structure to drive Malaysia's energy transition agenda.

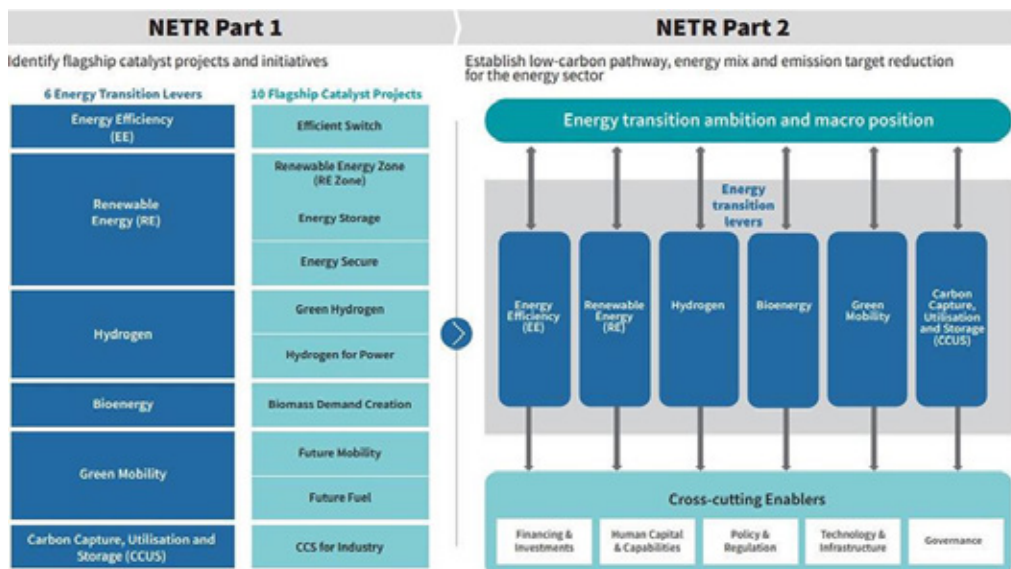
On the policy front, the Government introduced the **National Energy Policy 2022-2040 (NEP)**, which remains the overarching framework guiding Malaysia's energy transition. The NEP aims to position Malaysia to benefit from global energy transition megatrends by promoting

sustainable energy development while ensuring energy security, affordability, and environmental sustainability. The policy outlines key initiatives, including increasing renewable energy (RE) capacity and penetration, reducing reliance on coal in the power generation mix, accelerating electrification (including electric mobility) and enhancing energy efficiency across industrial, commercial, and residential sectors.

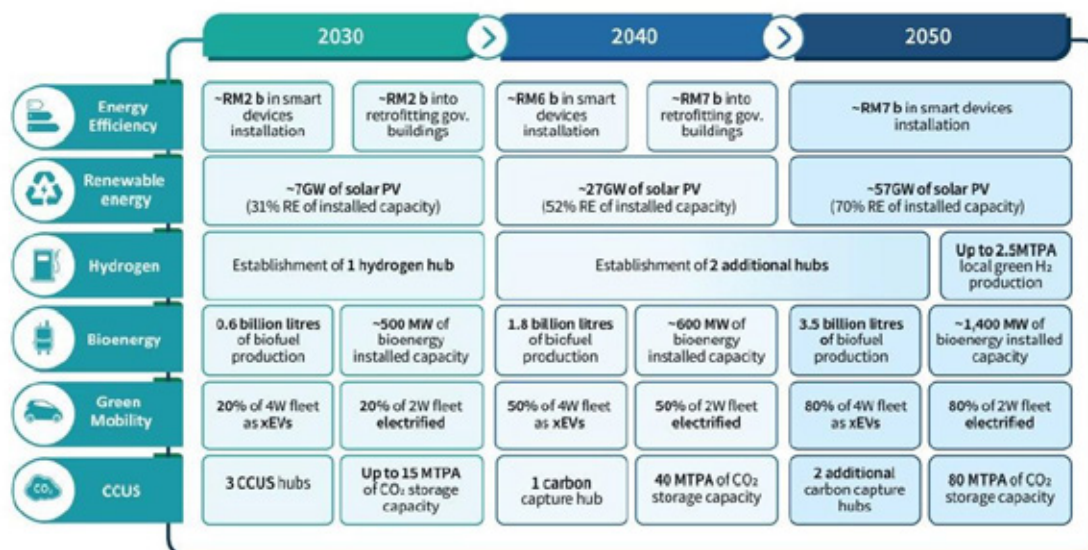
These initiatives are further reinforced under the National Energy Transition Roadmap (NETR), launched in 2023, which serves as Malaysia's primary implementation roadmap for energy transition. NETR provides clear flagship projects and investment pipelines, energy transition pathways across sectors and stronger emphasis on green economy development and financing.

In addition, the Malaysia Energy Transition Outlook (METO), a collaboration between International Renewable Energy Agency (IRENA) and the Malaysian Government, offers analytical insights to support long-term planning and scenario development for the energy transition.

National Energy Transition Roadmap (NETR)



National Energy Transition Roadmap 2023



Specifically on renewable energy, the Malaysia Renewable Energy Roadmap (MyRER), launched in 2021, remains a key reference framework. It aims to balance environmental ambitions with system reliability, affordability, and economic value, while addressing the challenges associated with integrating variable renewable energy (VRE) sources.

MyRER is underpinned by four strategic pillars and four enabling initiatives, targeting 31% renewable energy share in installed capacity by 2025 and 40% by 2035. These targets have since been further enhanced under NETR, which aspires to achieve 70% renewable energy capacity by 2050, signalling a more ambitious long-term trajectory for Malaysia's power sector.

Role of Public Utility in Renewable Energy Law

As a critical player in Malaysia's energy sector with an integrated presence across the value chain, Tenaga Nasional Berhad (TNB) continues to play a central role in delivering the nation's energy transition through renewable energy (RE) development, grid modernisation, and customer participation.

In support of its climate ambitions, TNB introduced its Energy Transition Plan in 2022, outlining a pathway towards achieving net zero emissions by 2050. This plan continues to evolve in alignment with national policies such as the National Energy Policy 2022-2040 (NEP) and the National Energy Transition Roadmap (NETR), reinforcing TNB's commitment to enabling a balanced,

secure, and affordable low-carbon energy system. TNB recognises that a well-managed energy transition is a key enabler for economic growth, resilience, and long-term sustainability in Malaysia.

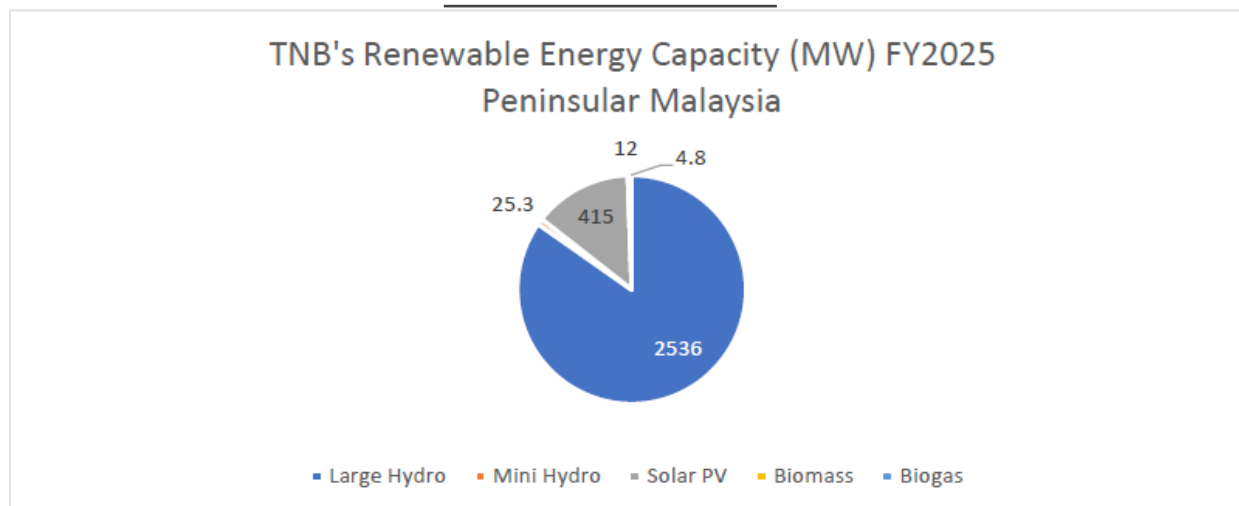
In advancing TNB's decarbonisation agenda, we are expanding our RE portfolio across both domestic and international markets. Currently, we have an installed RE capacity of 4.6 GW across our operations locally and internationally. Looking ahead, we are targeting a total operational capacity of around 10 GW by 2030, comprising 8 GW in Malaysia, 1 GW in the United Kingdom and Europe and 1 GW in Australia.

Our domestic RE growth is supported by Malaysia's National Energy Transition Roadmap (NETR) as TNB is identified as the champion for the 2.5 GW Hybrid Hydro Floating Solar (HHFS) and 750 MWp Centralised Solar Park (CSP), as well as the key partner for residential solar initiatives. Additionally, we are scaling our RE growth through the Large Scale Solar (LSS) programme, the Corporate Renewable Energy Supply Scheme (CRESS) and the Corporate Green Power Programme (CGPP), enabling greater adoption of RE among commercial and industrial customers.

We continue to strengthen our hydroelectric portfolio, currently operating 2.5 GW of large hydro capacity as a cornerstone of Malaysia's renewable mix. We are developing the 300 MW Nenggiri Hydroelectric Project and implementing the 648 MW Sungai Perak Life Extension Programme to advance our decarbonisation strategy.

Production of Renewable Energy (MWh) from Various Resources

TNB's Domestic RE Portfolio



***Non-equity basis. Including secured solar rooftop installation by GSPARX**

Environmental Problems of Electricity Supply and How It Is Being Addressed

Tenaga Nasional Berhad (TNB) continues to strengthen its commitment to minimising environmental impacts arising from its operations and value chain activities, in line with its ambition to progressively improve environmental performance and support Malaysia's energy transition agenda.

This commitment is embedded within a comprehensive environmental governance framework comprising key policies, standards, and operational guidelines, including:

- TNB Environmental Policy (2021, updated 2023)
- Scheduled Waste Management Guidelines (2nd Edition, 2021)
- Polychlorinated Biphenyls (PCB) Management Guidelines (2021)
- Environmental Impact Assessment (EIA) Guidelines (2022)
- e-Waste Management Policy (2022)
- Grid Division Green Code of Conduct

These instruments collectively establish the requirements for environmental compliance, pollution prevention, waste management, and sustainable operational practices across TNB's business divisions.

Environmental governance within TNB is overseen by the Health, Safety and Environment (HSE) department, Sustainability Division. Two(2) steering committees which are HSE Steering Committee, chaired by the Senior Chief Network Officer (SCNO) and HSE Working Committee, Chaired by the Head (OHSE), HSE department provides structured

oversight of environmental performance, regulatory compliance, and continuous improvement across the Group. In addition, the Environmental Performance Monitoring Committee and the Environmental Regulatory Compliance Monitoring Committee at divisional level ensures environmental performance, regulatory compliance, and continuous improvement at the operational level.

TNB's environmental management approach is further strengthened through the implementation of the Health, Safety and Environment Management System (HSEMS), which provides a standardised framework for identifying, assessing, and controlling environmental risks across all business units. The HSEMS has significantly enhanced governance consistency, accountability, and operational control of environmental risks across the organisation.



TNB's HSE Governance Structure

A special mention, TNB Grid Division has introduced its Grid Green Code Conduct with the aim to lead as a green responsible grid owner by reducing climate footprint and promoting sustainability throughout the supply chain. TNB Grid Division's initiatives and highlights include:

Category	Initiatives	Highlights
GHG Reduction	- Gas Insulated Switchgear (GIS) Refurbishment and sulphur hexafluoride (SF6) leak management to achieve <0.5% SF6 leak rate. - SF6 Gas Recycling and Reconditioning Centre to promote circular economy by reducing the needs to buy new SF6 gas or dispose the gas. It also minimizes the environmental impact of high-voltage equipment. - Replacement of old R22 air conditioner as part of energy efficiency and to reduce impact to environment. - Change of CO2 firefighting system to environmentally neutral inert gas system.	- In 2025, Grid has completed 100% GIS refurbishment. 8211 kg SF6 gas recycled. - Ongoing R22 air conditioner replacement at Eastern Region. - Completed GIS refurbishment for two (2) substations (PMU 132 kV SRYA and PMU 132 kV SBGS)
GHG Reduction	- Replacement of Oil Impregnated Paper (OIP) to Resin Impregnated Paper (RIP) transformer bushing replacement to reduce the impact of pollution from catastrophic bushing failure. - Usage of bio-based ester oil for new transformers beginning 2024 to reduce the impact of pollution. - Transformer oil-water separation pit for better effluent management.	- Completed replacement of OIP in 60 transformers. - Specification of ester oil for pilot project approved. - Completed oil-water separation pit.

The generation of electricity from our thermal plants inevitably leaves a large environmental impact, in the form of carbon release, water consumption for power plants and fuel consumption. Our environmental management is principally guided by TNB Environmental Policy which was reviewed in FY2023 to reflect the latest changes in environmental requirements and emerging issues.

Environmental Element

Key Highlights

Water Management

Implement rainwater harvesting at buildings to capture and store rainwater that falls on rooftops and other surfaces for environmental and economic benefits. Currently, these buildings have been installed with this system; Platinum Complex, Balai Islam, PSD Office, and control buildings of TNB Sepang (TSS) and TNB Bukit Selambau Solar (TBSS).

GHG Reduction

Emission Management The TNB Carbon Management Strategy established in 2024 guide TNB's journey towards Net Zero emissions in 2050. The strategy is built on three (3) pillars: Reduce, Capture and Offset and expansion of renewable energy. It targets a 5% annual reduction in GHG emissions intensity from 2024, supporting a 35% emissions intensity reduction by 2035 and Net Zero by 2050.

In 2025, TNB focused on heat rate improvement initiatives across its coal-fired plants to enhance fuel efficiency and lower emissions. These efforts include targeted operational enhancements, equipment upgrades and the adoption of digital performance monitoring tools.

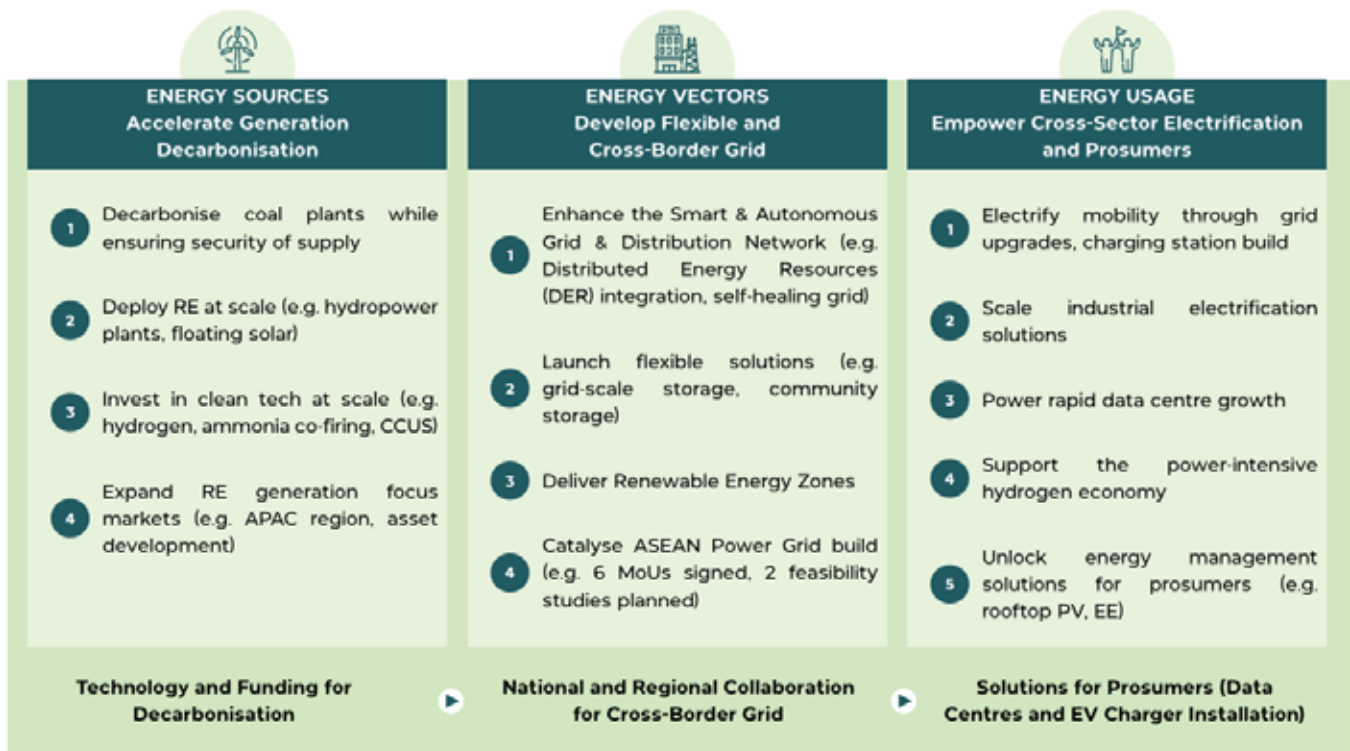
TNB is investing in a new combined-cycle gas-fired power plant in Paka, Terengganu, targeted to begin operations in December 2029. The 1,400 MW project will adopt hydrogen-ready technology, aiming to decarbonise through utilisation of emerging technology without jeopardising long-term energy reliability. Additionally, TNB is also developing a new 2,100 MW combined-cycle gas turbine (CCGT) facility at Kapar, which is expected to be completed by 2031. These projects reflect TNB's commitment to optimising gas utilisation as a transition fuel while preparing its generation assets for future integration of low-carbon fuels such as hydrogen

Complementing the above, TNB is advancing several initiatives to evaluate and deploy cleaner fuels in power generation. These include the exploration of low-carbon co-firing options such as hydrogen, ammonia, and biomass; the implementation of Project REACH (Renewable Energy-Assisted Clean Hydrogen), an initiative to demonstrate the integration of renewable energy and hydrogen production within an operational thermal power plant; and the development of a commercial-scale, end-to-end green hydrogen value chain—from the Kenyir Hydroelectric Hydrogen Facility (HHFS) to green hydrogen and derivatives production in Kertih, in partnership with Petronas.

Energy Management	We continue the campaign called “Drip by Drip, Watt by Watt” with the aim of promoting energy efficiency by reducing electricity and water consumption and involving 109 TNB-owned offices. This campaign has managed to reduce total electricity consumption reduction to 144,027 kWh as of FY2025.
Waste Management	TNB plans to adopt a category-based Waste Management Strategy (WMS), designed to ensure that each waste stream is managed according to its type, following regulatory requirements, if necessary. Rather than applying a uniform approach, waste is categorised and managed in accordance with the waste hierarchy principles of Reduce, Reuse, Recycle, Recovery and Safe Disposal. With the category-based WMS, TNB aims to reduce 20% of the total waste generated by 2030 and achieve a 50% annual recycling rate by 2030 from a 2023 baseline.
Biodiversity Management	The TNB Biodiversity Framework, established in 2024 provides a structured approach towards biodiversity management in TNB. Developed in alignment with the Kunming-Montreal Global Biodiversity Framework and Malaysia's National Policy on Biological Diversity 2022-2030, the framework outlines clear goals to embed biodiversity considerations throughout the asset lifecycle, conserve endangered species, enhance degraded ecosystems, promote urban biodiversity through spatial planning, and preserve forests as a nature-based solution to climate change. As part of TNB's biodiversity conservation strategy, we actively engage with diverse stakeholders across multiple platforms. These partnerships aim not only to conserve and rehabilitate natural habitats but also to promote sustainable livelihoods through activities such as ecotourism development and the mitigation of human-wildlife conflict. Through this integrated approach, TNB contributes to strengthening biodiversity outcomes, supporting local economies and building resilient communities that coexist harmoniously with the environment. TNB applies a site-specific biodiversity approach that considers local ecological characteristics, operational context and surrounding land-use conditions. This adaptive and science-based strategy supports TNB's long-term vision of integrating biodiversity stewardship into sustainable energy development. In 2025, we invested approximately RM3.4 million to implement biodiversity initiatives across key operational sites. For more reference, please refer our TNB Integrated Annual Report 2025

Plans and Strategies to Support Sustainability Agenda

The TNB Energy Transition Plan (ET Plan) is TNB's blueprint to achieve the Net Zero aspiration by 2050. In preparation for the target, we ensure our grid system reliability and long-term readiness through three (3) pillars that accelerate generation decarbonisation, develop a flexible and cross-border grid and empower cross-sector electrification and prosumers.



TNB Energy Transition Plan

Complimenting the ET Plan, the TNB Carbon Management Strategy aims to accelerate TNB's decarbonization by setting clear targets of:

1. Annual 5% emission intensity reduction from 2024
2. 35% emission intensity reduction by 2035
3. Net Zero by 2050

In addition, TNB has committed to cease further investments in coal-fired power plants, reduce coal generation capacity by 50% by 2035 and fully phase out coal by 2050, while ensuring that revenue from coal generation does not exceed 25% of total revenue.

Both ET Plan and Carbon Management Strategy supports the nation's climate commitments towards Net Zero by 2050. TNB will continue to play a crucial role in Malaysia's development in a sustainable manner while balancing the Energy Trilemma.

Policy on Energy Conservation and Efficiency

Empowering Customers Towards Energy Transition

Customers play a critical role in supporting energy transition by making smart choices and participating in the various Renewable Energy (RE) and Energy Efficiency (EE) programs. Our key initiatives, including those supporting government programs, for FY2025 include:

Solutions to Offset Scope 2 Emissions Through the Green Electricity Tariff (GET) and Malaysia Renewable Energy Certificates (mRECs)	▶ The GET and mRECs provide customers with a practical way to support renewable energy development and offset their Scope 2 emissions by purchasing green electricity without the need to install rooftop solar or other on-site renewable energy system. In 2025, a total of 1,580 customers procured 3,925 GWh out of the 6,600 GWh quota available of renewable-backed electricity through the GET, reflecting growing confidence and interest in credible green-power options.
Malaysia Green Attribute Trading System (mGATS)	▶ The mGATS serves as the central digital platform for trading, tracking and redeeming mRECs.
Net Energy Metering (NEM)	▶ NEM allows customers to export excess energy produced from their solar PV systems to the national grid. NEM 3.0 has been replaced with the Solar Accelerated Transition Action Programme (Solar ATAP). In 2025, 99,729 NEM participants made up a total installed capacity of around 1.9 GWp.
Solar Accelerated Transition Action Programme (Solar ATAP)	▶ Solar ATAP is Malaysia's new rooftop solar policy, providing customers with the option to export excess solar power to the grid at the energy rate of the System Marginal Price (SMP). Customers are able to install solar panels up to 100% of their maximum demand to encourage larger system adoption. The Solar ATAP scheme was announced in 2025 to replace NEM 3.0.
Solar for Rakyat Incentive Scheme (SolaRIS)	▶ SolaRIS encourages residential customers to embrace clean energy through the installation of solar PV systems at their homes. Under this initiative, eligible customers who submit their NEM <i>Rakyat</i> applications to the Sustainable Energy Development Authority (SEDA) from 1 April 2024 onwards and successfully commission their solar PV installations will receive a cash rebate of up to RM4,000. In 2025, TNB successfully disbursed RM179.3 million of SolaRIS rebates, benefiting 44,899 customers.

5 Advanced Metering Infrastructure (AMI)

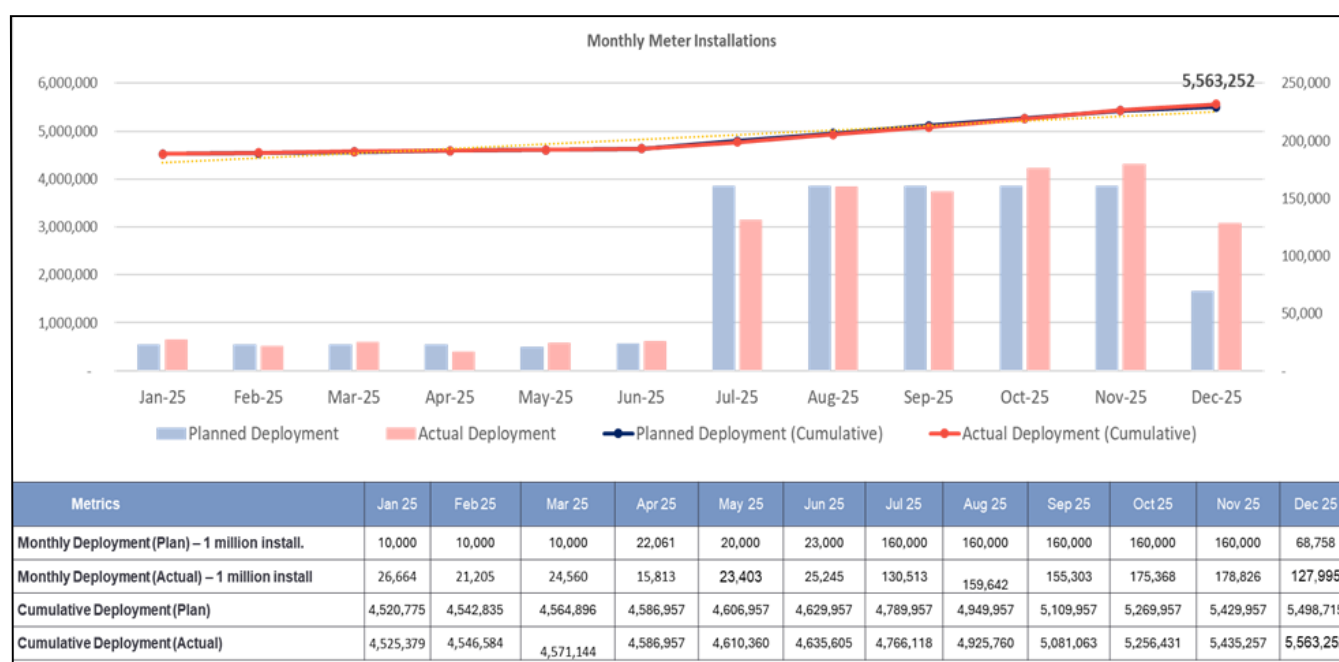
Latest AMI Installations

Tenaga Nasional Berhad (TNB)'s Advanced Metering Infrastructure (AMI) is an integrated system comprising smart meters, communication networks, and data management systems, which facilitates two-way communication between the utility provider (TNB) and customer-end smart meters. The AMI initiative aims to support the adoption of a smart energy lifestyle for customers, while simultaneously improving TNB's grid operational efficiency.

TNB AMI project is currently in the Regulatory Period 4 (RP4) phase (for 2025 – 2027). As of 31 December 2025, 5,563,252 units of smart meters have been installed across the country, including Melaka, Selangor, Kuala Lumpur, Penang, Johor, Perak, Kedah (Langkawi and Kuala Nerang) and Pulau Tenaga Hijau (includes Pulau Perhentian, Pulau Redang and Pulau Tioman).

Moving forward in 2026, AMI Project is targeting to install 1,400,000 smart meters, for a cumulative total of 6.96 million meters. Figure 1 below shows the actual deployment and achievement of TNB AMI project in year 2025:

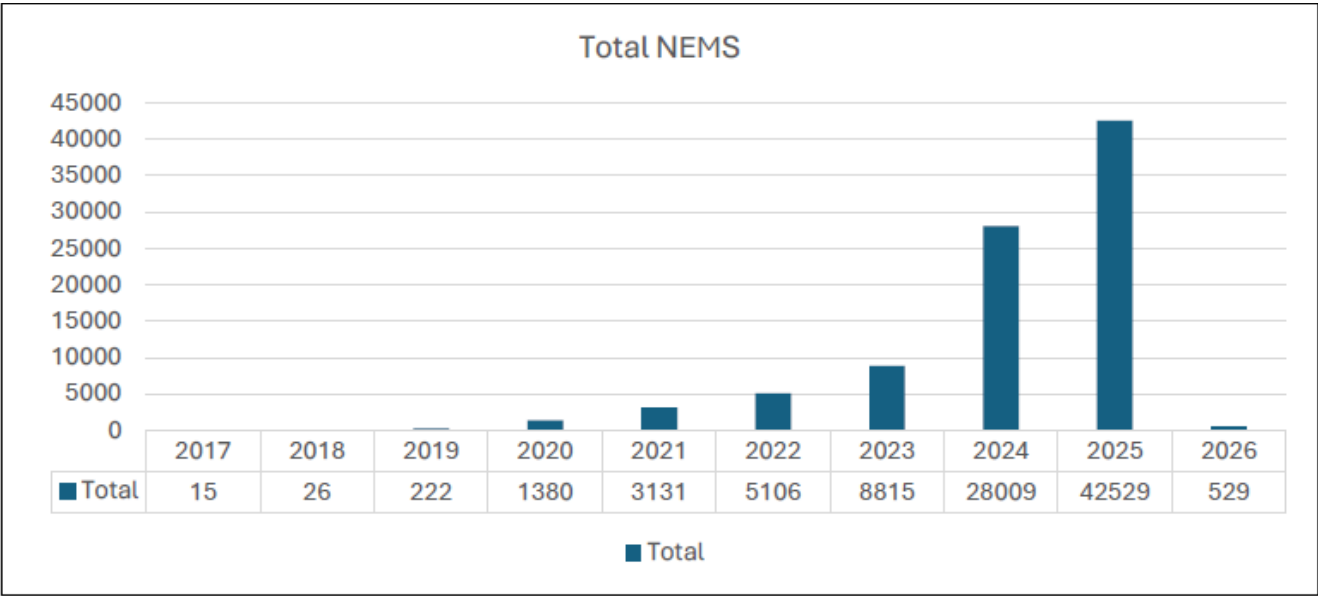
Figure 1: TNB AMI Project Deployment Progress, as of 31 Dec 2025



As one of the key enablers for Energy Transition, AMI provides valuable network visibility for TNB and enabling customers to participate in solar energy generation as prosumers through the Net Energy Metering

(NEM) scheme. As of March 2026, 89,762 customers participated in the NEM scheme program supplying excess solar energy to the grid. Figure 2 below shows the Net Energy Metering Yearly Installation:

Figure 2: TNB AMI Net Energy Metering Scheme (2017-2026)



* Data for the year 2026 is as of March 2026

TNB AMI Project Roll Out Plan

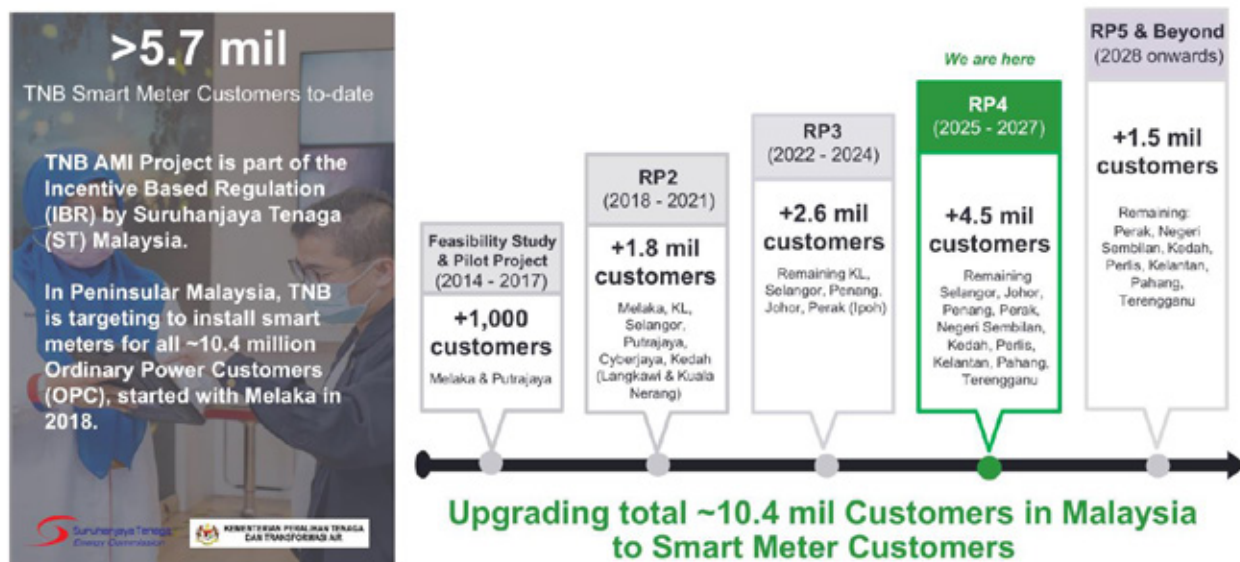
Overall, the TNB Advanced Metering Infrastructure (AMI) project aims to deploy approximately 10.4 million smart meters across Peninsular Malaysia, enhancing smart meter functionalities and services for Ordinary Power Customers (OPCs). The rollout is structured in alignment with the designated Regulatory Periods (RP) and allocated budget. For the current Regulatory Period 4 (RP4: 2025 - 2027) TNB is targeting

the installation of 9.0 million smart meters nationwide by the end of 2027, covering the remaining states, including Selangor, Johor, Perak, Kedah, Negeri Sembilan, Pahang and Terengganu. Figure 3 below illustrates the planned rollout schedule of the TNB AMI project beginning in 2025, encompassing all states within Peninsular Malaysia.



Figure 3: TNB AMI Project Roll Out Plan for Nationwide; “OPC” refers to Ordinary Power Customer

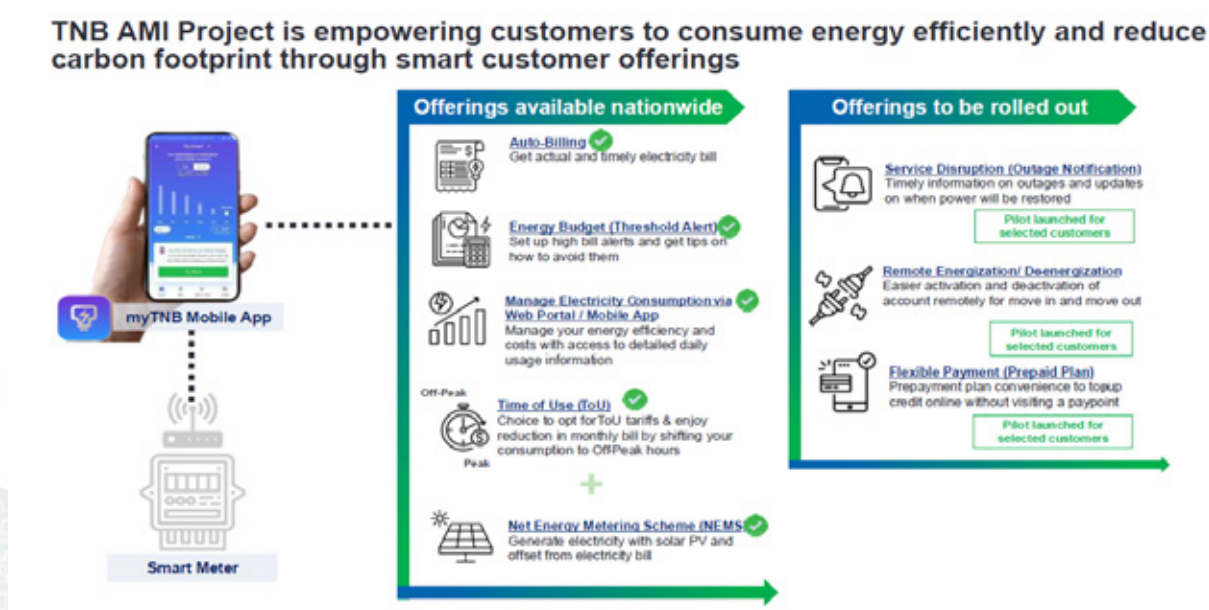
As part of Malaysia Digital and Sustainability plan, TNB is mandated to roll out AMI and Smart Meter for OPC customers nationwide



TNB AMI Smart Meter Customer Offerings

Smart meter customers can enjoy the customer offerings through myTNB Mobile Application and Web Portal. As of 31 December 2025, there are 7 offerings for smart meter customers, with 4 offerings rolled out to customers nationwide and the remaining are in progress of development/pilot. Furthermore, the smart meter also enables the customer to go solar through the NEM scheme program. Figure 4 below shows the customer offerings planned:

Figure 4: TNB AMI Customer Offerings



TNB Advanced Metering Infrastructure (AMI) is equipped with advanced digital features, providing near real-time consumption data accessible via the myTNB Web Portal and Mobile Application as well as providing the utility with near real time events and ability to control the meters remotely. This empowers customers to actively monitor and manage their electricity usage, potentially resulting in greater energy efficiency and cost savings and enable the utility to improve overall customers' experience.

To this end, TNB AMI is working closely with various stakeholders including the Energy Commission (ST), TNB ICT and TNB Retail divisions to deliver the customer offerings by year 2025. Below is the brief description of smart meter benefits intended in the project:

1. Actual and Timely Electricity Bill – Smart meter gives actual readings remotely on a daily basis; hence no site meter reading by meter reader is required. Timely electricity bill will be available in myTNB Web Portal and Mobile Application.

2. Energy Budget – Customers will receive alert notification via myTNB Mobile Application once they have reached the energy usage threshold limit set earlier. This feature allows customers to be more aware of and purposeful in their energy usage from time to time, which will contribute to savings on their electricity bills.

3. Manage Electricity Consumption – using daily profile via myTNB Web Portal and Mobile Application – Smart Meter customers could view detailed information on their energy consumption and electricity bills through myTNB Web Portal (down to 30 minutes' interval) and Mobile Application (daily), which empower customers to better manage their electricity usage and be more energy efficient.

4. Time of Use (ETOU) – This feature provides customers the choice to opt for ETOU scheme tariffs, which offer different pricing throughout various periods in a day e.g. peak, mid-peak and off-peak periods via Interval Billing Mechanism (IBM). As a result, customers could manage their energy usage according to the different tariffs offered in a day and enjoy reduction in monthly bills by shifting consumption to off-peak hours.

Besides, with ETOU implementation, the overall energy demand during peak-periods is reduced – resulting in lowered strain on grid infrastructure, increased energy efficiency and reduced customer impact.

5. Service Disruption (Outage Notification) – Customers will receive timely information on power outages and updates on when power will be restored via myTNB Mobile Application instead of calling TNB Careline to obtain update on the electricity supply situation. Hence, notification on supply disruption will reduce the complaints and enquiries from customers.

6. Remote Energisation and De-Energisation (Faster Power Reconnection (Move in / Move Out)) – with this feature, customers could request to connect and disconnect power supply remotely when moving in or out of their houses and hence reducing the time for supply reconnection from up to 3 days by conventional method to within a day using smart meters – greatly enhancing existing customer experience. Customer could also choose the time and date at their convenience for connection and disconnection.

7. Flexible payment scheme (Prepayment) – this feature helps customers to control their energy consumption by empowering them to manage their energy budget and spending. The flexible Payment Scheme also enables the premise owner to control energy usage at rented premises and reduce the risk of unpaid bills by the tenant. It allows customers to opt for flexible payment options such as post-paid or prepaid plans. Besides, customers could top-up credit online without visiting a pay-point.

Customer offerings on the Actual and Timely Electricity Bill (Auto-Billing), Manage Electricity Consumption using Daily Profile, Energy Budget, Time of Use (eTOU) as well as the Net Energy Metering (NEM) scheme which allows customers with solar panels to export energy back to the grid are enabled for smart meter customers nationwide.

The other three customer offerings on Service Disruption (Outage Notification) of Electricity Supply Interruption & Restoration, Remote Energization / De-energization, and Flexible Payment Scheme (Prepaid Plan) are currently in development and pilot stage.

TNB AMI Project Challenges and Initiatives

Throughout the TNB AMI project implementation, the challenges faced are mitigated through various action plans and initiatives. The key challenges and initiatives taken in TNB AMI project implementation are shared in the table below:

No Challenges

1 Public concern on safety and health due to perceived radiation harm by the radio frequency (RF) equipment on the communication network poles

Initiatives to Mitigate the Challenges

- University Technology Malaysia (UTM) has conducted an independent study on radiation and found that the RF radiation from the communication network pole is safer than microwave oven and Wi-Fi.

2 Misconception that Smart Meters will increase the customers' electricity bill

- Public engagements with community leaders, Members of Parliament (MP), Members of State Assembly, Resident Associations (RAs) Consumer Association were done to share the findings from the UTM study.

- All smart meters are tested and certified by Standards and Industrial Research Institute of Malaysia (SIRIM⁴) for accuracy.

- During installation, there is a Standard Operating Procedure (SOP) in place by the regulator (Energy Commission) to test the accuracy of the old meter (before meter removal) and the new meter (after installation).

- Engagements with Resident Associations and customers were also conducted to educate them on how to use the MyTNB Mobile Application and Smart Meters to improve their energy efficiency.

- Public education was also done through mass media (TV, newspapers) and social media on energy efficiency.

3 Customers are not available at home during meter installations

- Notice about appointments is left behind at the customer premises for them to set an appointment with TNB through the call centre - TNB Careline.

4 Customers do not understand the benefit of smart meters

- Public engagement (physical and virtual meetings) with community leaders, Members of Parliament (MP), Members of State Assembly, Resident Associations (RA) and Consumer Associations were conducted to share the benefits of smart meters.

- TNB Careline also call back the customers after smart meter installation to explain about the smart meter benefits.

5 Customers are hesitant to allow TNB contractors to enter their premise because they are concern over their safety

- Customers will be given early notice notifying them that TNB will be installing smart meters in their area.

- All TNB trained contractors are equipped with standard uniform and identification Card.

- Customers can also call TNB Careline to verify the Identification Card of the installers.

⁴Standards and Industrial Research Institute of Malaysia (SIRIM), is a government-owned organization in Malaysia focusing on industrial research, technology development, and standards

One of the significant challenges encountered was the disruption in the global supply chain, which affected the timely availability of critical components for the AMI project. To address this, TNB has strengthened its supply chain resilience by engaging strategic suppliers to ensure continuity and reliability. Proactive measures

were taken to diversify sourcing options, establish buffer stocks, and improve forecasting accuracy, thereby minimizing potential delays and ensuring smoother project implementation. This has enabled TNB to continue building momentum in the installation of smart meters as it expands nationwide.

6

The Latest Technologies or innovations deployed by Tenga Nasional Berhad

Strategy Pillar: Dynamic Energy Solutions

- EV Charging platform
- Customer digital platform (myTNB) offering integrated energy-related products and services, including digital billing, energy budgeting tools, and Time-of-Use (TOU) features.
- Energy monitoring system for smart building (Virtual Energy Maager - VEM)
- Behind-the-meter solar-BESS hybrid system

Strategy Pillar: Develop Energy Transition Network

- Feasibility Study stage
 - ▶ Regional interconnection
 - ▶ Grid flexibility to better manage VRE integration, optimise electricity infrastructure, and enhance demand response capabilities.
 - ▶ Investments in Community Energy Storage System (CESS) and VDRT to enable greater DER participation in line with the growth of CRESS and CREAM.

- Technology solutions & pilots stage
 - ▶ Battery Energy Storage System
 - ▶ AI-based Predictive Maintenance for Underground Cable
 - ▶ Smart Green Island

Strategy Pillar: Deliver Clean Generation

- Feasibility Studies stage
 - ▶ Green Hydrogen
 - ▶ Carbon Capture Utilisation and Storage (CCUS)
 - ▶ Nuclear Power
- Technology solutions & pilots
 - ▶ AI-based predictive and analytical tools for power machineries and cables
 - ▶ Onshore and offshore floating solar
 - ▶ Renewable Natural Gas (RNG) production
 - ▶ Green hydrogen production and application
 - ▶ Carbon capture and utilization technologies

7

Major Issues or Transformation Impacting the Electricity Supply Industry in Malaysia

The electricity supply industry in Malaysia and across ASEAN is undergoing a period of significant transformation, shaped by global energy trends, regional cooperation efforts, and country-specific development goals. As ASEAN member states move toward more sustainable, secure, and resilient energy systems, several key issues and structural shifts are reshaping the landscape, especially in Malaysia:

a. Renewable Energy Integration

The growing adoption of renewable energy (RE) technologies, particularly solar and hydro, is accelerating across the region in response to decarbonisation targets and climate commitments. In Malaysia, the government has set a target of achieving 70% renewable energy (RE) capacity in the national power mix by 2050 under the National Energy Transition Roadmap (NETR). However, integrating variable RE sources poses technical challenges to grid stability and reliability. This necessitates investments in flexible generation, energy storage systems, and enhanced forecasting capabilities to ensure secure and efficient operations.

b. Regional Power Grid Integration

The ASEAN Power Grid (APG) initiative aims to promote multilateral power trading and enhance energy security through interconnected national grids. While significant progress has been made through cross-border electricity initiatives—including multilateral trading arrangements such as the Laos–Thailand–Malaysia–Singapore Power Integration Project (LTMS-PIP) and bilateral renewable electricity export mechanisms such as Energy Exchange Malaysia (ENEGEM) – scaling these efforts into a truly integrated regional grid remains a complex challenge. Technical harmonisation, regulatory alignment, and investment mobilisation are essential to advance cross-border electricity trade.

c. Cybersecurity in a Digital Era

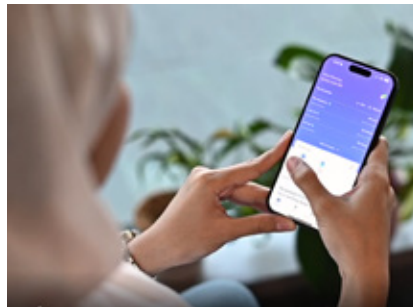
As digitalisation accelerates, cybersecurity has emerged as a critical concern for the electricity supply industry. Utilities are increasingly exposed to cyber threats targeting operational technologies (OT) and information systems (IT). Ensuring the resilience of critical infrastructure requires enhanced security protocols, regional cooperation, and alignment with international best practices.

d. Decarbonisation and the Energy Transition

Driven by national commitments under the Paris Agreement and increasing investor focus on ESG metrics, ASEAN utilities are transitioning away from fossil fuels toward cleaner energy sources. In Malaysia, the shift from coal to natural gas and renewables is central to the country's low-carbon roadmap. Malaysia, with the support of TNB, is also considering alternative energy sources such as nuclear, which would introduce a different set of technical, regulatory, and societal challenges that would require strong national commitment supported by international collaboration and knowledge exchange.

Electrification of the transport and industrial sectors, as well as exploration of green hydrogen and carbon capture technologies, are expected to play a growing role. Ensuring a just and equitable transition—especially for communities dependent on legacy energy systems – remains a priority. In Malaysia, there is strong legislative and policy support to enable the adoption of these nascent technologies, such as the introduction of the Carbon Capture, Utilisation & Storage (CCUS) Bill 2025, Low Carbon Mobility Blueprint 2021 – 2030 and National Energy Transition Roadmap (NETR) 2023.

In conclusion, the electricity supply industry in Malaysia and the ASEAN region is at a pivotal juncture. Regional cooperation, innovative technologies, and progressive policy frameworks will be key to navigating the energy transition while ensuring secure, affordable, and sustainable electricity for all.



NEPAL



1

Country Information

General Information (as of Dec 2024 or latest data)

Capital:	Kathmandu
Area:	147,516 km ²
Population:	29 millions
GDP:	US\$ 50.08 billion (nominal, 2025 est) US\$ 180.64 billion (PPP, 2025 est)
Currency:	Nepali Rupees (NPR)
Installed Capacity:	4.3 GW
Percentage of population electrified:	99.0%

Power Demand by Sector

Sector	Total Power Consumption as of 2025 (MWh)	Projected Change (%) 2024/2025	Projected Power Consumption as of end 2025 (MWh)
Industrial	4,136,000	33.44%	36.54%
Commercial	880,000	6.89%	7.77%
Residential	4,782,000	36.44%	42.24%
Other	1,521,000	10.63%	13.43%
Export (to India & Bangladesh)	2,380,000	45%	22%
TOTAL	11,319,000	13%	15,641,000

2

Energy Policy and Electricity Market (as of end 2025 or latest data)

National Energy Policy

- Article 51 (f) 3 of the constitution of Nepal has mentioned "ensuring reliable and affordable energy supply and proper utilization of energy by generation and development of renewable energy for the fulfillment of citizen's basic needs."
- Present Condition & Future Roadmap of Energy, Water Resources & Irrigation sector (White Paper by Ministry of Energy, Nepal), 2018
- Electricity Development Roadmap, 2023 Ministry of Energy, Water Resources and Irrigation for energy target (www.moewri.gov.np)
- New Hydropower Bill in Parliament for Approval

Supply and Demand of Primary Natural Resources = 640 PJ

- Traditional (Firewood, Agricultural, Biomass) = 410.688 PJ (64.17 %)
- Commercial fuels (Petroleum, Coal) = 181.44 PJ (28.35%)
- Grid Electricity = 31.744 PJ (4.96%)
- Renewable Energy = 16.13 PJ (2.52%)

Laws and Governmental Regulations

- Electricity Act, 1992
- Electricity Regulation, 1993
- Nepal Electricity Authority Act, 1984
- Electricity Regulatory Commission Act, 2018
- Hydropower Development Policy, 2001

Total amount of energy consumption by sectors (Industrial/ Commercial/ Residential) (in MWh and %)

Sector	Total Power Consumption as of 2025 (MWh)	% in Total
Industrial	4,136,000	33.44%
Commercial	880,000	6.89%
Residential	4,782,000	36.44%
Other	1,521,000	10.63%
Import (Purchase from India)	1,681	14%
Export	2,380,000	22.00%
TOTAL	11,319,000	100.00%

3

Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (coal, LNG, Gas, Nuclear-fired and hydro) including total installed capacity of generation (in MWe) and total production (in MWh) by fuel types (in MWe)

	Installed Capacity (GW)	Year-on-year Growth (%)		Generation (TWh)	Year-on-year Growth (%)
Total	4.302	22.59%	Total	12.64	14.22
Hydropower	4.0352 (93.84%)	21.93%	Hydropower	10.786	13.25
Thermal Power	0.05341 (1.23%)	0.00%	Thermal Power	0.000013	-59.77
Alternative other	0.090 (2.09%)	20.00%	Nuclear Power	0.00	0.00
Solar Power	0.117 (2.71%)	100.00%	Wind Power	0.00	0.00
			Solar Power	1.85	1.54
			Other	N/A	N/A

4

Reduction of Carbon Emission

Policy on Renewable Energy and major regulatory mechanism of Renewable Law

- Nepal plans to reach 28,500 MW total installed capacity by 2035 of which Storage Hydropower: 30-35%, Peaking Run of River Hydropower: 25-30%, Run of River Hydropower: 30-35% & Alternative Sources: 5-10% (Energy Development Roadmap and Working Plan, 2023 - Ministry of Energy, Nepal)
- Policy of the government to provide tax rebate on replacement of fossil fuel equipment by electrical equipment (such as electric vehicles, induction cookers, etc.)
- Hydroelectric Investment and Development Company Limited (HIDCL) established for investment on renewable energy sources and hydropower projects.
- Water and Energy Commissions Secretariat (WECS) established under the Ministry of Energy, Nepal for analysis of electricity demand & consumption, formulation of electricity master plan, etc.

Production of Renewable Energy (MWh) from various resources (wind, solar, biomass, landfill gas, etc)

- Nepal has 118.94 MWp of solar PV plants installed which generates around 2000 GWh of electricity annually.
- 4035 MW of hydropower plants are installed in Nepal, which serves as the major source of electricity generated in the country.
- 6 MW of Bagasse power generation (co-generation). No wind, biomass or landfill gas power plants.

Environmental Problems of Electricity Supply (including generation and T&D sectors) and How It Is Being Addressed

- Trees cutting & Environmental Study reports to be approved by the Ministry of Forest that takes huge time and resources creating hindrance to development of hydropower and transmission projects.
- Import, storage & use of explosives restricted by law, requires special permission.

Plans and Strategies to Reduce Greenhouse Gas Emissions

- Nepal aspires to reach Net Zero emissions by 2045 AD
- Increasing the use of clean and renewable energy in all sectors
- Increasing the efficiency
- Increase the share of clean energy in the country's energy demand to 15% and forest cover to 45% by 2030.
- Increase the per capita electricity consumption to 1500 kWh from current 450 kWh.

Policy on Energy Conservation and Efficiency

- National Energy Efficiency Strategy, 2019 (Ministry of Energy, Nepal) formulated for the promotion of energy efficiency, increase energy security, and increase energy access in order to double the average improvement rate of energy efficiency in Nepal from 0.84% per year (2015) to 1.68% per year (2030).

Latest AMI Installations and Smart Grid Projects

- Nepal Electricity Authority (NEA), intends to set up 5 million smart meters in Nepal by May 2025. Only around 60,000 smart meters have been deployed in the capital city Kathmandu till now.
- NEA is implementing a Grid substation automation system in all of its (almost 70) Grid Substations throughout the country to make the grid operation digital and more reliable.
- Nepal has experienced a dramatic surge in electric vehicle (EV) adoption, with over 70% of new car sales in 2025 being electric, making it one of the top global markets for EV adoption behind only Norway. NEA installed 62 EV charging stations across the country, and 510 charging stations will be installed all over the country.

Challenges Faced and How It Was Solved / Addressed:

- EV imports in FY 2023/24 reached with over 80% coming from China, marking a 20% increase over the previous year.
- Land availability issues for EV charging stations. Utilized government, public lands where available and collaborated with public companies, government agencies to utilize their land.
- Favorable tax rates, reduced import duties, and promotion of charging infrastructure have fueled this growth, with an aim to make 90% of new private vehicles electric by 2030.

AMI Roll out plan

- AMI to be gradually implemented for domestic and commercial users.
- Issue of frequency license from Nepal Telecommunication Authority. On September 29, 2022 the Ministry of Communication and Information Technology, Nepal issued notice in the Nepal Gazette permitting the NEA to use three frequency bandwidths from 396.5 MHz to 399.5 MHz to install the smart meters. For this purpose, the NEA needs to pay Rs 12 million per MHz to the government.

AMI offerings to customers and its benefit

- Implementing AMI reduces meter reading expenditure and increases efficiency & revenue collection.
- Integration with payment gateway providers, banks for digital payment of electricity bills.

6

Latest Technologies or Innovations

- Nepal government (NEA) will install 100 MW BESS this year is under feasibility study stage.
- NEA has installed Solar PV power plants with battery energy storage system (AC bus) at rural districts of Karnali province of Nepal in 4 sites totaling 2,925 KW PV and 11,000 kWh BESS.
- Grid Resilience through Intelligent Photovoltaic Storage (GRIPS) project introduced a smart storage system that seamlessly switches between grid, battery, and solar power during outages, to build smarter grids and ensure reliability through intelligent off-grid storage.
- 3 pump (appr. 1000 MW) storage project under the process of feasibility study on Shyarpudaha, Kulekhani and Hunilgtar.
- Planned to install 25 MW solar power annual.

7

Major Issues or Transformation Impacting the Electricity Supply Industry

- Transmission line bottlenecks impacting supply and delivery of generated electricity to load centres, resulting in spill of hydroelectricity at the same time there is import of electricity to meet domestic demand.
- Aged infrastructure and insufficient capacity of transmission, distribution network hampering the reliable supply of electricity.
- Various capacity of 12 Cross border Transmissions with India.
- Nepal- China Cross Border 220 kV Transmission line project in implementation stage.

NEW ZEALAND



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Wellington
Area:	268,838 km ²
Population:	5.34 million
GDP:	\$176 billion USD
Currency:	New Zealand dollar (NZD)
Installed Capacity:	10.6 GW
Percentage of population electrified:	100%

Power Demand by Sector

Sector	Total Power Consumption as of 2025 (MWh)	Projected Change (%) 2024/2025	Projected Power Consumption as of end 2025 (MWh)
Industrial	12,469,000	-3.4%	12,045,106
Commercial	9,356,000	-1.0%	9,262,440
Residential	14,008,000	1.9%	14,274,152
TOTAL	35,833,000	-0.70%	35,581,698

(Data in the table sourced from:
<https://www.mbie.govt.nz/building-and-energy/energy-and-natural-resources/energy-statistics-and-modelling/energy-statistics/electricity-statistics/>)

National Energy Policy¹⁰

Supply and Demand of Primary Natural Resources

Full data tables may be downloaded from the Energy in New Zealand webpage¹¹

Law and Government Regulations

The main laws governing the energy sector are, **The Crown Minerals Act 1991** (The Act) sets out the management regime for Crown-owned minerals. The Act covers the efficient allocation of the Crown's minerals, the effective management of those rights, and for obtaining a fair financial return from the development of Crown-owned minerals. The purpose of the Act is to promote the prospecting for, exploration for, and mining of Crown-owned minerals for the benefit of New Zealand. 'The benefit of New Zealand' is defined as best achieved by increasing New Zealand's economic wealth through maximising the economic recovery of New Zealand's Crown-owned minerals.

The Electricity Industry Act 2010 provides a framework for the regulation and governance of the electricity industry. It sets out the Electricity Authority's functions, objectives and monitoring, and enforcement powers, and provides for the Electricity Industry Participation Code, which are the industry "rules". The Act also sets out requirements relating to the separation of distribution and certain generation and retail activities, places an obligation on distributors to maintain supply in certain circumstances, and contains a range of regulation-making powers. The Act provides for the industry consumer dispute resolution scheme, but the Minister of Commerce and Consumer Affairs is responsible for these provisions.

The Electricity Act 1992 deals with operational matters, setting out the regulatory framework for the supply and use of electricity. The Act contains wide-ranging regulation-making powers for matters of electrical safety (including for the purposes of public health and

safety and prevention of property damage). The Act confers powers and duties on electricity operators and other owners of electricity works; provides for the registration and licensing of electrical workers and the administration of the Electrical Workers Registration Board; and sets out the functions of WorkSafe New Zealand as regulator of the safe supply and use of electricity.

The Energy Efficiency and Conservation Act 2000 established the Energy Efficiency and Conservation Authority (EECA), and forms the legislative basis for promoting energy efficiency, energy conservation and renewable energy. The act includes regulation-making powers for product energy efficiency standards and labelling, as well as the disclosure of information in order to allow for the compilation of statistics on energy efficiency, energy conservation and renewable energy.

The Gas Act 1992 sets out the regulatory framework for the supply and use of gas. It confers powers and duties on gas operators and other owners of gas fittings; provides for the governance of the gas industry, including providing for co-regulation with a gas industry body, currently the Gas Industry Company (GIC), and; sets out the functions of WorkSafe New Zealand as regulator of the safe supply and use of gas.

Total Amount of Energy Consumption by Sectors (Industrial/ Commercial/ Residential) (in MWh and %)

Sector	Total Power Consumption as of 2025 (MWh)	% in Total
Industrial	12,498,000	31.27%
Commercial	9,219,000	23.07%
Residential	14,068,000	35.20%
Other	4,182,000	10.46%
TOTAL	39,967,000	100%

¹⁰ <https://www.mbie.govt.nz/building-and-energy/energy-and-natural-resources/energy-strategies-for-new-zealand/new-zealand-energy-strategy>

¹¹ <https://www.mbie.govt.nz/building-and-energy/energy-and-natural-resources/energy-statistics-and-modelling/energy-publications-and-technical-papers/energy-in-new-zealand>

3

Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (coal, LNG, Gas, Nuclear-fired and hydro) including total installed capacity of generation (in MWe) and total production (in MWh) by fuel types (in MWe)

	Installed Capacity (GW)	Year-on-year Growth (%)		Generation (TWh)	Year-on-year Growth (%)
Total	10.662	5.5%	Total	42.235	1.0%
Hydropower	5.581	0.4%	Hydropower	23.49	-10.7%
Thermal Power	1.275	21.1%	Thermal Power	8.719	13.1%
Nuclear Power	0	0%	Nuclear Power	0	0%
Wind Power	1.265	10.4%	Wind Power	3.919	22.3%
Solar Power	0.568	51.1%	Solar Power	0.595	60.2%
Other	1.974	0.20%	Other	5.512	30.60%

4

Reduction of Carbon Emission

Policy on Renewable Energy and major regulatory mechanism of Renewable Law

New Zealand is on the way toward a greener future. The Government has passed the Climate Change Response (Zero Carbon) Amendment Act and set a target of net-zero carbon emissions by 2050 (except biogenic methane). Future demand for electricity will depend heavily on the path of decarbonisation and the extent of changes in electricity using technologies and supply technologies over the next 30 years. Process heat is the energy used as heat mainly by industrial and commercial sectors for industrial processes, manufacturing, and warming spaces. Process heat contributes about 8 per cent of New Zealand's total greenhouse gas emissions. Therefore, the ability of organisations to switch to lower-emission fuel is crucial for New Zealand to meet the target of net-zero carbon emissions. However, the ability and incentives for organisations to switch to electricity is dependent on both the relative price of electricity and the future of process heat technology.¹²

Production of Renewable Energy

In New Zealand, around 80% of our electricity is generated from clean, renewable resources such as hydro, geothermal and wind. Switching to electric technology over coal and oil — and using all energy efficiently — are our best opportunities to reduce carbon emissions.

For more information, please refer to the link.¹³

Environmental Problems of Electricity Supply:

About 40% of New Zealand's total greenhouse gas emissions come from our energy use. We create these greenhouse gases when we burn fossil fuels — whether driving a petrol car or diesel truck, travelling by plane, burning gas for manufacturing or burning coal to create electricity. Reducing our energy-related emissions will be challenging. Although New Zealand is investing resources and effort into decarbonisation and renewable energy, much more needs to happen.¹⁴

¹² <https://www.legislation.govt.nz/act/public/2019/0061/latest/LMS183736.html>

¹³ <https://www.eeca.govt.nz/why-energy-matters/climate-change-and-energy/energy-and-emissions/#:~:text=In%20New%20Zealand%2C%20around%2080,opportunities%20to%20reduce%20carbon%20emissions.>

¹⁴ <https://www.eeca.govt.nz/insights/energys-role-in-climate-change/new-zealands-energy-related-emissions/>

Plans and Strategies to Reduce Greenhouse Gas Emissions

Our strategic focus areas can be found here¹⁵ and include Efficient and low emissions transport, energy efficient homes and productive and low emissions businesses.¹⁶

Policy on Energy Conservation and Efficiency

Policy proposals can be found here.¹⁷

5

Advanced Metering Infrastructure (AMI/Smart Grid Deployment)

Latest AMI Installations and Smart Grid Projects

Over last couple of years network operational data has been made available to Distribution Network companies to create visibility and better management of low voltage networks:¹⁸

Challenges faced and how it was solved / addressed

Smart meter rollout was not a mandate in New Zealand. It was driven by electricity retailers and the customers had their discretion on installing smart meters. Therefore, it was not easy to implement the plan.

Since this was deployed by retailers, there were some inconsistencies between different retailers in the early phases.

Perceived health risks: Some people have concerns about how smart meters may affect their health. Concerns may be about the radiofrequency energy (RF) or electromagnetic frequency (EMF) a smart meter emits when collecting or sending data. Smart meters must comply with maximum exposure limits set out in New Zealand Standard 2772.1:1999 for exposures to radiofrequency fields. It is an individual's right not to have a smart meter if they so choose, and the concerns can be very real.

Data Access and Consumer Control: Consumers initially had limited access to their own real-time usage

data. This reduced the ability of consumers to actively engage in energy-saving behaviours, which is one of the key benefits of smart meters.¹⁹

However, newer regulations and APIs have been introduced to increase transparency and give consumers better control.

Interoperability and Proprietary Systems: Different power companies and meter owners used proprietary platforms. This made it hard to switch retailers or integrate systems, creating vendor lock-in and reducing competition in the market. As a remedy, an Open AMI System/Services Access Interface was implemented by EA through the Advanced Metering Policy.²⁰

Privacy and Security Concerns: There were concerns around data privacy, cybersecurity, and how usage data could be shared or misused. This slowed public acceptance in certain demographics and required policy clarification and new privacy standards.

Low Awareness and Engagement: Some consumers were unaware of the benefits or didn't trust the technology. This affected customer buy-in and undermined energy-saving goals.

AMI Roll out plan

The natural maximum penetration is expected to be around 90%, given some consumers are in rural areas where the benefits of smart meters are not as great, and some consumers do not want a smart meter.

¹⁵ <https://www.eeca.govt.nz/strategic-focus-areas/>

¹⁶ <https://www.eeca.govt.nz/strategic-focus-areas/>

¹⁷ <https://www.eeca.govt.nz/insights/eeca-insights/new-proposal-to-boost-voluntary-climate-action-in-aotearoa-new-zealand/>

¹⁸ <https://bluecurrent.co.nz/our-products/data-services/network-operations-data-service>

¹⁹ <https://www.mbie.govt.nz/dmsdocument/30317-consumer-nz-exploring-a-consumer-data-right-for-the-electricity-sector-submission-pdf>

²⁰ https://www.ea.govt.nz/documents/217/Guidelines_on_advanced_metering_infrastructure.pdf

AMI offerings to customers and its benefit

At the moment, there are 30 retailers in the market. Most of these retailers have built their business models to use smart meter data to reduce costs, increase efficiency, improve customer service and offer new services and tariffs. Some of these new retailers are growing their market share rapidly, encouraging other, existing retailers to create new services to retain customers. Retailers are now using smart meters to provide a gateway for services such as Contact's "Good Nights" plan, Electric Kiwi's free 'hour of power'

plan, Flick Electric's wholesale price following tariff and Glo-bug's web-based prepay service.

Further, Electricity Distribution Companies have started using smart meter data in their day-to-day activities such as providing new connections, fault locating, short/long term planning. Hence, customers are provided with more efficient and cost-effective service, resulting in higher satisfaction.

6

Latest Technologies or Innovations

Solar generation has transitioned from small-scale pilot projects to a growing portfolio of utility-scale developments across New Zealand, with multiple grid-connected projects now commissioned and others in development.²¹

Battery Energy Storage Systems (BESS) are also emerging as a key enabler of system flexibility and peak demand management. Utility-scale BESS projects are increasingly being deployed to support renewable integration, enhance grid stability, and improve overall energy resilience, particularly as the share of intermittent generation continues to grow.

7

Major Issues or Transformation Impacting the Electricity Supply Industry

Key transformations impacting New Zealand's electricity supply industry include the increasing penetration of intermittent renewable generation, and the growing impacts of climate change on network resilience.

As renewable generation (particularly wind and solar) continues to grow, there is an increasing need for firming capacity and energy storage solutions to maintain system reliability and manage peak demand. This is driving investment in flexible resources such as battery storage, demand response, and fast-start generation.

At the same time, climate change is contributing to more frequent and severe weather events, including storms, cyclones, and flooding. Recent analysis highlights the likelihood of more intense rainfall events and increased weather variability across New Zealand, reinforcing risks to electricity infrastructure.^{22 23}

These factors are accelerating the need for grid modernisation, resilience investment, and more flexible system operation.

²¹ <https://www.transpower.co.nz/connections/whats-latest-grid-connections>

²² <https://www.rnz.co.nz/news/weather/591356/bigger-storms-more-often-new-study-projects-likely-future-rainfall-impacts-on-nz>

²³ <https://environment.govt.nz/publications/our-environment-2025/atmosphere-and-climate/>





PEOPLE'S REPUBLIC OF CHINA



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Beijing
Area:	9.6 million km ²
Population:	1400 millions
GDP:	US\$ 19.6 trillion
Currency:	RMB
Installed Capacity:	3891 GW
Percentage of population electrified:	100%

Power Demand by Sector

Sector	Total Power Consumption as of 2024 (MWh)	Projected Change (%) 2024/2025	Projected Power Consumption as of end 2025 (MWh)
Industrial	6,389,600,000	+3.9%	6,636,600,000
Commercial	1,835,000,000	+8.7%	6,636,600,000
Residential	1,493,600,000	+6.3%	1,588,000,000
TOTAL	9,854,000,000	+5.2%	10,368,200,000

2

Energy Policy and Electricity Market (as of end 2025 or latest data)

General

- Environmental Impact Assessment Law of PRC
- Special Equipment Safety Law of PRC
- Renewable Energy Law of PRC
- Mineral Resources Law of PRC
- Environmental Protection Law of PRC
- Safety Production Law of PRC

Electricity

- Electricity Law of PRC
- Power Grid Dispatching Regulation
- Power Equipment Protection Regulation
- Power Supply and Use Regulation
- Power Supervision Regulation
- Power Accident Emergency Treatment and Investigation Regulation

Energy Saving

- Energy Saving Law of PRC
- Energy Saving Regulation for Public Agencies
- Energy Saving Regulation for Civil Buildings

Nuclear Saving

- Civil Nuclear Equipment Safety Supervision and Management Regulation of PRC
- Nuclear Material Control Regulation of PRC
- Civil Nuclear Safety Equipment Supervision and Management Regulation
- Nuclear Power Station Accident Emergency Control Regulation

Total amount of energy consumption by sectors (Industrial/ Commercial/ Residential) (in MWh and %)

Sector	Total Power Consumption as of 2024 (MWh)	% in Total
Industrial	6,389,600,000	64.84%
Commercial	1,835,000,000	18.62%
Residential	1,493,600,000	15.16%
Other	135,800,000	1.38%
TOTAL	9,854,000,000	100%

3

Power Generation and T&D Sectors (as of end 2024 or latest data)

Total installed capacity of generation by fuel types (Coal, LNG, Gas, Nuclear-fired and Hydro) including total installed capacity of generation (in GW) and total production (in TWh) by fuel types (in MWe)

	Installed Capacity (GW)	Year-on-year Growth (%)		Generation (TWh)	Year-on-year Growth (%)
Total	3349.22	14.6	Total	10,086.9	6.7
Hydropower	435.52	3.1	Hydropower	1,425.7	10.9
Thermal Power	1,444.62	3.9	Thermal Power	6,374.3	1.7
Nuclear Power	60.83	6.9	Nuclear Power	450.9	3.7
Wind Power	520.78	18.0	Wind Power	5,207.8	12.5
Solar Power	887.42	45.4	Solar Power	8,874.2	43.6
Other	/	/	Other	/	/

4

Reduction of Carbon Emission

Renewable Energy Policies and Regulations

Energy Law of the People's Republic of China (Adopted in 2024, effective 2025): Clarifies the state's mandate to establish mechanisms promoting green energy consumption—such as the Renewable Portfolio Standards (RPS) and the Green Electricity Certificate (GEC) system. It serves as the overarching legal basis for recent renewable energy policies.

14th Five-Year Plan for Renewable Energy Development (Issued in 2022, currently under implementation): Sets key targets for 2025, including a non-fossil energy consumption share of approximately 20% and a doubling of renewable electricity generation. It acts as the master plan guiding all subsequent sub-sectoral policies.

Renewable Energy Generation (2025 Total: 3990 TWh)

Total renewable generation reached 3990 TWh in 2025, accounting for 38% of national electricity output—comprising hydro (≈1470 TWh), wind (≈1130 TWh), solar PV (≈1170 TWh), and biomass & refuse incineration power generation (0.2247 TWh).

Power Industry Environmental Issues and Treatment

Key environmental impacts include air pollutant emissions from thermal power plants, soil erosion caused by transmission and substation construction, and ecological disturbance from wind/solar farm land use. Measures taken: full completion of ultra-low emission retrofits for coal-fired power units; eco-optimized design of transmission projects; post-construction vegetation rehabilitation at renewable energy sites; and full-process online pollutant discharge monitoring.

Greenhouse Gas Mitigation Planning

Committed to peaking carbon emissions by 2030 and achieving carbon neutrality by 2060. Key actions include scaling up wind, solar, and energy storage; transmitting renewable power via UHV (ultra-high voltage) lines; advancing CCUS (Carbon Capture, Utilization and Storage) demonstration projects; implementing flexibility retrofits for coal-fired units; and operating the national ETS (Emissions Trading Scheme) to constrain power generators.

Energy Conservation Efficiency Policies

A dual-control system targeting both energy consumption intensity and carbon emission intensity is enforced. Initiatives include energy-saving retrofits for coal-fired power plants, reduction of grid line losses, promotion of high-efficiency electrical equipment, time-of-use (TOU) pricing and demand response programs, as well as special action plans for industrial electricity conservation.

5

Advanced Metering Infrastructure (AMI/Smart Grid Deployment)

Latest AMI Installations and Smart Grid Projects

China has deployed over 530 million smart meters with 98% nationwide coverage. New HPLC IoT smart meters support high-frequency data collection and two-way communication. Supporting smart grid schemes cover digital distribution networks, virtual power plants and distributed PV monitoring.

Challenges Faced and How It Was Solved / Addressed:

Challenges: Inconsistent device protocols, data security risks and aging meters.

Solutions: Adopt HPLC+5G hybrid communication, unify national technical standards, apply domestic encryption and carry out phased meter replacement.

AMI Roll out plan

Short-term: Complete upgrading of urban old-generation meters, raise two-way communication ratio in rural areas to over 90%. **Medium-term:** Finish full IoT AMI renovation nationwide, equip all meters with distributed energy metering modules.

Long-term: Build integrated AMI data cloud platform to coordinate source-grid-load-storage operation, supporting nationwide power spot market settlement.

AMI offerings to customers and its benefit

AMI could provide real-time power check, remote payment, time-of-use pricing and automatic fault alerts. Residents cut bills via off-peak power use; industrial users optimize loads for demand response. Power utilities reduce manual reading costs and grid losses, facilitating renewable energy integration.

6

Latest Technologies or Innovations

In recent years, China's power industry has achieved world-class breakthroughs in multiple core technologies. In the wind power sector, the world's largest 26-megawatt offshore wind turbine has been officially launched, while technologies for deep-sea wind power continue to advance rapidly. In the photovoltaic sector, crystalline-perovskite tandem solar cells have achieved a record-breaking photoelectric conversion efficiency of 34.6%. Over the past decade, the development costs of wind and photovoltaic power have dropped by 60% and 80% respectively, greatly improving economic efficiency. In terms of nuclear power, China possesses the mature and

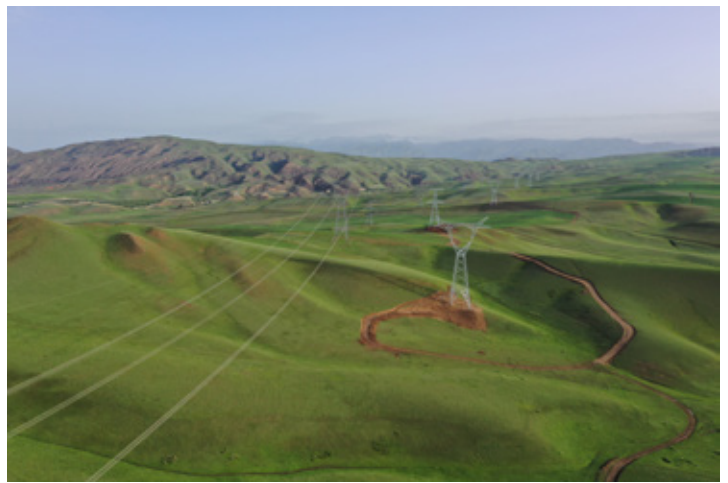
advanced third-generation Hualong One technology, while major breakthroughs have been made in fourth-generation nuclear power technology. New reactor types such as thorium-based molten salt reactors have completed technical verification, delivering significantly enhanced safety and fuel utilization efficiency and leading global research. Furthermore, flexible DC ultra-high-voltage transmission, new energy storage, smart grid systems and ultra-supercritical thermal power technologies work collaboratively to effectively solve renewable energy consumption challenges and support the construction of a new modern power system.

7

Major Issues or Transformation Impacting the Electricity Supply Industry

China's power sector is undergoing profound green transformation while confronting multiple core challenges. First, integrating high shares of renewables poses great difficulties. Wind and solar power are intermittent and volatile, and energy resources are geographically mismatched with power loads. Insufficient peak-shaving capacity and energy storage infrastructure, plus massive power electronic devices lowering grid inertia, bring heavy pressure on grid stability. Second, cybersecurity risks are prominent. Widespread digitalized facilities and smart grids expose

power industrial control systems to cyberattacks, and extreme weather also threatens grid resilience. Third, the power market mechanism needs further refinement, as pricing systems for flexible resources like energy storage and virtual power plants remain incomplete. Meanwhile, the industry accelerates coordinated development of generation, grid, load and storage, ultra-high voltage transmission and flexible power unit retrofits to address transition hurdles and build a secure, reliable new power system.





PHILIPPINES



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Manila
Area:	343,448 km ²
Population:	112.729 million ²⁴ (as of July 2024)
GDP:	497.5 B ²⁵ (as of April 2025)
Currency:	Philippine Pesos (Php)
Installed Capacity (in GW):	29.706 ²⁶ (as of 2024)
Percentage of population electrified:	99.99% Barangay Electrification

Electricity Sales and Consumption (in GWh and %)²⁷

PHILIPPINES	2023	2024	% Inc/(Dec)
Residential	36,968	41,205	11.46
Commercial	26,236	28,033	6.85
Industrial	29,493	31,074	5.36
Others	3,112	3,596	15.55
Electricity Sales	95,808	103,908	8.45
Own-Use	10,403	10,558	1.49
System Loss	11,793	12,475	5.78
Electricity Consumption	118,004	126,941	7.57

²⁴Official results of the 2024 Census of Population (POPCEN) - https://pco.gov.ph/news_releases/pbbm-declares-112-7m-philippine-population-count-as-official/

²⁵International Monetary Fund, https://pco.gov.ph/news_releases/pbbm-declares-112-7m-philippine-population-count-as-official/

²⁶2024 Power Statistics, Department of Energy, https://legacy.doe.gov.ph/sites/default/files/pdf/energy_statistics/02_Summary.pdf

²⁷2024 Power Statistics, https://legacy.doe.gov.ph/sites/default/files/pdf/energy_statistics/02_Summary.pdf

National Energy Policy²⁸

The Philippine government is leading a strategic transition toward a more sustainable, inclusive, and secure energy future. Central to this vision is the deployment of modern, clean energy technologies to drive economic growth, improve community well-being, and enhance energy resilience. With the Department of Energy at the helm, the government is advancing ambitious targets for renewable energy expansion, energy efficiency improvements, and the diversification of energy sources. Building climate-resilient infrastructure and integrating indigenous energy resources, such as solar, wind, geothermal, and hydropower, are key priorities, aligning the country with global efforts toward a low-carbon future.

To support these goals, comprehensive policy reforms, regulatory enhancements, and investment incentives are being implemented to attract private sector participation and accelerate the adoption of clean energy. Parallel efforts focus on upgrading transmission infrastructure to accommodate increased renewable energy capacity while ensuring energy reliability and security. Through these coordinated actions, the Philippines is laying the foundation for a just and sustainable energy transition that supports long-term national development.

(source: PEP 2023-2050 Volume 1, page 1 "Overview", <https://doe.gov.ph/sites/default/files/pdf/pep/PEP%202023-2050%20Vol.%20I.pdf>)



ACCESS TO AFFORDABLE ENERGY

- Achieve 100 percent household electrification by 2028 (based on 2020 PSA Census)
- through the Total Electrification Program
- Interconnection of off-grid areas to the national grid
- Hybridization and DES in off-grid areas
- Access to modern fuels and productive uses of RE for household and other social and economic activities
- Compliance with the Competitive Selection Process (CSP) Policy
- Rationalization and graduation of Universal Charge for Missionary Electrification (UCME) subsidy
- Review energy taxation
- Power of choice and consumer welfare protection

RELIABLE AND RESILIENT ENERGY SUPPLY

- Resiliency and security of energy infrastructure
- Increase energy self-sufficiency level
- LNG as a transition fuel to complement variable RE (VRE)
- Timely completion of energy infrastructure projects
- Smart and Green Grid Plan (SGGP)
- Strict compliance with the Grid Operating and Maintenance Program
- Adequate levels of Ancillary Services

²⁸Philippine Energy Plan 2023 to 2050 <https://doe.gov.ph/pep/philippine-energy-plan-2023-2050-0>

CLEAN, SUSTAINABLE, AND CLIMATE-CENTERED ENERGY RESOURCES

- Increase RE share in the power generation mix up to 35.0 percent by 2030, 50.0 percent by 2040, and more than 50.0 percent by 2050
- Increase port infrastructures to support the development of offshore wind projects
- 4,800 MW of nuclear capacity by 2050
- Strengthen energy efficiency and conservation (EEC) programs and initiatives
- Pursue alternative fuels and emerging technologies: Electric Vehicles (EVs), hydrogen and its derivatives, Energy Storage System (ESS), Small Modular Reactor (SMR)
- Strategic placements of ESS in the power systems including integration of ESS in Variables REs

(source: PEP 2023-2050 Volume 1, page 2 to 5,
<https://doe.gov.ph/sites/default/files/pdf/pep/PEP%202023-2050%20Vol.%20I.pdf>)

Supply and Demand of Primary Natural Resources²⁹

Oil remains the country's main energy source, albeit with a slightly lower share of the energy mix in 2050 at 31.0 percent vis-à-vis 32.2 percent share in 2022. The bulk of oil supply supports the steady demand for oil and oil products, particularly from the transport sector. Coal supply expands slowly by 0.6 percent as its utilization for power generation diminishes across the planning period because of decarbonization targets. LNG imports augment natural gas supply and result in an 8.4 percent increase in levels between

2022 and 2050. The transition towards cleaner fuels, especially in power generation, sustains the 3.3 percent expansion in aggregate RE supply. Wind and solar post double-digit growths of 17.4 percent and 14.1 percent, respectively, while geothermal and hydro also add up to the increase in RE shares from 32.6 percent in 2022 to 35.5 percent by the end of the planning period. On the other hand, biomass supply contracts by 1.2 percent as demand for clean fuel for cooking improves, particularly in the household sector.

Fuel Type	2022		2030		2040		2050		AAGR (%) *
	Levels	% Shares	Levels	% Shares	Levels	% Shares	Levels	% Shares	
Coal	19.08	30.99	22.63	28.40	22.43	20.83	22.28	15.86	0.56
Natural Gas	2.61	4.24	4.92	6.18	10.83	10.06	24.82	17.67	8.37
Oil-based	19.83	32.22	25.58	32.10	34.50	32.05	43.50	30.96	2.84
Renewable	20.04	32.55	26.56	33.33	39.89	37.06	49.89	35.51	3.31
Geothermal	8.96	14.56	12.36	15.51	16.23	15.08	18.30	13.03	2.58
Hydro	2.51	4.08	2.70	3.39	6.65	6.18	10.59	7.54	5.27
Wind	0.09	0.14	1.34	1.69	4.21	3.91	7.98	5.68	17.44
Solar	0.16	0.25	1.54	1.94	4.30	3.99	6.34	4.51	14.13
Biomass	7.73	12.56	7.88	9.89	7.56	7.02	5.60	3.98	-1.15
Biofuels	0.59	0.95	0.73	0.91	0.94	0.87	1.09	0.78	2.24
Total	61.56	100	79.69	100	107.65	100	140.50	100	2.99
Self-Sufficiency (%)	49.4		49.5		41.6		38.9		

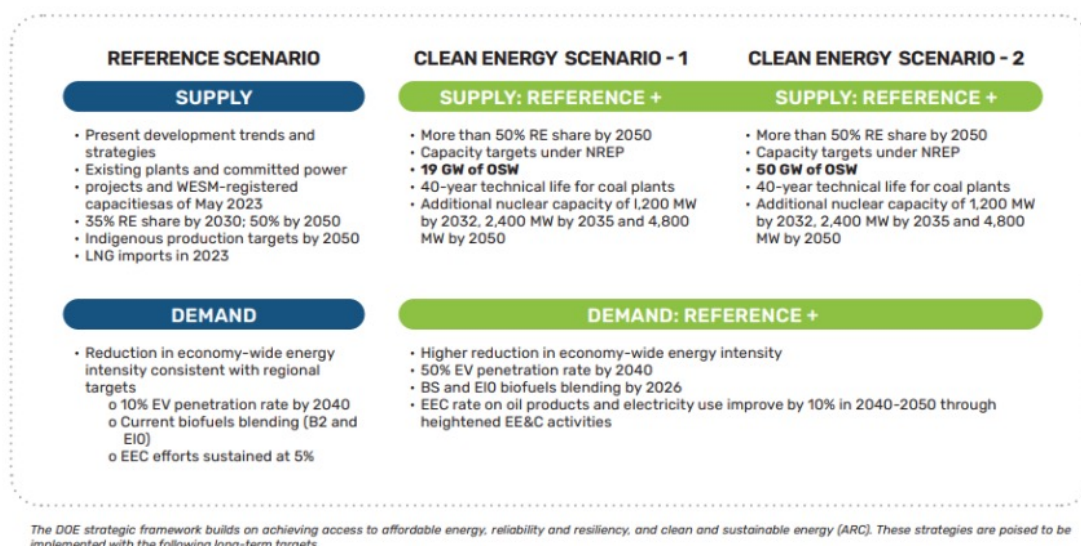
*Average annual growth rates

²⁹ PEP 2023-2050 Volume 1, page 75, <https://doe.gov.ph/sites/default/files/pdf/pep/PEP%202023-2050%20Vol.%20I.pdf>

Electric Energy and Demand Supply Outlook³⁰

The Energy Outlook takes into account the country's growth trajectory as indicated in the PDP 2023-2028 and aligns with both AmBisyon Natin 2040 and President Marcos Jr. Administration's 8-point socio-economic agenda with a focus on protecting the purchasing power of families by reducing energy cost and creating more jobs with indigenous resources vis-à-vis energy security. The country sustained its 7.6 percent expansion in real gross domestic product (GDP) in 2022 with projected peak growth of 8.0 percent by 2028.

Towards the end of the planning horizon, the Philippine economy registers annual increments of 7.1 percent. The Energy Outlook also analyzes two (2) possible energy scenarios for the country's pathway to energy transition – Reference Scenario (REF) and Clean Energy Scenario (CES). The former primarily paints a situation where current energy policies are maintained, while the latter represents the energy sector's aggressive targets pursued within the planning horizon.



Laws and governmental regulations

- **RA 7638 – Department of Energy (DOE) Act of 1992³¹**
This Act mandates the Department to prepare, integrate, coordinate, supervise and control all plans, programs, projects and activities of the Government relative to energy exploration, development, utilization, distribution and conservation.

- **RA 9136 – Electric Power Industry Reform Act of 2001³²**
This Act shall provide a framework for the restructuring of the electric power industry, including the privatization of the assets of NPC, the transition to the desired competitive structure, and the definition of the responsibilities of the various government agencies and private entities.

³⁰PEP 2023-2050 Volume 1, page 11, <https://doe.gov.ph/sites/default/files/pdf/pep/PEP%202023-2050%20Vol.%201.pdf>

³¹Source: Section 2. Declaration of Policy, https://doe.gov.ph/sites/default/files/pdf/ocsp/ra_7638.pdf

³²Source: Section 3. Purpose, <https://www.officialgazette.gov.ph/2001/06/08/republic-act-no-9136/>

- **RA 9513 – Renewable Energy Act of 2008³³**

This Act shall establish the framework for the accelerated development and advancement of renewable energy resources, and the development of a strategic program to increase its utilization.

- **RA 12120 – Philippine Natural Gas Industry Development Act³⁴**

This Act shall provide a framework for the development of the PDNGI, and its transition from an emerging industry into a mature industry within a competitive natural gas market. It shall also define the responsibilities of various government agencies and private entities in furtherance of the national goal.

It shall apply to the permitting of siting, construction, operation and maintenance, expansion, modification, rehabilitation, decommissioning, and abandonment of PDNGI Facilities for Own-Use or Third-Party Access (TPA). It shall likewise apply to the accreditation of Participants in the trade of natural gas, including but not limited to the purchase, supply, aggregation, debunkering, reselling, and export of natural gas, and any other activities related to the PDNGI.

All activities of the Department of Energy (DOE), Energy Regulatory Commission (ERC), and other concerned government agencies relevant to the development and regulation of the PDNGI shall be in accordance with this Act.

- **DC2017-12-0015 - Promulgating the Rules and Guidelines Governing the Establishment of the Renewable Portfolio Standards For On-Grid Areas³⁵**

The RPS On-Grid Rules are adopted to: (a) Mandate the electric power industry participants to source or produce a specified portion of their electricity requirements from eligible RE resources in order to develop indigenous and environmentally friendly energy sources; (b) Establish a minimum annual RPS requirement and minimum annual incremental RE percentage; (c) Prescribe the eligible RE facilities and the compliance mechanisms; (d) Monitor compliance of the mandated industry participants; and (e) Provide the penalties for non-compliance.

- **DC2018-08-0024 - Promulgating the Rules and Guidelines Governing the Establishment of the Renewable Portfolio Standards For Off-Grid Areas³⁶**

This RPS Off-grid Rules is adopted in order to contribute to the growth of the RE industry in the Off-Grid and Missionary Areas by mandating electric power industry participants to source or produce a specified portion of their electricity requirements from eligible RE sources.

The RPS Off-Grid Rules is intended to rationalize the efficient use of the Universal Charge for Missionary Electrification (UC-ME) and improve self-sufficiency in power generation through integration of RE in the supply mix in Off-Grid Areas.

- **DC2023-12-0035 – Prescribing the Policy and General Framework on Expanded Roof-Mounted Solar Program (ERSP) in the Philippines³⁷**

This Circular shall prescribe the rules and guidelines for all roof-mounted solar energy generating facilities with a capacity of above one hundred kilowatt peak (100 kWp), intended for own-use and/or export of energy to its host distribution utility (DU) or to the Grid. The ERSP shall be distinct from the Net-Metering Program, Distributed Energy Resources Program, and Green Energy Option Program, and shall apply for both on-grid and off-grid areas.

- **DC2017-11-0012 – Philippine Downstream Natural Gas Regulation³⁸**

This Circular shall apply to the permitting, securing of location, construction, operation and maintenance, modification, expansion, rehabilitation, decommissioning and abandonment of Downstream Natural Gas Facilities. It shall likewise apply to the permitting of commercial activities in the PDNG Industry Value Chain, including the purchase, Supply, Aggregation, blending, storage, trading, transshipment, sale, import and/or export of Natural Gas and/or Liquefied Natural Gas (LNG), and any other activities related to the PDNG Industry.

The State shall ensure the development of the PDNG Industry and its transition from emerging into mature industry within a competitive Natural Gas market to gain greater energy security and sustainability.

³³Source: Section 3. Purpose, <https://www.officialgazette.gov.ph/2008/12/16/republic-act-no-9513/>

³⁴Source: Section 3. Purpose, <https://www.officialgazette.gov.ph/2025/01/08/republic-act-no-12120/>

³⁵Source: Section 2. Purpose, <https://doe.gov.ph/laws-and-issuances/department-circular-no-dc2017-12-0015>

³⁶Source: Section 2. Purpose, <https://doe.gov.ph/laws-and-issuances/department-circular-no-dc2018-08-0024>

³⁷Source: Section 3. Coverage, <https://doe.gov.ph/laws-and-issuances/department-circular-no-dc2023-12-0035-0>

³⁸Source: Section 2. Scope, <https://doe.gov.ph/laws-and-issuances/department-circular-no-dc2017-11-0012>

- **DC2021-11-0036 – Providing the Revised Guidelines for the Green Energy Auction Program in the Philippines³⁹**

This GEAP Guidelines are promulgated to: (a) establish the implementation framework for the GEAP and provide clarity on the roles of implementing agencies/entities; (b) Support and/or facilitate immediate and timely investments in new or additional RE capacities to ensure provision of adequate supply and competitive rates of electricity in the country; (c) Support the development of new RE projects under a competitive process, together with long-term contracts; and (d) Implement the mandate of giving preference to RE sources for a cleaner and sustainable environment.

- **DC2026-01-0002 – Promulgating the Revised Rules and Guidelines Governing the Implementation of the Green Energy Option Program (GEOP)⁴⁰**

This Circular shall harmonize existing rules and set transparent guidelines to address current challenges of the GEOP and expand the coverage of the GEOP to enable broader participation of End-Users, RE Suppliers, and other industry players, consistent with the objectives of the RE Act to promote greater utilization and broader access to renewable energy resources.

- **DC2026-02-0005 – Prescribing the Policy for Categorization of Energy Resources and Technologies in the Energy Mix to Enhance Sufficiency of Generation, Adequacy and Reliability of Supply for Electric Power⁴¹**

To promote a balanced dispatch among various generation technologies, the DOE found it necessary to issue guidance on classifying resources according to baseload, mid-merit, and peaking requirements, taking into account the actual load profile (LP), load duration curve (LDC), and other relevant parameters. This categorization is further intended to enable generation companies to operate more efficiently and to ensure the proper maintenance of generating units.

Total amount of energy consumption by sectors (Industrial / Commercial / Residential) (in GWh and %)⁴²

PHILIPPINES	2024	% Share
Residential	41,205	32.46
Commercial	28,033	22.08
Industrial	31,074	24.48
Others	3,596	2.83
Electricity Sales	103,908	81.86
Own-Use	10,558	8.32
System Loss	12,475	9.83
Electricity Consumption	126,941	100.00

³⁹Source: Section 2.1. Purposes, <https://doe.gov.ph/laws-and-issuances/department-circular-no-dc2021-11-0036>

⁴⁰Source: Section 2. Purpose, <https://prod-cms.doe.gov.ph/documents/d/guest/geop-dc2026-01-0002-pdf>

⁴¹Source: <https://prod-cms.doe.gov.ph/documents/d/guest/dc2026-02-0008-pdf>

⁴²2024 Power Statistics, https://prod-cms.doe.gov.ph/documents/d/guest/02_summary-pdf

3

Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (coal, LNG, Gas, Nuclear-fired and hydro) including total installed capacity of generation and total production by fuel types

	Installed Capacity ⁴³ (GW) as of end of 2024 (GW)	Year-on-year Growth (%) 2024 vs 2023		Generation ⁴⁴ (TWh) as of end of 2024	Year-on-year Growth (%)
Total	29.71	5.02	Total	126.94	7.57
Hydropower	3.84	1.05	Hydropower	10.91	6.05
Thermal Power	20.19	0.60	Thermal Power	98.75	7.66
Wind Power	0.43	-	Wind Power	1.24	-5.28
Solar Power	2.71	64.24	Solar Power	3.81	49.80
Other	2.55	0.39	Other	12.24	0.79

4

Reduction of Carbon Emission

Policy on Renewable Energy and Major Regulatory Mechanism of the Renewable Energy Act

- **Republic Act No. 9513⁴⁵, "Renewable Energy Act of 2008"** declares it the policy of the State to: a) Accelerate the exploration and development of renewable energy resources such as, but not limited to, biomass, solar, wind, hydro, geothermal and ocean energy sources, including hybrid systems, to achieve energy self-reliance, through the adoption of sustainable energy development strategies to reduce the country's dependence on fossil fuels and thereby minimize the country's exposure to price fluctuations in the international markets, the effects of which spiral down to almost all sectors of the economy; (b) Increase the utilization of renewable energy by institutionalizing the development of

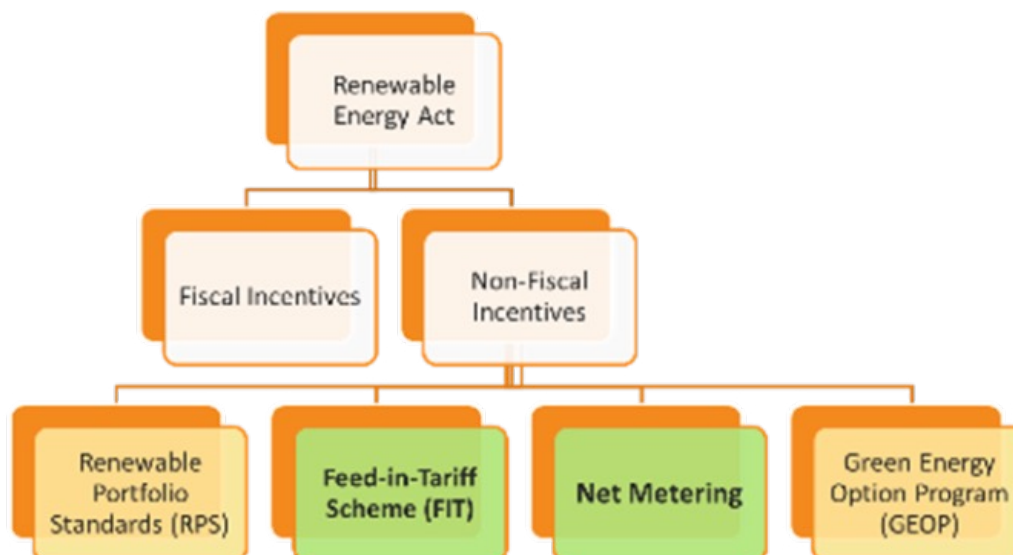
national and local capabilities in the use of renewable energy systems, and promoting its efficient and cost-effective commercial application by providing fiscal and nonfiscal incentives; (c) Encourage the development and utilization of renewable energy resources as tools to effectively prevent or reduce harmful emissions and thereby balance the goals of economic growth and development with the protection of health and the environment; and (d) Establish the necessary infrastructure and mechanism to carry out the mandates specified in this Act and other existing laws.

⁴³2024 Power Statistics, https://prod-cms.doe.gov.ph/documents/d/guest/02_summary-pdf

⁴⁴2024 Power Statistics, https://prod-cms.doe.gov.ph/documents/d/guest/02_summary-pdf

⁴⁵<https://www.officialgazette.gov.ph/2008/12/16/republic-act-no-9513/>

Coverage of RA 9513, or the RE Act of the Philippines



The RE Law provides for four (4) major non-fiscal policy mechanisms: Renewable Portfolio Standards (RPS), Feed-in-Tariff (FIT) Scheme, Net Metering Program, Green Energy Option Program (GEOP).

The following mechanisms of RA 9513 have so far been implemented:

- **Net Metering.** In July 2013, the ERC issued the Net Metering Rules to govern the implementation of the Net Metering Program in the country. Under the Net Metering Program, customers of Distribution Utilities are allowed to install on-site renewable energy facilities not exceeding 100 kilowatts (kW) in capacity so that they can generate electricity for their own use and export excess generation to the host DU.
- **Renewable Portfolio Standards (RPS).** RPS is a policy mechanism that requires electricity suppliers to source a portion of their energy supply from eligible RE resources. The DOE issued Department Circulars promulgating the RPS Rules for On-Grid Areas and the RPS Rules for Off-Grid Areas, on 22 December 2017 and 24 August 2018, respectively.
- **Renewable Energy Market (REM).** As part of the implementation of the RPS, the DOE issued Department Circular No. 2019-12-0016, "Promulgating the Renewable Energy Market Rules (REM Rules)" in December 2019. The REM is a market for the trading of Renewable Energy Certificates (RECs) pursuant to Section 8 of the RE Act. REC refers to a certificate issued for every 1MWh generated from eligible RE facilities. RECs may be traded in the REM in complying with the RPS.
- **Green Energy Auction Program (GEAP).** On 14 July 2020, the DOE promulgated Department Circular No. 2020-07-0017, "Promulgating the Guidelines Governing the Policy for the Conduct of Green Energy Auction in the Philippines". The GEAP provides an avenue for the RPS Mandated Participants to meet their RPS requirements. Below are the summaries for GEA-1, GEA-2 and GEA-4 winning bidders and the approved GEAR rate for each technology type.

Winning Bidders for the GEA-1⁴⁶

Technology Type	Capacity (MW)			Approved Rate (PhP/kWh)
	Luzon	Visayas	Mindanao	
Hydro	80.00	–	19.15	5.4913
Solar	1,070.38	300.00	120.00	3.6779
Wind	360.80	13.20	–	6.0584
Biomass	–	–	3.40	5.0797

Winning Bidders for the GEA-2⁴⁷

Technology Type	Capacity (MW)			Approved Rate (PhP/kWh)
	Luzon	Visayas	Mindanao	
Ground-mounted Solar	1,657.98	173.00	48.00	4.4043
Roof-mounted Solar	458.51	0.25	0.18	4.8738
Floating Solar	90.00	–	–	5.3948
On-Shore Wind	960.38	502.00	–	5.8481

Winning Bidders for the GEA-4⁴⁸

Technology Type	Capacity (MW)			Approved Rate (PhP/kWh)
	Luzon	Visayas	Mindanao	
Ground-mounted Solar	2,629.43	1,214.65	335	4.4832
Roof-mounted Solar	24.82	–	–	5.6800
Floating Solar	2,150.00	134.00	–	6.5258
On-Shore Wind	1801.29	566.00	151.00	6.0859
Solar +BESS (IRESS)	715.00	307.51	166.78	5.4028

- **Green Energy Option Program (GEOP).** The GEOP will empower end-users to choose renewable energy in meeting their energy requirements. On 18 July 2018, the DOE promulgated the “Rules and Guidelines Governing the Establishment of the Green Energy Option Program (GEOP) in the Philippines”. Related regulations and a regulatory framework are currently being drafted and will become the basis for implementing GEOP.

⁴⁶<https://doe.gov.ph/geap/notice-award-list-winning-bidders-gea-1>

⁴⁷<https://doe.gov.ph/geap/notice-award-list-winning-bidders-gea-2>

⁴⁸<https://doe.gov.ph/articles/3141087--gea-4-notice-of-award?title=GEA-4%3A%20Notice%20of%20Award>

Production of Renewable Energy (GWh) from various resources (wind, solar, biomass, landfill gas, etc)

Total 2024 Philippine RE Power Generation by Source in GWh⁴⁹

Energy Source	Capacity in GWh	% Share
RE	28,193	22.21
Geothermal	10,789	38.27
Hydro	10,909	38.69
Biomass	1,446	5.13
Solar	3,811	13.52
Wind	1,239	4.39

Plans and Strategies to Reduce Greenhouse Gas Emissions

of the enabling legislations and government initiatives contributing to the country's programs towards reducing greenhouse gas emissions include:

I. Republic Act No. 8749, "Philippine Clean Air Act of 1999". The Clean Air Act aims to provide a comprehensive pollution control policy for the country. It sets fines and penalties as much as P100,000 per day for violating operators of facilities and as much as P6,000 and suspension of registration for violating motor vehicle owners.

II. Executive Order No. 320, Series of 2004. The EO designates the Department of Environment and Natural Resources (DENR) as the National Authority for Clean Development Mechanisms. Projects to be implemented under CDMs are aimed at preventing or absorbing emitted GHGs.

III. Republic Act No. 9729, "Climate Change Act of 2009". Enacted in July 2009, it aims to mainstream climate change into government policy formulations, establish the framework strategy and program on climate change that led to the creation of the Climate Change Commission (CCC). The CCC is the sole policy-making body responsible for coordinating, monitoring and evaluating government programs toward mitigating climate change.

IV. Executive Order No. 174, Series of 2014. This institutionalizes the Philippine Greenhouse Gas inventory management and reporting system. This aims to enable the country's transition towards a climate-resilient road to sustainable development.

V. National Framework Strategy on Climate Change (NFSCC)

The NFSCC envisions a "Climate-resilient Philippines" with healthy, safe, prosperous and self-reliant communities, and thriving and productive ecosystems". The NFSCC has two (2) main pillars of action, adaptation and mitigation.

Under the **Mitigation Pillar**, the Climate Change Commission (CCC) cites the following guiding strategies to reduce GHG emissions:

- **Energy Efficiency and Conservation:** Develop and enhance clean energy sources, utilization, and other efficiency measures towards a low carbon economy in the energy sector.
- **Renewable Energy:** Realize the full potential of the country's renewable energy capacity so as to further contribute to energy security and promote low-carbon growth in the energy sector.
- **Environmentally Sustainable Transport:** Improve the efficiency of the transport sector through increased uptake of alternative fuels and expansion of mass transport systems.
- **Sustainable Infrastructure:** Reduce carbon footprint through energy-efficient design and materials for public infrastructure and settlements.
- **National REDD + Strategy:** Reduce emissions from deforestation and forest degradation (REDD) through the sustainable management of forests and the protection and enhancement of carbon stocks in watersheds, forests and other terrestrial ecosystems.
- **Waste Management:** Full implementation of proper waste management.

vi. Republic Act (RA) No. 9367, "Biofuels Act of 2006", declares it the policy of the State to reduce dependence on imported fuels with due regard to the protection of public health, the environment and the natural ecosystems consistent with the country's sustainable economic growth that would expand opportunities for livelihood by mandating the use of biofuels.

⁴⁹2024 Power Statistics, https://prod-cms.doe.gov.ph/documents/d/guest/02_summary-pdf

Policy on Energy Conservation and Efficiency

It is a declared policy of the government to promote the judicious conservation and efficient utilization of energy resources through adoption of the cost-effective options toward the efficient use of energy to minimize environmental impact.

“Republic Act No. 11285, “Energy Efficiency and Conservation Act”⁵⁰

With the signing into law of Republic Act No. 11285 on 12 April 2019, the government recognizes the need to “institutionalize energy efficiency and conservation as a national way of life geared towards the efficient and judicious utilization of energy by formulating, developing, and implementing energy efficiency and conservation plans and programs.”

The law strengthens implementing institutions by clarifying roles of government agencies both at the national and local levels. The DOE, as the lead agency in the implementation of the law, is tasked to develop and monitor the National Energy Efficiency and Conservation Plan (NEECP); develop and maintain the National Energy Efficiency and Conservation Database (NEECD); ensure compliance with the Government Energy Management Program (GEMP); develop, impose, review, and implement the Minimum Energy Performance Standards (MEPS), among others.

The law also defines obligations of designated establishments, and expands the pool of certified energy conservation officers (CECO) and energy managers (CEM).

Philippine Energy Efficiency and Conservation Roadmap 2020–2040

Energy efficiency, sometimes referred to as the fifth fuel as reduced demand means a decrease in additional supply requirements, provides the most cost-effective and environment-friendly fuel to help build greener and more vibrant economies. To reach the country's EEC goals, the DOE will not only sustain but accelerate efforts and initiatives to bring in real and positive changes in the lives of energy consumers and stakeholders. The EEC Roadmap was designed to be a working document, serving as a reference tool for the formulation and implementation of policies and programs, technology and standards development, and information/promotional activities.

The objective of this Roadmap is to build on the collective priorities and strengths of the country's EEC goal of reducing energy intensity and consumption, as well as achieve the Clean Energy Scenario (CES) of the Plan Update. It identifies a variety of paths to help all energy consumers and stakeholders seize the EEC opportunities available in the country.



⁵⁰<https://doe.gov.ph/laws-and-issuances/republic-act-no-11285>

MERALCO's Smart Grid Program is built on two foundational pillars: Advanced Network Automation (ANA) and Advanced Metering Infrastructure (AMI). Together, these technologies are designed to transform the electric distribution network into a resilient, intelligent, integrated, and highly efficient system. This transformation is driven by a commitment to deliver enhanced reliability, operational excellence, renewable energy integration and superior service—all for the benefit of MERALCO's customers.

a. Advanced Network Automation (ANA)

The first pillar, Advanced Network Automation (ANA), focuses on modernizing the grid through the integration of three core components: smart field devices, Operational Technology (OT) Systems, and a robust communication network. These elements work together to enable real-time monitoring, intelligent control, and optimization of network operations, assets and other distributed energy resources.

To strengthen visibility and control across its distribution network, MERALCO has deployed over 3,000 smart field devices—including remote-controlled reclosers, capacitors, and voltage regulators. These devices support rapid fault isolation, faster service restoration, and improved system reliability. In parallel, MERALCO continues to roll out Smart Substations compliant with the IEC 61850 communication standard, enabling more efficient and integrated grid operations. The company is also exploring the use of asset condition sensors and environmental intelligence sensors to monitor critical parameters and other conditions that may affect equipment performance and grid stability.

Supporting these field devices are MERALCO's Operational Technology Systems, which serve as the brain of the smart grid. The ongoing implementation of Advanced Distribution Management System (ADMS) 2.0 will unify SCADA, Energy Management, Outage Management, and Distribution Management into a single platform. ADMS 2.0 will enable advanced grid functionalities such as Fault Location, Isolation, and Service Restoration (FLISR), Integrated Volt/VAR Optimization (IVVO), and DER-aware operations with enhanced cybersecurity. To complement this, MERALCO will be deploying a Distributed Energy Resources Management System (DERMS) to effectively integrate, forecast, coordinate, and optimize DERs—ensuring grid reliability, efficiency and empowering consumers.

With the roll out of AMI, increasing penetration of distributed energy resources and the widespread adoption of Electric Vehicles or (EV), MERALCO will be implementing a low voltage Operational Technology system which we will call the "Grid Edge Operation Control Center" or GEOCC. This system will complement ADMS 2.0 to monitor, control and optimize the low voltage system focusing on edge devices like smart meters, roof top solar, and household electric vehicle charging pods. This is important to manage millions of smart meters, increase renewable energy penetration on the low voltage distribution level, ensure power quality, and prevent constraints.

To further enhance operational intelligence, MERALCO has implemented key systems such as Enterprise Asset Management (EAM) for asset and work order tracking, Enterprise Geographic Information System (EGIS) for network topology and connectivity, and a Data Historian for time-series data collection and visualization. Leveraging this data, MERALCO will soon deploy an Asset Performance Management (APM) system to predict asset health and risks, enabling proactive maintenance and extending asset life. Additionally, a Protection Management System (PMS) will be introduced to remotely manage substation-based protective equipment. MERALCO has also launched its Mobile Workforce Management System (MWMS) to streamline field operations through automation and cloud-based tools—enhancing job dispatch, optimizing routes, and resource allocation, and real-time navigation.

In alignment with global energy transition goals, MERALCO has energized its Cagbalete Island Microgrid, a pioneering initiative and benchmark for Distribution Utility (DU) Operated Microgrids. It is powered by Renewable Energy (RE) and Battery Energy Storage Systems (BESS) with back-up gensets to provide electricity to unserved and underserved communities. Building on this milestone, MERALCO now moves forward with the expansion of its Isla Verde Microgrid. These microgrids aim to provide 24/7 access to reliable and affordable power, fostering inclusive growth. In addition, MERALCO plans to deploy innovative BESS projects across strategic locations, including grid-scale, mobile, and pole-mounted solutions, to support renewable integration and ensure a stable, flexible power supply.

A critical enabler of ANA is the communications infrastructure, which facilitates the two-way flow of data and control signals between field devices and back-end systems. A fast, secure, and reliable communication network is essential for real-time grid operations and the success of the Smart Grid. MERALCO is implementing its dedicated wireless private network coupled by its extensive fiber network and Microwave systems to support various smart grid initiatives.

In essence, Advanced Network Automation (ANA) marks a transformative leap toward building a smarter, more resilient, and sustainable power grid. Through real-time monitoring, intelligent control, and rapid response capabilities, ANA not only enhances system reliability and operational efficiency but also supports the seamless integration of renewable energy and electric mobility for a cleaner future.

b. Advanced Metering Infrastructure (AMI)

The second pillar of the MERALCO Smart Grid Program, Advanced Metering Infrastructure (AMI), is designed to empower end-users through an integrated system of smart meters, communication networks, and back-end platforms that enable two-way communication between MERALCO and its customer meters. This infrastructure allows for real-time data exchange, enhancing transparency, operational efficiency, and customer engagement.

MERALCO began its AMI journey in 2014 with the launch of its prepaid electricity service branded as “Kuryente Load” (KLoad)—with “Kuryente” meaning electricity—piloted in Angono, Rizal.

Since then, MERALCO has progressively expanded the use of AMI-capable meters across various use cases including AMI postpaid services, elevated metering transformation (EMX), compliance with the Customer Choice Program (CCP) and Distribution Transformer Monitoring (DTME). To date, MERALCO has deployed a total of 196,128 smart meters across these use cases, primarily in the most densely populated municipalities within its franchise area, including Manila, Makati, Mandaluyong, and Pasig. Additional installations have been made in San Juan, Quezon City, Parañaque, Novaliches, Caloocan, and several towns in Rizal (Taytay, Cainta, Binangonan), Cavite, and Bulacan.

To maximize both customer and utility benefits,

MERALCO's AMI deployment strategy is focused on rapid and efficient scaling. The rollout will follow a phased approach aligned with the regulatory cycle, ensuring compliance while progressively expanding coverage and functionality.

i. AMI Roll-out Plan

MERALCO's ultimate goal is to deploy Advanced Metering Infrastructure (AMI) across its entire franchise area, ensuring that all existing and new customer premises are equipped with smart meters to maximize both utility and customer benefits. Over the next four years, MERALCO plans to install approximately 3.41 million smart meters. The prioritization of use cases for this initial deployment is guided by several key considerations. First, it supports compliance with Department of Energy (DOE) and Energy Regulatory Commission (ERC) programs that promote customer choice, such as Retail Competition and Open Access (RCOA), Green Energy Option Program (GEOP), and Retail Aggregation (RA). Second, it enhances operational efficiency through solutions like Elevated Metering Centers (EMC), distribution transformer monitoring, and improved access to meters in physically hard-to-reach locations. Third, deployment efficiency is addressed by supporting customer load growth and enabling proactive meter replacement, particularly for entire meter reading routes. Lastly, service enhancements are targeted through support for net metering and time-of-use or peak/off-peak rate structures. The deployment will begin with RCOA, GEOP, and RA customers in line with DOE and ERC regulations, followed by new customers to allow them to immediately benefit from AMI. After the initial four-year rollout, MERALCO will continue expanding its AMI deployment until all customers within the franchise area are equipped with smart meters by 2035.

ii. AMI Offerings to Customers and its Benefits

In accordance with the Philippines' Energy Regulatory Commission's (ERC) Amended AMI Rules, MERALCO's Advanced Metering Infrastructure (AMI) will deliver essential services such as remote meter reading, remote connection and reconnection, remote disconnection, consumption monitoring, outage detection, meter irregularity detection (MID) and Meter Installation Inquiry. To support these services, customers will be provided with mechanisms to receive notifications and alerts through various channels including web portals, email, SMS, mobile applications,

and other platforms that will be introduced over time. Beyond these basic functionalities, MERALCO also plans to expand the use of smart meters and AMI to offer supplemental services. These include advanced outage management, power quality management, demand response, net metering and distributed energy resource (DER) management, prepaid retail electricity service, and smart street lighting solutions.

iii. Utility Benefits

Advanced Metering Infrastructure (AMI) enhances utility operations through its remote capabilities, enabling the avoidance of manual field activities and resulting in cost reductions, improved productivity, and new revenue opportunities. In terms of cost reduction and avoidance, AMI minimizes labor, transportation, and material expenses associated with field tasks such as meter reading (e.g., monthly energy, off-cycle, load profile), out-of-schedule bill deliveries due to unread or delayed readings, and disconnection or reconnection services. Additionally, AMI's proactive approach to installing and replacing meters and distribution transformers (DTs) helps avoid capital expenditures, as smart meters replace regular ones and DT failures are reduced.

Utility productivity also benefits from AMI, as remote meter reading and outage detection streamline operations. Manual reading errors are eliminated, reducing the time spent resolving billing discrepancies and handling related complaints. With last gasp detection, outages can be pinpointed down to the household level, accelerating restoration times and decreasing outage-related customer calls.

Moreover, AMI opens opportunities for additional savings and revenue. Operational efficiencies such as remote reconnection and outage detection shorten service restoration times, easing customer

inconvenience. AMI meters also support Distribution Transformer Monitoring, enhancing asset availability and enabling proactive resolution of transformer issues before they escalate into failures requiring extended downtime. Finally, AMI helps reduce the number of unbilled customers and minimizes billing adjustments caused by manual meter reading errors, further contributing to MERALCO's operational and financial performance.

iv. Customer Benefits

Advanced Metering Infrastructure (AMI) offers significant benefits to customers. With its ability to retrieve meter data at frequent intervals (e.g., at 15-minute or 5-minute intervals), customers gain access to more detailed consumption information. This data empowers them to monitor and manage their energy usage more effectively. In fact, MERALCO's prepaid customers have reported savings of 13% to 15% compared to their previous postpaid service, highlighting AMI's potential to reduce electricity bills. Beyond energy savings, AMI also helps customers avoid expenses related to service inquiries, such as phone calls or trips to business centers, by providing real-time access to meter information, including outage status. These enhancements in MERALCO's operations and the availability of consumption data contribute to an improved customer experience, minimizing service disruptions and complaints. Additionally, AMI supports efforts to reduce non-technical system loss. Through its capability to detect physical tampering and analyze consumption patterns, MERALCO can promptly identify and address violations, thereby lowering the amount of system loss that would otherwise be recovered through customer billing.

The rapid advancement in energy technology and rise of digitalization has fueled the transformation of the global energy industry – shifting from a centralized one-way power flow to a decentralized two-way power system capable of supporting both the flow of power and data. Adoption of these new technologies in the Philippine setting will enable the country to achieve a more secure, efficient and sustainable energy future. a. Developments in Generation & Distribution Technology

a. Development in Generation & Distribution Technology

The Philippine energy sector is undergoing a profound transformation, driven by rapid technological advancements and the accelerating pace of digitalization. This shift marks a departure from traditional, centralized power systems toward decentralized, intelligent networks capable of supporting two-way flows of electricity and data. By embracing cutting-edge innovations in generation, distribution, and information technologies, the country is laying the foundation for a more secure, efficient, and sustainable energy future. The following sections highlight the latest technologies and strategic initiatives deployed by Philippine utilities to modernize infrastructure, enhance customer experience, and future-proof operations.

i. Supercritical Generation Technology

Supercritical generation plants are gradually becoming the standard for coal-fired power facilities in the Philippines due to their superior efficiency and reduced emissions of greenhouse gases, pollutants, and particulate matter. Compared to older coal plants that operate at around 33% efficiency, supercritical (SC) and ultra-supercritical (USC) technologies can achieve approximately 46–47% and 50% cycle efficiency (LHV basis), respectively, resulting in a 20% (SC) and 30% (USC) reduction in CO₂ emissions for the same megawatt output. Several generation companies in the country have already invested in these advanced technologies, including San Buenaventura Power Ltd. Co. (SBPL) with its 455 MW plant in Mauban, Quezon, which began commercial operations on 26 September 2019; Masinloc Power Co. Ltd. (MPCL) with its expansion units—Unit 3 (300 MW), Unit 4 (350 MW), and Unit 5 (350 MW)—at the Masinloc Power Station in Zambales, with Unit 3 operational since 17 October 2020; and GN Power Dinginin (GNPD), which operates two 668 MW supercritical units using

Alstom steam turbine generators, with Unit 1 and Unit 2 commencing operations on 2 December 2021 and 11 October 2022, respectively.

ii. Liquefied Natural Gas (LNG) Regasification Technology

Liquefied natural gas (LNG) is emerging as the primary fuel for natural gas-fired power facilities in the Philippines, driven by the declining output from the Malampaya gas field. Existing Malampaya-based plants have begun transitioning to LNG as their main fuel source. Unlike indigenous natural gas, LNG is stored in liquid form, making it easier to transport and store. Environmentally, LNG offers significant advantages, emitting approximately 40% less carbon dioxide than coal and 30% less than oil. In response to this shift, several new generation companies in the Philippines are investing in LNG regasification technology. Excellent Energy Resources Inc. (EERI) is developing a facility in Batangas City with two phases: Phase 1 includes three 440 MW units begun commercial operations in the first half of 2025, followed by a single 440 MW unit in Phase 2. Energy World Corporation is building a 650 MW gas-fired combined cycle power plant in Pagbilao, Quezon.

iii. Solar Electricity Technology

Driven by a supportive regulatory environment, solar power generation in the Philippines has grown significantly in recent years. In 2024 alone, the country added 794.34 megawatts (MW) of new renewable energy capacity, exceeding the combined 759.82 MW installed in 2021 (230.10 MW), 2022 (328.18 MW), and 2023 (201.54 MW). At the household level, the Net Metering Program has further encouraged solar adoption. As of December 2025, DOE data showed 17,141 qualified end-users in the Luzon grid with a combined rated capacity of about 151 MWp, accounting for 73.43% of total installations. The Visayas grid recorded 3,179 qualified end-users with around 44 MWp (21.21%), while Mindanao had 904 qualified end-users with 11 MWp (5.36%). Looking ahead, the implementation of the Renewable Portfolio Standards (RPS) and the Green Energy Option Program (GEOP), along with declining solar technology costs, are expected to drive continued growth. As the global oil market remains impacted by developments in the Middle East, the Department of Energy (DOE) accelerates the grid entry of around 1,471 megawatts (MW) of committed new power capacity from twenty-

two (22) power projects targeted for operation by April 2026. These incoming capacities will come from renewable energy sources, with solar leading the way at twelve (12) solar projects accounting for approximately 1,284 MW. This expansion supports the country's goal of increasing the share of renewables in the energy mix to 35% by 2030, up from the current 25% as of 2025. Additionally, as of March 2025, the DOE projects 19,190 MW of committed capacity from 2025 to 2030, with 5,998 MW expected in 2026 and additional projects through 2030.

iv. Energy Storage System (ESS) Technology

Battery technology is becoming increasingly integrated into power networks worldwide, driven by improvements in reliability, cost efficiency, and terminal value. Batteries now play a vital role in helping utilities manage intermittent renewable energy sources such as solar and wind by storing excess energy and supporting grid stability. In the Philippines, investment in battery energy storage systems (BESS) is largely motivated by the need to enhance grid reliability through fast-response ancillary services. A landmark development in this space is the MTerra Solar Project, launched in November 2024 by Terra Solar Philippines, Inc. (TSPI), MERALCO PowerGen Corporation (MGEN), and SP New Energy Corporation (SPNEC). Located across Nueva Ecija and Bulacan, this project is set to become the world's largest integrated solar and battery storage facility, featuring 3,500 MWp of photovoltaic capacity and 4,500 MWhr of BESS. In March 2026, the MTerra Solar Project was successfully energized as a generator – marking the moment the facility went live and began delivering electricity to the grid. This phase is set to supply up to 85-megawatts (MW) of mid-merit power to the grid, supporting energy demand during peak daytime hours and marking the initial delivery to MERALCO. Following its energization as a generator, the project will continue to ramp up its output in the coming months in close coordination with the National Grid Corporation of the Philippines (NGCP), in line with its System Impact Study (SIS) and operational plan.

According to the Department of Energy (DOE), Luzon is expected to generate 230 MW of BESS and energy storage systems (ESS), while Mindanao will contribute 100 MW by 2025. The country anticipates nearly 3,800 MW of installed BESS capacity by 2033, with 1,954 MW of committed projects by 2030. Recognizing the strategic importance of ESS, the DOE released a policy

framework on August 1, 2019, to regulate and facilitate its integration into the power sector. This framework was further enhanced on April 20, 2023, to support the growing need for ESS in stabilizing the grid amid rising variable renewable energy sources.

v. Microgrid Technology

As advancements in solar and battery technologies continue, interest in microgrids is gaining momentum in the Philippines. A microgrid is a small-scale power system capable of delivering stable electricity to off-grid areas by integrating distributed energy sources—such as diesel generators, solar, and wind—with energy storage systems like batteries. The Microgrid Systems Act (RA11646), signed into law in January 2022 and followed by its implementing rules and regulations revised in May 2025, promotes the deployment of microgrid systems in unserved and underserved areas, enabling qualified entities to provide integrated generation and distribution services with a focus on low-cost and renewable energy. In support of this, the Department of Energy (DOE) released the 2024–2033 National Total Electrification Roadmap (NTER) in June 2025, outlining the selection of Microgrid System Providers (MGSPs) to serve designated areas under Microgrid Service Contracts (MSCs) with access to government subsidies. In April 2024, an MGSP was awarded for the development of microgrid systems in eight unserved areas across Cebu, Quezon, and Palawan. MERALCO successfully energized its first microgrid, the Cagbalete Microgrid in Mauban, Quezon under RA11646 last December 2025. It is also actively supporting microgrid development to electrify another island in Isla Verde, Batangas. Both these service areas are utilizing a combination of solar PV, energy storage systems, and diesel generators to ensure 24/7 power availability. These microgrids will operate under a 20-year contract with a guaranteed renewable energy mix of at least 50%.

b. Developments in Information and Communication Technology

Utilities are increasingly navigating both challenges and opportunities brought about by the rapid evolution of information and communication technology. The emergence of the third IT platform—comprising social, mobile, cloud, and big data—has disrupted traditional industries and fueled the rise of tech-driven companies like Uber, Airbnb, Amazon, and Facebook, while contributing to the decline of conventional

businesses such as banks, retailers, and oil firms. This digital revolution has also impacted the power sector, prompting major players in the Philippines to craft strategic roadmaps that leverage these technologies. Looking ahead, the next wave of disruption—driven by the proliferation of IoT devices and sensors, artificial intelligence, and quantum computing—is poised to further transform the energy landscape, with key industry leaders already exploring these innovations to unlock their full potential.

i. Transforming Customer Experience

Digitalization has reshaped global business culture, compelling utilities to rethink customer engagement strategies in response to rising expectations for transparency, convenience, and multi-channel access. In the Philippines, energy companies are modernizing their customer interaction platforms, moving beyond traditional contact centers and social media to adopt integrated customer engagement systems powered by mobile apps and big data analytics. MERALCO, for instance, launched the My MERALCO app in 2018 to empower customers with 24/7 account access, bill payment, consumption tracking, and outage reporting. Complementing this, MERALCO introduced a chatbot for common inquiries and upgraded its billing, CRM, and analytics platforms to enhance customer experience. Under its Customer Centricity Transformation Program + Digital (CCTP+D), MERALCO accelerated innovation, launching high-impact solutions such as the Certified by MERALCO (CBM) platform, which connects customers with certified electrical contractors. Since its October 2022 launch, CBM has significantly reduced service application processing time from 28 to 13 days and lowered rework rates, with over 61,000 applications processed in 2023 and 94 contractors onboarded—demonstrating strong customer adoption and MERALCO's commitment to digital transformation.

ii. Transforming the Distribution Network

Modern technologies and solutions are revolutionizing grid operations and management, with Philippine utilities actively adopting innovations such as Distribution Automation (DA) devices, Advanced Distribution Management System (ADMS) with integrated SCADA, Advanced Metering Infrastructure (AMI), and Enterprise Geographic Information System (EGIS) to enhance network visibility and support renewable energy integration. The Department of

Energy (DOE) has laid out the Smart Distribution Utility Roadmap (SDUR) to guide utilities toward a smart future, recommending gradual adoption of advanced technologies like Demand Response, Distributed Energy Resource Management System (DERMS), and EV Management. Supporting these efforts, MERALCO is upgrading its field communication system by replacing legacy narrowband radios with a Wireless Private Network (WPN), enabling real-time data access and improving operational responsiveness. MERALCO is also embracing mobility solutions for field crews using its Mobile Work Management System (MWMS) and exploring drone technology for safer and more efficient line inspections and maintenance. Artificial Intelligence (AI) is being tested for vegetation management, asset monitoring, and fault prediction to boost service reliability. MERALCO's AMI initiative aims to improve billing services, customer experience, and outage management through smart meters and last-mile communication technologies such as RF-Mesh, High-speed Power Line Communication, and cellular networks. AMI also enhances net metering accuracy and supports faster adoption of renewables. By integrating these technologies, MERALCO can remotely detect outages, manage pilferage, and expand system capabilities. To meet the demands of smart grid transformation, MERALCO is modernizing its private telecommunications infrastructure, ensuring secure, resilient, and future-proof connectivity for remote device control and efficient fault isolation.

iii. Enabling the Digital Workforce and Workplace

Developing and empowering the workforce for the digital age is now a strategic priority for utilities, as digital technologies reshape how work is done and how employees collaborate. In the Philippines, utilities like MERALCO have embraced productivity platforms such as Microsoft Office 365, and rapidly adopted video conferencing tools during the pandemic, with Microsoft Teams usage jumping from 15% to 90% by March 2020. To support mobility and sustainability, MERALCO transitioned from desktops to energy-efficient laptops and upgraded its wireless LAN infrastructure for secure, stable connectivity. Utilities have also digitized employee services, using chatbots for health monitoring and automating back-office tasks through Robotic Process Automation (RPA), which streamlines processes like smart meter management and billing reconciliation. Conversational platforms like MERALCO's TRACI and Ding enhance recruitment and IT support, while AI tools such as MERALCO Interactive

Data Assistant (MIDAS) enable intuitive, secure, and self-service access to enterprise data. MERALCO is further exploring customer-facing AI solutions and upgrading its Enterprise Data Platform to strengthen analytics capabilities. To ensure system resilience, utilities are enhancing ICT infrastructure, with MERALCO establishing the country's largest ICT Command Center and upgrading its data centers. Innovation labs like MERALCO's Powertech Innovation and Experience Lab (PIXL) showcase emerging technologies such as VR, smart helmets, and digital twins for training and remote operations. Recognizing that digital transformation depends on people, MERALCO fosters a culture of innovation through DevOps, PowerTech training facilities, and collaborative communities like Centers of Excellence and Communities of Practice. These initiatives culminate in MERALCO's annual Innovation Month, celebrating creativity and technological advancement across the organization.

iv. Securing the Grid

The establishment of the Department of Information and Communications Technology (DICT) in 2016 marked a significant milestone in the Philippines' national ICT development agenda. The DICT was mandated to protect Philippine cyberspace and to formulate comprehensive cybersecurity strategies, with particular emphasis on safeguarding critical infrastructure, including the electric power grid.

This mandate gained increased urgency as global cyber threats to energy systems escalated. Notable incidents—such as the 2012 cyber breach of a major fuel supplier in Saudi Arabia, the 2016 cyberattack on Ukraine's power grid, and the 2019 cyber incident affecting a Norwegian power company—demonstrated the operational and systemic vulnerabilities of power sector infrastructure.

In response, the DICT's National Cybersecurity Plan of 2022 required critical information infrastructure operators to align with internationally recognized cybersecurity standards, including the ISO/IEC 27000 series. This policy direction prompted electric utilities to strengthen cybersecurity governance through the establishment of Computer Emergency Response Teams (CERTs) and the development of formal incident detection and response playbooks.

MERALCO, for example, has adopted a Defense-in-Depth strategy in conjunction with a Zero Trust framework, implementing layered security controls and strict access management. The utility has also invested in improved detection and response to address cyber threats, enhanced enterprise-wide risk and vulnerability management, and conducted regular internal security assessments.

MERALCO's cybersecurity program is aligned with globally accepted standards and best practices, including ISO/IEC 27001, the NIST Cybersecurity Framework, CIS Critical Security Controls, IEC 62351 for power system communications security, and IEC 62443 for the security of industrial automation and control systems. Together, these frameworks support an end-to-end cybersecurity lifecycle encompassing asset identification, protection, threat detection, incident response, and recovery. Knowledge-sharing initiatives—drawing lessons from events such as the Ukraine power grid attack—have reinforced operational controls, including the use of two-factor authentication.

As cyber threats continue to evolve in scale and sophistication, MERALCO and other power utilities recognize the need for continuous capability improvement, resilient governance structures, and proactive adaptation to emerging risks. Maintaining a dynamic and future-ready cybersecurity posture remains essential to ensuring the reliability, safety, and integrity of the electric power grid.

v. Future Proofing the Energy Sector

Digital Customer Experience

Utilities today must strengthen their digital customer experience by understanding how customers interact with their brand across online channels—such as websites, mobile apps, and self-service platforms. As most interactions now occur digitally, mapping the critical customer journey becomes essential to shaping perceptions and satisfaction. Customers increasingly value seamless experiences across devices and prefer resolving issues through web or mobile self-service rather than traditional phone support. To meet these expectations, utilities must leverage digital touchpoints to deliver consistent, responsive, and personalized service that builds deeper connections and trust.

Digital Grid

To meet the evolving energy demands of customers, the grid must become smarter, more resilient, and highly connected. Utilities must prioritize investments in robust communications infrastructure that can support emerging technologies such as IoT devices, Smart Meters, Streetlights, and Distribution Automation systems. As customers embrace digital lifestyles, the grid must adapt accordingly by integrating distributed energy resources like Solar PV, Battery Storage, Electric Vehicles (EV), and EV Charging Stations. Future-ready smart grid networks must be flexible and high-capacity, capable of supporting interoperability and multiple network functions rather than single-purpose technologies. Strategic investment in high-quality communication systems will be essential to ensuring grid stability, scalability, and readiness for Smart City and Smart Home services.

Digital Enterprise

As customer expectations continue to rise, utilities can no longer afford siloed ICT systems and disconnected business functions. What happens in the back-end infrastructure directly impacts customer experience and the company's bottom line—downtime in a website or customer portal, for instance, can immediately lead to dissatisfaction. To deliver a truly digital experience and future-proof operations, utilities must proactively digitalize the back office, recognizing its growing interdependence with front-end services. A seamless connection between back and front office systems is now essential to delivering comprehensive, responsive, and customer-centric service.

Data & AI

Addressing the challenges of delivering a total customer experience hinges on the strategic use of data, technologies, and processes—not just their presence, but their seamless integration and alignment. Systems must work together to enable innovative capabilities, such as real-time customer notifications during power outages, whether planned or unplanned, including estimated restoration times without requiring contact center calls. This demands backend integration across workforce management, enterprise asset management, and customer-facing platforms like portals, SMS, and social media. To further enhance operational efficiency, MERALCO is adopting AI tools like Robotic Process Automation (RPA), which

automates repetitive, rule-based tasks such as meter reading validation, billing, and account setup. RPA reduces manual workload, minimizes errors by over 50–60%, and operates 24/7 without disrupting existing IT systems—making it accessible even to non-IT staff. MERALCO is also leveraging AI through computer vision models to boost asset management and grid resilience, focusing on insulator monitoring, roof area identification for solar panels, and vegetation mapping near power lines. These technologies enable predictive maintenance, improve customer service, and support sustainable operations.

Agile

To succeed in the digital era, MERALCO embraces adaptability and agility, enabling rapid responses to emerging opportunities and threats. Agile methodology, increasingly adopted across organizations, supports this need by promoting iterative development through collaborative sprints that yield working products for immediate stakeholder feedback. While many companies have digital transformation strategies in place, execution remains a challenge—Agile helps bridge this gap by fostering a mindset of flexibility and continuous improvement. It encourages teams to operate within shorter planning cycles, optimizing stakeholder experiences through the right mix of technologies and data. This environment nurtures innovation, allowing ideas to evolve into prototypes and ultimately into customer-centric solutions. By embedding Agile thinking into their culture, utilities can better meet escalating customer expectations and continuously enhance service delivery in a dynamic landscape.

Sustainable Enterprise

Sustainability is no longer just MERALCO's corporate responsibility, it is a strategic imperative navigating the future of energy. Environmental risks are reshaping business models, with sustainability now influencing core strategies and operational decisions. Customers are increasingly eco-conscious, demanding that service providers, including utilities, adopt greener practices. As utilities advance their digital transformation journeys, they must embed environmental sustainability into every layer of their operations to drive positive outcomes for society, the economy, and the planet. Technologies such as AI, 5G, IoT, and cloud computing not only accelerate this shift but also generate valuable data that can be transformed into actionable insights.

The Philippine electricity supply industry is undergoing a system-wide transformation driven by decarbonization, digitalization, and electrification, set against a backdrop of climate exposure and geopolitical uncertainty. These forces are redefining how utilities plan capacity, operate the grid, manage risk, and deliver customer value. The pace and scale of change are unprecedented, and execution risk is now a central management concern.

A defining shift is the accelerated move toward renewable energy integration, supported by aggressive national targets. This transition is structurally altering system dynamics. Intermittency, voltage variability, and bidirectional flows are placing sustained pressure on transmission and distribution networks, while interconnection delays and permitting constraints continue to limit delivery timelines. Utilities are being required to build capabilities—particularly in forecasting, distributed energy resource (DER) management, and system balancing—that did not exist at scale even a few years ago.

At the same time, energy storage is becoming a core requirement for system flexibility. However, regulatory clarity on ownership, tariffs, and safety standards remains a work in progress, while capital intensity and supply chain concentration continue to constrain deployment. Storage economics are improving, but bankability remains highly dependent on policy alignment and market design.

The generation mix transition introduces parallel risks. The shift away from coal toward LNG and renewables must be executed without compromising supply adequacy. Exposure to global fuel markets and geopolitical developments is increasing, while timing mismatches between retiring capacity and new build-outs present real grid stability concerns. The industry remains tightly constrained by the need to balance reliability, affordability, and sustainability.

Running alongside these shifts is the digitalization of the grid, including smart metering, automation, and AI-enabled operations. While these initiatives are foundational to modern grid performance, they also introduce structural challenges—high capital requirements, interoperability gaps, and limited technical talent. More critically, digitalization is materially expanding the sector's cyber attack surface.

Cybersecurity is now a strategic imperative. The convergence of physical infrastructure and digital systems has made the energy sector a high-value target. Threat actors are increasingly focused on industrial control systems and operational technology environments, where disruption directly translates into outages and economic impact. Cyber risks are no longer confined to IT—they reflect broader geopolitical realities and must be managed as such.

Customer expectations are shifting toward digital-first engagement, requiring real-time, mobile-enabled, and integrated services. Simultaneously, the electrification of transport is introducing new load dynamics that existing planning models must quickly adapt to. These developments are further complicated by the Philippines' exposure to climate-related risks, which necessitates embedding resilience into grid design and operations. External pressures, including supply chain disruption and currency volatility, continue to affect the pace and cost of infrastructure delivery.

Across all these transformations, talent and capability constraints remain a limiting factor, particularly in cybersecurity, AI, and grid automation. Building and sustaining these capabilities is now a strategic priority for the sector.

One MERALCO Group Response and Contributions

The One MERALCO Group is responding to this transformation with a coordinated, enterprise-wide strategy focused on clean energy scale-up, digital grid capability, and resilience by design.

In renewable energy integration, the Group, through MERALCO PowerGen (MGen), is advancing large-scale developments such as the MTerra Solar project, which integrates utility-scale solar with 4,500 MWh of battery storage capacity—one of the largest solar-plus-storage developments in the region. This is complemented by MGreen's renewable portfolio expansion and MSpectrum's nationwide rollout of distributed solar solutions for commercial and industrial customers. This combined approach—utility-scale plus distributed—directly supports grid decarbonization while easing pressure on centralized generation and transmission infrastructure.

In parallel, the Group is strengthening its position in energy security and generation diversification through strategic investments in both conventional and transition fuels. These include stakes in major generation assets such as South Premiere Power Corporation and Excellent Energy Resources, Inc., alongside participation in LNG infrastructure that supports a significant portion of Luzon's power demand. This balanced portfolio ensures supply adequacy while enabling a measured transition toward cleaner energy sources.

In grid digitalization, MERALCO is executing a structured smart grid program that includes the deployment of Advanced Metering Infrastructure (AMI) in partnership with Itron, distribution automation systems, and customer platform modernization. These investments are supported by AI-enabled geospatial analytics and DER visibility tools, enabling more precise load forecasting, improved outage response, and better integration of distributed energy resources. The objective is not incremental improvement, but a step-change in grid observability and operational control.

On cybersecurity, the Group has institutionalized a comprehensive and forward-looking strategy anchored on ARSENAL, its long-term cybersecurity architecture and investment program. ARSENAL operationalizes a multi-year approach to achieving verifiable cybersecurity maturity, grounded on an "assumed breach" framework and designed to ensure continuous resilience across IT and OT environments. This is supported by layered controls, including IT-OT segmentation, identity and access management (IAM), web application firewalls, and secure configuration and patch management, all implemented at enterprise scale.

Governance is reinforced through alignment with global standards such as ISO/IEC 27001, IEC 62443, NIST frameworks, and DOE's Cybersecurity Capability Maturity Model, as well as through internal governance mechanisms such as the CARE Standard (Consistency, Adequacy, Reliability, Effectiveness), which ensures that security controls are uniformly enforced and outcome-driven. Cybersecurity is therefore managed not as a support function, but as a core enabler of operational continuity and stakeholder trust.

In customer digitalization, the Group continues to expand its integrated digital ecosystem, including the My MERALCO mobile application, digital payments through MERALCO Pay (with GCash integration), digital official receipts, and end-to-end online service applications. These platforms are designed to deliver real-time engagement, improve transparency, and align utility service delivery with evolving customer expectations.

To address the electrification of transport, the Group, through Movem Electric, is deploying a nationwide EV charging network supported by strategic partnerships, including collaboration with international technology providers. At the same time, distribution planning frameworks are being recalibrated to incorporate EV-driven load growth, ensuring that infrastructure development keeps pace with adoption.

In building climate resilience, the Group is investing in programs that strengthen both infrastructure and operations, including resilient communications systems, enhanced mobile workforce capabilities, and data-driven post-event analytics. These initiatives are designed to improve response time, reduce service interruptions, and institutionalize learning from extreme weather events.

Finally, in addressing talent and capability development, the Group is investing in enterprise-wide digital transformation programs, including AI productivity initiatives, capability-building efforts across cybersecurity and data analytics, and Project LIGHT, a digital literacy program developed with the Cyber Security Group and One MERALCO Foundation. These initiatives extend capability-building beyond the organization, contributing to broader ecosystem readiness in cybersecurity and responsible technology use.



SINGAPORE



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Country Information

General Information (as of Dec 2025 or latest data)

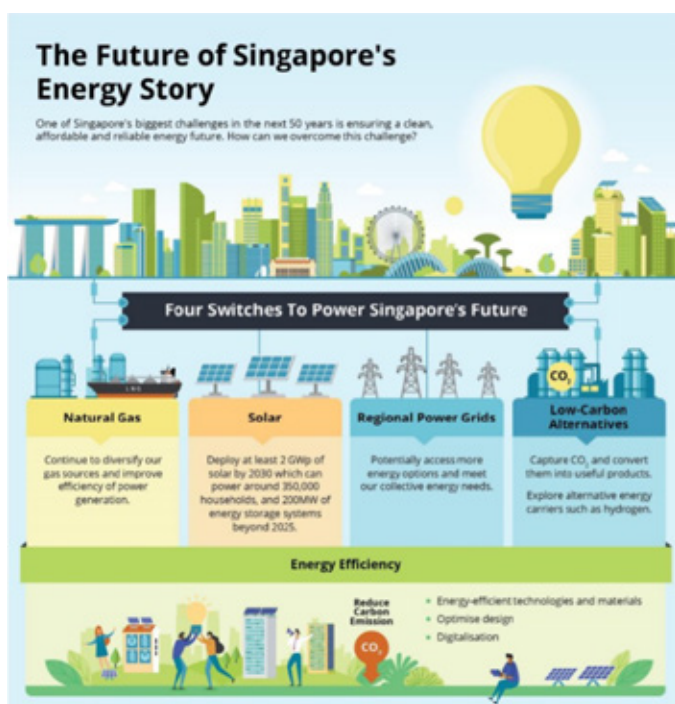
Capital:	Singapore
Area:	736.3 km ²
Population:	6.11 million
GDP (in S\$):	S\$ 790 Billion
Currency:	Singapore Dollar (S\$)
Installed Capacity (in GW):	13.3
Percentage of population electrified:	100%

Power Demand by Sector

Sector	Total Power Consumption as of 2024 (MWh)	Projected Change (%) 2024/2025	Projected Power Consumption as of 2025 (MWh)
Industrial	22,704,000	-2.7%	22,999,600
Commercial	23,195,000	4.3%	23,673,000
Residential	8,422,300	1.2%	7,987,600
Transport	3,199,800	4.5%	3,181,800
Others	118,600	-18.0%	1,600
TOTAL	57,639,700	0.9%	57,843,600

National Energy Policy

Singapore's national energy policy focuses on transforming the way the country uses energy and adopting clean sources to mitigate emissions. The policy aims to achieve a sustainable, affordable and reliable energy supply through four key strategies, known as the "Four Switches", together with the more efficient use of energy.



Natural Gas:

- As Singapore advances its clean energy agenda, natural gas continues to play a vital role in our energy landscape. This reliable energy source provides the reliability needed as we scale up our other energy switches.
- In April 2025, a milestone was reached with the establishment of Singapore GasCo. The new entity will centralise the procurement and supply of natural gas to the power sector in Singapore, achieving economies of scale through more favourable gas contracting terms. It will also procure natural gas from diverse sources and secure longer-term gas contracts to provide more stable supply and prices.

Solar:

- Solar energy is Singapore's most promising renewable energy source. In the last five years, the installed solar capacity has more than tripled – from under 0.4 GWp in 2019 to over 1.5 GWp in 2024. In doing so, Singapore has achieved our 2025 target of 1.5 GWp a year ahead of schedule. This milestone reinforces Singapore's position as a leading solar-dense city and puts us on track to achieve our 2030 target of at least 2 GWp.



Regional Interconnectivity:

- To better meet Southeast Asia's growing energy demand, the Lao PDR-Thailand-Malaysia-Singapore Power Integration Project (LTMS-PIP) was enhanced in September 2024. Under phase 2 of the project, the project's electricity trading capacity doubled from up to 100 MW to a maximum of 200 MW.
- Singapore is scaling up our electricity import ambition to support our decarbonisation goals. From the initial target of 4 GW, we now aim to import around 6 GW of low-carbon electricity by 2035. EMA has granted Conditional Approvals and Conditional Licences to electricity import projects from Australia, Cambodia, Indonesia and Vietnam.



Supply and Demand of Primary Natural Resources

Singapore's electricity demand is driven by factors such as weather patterns, population changes and economic activities, based on investment and development outlooks from economic and planning agencies. The system peak demand has been increasing steadily and is projected to grow at a Compound Annual Growth Rate ("CAGR") per annum of 2.4% - 4.8% (or 2.0GW – 4.3GW) from 2025 to 2034.

The majority of Singapore electricity is generated using natural gas with a small percentage coming from other sources. Given its small geographical size and lack of renewable energy resources, Singapore is alternative energy-disadvantaged and heavily dependent on external sources of supply, making it vulnerable to geopolitical and economic shifts in the energy supply chain. Any rising demand for energy or related technology supply could potentially make Singapore vulnerable to volatility and price shocks.

Laws and Governmental Regulations

The Electricity Act (along with the subsidiary regulations) is the main legislation governing the electricity market, which aims to:

- (i) create a competitive market framework for the electricity industry; and
- (ii) make provision for the safety, technical and economic regulation of the generation, transmission, supply and use of electricity.

All Market Participants must also comply with the market rules and codes of practice issued or approved by Energy Market Authority (EMA).

In 2019, the Carbon Pricing Act came into operation. The Act imposes a carbon tax on greenhouse gas (GHG) emissions and obligations on (a) facilities which emit 25,000 tons or more of carbon dioxide equivalent (tCO₂e) of GHG emissions annually (taxable facilities) and (b) facilities which emit at least 2,000 but less than 25,000 tCO₂e of GHG emissions annually (reportable facilities).

Low-Carbon Alternatives for Longer-Term Decarbonisation

- EMA is also exploring a range of emerging low-carbon alternatives to ensure a secure, reliable and sustainable power system in the longer term. Ammonia is currently one of the most technologically ready hydrogen carriers, with an established international supply chain for industrial use. In July 2024, EMA and the Maritime and Port Authority of Singapore shortlisted two consortia to proceed to the next evaluation stage for proposals to provide a low-carbon ammonia solution on Jurong Island for power generation and bunkering.
- In April 2024, Singapore launched a consultancy study to assess Singapore's geothermal energy potential. EMA has commenced the non-invasive geophysical survey activities, which will help us ascertain the viability of this energy source for power generation.

Harnessing Demand-Side Resources

- EMA's Demand Response programme plays a key role in managing system costs and improving grid efficiency. Between 2023 and mid-2024, the programme delivered over \$700 million in savings for electricity buyers in the Singapore Wholesale Electricity Market.
- Building on this success, EMA enhanced the programme in October 2024. This included retaining the lower compliance threshold and initial penalty waivers introduced in the 2023-2024 sandbox, as well as allowing businesses with Battery Energy Storage Systems to participate.

The carbon tax is set at a rate of S\$5/tCO₂e from 2019 to 2023. This will be progressively raised to S\$25/tCO₂e in 2024 and 2025, and S\$45/tCO₂e in 2026 and 2027, with a view to reach S\$50-S\$80/tCO₂e by 2030.

The carbon tax will incentivise emissions reductions across all sectors and support the transition to a low-carbon economy. There are no exemptions for covered facilities, to maintain a transparent, fair, and consistent price signal across the economy.

Total amount of energy consumption by sectors (Industrial/ Commercial/ Residential) (in MWh and %)

Sector	Total Power Consumption as of 2024 (MWh)	% in Total
Industrial	22,704,000	39.4%
Commercial	23,195,000	40.2%
Residential	8,422,300	14.6%
Transport	3,199,800	5.6%
Others	118,600	0.2%
TOTAL	57,639,700	

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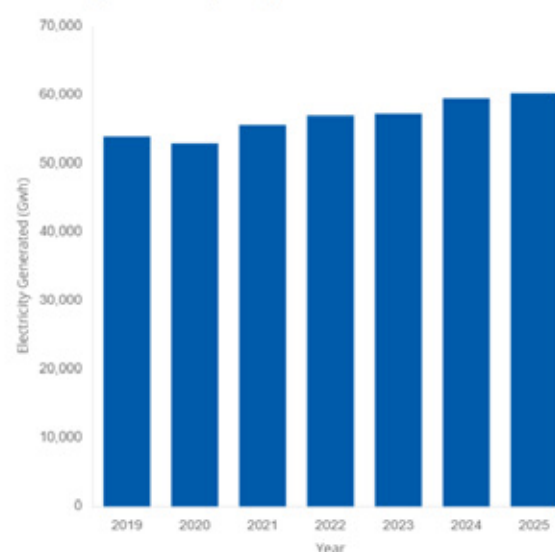
Power Generation and T&D Sectors (as of end 2025 or latest data)

Total installed capacity of generation by fuel types (coal, LNG, Gas, Nuclear-fired and hydro) including total installed capacity of generation (in MWe) and total production (in MWh) by fuel types (in MWe)

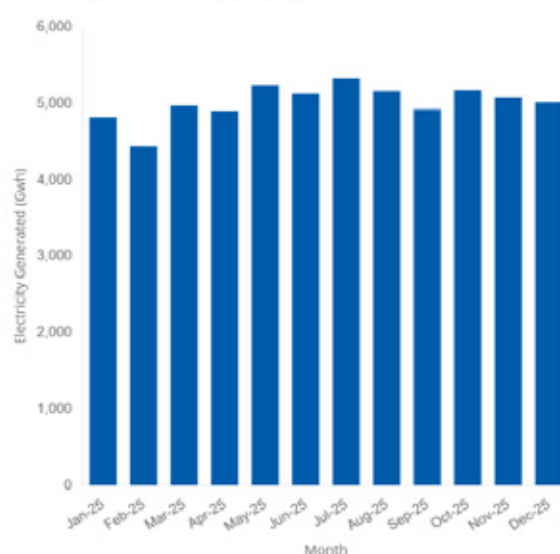
In 2024, 10,533 ktoe (123 TWh) of energy inputs generated a gross electricity output of 5,126 ktoe (60 TWh). This was a 2.5% increase from 2023, reflecting continued growth in electricity demand.

As of 1H 2025, 5,163 ktoe (60 TWh) of energy inputs generated a gross electricity output of 2,546 ktoe (30 TWh).

Electricity Generated (Yearly)



Electricity Generated (Monthly)



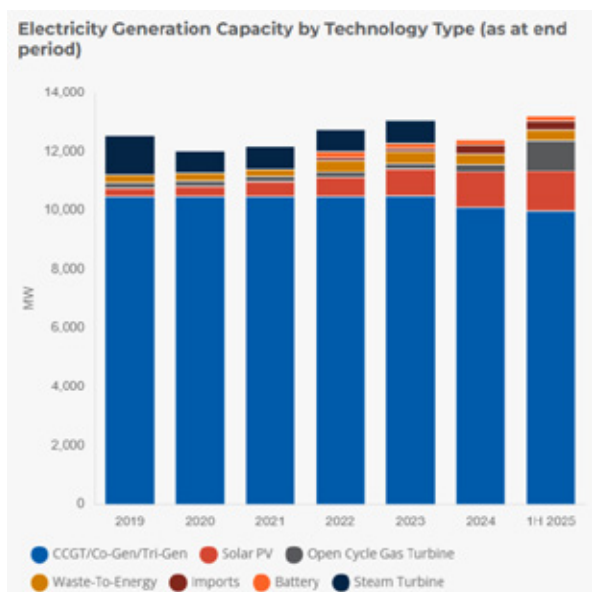
Installed Capacity (GW)⁵¹ Year-on-year Growth (%)

Total	12.44	-4.8%
CCGT/Co-Gen/Tri-Gen	10.11	-3.8%
Steam Turbine	0.01	-98.2%
Open Cycle Gas Turbine	0.26	44.4%
Waste-To-Energy	0.35	-12.2%
Solar PV ⁵²	1.21	31.9%
Energy Storage System ⁵³	0.20	0.0%
Imports	0.30	200%

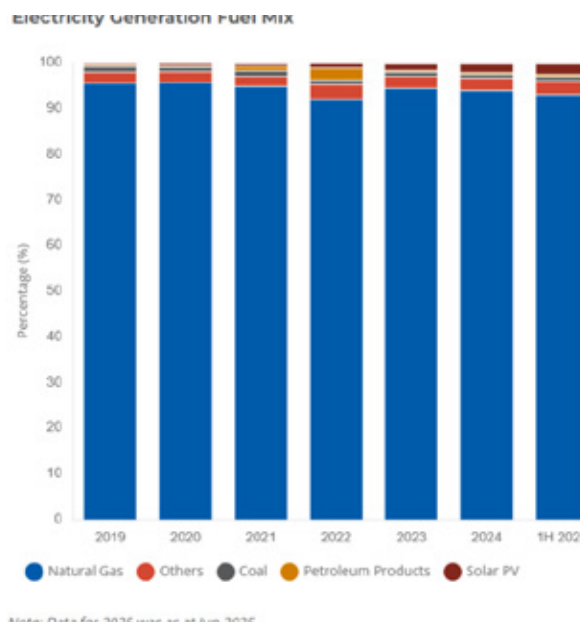
⁵¹ As at 2024

⁵² Solar PV figures are presented in GWac

⁵³ Same as the previous year value. Hence, growth rate is zero



In the first half of 2025, Natural Gas accounted for 93.1% of Singapore's fuel mix, of which Other Energy Products accounted for 3.0%, and Solar PV accounted for 2.5%.



In 2024, Natural Gas accounted for 94.0% of Singapore's fuel mix. Other Energy Products (e.g., Municipal Waste, Biomass, and Imports) accounted for 2.7%, Solar PV accounted for 2.1%, while Coal and Petroleum Products (1.2%) made up the balance.

4 Reduction of Carbon Emission

Policy on Renewable Energy and Major Regulatory Mechanism of Renewable Law

Singapore's policy on renewable energy focuses on transitioning to cleaner energy sources, with a goal to reduce emissions and achieve sustainability. Key initiatives include:

- a. Solar Energy:** Singapore has reached its 2030 solar deployment target of 2 GW peak ahead of schedule, and will thus raise the target to 3 GW peak by 2030. 2 GW-peak can meet around 3% of the country's projected total electricity demand in 2030.
- b. Energy Storage Systems (ESS):** To address solar intermittency and enhance grid resilience, Singapore is deploying ESS. A 285 megawatt-hour ESS on Jurong Island, launched in February 2023, is the largest in Southeast Asia.
- c. Regional Power Grids:** Singapore has granted conditional approvals and licences to import greener electricity from Malaysia, Indonesia, Cambodia, Vietnam and Australia. The 11 projects across the countries cover solar, wind and hydropower, with a combined capacity of around 8GW.

d. Hydrogen Strategy: Singapore is exploring hydrogen as a major decarbonisation pathway, and has developed a national hydrogen strategy to progressively build capabilities in industry, workers and the Government in areas that are critical to unlocking hydrogen adoption in Singapore.

e. Green Transport: The vision is for all vehicles to run on cleaner energy by 2040,

with targets to roll out charging points in public and private carparks in tandem with EV adoption. Singapore also aims to electrify 50% of our public bus and taxi fleet.

Major Regulatory Mechanisms Include:

- a. Energy Conservation Act 2012:** Introduced measures to reduce energy consumption, including requirements for consumers of 15 GWh of energy per year.
- b. Renewable Energy Integration Demonstrator (REIDS) Microgrid Project:** Aims to test alternative renewable sources and ensure efficient energy storage systems are in place.
- c. International Renewable Energy Certificate (I-REC) Mechanism:** Supports the development of renewable energy projects and tracks renewable energy production.

5

Advanced Metering Infrastructure (AMI/Smart Grid Deployment)

Latest AMI Installations and Smart Grid Projects

All 1.58 million household accounts in Singapore can expect to have smart meters installed at their premises by 2026.

Challenges Faced and How It Was Solved / Addressed

The key challenge is customers' outright refusal to change. To overcome this issue, SP Group is actively engaging customers and updating the regulator to demonstrate efforts to engage and convince.

AMI Offerings to Customers and Its Benefit

AMI meters allow customers to view their half-hourly electricity consumption using the SP app. This will, in turn, enable customers to track their consumption in a more timely manner and facilitate accurate monthly billing. With almost real-time monitoring of their consumption usage, consumers would be incentivised to better manage their demand and are empowered to use electricity more efficiently.

6

Latest Technologies or Innovations

SP Group has deployed several innovative technologies to drive sustainability and efficiency in the energy sector. Some of the notable ones include:

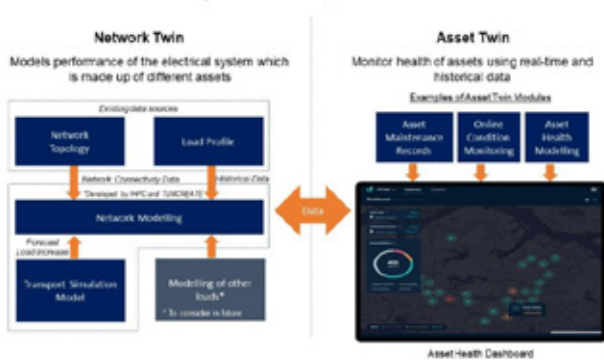
- Distributed Energy Resource Management System (DERMS) pilot:** SP has embarked on DERMS pilots to monitor and manage DERs and to enable effective network design and planning. By using real-time monitoring and control capabilities, SP aims to manage the impact from the growing number of DER connections in the grid. This will address challenges arising from fluctuations caused by solar intermittency and mitigate potential grid constraints from rising EV charging demand. In addition, SP and EMA will explore communication standards and solutions to facilitate reliable and secure information exchange between DERMS platform and DERs to enable real-time coordination.
- Virtual Power Plants (VPPs):** VPPs are also an important enabler to harness the flexibility of DERs to provide useful grid services to manage solar intermittency, system peaks and grid balancing. This is done through controlling, optimising, and aggregating a network of DERs across various locations to operate akin to a single generator. DERs may not be able to individually provide services to the grid, due to reliability issues and the cost-effectiveness of control systems such as energy management systems and pilot wires.



(Source: Energy Market Authority)

A VPP registers itself as a single power generator in the Singapore Wholesale Electricity Market and performs these services: (i) supply electricity to meet demand; (ii) provide regulation services by adjusting its electricity use in real-time to balance the grid for stability; and (iii) provide contingency services by injecting electricity into the grid when there is a sudden loss of supply (e.g. power plant outages) in the power system.

- **Grid Digital Twin:** EMA and SP have initiated the development of a grid digital twin to maintain grid reliability and optimise grid investments for the future. This includes an asset twin which monitors the health of critical grid assets, and a network twin to model and simulate how DERs affect the grid. With decreasing cost of sensors and the technological breakthrough in computation, SP also plans to advance the digital capabilities of digital twins by deploying sensors and Artificial Intelligence (AI) to provide strategic insights into asset and grid conditions.



- **Grid monitoring capabilities in the distribution network:** For example, in areas with high concentration of DERs, a surge in solar PV generation or EV charging demand can potentially overload transformers and distribution cables, causing network congestion. The network may also have insufficient capacity to accommodate new DER connections. SP will deploy sensors and collect data at the distribution network to forecast the power flow from DERs and project available network capacity. This will allow SP to identify potential constraints ahead of time and make informed decisions on network planning and grid infrastructure upgrades.
- **Digital tools adoption:** SP's workforce will be equipped with digital tools that are integral to augmenting and optimising manpower resources for essential tasks in substations such as inspection and maintenance. SP has piloted the use of digital applications, inspection drones and robots, and Augmented Reality tools to assist in daily inspection and critical maintenance procedures. For the next phase of these pilots, SP will use AI to increase the efficiency and effectiveness of field operations and maintenance tasks and optimise resources.

- **Distributed district cooling network:** In March 2025, SP Group launched Singapore's first distributed district cooling network at Tampines. The sustainable cooling solution strengthens Tampines' Eco Town ambitions, by reducing carbon emissions by 1,000 tonnes, and achieving energy savings of more than 2,300,000 kilowatt-hours (kWh) annually.
- **Green Mobility Network:** In February 2025, SP Group launched the first electric vehicle (EV) charging points in Tengah estate, Singapore's first and largest smart and sustainable town, planned with green and sustainable features, and smart technologies from the outset. This milestone event was graced by Dr Amy Khor, Senior Minister of State for Ministry of Transport & Ministry of Sustainability and the Environment, and Adviser to Hong Kah North Grassroots Organisations. The launch supports sustainable mobility and smart energy solutions for residents.
- In July 2024, as part of SingHealth's Decarbonisation Plan to reduce carbon emissions and promote sustainable practices throughout its operations, SingHealth has taken a significant step towards environmental sustainability by installing its first batch of 12 EV charging points at SingHealth Tower, located within the Singapore General Hospital (SGH) Campus. These chargers will be available for use by staff and visitors. SingHealth, together with its sustainability partner, SP Group (SP), plans to progressively install up to 300 EV charging points by 2028 across its institutions, with SingHealth set to become the public healthcare cluster with the largest number of EV chargers in Singapore upon completion.

These innovations demonstrate SP Group's commitment to driving sustainability and efficiency in the energy sector through digitalization and technology.

Solar Energy Expansion

Singapore's solar energy capacity has tripled in the last five years, with a target of 3 GWp by 2030. The Enhanced LTMS-PIP project aims to enhance electricity trading capacity and support low-carbon electricity imports.

Net-Zero Emissions:

Singapore aims to reach net-zero emissions by 2050, despite the challenges posed by global gas market volatility and geopolitical instability. The country is investing heavily in emerging technologies to supply carbon-free electricity.

Demand-Side Resources;

The introduction of electric vehicles and renewable energy sources requires the power grid to become more flexible and responsive. EMA announced in October 2025 that they are exploring a demand-side flexibility roadmap aimed at allowing the grid to tap “underutilised” distributed energy resources such as battery energy storage systems and backup generators. The roadmap by EMA outlines measures to future-proof the grid, focusing on inter-connectivity, resilience, and flexibility. They will explore

THAILAND



1

Country Information

General Information (as of Dec 2025 or latest data)

Capital:	Bangkok
Area:	513,120 km ²
Population:	71.5 million
GDP (US\$):	577 billion (2025 estimate)
Currency:	Thai Baht (THB)
Contracted Capacity (in GW):	49,962 GW
Percentage of population electrified:	99.8%

Power Demand by Sector

Sector	Total Power Consumption as of 2025 (MWh)	Projected Change (%%) 2024/2025
Industrial	86,786,000	-5.35
Commercial	52,811,000	-0.70
Residential	58,863,000	-1.63
Other	5,119,000	-17.57
TOTAL	203,579,000	-2.82

National Energy Policy

Thailand's national energy policy aims to balance energy security, affordability, and sustainability while steering a transition toward lower-carbon energy systems. The government has established a National Energy Plan (NEP) as a framework to integrate and align the various sectoral energy plans (Power Development Plan, Alternative Energy Development Plan, Energy Efficiency Plan, Oil & Gas Plan) into a coherent strategy.

EGAT's Strategic Plan 2026 – 2030 emphasizes “Energy Security for Energy Transition.” The plan focuses on developing innovative and modernized power systems, supporting national power policy for economic efficiency, and promoting green innovation for sustainability. EGAT's mission is to secure power reliability and enhance national competitiveness through innovation. This includes investment in clean energy, modernization of the transmission grid, flexible operation of baseload power plants, and establishment of key systems such as the Renewable Energy Forecast Center (REFC) and Demand Response Control Center (DRCC) to balance supply and demand and integrate higher shares of renewable energy.

Supply and Demand of Primary Natural Resources

In 2025, Thailand's electricity system had a total contracted generating capacity of 49,962 MW, with 63% of electricity generated from natural gas, 11.8% from imported power, 11.42% from coal/lignite, and around 13.3% from domestic renewables (hydro, biomass, solar, wind, geothermal). Natural gas remains the dominant fuel but faces declining domestic production, prompting increased LNG imports under ERC-approved short-term contracts. Coal production from the Mae Moh mine reached 12.5 million tons, meeting over 89% of its annual plan. EGAT also maintains oil and diesel reserves for emergencies and peak demand periods. Rising demand—driven by industrial recovery, tourism growth, and the surge in EV adoption but the average weather patterns and climate variability affect to average temperature in 2025 lower than 2024 — pushed 2025 net peak demand to 34,568 MW, down 5.23% from 2024.

Laws and Governmental Regulations

EGAT operates under the EGAT Act B.E. 2511 (1968) and is supervised by the Ministry of Energy and Ministry of Finance. The sector is regulated by the Energy Regulatory Commission (ERC), established under the Energy Industry Act B.E. 2550 (2007), which oversees licensing, tariffs, and market rules. EGAT must comply with cabinet resolutions, laws, and announcements relevant to power system operations. In 2024, EGAT's Audit Committee confirmed full compliance with laws and regulations, transparent corporate governance, and adherence to Thai Financial Reporting Standards. Regulations also cover LNG market liberalization, renewable energy incentives (Feed-in Tariff, Adder), and demand-side management programs, reflecting the government's commitment to energy security, affordability, and decarbonization.

Total amount of energy consumption by sectors (Industrial/ Commercial/ Residential) (in MWh and %)

Sector	Total Power Consumption as of 2025 (MWh)	% in Total
Industrial	86,786,000	28.91
Commercial	52,811,000	25.94
Residential	58,863,000	42.63
Other	5,119,000	2.51
TOTAL	203,579,000	

Total installed capacity of generation by fuel types (coal, LNG, Gas, Nuclear-fired and hydro) including total installed capacity of generation (in MWe) and total production (in MWh) by fuel types (in MWe)

	Contracted Capacity (GW)	Year-on-year Growth (%)
Total	49.96	-2.82
Hydropower	7.52	0
Thermal Power	37.50	-4.64
Nuclear Power	-	-
Wind Power	1.45	0
Solar Power	0.91	35.82
Other	2.58	1.93

	Generation (TWh)	Year-on-year Growth (%)
Total	212.72	-3.57
Hydropower	32.12	13.94
Thermal Power	166.04	-7.20
Nuclear Power	-	-
Wind Power	3.36	1.51
Solar Power	1.92	68.42
Other	9.28	2.77

4

Reduction of Carbon Emission

Policy on Renewable Energy and major regulatory mechanism of Renewable Law

EGAT's Strategic Plan 2026 – 2030 explicitly targets increased renewable penetration with a goal of up to 50% renewable share in electricity generation in the long term.

The Energy Regulatory Commission (ERC) plays a central role by approving renewable procurement frameworks, setting tariffs, and facilitating grid connection rules to enable VSPP/SPP participation.

Production of Renewable Energy (MWh) from various resources (wind, solar, biomass, landfill gas, etc)

In 2025, renewable energy generation accounted for 46,678 GWh (21.59% of total generation), comprising hydropower, biomass, solar, wind, geothermal sources and waste. Notable milestones include the commissioning of the Hydro-Floating Solar Hybrid Project at Ubol Ratana Dam (24 MW), which combines solar, hydro, and battery storage technologies to deliver firm clean energy.

Plans and strategies to reduce greenhouse gas emissions

- Increasing renewable share through solar (including floating), wind, biomass, and cross-border hydropower imports.
- Deploying Grid Scale Battery Energy Storage Systems and pumped-storage hydropower to stabilize renewable output.
- Promoting demand-side management via Energy Label No.5 and the ENZY Platform for energy management.
- Supporting electrification of transport by installing 289 EV charging stations nationwide under "EleX by EGAT."
- Conducting carbon reduction projects such as the "10 Projects Continuing the Royal Aspirations of the King," which save 5.3 GWh and reduce CO₂ emissions by 2,800 tons annually.

Policy on Energy Conservation and Efficiency

Energy efficiency is promoted through the Energy Conservation Promotion Act B.E. 2535 (1992).

Key elements of the policy include:

1. Mandatory Energy Management: Designation of "Controlled Factories and Buildings" that are required to implement energy management systems, conduct energy audits, and report performance annually.
2. Promotion and Incentives: Operation of the Energy Conservation Promotion Fund, which finances energy efficiency projects, R&D, and technology adoption.
3. Appliance and Equipment Standards: Implementation of the Energy Label No. 5 Program, which identifies and promotes high-efficiency products. Currently, EGAT expanded labeling to include distribution transformers, solar panels, and inverters, encouraging cost reduction for consumers while improving energy performance.
4. Public Awareness and Capacity Building: Nationwide campaigns and training programs to promote behavioral change, energy literacy, and adoption of efficient technologies in households and businesses.
5. Demand-Side Management and Smart Energy Solutions: Deployment of ENZY Platform and EGAT Smart Energy Solutions to enable real-time energy monitoring, data analytics, and optimal energy use.

5

Advanced Metering Infrastructure (AMI/Smart Grid Deployment)

Latest AMI installations and smart grid projects

End Year:	2020
Meter Type:	Grid Revenue Meter (Active Energy Class 0.2S)
Number of Meters:	3000 EA (Landis+Gyr, Schneider, CEWE)
Communication:	Ethernet via EGAT WAN (Fiber Optic, SD WAN), IPsec VPN via Cellular Network
AMR:	2-AMR Systems at HQ, 15-minutes Reading Intervals

Challenges faced and how it was solved / addressed

Failure in Measurement Transformers and
Loss of Communication
> Developing the METER Monitoring System

AMI offerings to customers and its benefit

Periodical Measurement Data Service
at 1 or 15-Minutes.

AMI Roll out plan

During the proof-of-concept phase, the revision of
the AMR is targeted for completion by 2029.

6

Latest Technologies or Innovations

Digital Substation:

Develop a substation inspection robot system capable of capturing high-frequency visual and thermal images, with integrated image processing, data storage, reporting, alerting, and trend analysis functions. The system supports systematic condition-based and predictive maintenance, reduces workload, and enhances the ability to prevent potential equipment failures. The project aims to be implemented as a joint research initiative with institutions during 2025 – 2027, resulting in one system designed for indoor substation inspections.

Hydro Floating Solar:

The Floating Solar Power Plant projects will be developed in accordance with the Power Development Plan (PDP) 2018 Revision 1. In the near future, large-scale floating solar projects will be implemented at Bhumibol Dam (158 MWac), Srinagarind Dam (140 MWac), and Vajiralongkorn Dam (50 MWac) under Phase 1, with expansions in Phase 2 and additional dam sites in the following years. These floating solar initiatives will be a key contributor to Thailand's to achieve net-zero greenhouse gas emissions by 2050.

Grid modernization refers to the strategic transformation of electricity transmission system to make it more reliable, flexible, efficient, and resilient in the face of changing energy landscapes. For Thailand, this transformation is essential to meet growing electricity demand, integrate increasing shares of renewable energy, and support the national targets of net-zero greenhouse gas emissions by 2050.

There are 5 key areas of EGAT's Grid Modernization in the following:

1. Digital Control Platform

EGAT is upgrading its National and Regional Control Centers (NCC/RCC) with advanced SCADA/EMS, WAMPAC, and the EGAT Master Control system. These platforms integrate the Renewable Energy Forecast Center (REFC), EV Forecasting, and the Demand Response Control Center (DRCC), enabling accurate renewable forecasts, demand-side management, and operation of Virtual Power Plants (VPPs).

2. Infrastructure Expansion Planning and Development

The modernization plan invests in new grid-scale Battery Energy Storage Systems (BESS), floating solar-hydro hybrid projects, and pumped-storage hydropower (PSH) at many sites. Expansion of transmission infrastructure and deployment of FACTS devices (e.g., STATCOMs) enhance stability, system flexibility, and the capacity to integrate variable renewable energy.

3. Transmission Protection Control, Monitoring and Asset Management

Modern technologies such as EGAT SCADA X, Automatic Fault Analysis (AFA), Protection Relay Centers, and Wide-Area Monitoring Systems (WAMS) are being introduced to improve grid reliability. Predictive maintenance tools — such as health monitoring of transformers and circuit breakers — together with GIS and unmanned aircraft systems (UAS) support asset management.

4. New Agreements and New Business

EGAT has new renewable energy PPAs with provisions for curtailment and revises the Grid Code to reflect system requirements. These measures create a regulatory environment that enables more renewable integration, and prepares the sector for emerging business opportunities in a market.

5. Support Systems

To ensure secure and efficient operations, EGAT is strengthening cybersecurity for operational systems and deploying cloud-based platforms for data-driven grid management. These support functions enhance operational resilience, enable real-time analytics through APIs, and build a foundation for digital transformation across the electricity supply chain.



1 Country Information

01 Country Information

01 Country Information

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01 Country Information



VIETNAM



Power Demand by Sector (as of end 2024)

Total power production and purchase by EVN reached 308.732 billion kWh and total power sales by reached 276.029 billion in 2024

Total power sales (billion kWh)	276.029
Agriculture	10.035
Industry	142.416
Commercial & Hotels, Restaurants	14.976
Administration & Residential	97.11
Others	11.491

Power Generation

Vietnam's overall installed electricity generation capacity was 82,387 MW by the end of 2024. Hydropower, coal, natural gas and RE are the dominant energy sources for power production.

EVN contributes up to 38% of the national power generation system with large-scale hydropower, coal fires, and gas-fired power.

Total installed capacity (MW)	82,387
Hydropower	23,664
Coal fired	26,757
Gas + Oil fired	8,653
Renewable energy	21,447
Others	644
Imported	1,222

Power Transmission and Distribution

Vietnam's power system is operating at a high voltage of 500kV, 220kV and 110kV and medium and low voltage. The power transmission lines of 500kV and 220kV are managed by the National Transmission Power Corporation (EVNNPT) while the lines of 110 kV, medium and low voltage are managed by 5 Power Corporations.

EVN owns the entire national power transmission and distribution system which covers all provinces and cities.

In power exchanges between neighbor countries in the period 2020–2024, EVN imported 5.623 billion kWh from China and 11.482 billion kWh from Laos and exported more than 4.98 billion kWh to Cambodia and nearly 235 million kWh to Laos.

The Outlook Plan

Amendment to National Electricity Development Planning of 2021 - 2030 period and vision to 2050 has been approved by the Prime Minister of Vietnam Government through decision no. 768/QĐ-TTg dated April 15, 2025.

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China Southern Power Grid Co., Ltd. (CSG) is an internationally advanced builder and operator of digital power grid infrastructure, which ensures reliable power supply to Southern China, Hong Kong SAR, and Macao SAR. The company's service area encompasses over one million square kilometers, providing electricity to more than 273 million residents. In 2025, CSG sold 1,535.4 TWh of electricity, generating an operating revenue of USD 126.2 billion and holding total assets worth USD 213.5 billion. Currently, CSG ranks 86th on the Fortune Global 500 list and has received sovereign-level credit ratings from the three credit rating agencies: S&P, Moody's, and Fitch.

Focusing on power transmission and distribution, CSG boasts extensive expertise in ultra-high-voltage transmission, digital grid operation, intelligent O&M, energy storage, as well as energy conservation and carbon reduction. The West-to-East Power Transmission Project spanning over 2,000 kilometers comprises 8 AC and 11 DC lines of 500 kV and above, with a maximum transmission capability of over 58 GW. CSG has established interconnections with Vietnam, Laos, and Myanmar through 17 circuits of 110 kV and above lines, with accumulated electricity traded exceeding 84 TWh. The Company takes the lead in establishing the Global South Power Partnership Development Platform for power enterprises, focusing on energy and power investment and development in regions including Southeast Asia, Latin America, Central Asia and Europe. At present, it holds equity stakes in 13 energy and power projects across 8 Belt and Road countries such as Laos, Vietnam, Chile and Peru.



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The partners span major regions worldwide: Africa, Southeast Asia, South Asia, the Middle East, Central Asia, Latin America, Europe, Hong Kong & Macau. We have also established seven national-level dispatch centers, including in Nicaragua and Azerbaijan, further strengthening our global footprint.

Since 2024, DFE's new-generation CRTU has been deployed across hundreds of CLP sites, advancing substation digitalization and renewable integration while improving grid automation, reliability, and energy efficiency.



In 2024, DFE partnered with CEM and the University of Macau to establish a Joint Laboratory for New Distribution and Consumption Technologies, supporting Macau's digital-intelligent power transformation and smart grid development.

7

National-Level
SCADA/EMS/DMS

150+

City-Level
SCADA/EMS/DMS

500,000+

Power Automation
Terminals

300,000+

Distribution
transformers and RMU

1,000,000+

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1st

Ranked 1st in 2025 Global Newly Wind Power Installed Capacity



6

Continents

49

Countries

Global Business Network



169⁺ GW

Global Total Installed Capacity



560⁺

Term

Net-Zero Projects Implemented



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Its wind power subsidiary Siemens Gamesa makes Siemens Energy a global market leader for renewable energies. An estimated one-sixth of the electricity generated worldwide is based on technologies from Siemens Energy. Siemens Energy employs around 105,000 people worldwide in more than 90 countries and generated revenue of €39.1 billion in fiscal year 2025.



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 Already connecting millions of devices and sensors.
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